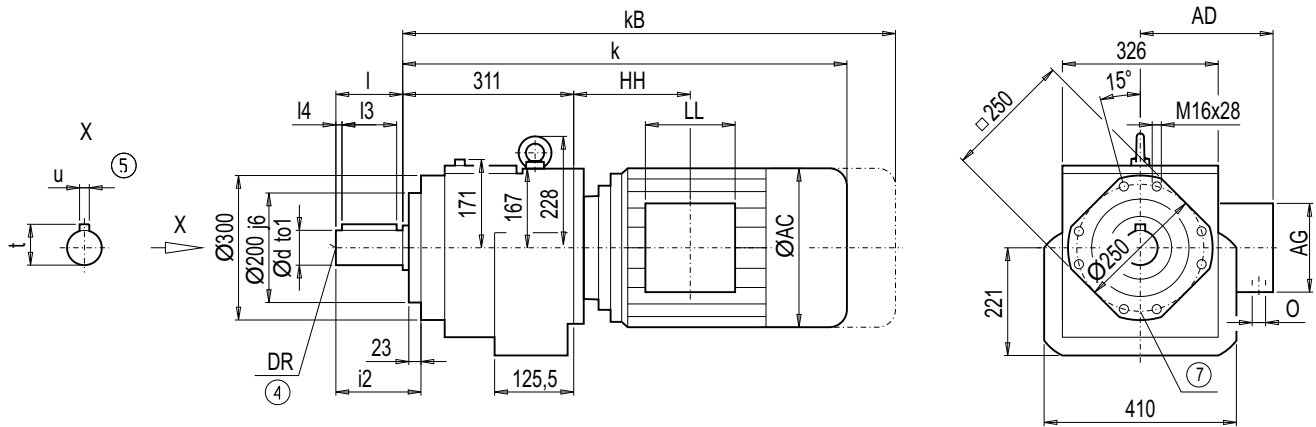


Gearbox DZ/ZZ108 (3- / 2-stage), housing-flange-mounted design (C-type)

DZZ011



d	to1	l	l4	l3	t	u	i2	DR
60 *)	m6	120	10	100	64.0	18	148	M20x42
70	m6	140	15	110	74.5	20	168	M20x42

*) Preferred series

Motor	ZZ108		DZ108		AC	AD	AG	LL	ZZ108		O	Weight	
	k	kB	k	kB					HH	HH		ZZ108	DZ108
LA80	–	–	599.0	662.5	156.5	155.0	90	90	–	106.5	M20x1.5/M25x1.5	–	121
LA80Z	–	–	621.5	685.0	156.5	155.0	90	90	–	179.5	M20x1.5/M25x1.5	–	125
LA90S/L	599.5	670.5	630.0	701.0	174.0	163.0	90	90	76.0	106.5	M20x1.5/M25x1.5	124	126
LA90ZL	644.5	715.5	675.0	746.0	174.0	163.0	90	90	200.0	230.5	M20x1.5/M25x1.5	130	132
LA100L	642.5	723.5	676.0	757.0	195.0	168.0	120	120	113.5	147.0	2xM32x1.5	132	135
LA100ZL	712.5	793.5	746.0	827.0	195.0	168.0	120	120	245.5	279.0	2xM32x1.5	142	145
LA112M	669.0	750.0	700.5	781.5	219.0	181.0	120	120	116.0	147.5	2xM32x1.5	144	147
LA112ZM	697.0	778.0	728.5	809.5	219.0	181.0	120	120	220.0	251.5	2xM32x1.5	151	154
LA132S/M	728.0	830.0	760.5	862.5	259.0	195.0	140	140	155.5	188.0	2xM32x1.5	154	159
LA132ZM	774.0	876.0	806.5	908.5	259.0	195.0	140	140	263.5	296.0	2xM32x1.5	175	180
LA160M/L	833.5	952.0	863.0	981.5	313.5	227.0	165	165	184.0	213.5	2xM40x1.5	189	196
LA160ZL	881.5	1 000.0	911.0	1 029.5	313.5	227.0	165	165	337.0	366.5	2xM40x1.5	228	235
LG180M/L	890.0	1 012.0	–	–	348.0	322.5	260	192	198.0	–	2xM40x1.5	285	–
LG180ZM/ZL	941.0	1 063.0	–	–	348.0	322.5	260	192	198.0	–	2xM40x1.5	315	–
LG200L	946.0	1 072.0	–	–	385.0	301.0	260	192	228.0	–	2xM50x1.5	365	–
K4-LGI225S	1 206.5	1 445.5	–	–	442.0	325.0	260	192	443.0	–	2xM50x1.5	521	–
K4-LGI225M	1 206.5	1 445.5	–	–	442.0	325.0	260	192	443.0	–	2xM50x1.5	509	–
K4-LGI225ZM	1 266.5	1 505.5	–	–	442.0	325.0	260	192	443.0	–	2xM50x1.5	567	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 2/192

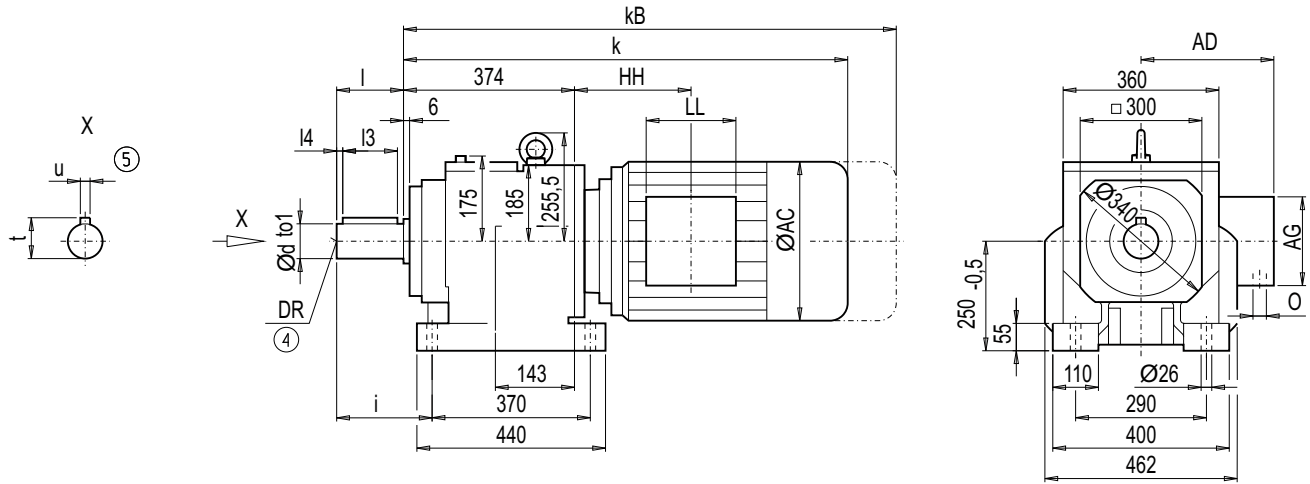
MOTOX Geared Motors

Helical geared motors

Dimensions

Gearbox D/Z128 (3- / 2-stage), foot-mounted design

DZ011



d	to1	l	l4	l3	t	u	i	DR
70 *)	m6	140	15	110	74.5	20	186	M20x42
90	m6	170	15	140	95.0	25	216	M24x50

*) Preferred series

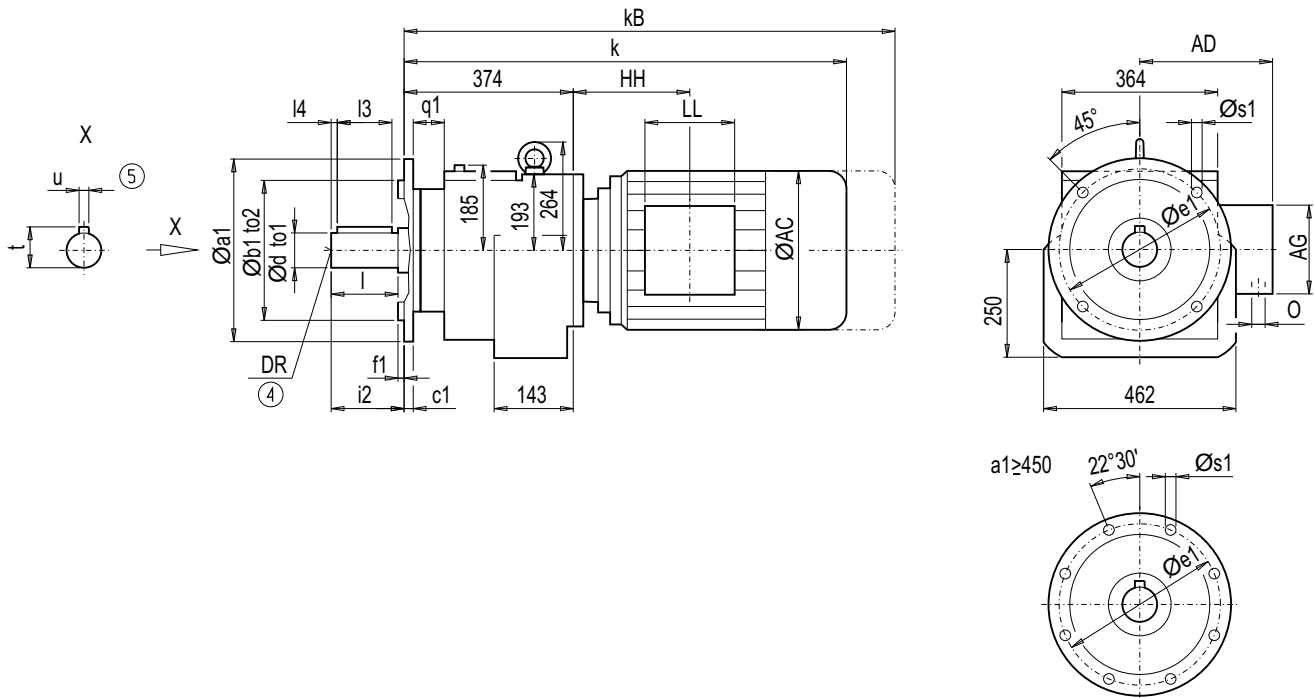
Motor	Z128		D128		AC	AD	AG	LL	Z128	D128	O	Weight	
	k	kB	k	kB					HH	HH		Z128	D128
LA90S/L	–	–	686.0	757.0	174.0	163.0	90	90	–	99.5	M20x1.5/M25x1.5	–	212
LA90ZL	–	–	731.0	802.0	174.0	163.0	90	90	–	223.5	M20x1.5/M25x1.5	–	218
LA100L	696.0	777.0	732.0	813.0	195.0	168.0	120	120	104.0	140.0	2xM32x1.5	214	221
LA100ZL	766.0	847.0	802.0	883.0	195.0	168.0	120	120	236.0	272.0	2xM32x1.5	224	231
LA112M	721.5	802.5	755.5	836.5	219.0	181.0	120	120	105.5	139.5	2xM32x1.5	226	233
LA112ZM	749.5	830.5	783.5	864.5	219.0	181.0	120	120	209.5	243.5	2xM32x1.5	233	240
LA132S/M	780.5	882.5	814.5	916.5	259.0	195.0	140	140	145.0	179.0	2xM32x1.5	235	246
LA132ZM	826.5	928.5	860.5	962.5	259.0	195.0	140	140	253.0	287.0	2xM32x1.5	256	267
LA160M/L	880.0	998.5	917.0	1 035.5	313.5	227.0	165	165	167.5	204.5	2xM40x1.5	274	282
LA160ZL	928.0	1 046.5	965.0	1 083.5	313.5	227.0	165	165	320.5	357.5	2xM40x1.5	313	321
LG180M/L	939.5	1 061.5	976.5	1 098.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	365	378
LG180ZM/ZL	990.5	1 112.5	1 027.5	1 149.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	395	408
LG200L	995.5	1 121.5	1 032.5	1 158.5	385.0	301.0	260	192	214.5	251.5	2xM50x1.5	445	458
LG225S	1 066.5	1 305.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	522	–
LG225M	1 066.5	1 305.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	510	–
LG225ZM	1 126.5	1 365.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	568	–
K4-LGI250M	1 353.5	1 578.5	–	–	495.0	392.0	300	236	469.5	–	2xM63x1.5	689	–
K4-LGI250ZM	1 423.5	1 648.5	–	–	495.0	392.0	300	236	469.5	–	2xM63x1.5	792	–

© DIN 332

© Feather key / keyway DIN 6885

Gearbox DF/ZF128 (3- / 2-stage), flange-mounted design (A-type)

DZF011



Flange	a1	b1	to2	c1	e1	f1	q1	s1	d	to1	l	l4	l3	t	u	i2	DR
A350 ¹⁾	350	250	h6	18	300	5	60	17.5	70 ^{*)}	m6	140	15	110	74.5	20	140	M20x42
									90	m6	170	15	140	95.0	25	170	M24x50
A450	450	350	h6	22	400	5	56	17.5	70 ^{*)}	m6	140	15	110	74.5	20	140	M20x42
									90	m6	170	15	140	95.0	25	170	M24x50
A550	550	450	h6	22	500	5	56	17.5	70 ^{*)}	m6	140	15	110	74.5	20	140	M20x42
									90	m6	170	15	140	95.0	25	170	M24x50

¹⁾ If torque > 3500 Nm, the flange must be pinned. We recommend you use 2 pins with a 12 mm diameter

^{*)} Preferred series

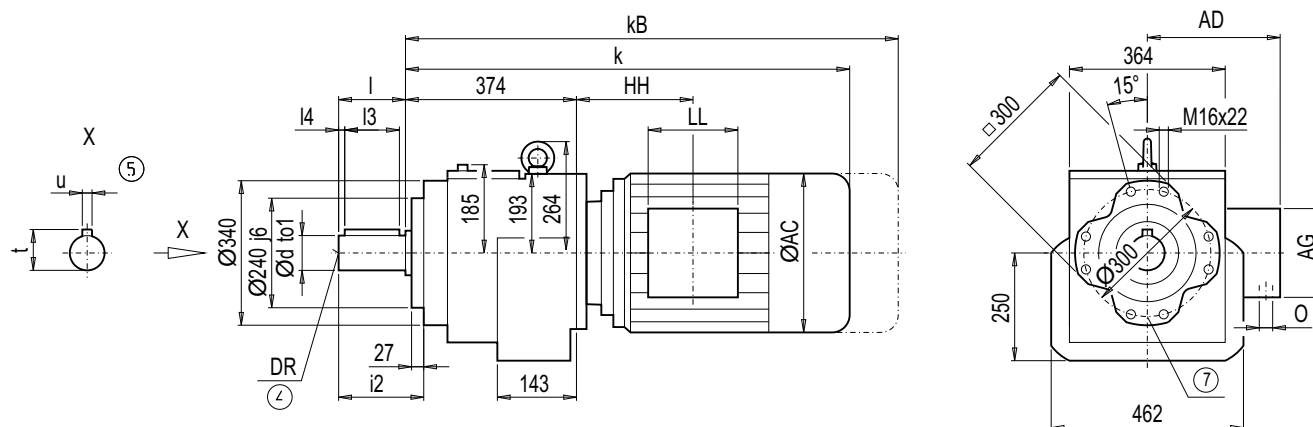
Motor	ZF128		DF128		AC	AD	AG	LL	ZF128		DF128		Weight	
	k	kB	k	kB					HH	HH	O		ZF128	DF128
LA90S/L	—	—	686.0	757.0	174.0	163.0	90	90	—	99.5	M20x1.5/M25x1.5	—	206	—
LA90ZL	—	—	731.0	802.0	174.0	163.0	90	90	—	223.5	M20x1.5/M25x1.5	—	212	—
LA100L	696.0	777.0	732.0	813.0	195.0	168.0	120	120	104.0	140.0	2xM32x1.5	209	216	—
LA100ZL	766.0	847.0	802.0	883.0	195.0	168.0	120	120	236.0	272.0	2xM32x1.5	219	226	—
LA112M	721.5	802.5	755.5	836.5	219.0	181.0	120	120	105.5	139.5	2xM32x1.5	220	227	—
LA112ZM	749.5	830.5	783.5	864.5	219.0	181.0	120	120	209.5	243.5	2xM32x1.5	227	234	—
LA132S/M	780.5	882.5	814.5	916.5	259.0	195.0	140	140	145.0	179.0	2xM32x1.5	230	240	—
LA132ZM	826.5	928.5	860.5	962.5	259.0	195.0	140	140	253.0	287.0	2xM32x1.5	251	261	—
LA160M/L	880.0	998.5	917.0	1 035.5	313.5	227.0	165	165	167.5	204.5	2xM40x1.5	269	276	—
LA160ZL	928.0	1 046.5	965.0	1 083.5	313.5	227.0	165	165	320.5	357.5	2xM40x1.5	308	315	—
LG180M/L	939.5	1 061.5	976.5	1 098.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	360	372	—
LG180ZM/ZL	990.5	1 112.5	1 027.5	1 149.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	390	402	—
LG200L	995.5	1 121.5	1 032.5	1 158.5	385.0	301.0	260	192	214.5	251.5	2xM50x1.5	440	452	—
LG225S	1 066.5	1 305.5	—	—	442.0	325.0	260	192	250.5	—	2xM50x1.5	517	—	—
LG225M	1 066.5	1 305.5	—	—	442.0	325.0	260	192	250.5	—	2xM50x1.5	505	—	—
LG225ZM	1 126.5	1 365.5	—	—	442.0	325.0	260	192	250.5	—	2xM50x1.5	563	—	—
K4-LGI250M	1 353.5	1 578.5	—	—	495.0	392.0	300	236	469.5	—	2xM63x1.5	684	—	—
K4-LGI250ZM	1 423.5	1 648.5	—	—	495.0	392.0	300	236	469.5	—	2xM63x1.5	787	—	—

MOTOX Geared Motors

Dimensions

Gearbox DZ/ZZ128 (3- / 2-stage), housing-flange-mounted design (C-type)

DZZ011

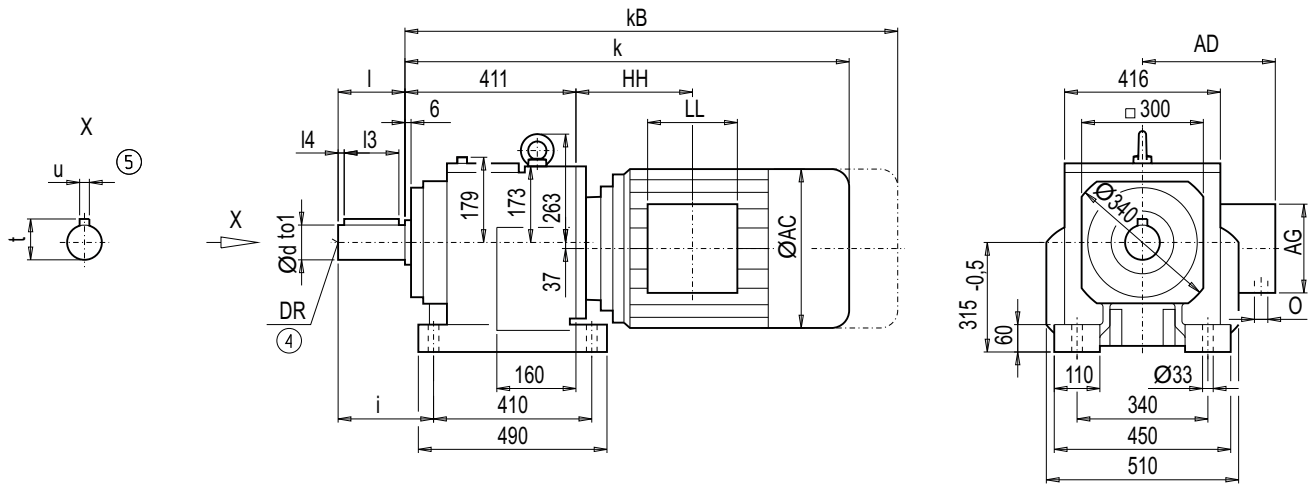


d	to1	l	l4	l3	t	u	i2	DR
70 *)	m6	140	15	110	74.5	20	173	M20x42
90	m6	170	15	140	95.0	25	203	M24x50

*) Preferred series

Motor	ZZ128		DZ128		AC	AD	AG	LL	ZZ128	DZ128	O	Weight	
	k	kB	k	kB					HH	HH		ZZ128	DZ128
LA90S/L	–	–	686.0	757.0	174.0	163.0	90	90	–	99.5	M20x1.5/M25x1.5	–	190
LA90ZL	–	–	731.0	802.0	174.0	163.0	90	90	–	223.5	M20x1.5/M25x1.5	–	196
LA100L	696.0	777.0	732.0	813.0	195.0	168.0	120	120	104.0	140.0	2xM32x1.5	192	199
LA100ZL	766.0	847.0	802.0	883.0	195.0	168.0	120	120	236.0	272.0	2xM32x1.5	202	209
LA112M	721.5	802.5	755.5	836.5	219.0	181.0	120	120	105.5	139.5	2xM32x1.5	203	210
LA112ZM	749.5	830.5	783.5	864.5	219.0	181.0	120	120	209.5	243.5	2xM32x1.5	210	217
LA132S/M	780.5	882.5	814.5	916.5	259.0	195.0	140	140	145.0	179.0	2xM32x1.5	213	223
LA132ZM	826.5	928.5	860.5	962.5	259.0	195.0	140	140	253.0	287.0	2xM32x1.5	234	245
LA160M/L	880.0	998.5	917.0	1 035.5	313.5	227.0	165	165	167.5	204.5	2xM40x1.5	252	259
LA160ZL	928.0	1 046.5	965.0	1 083.5	313.5	227.0	165	165	320.5	357.5	2xM40x1.5	291	298
LG180M/L	939.5	1 061.5	976.5	1 098.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	343	355
LG180ZM/ZL	990.5	1 112.5	1 027.5	1 149.5	348.0	322.5	260	192	184.5	221.5	2xM40x1.5	373	385
LG200L	995.5	1 121.5	1 032.5	1 158.5	385.0	301.0	260	192	214.5	251.5	2xM50x1.5	423	435
LG225S	1 066.5	1 305.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	500	–
LG225M	1 066.5	1 305.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	488	–
LG225ZM	1 126.5	1 365.5	–	–	442.0	325.0	260	192	250.5	–	2xM50x1.5	546	–
K4-LG1250M	1 353.5	1 578.5	–	–	495.0	392.0	300	236	469.5	–	2xM63x1.5	667	–
K4-LG1250ZM	1 423.5	1 648.5	–	–	495.0	392.0	300	236	469.5	–	2xM63x1.5	770	–

Gearbox D/Z148 (3- / 2-stage), foot-mounted design

DZ011


d	to1	l	l4	l3	t	u	i	DR
90 *)	m6	170	15	140	95	25	220	M24x50
100	m6	210	15	180	106	28	260	M24x50

*) Preferred series

Motor	Z148		D148		AC	AD	AG	LL	Z148 HH	D148 HH	O	Weight	
	k	kB	k	kB								Z148	D148
LA100L	–	–	764.0	845.0	195.0	168.0	120	120	–	135.0	2xM32x1.5	–	313
LA100ZL	–	–	834.0	915.0	195.0	168.0	120	120	–	267.0	2xM32x1.5	–	323
LA112M	–	–	789.5	870.5	219.0	181.0	120	120	–	136.5	2xM32x1.5	–	324
LA112ZM	–	–	817.5	898.5	219.0	181.0	120	120	–	240.5	2xM32x1.5	–	331
LA132S/M	809.5	911.5	847.5	949.5	259.0	195.0	140	140	137.0	175.0	2xM32x1.5	325	336
LA132ZM	855.5	957.5	893.5	995.5	259.0	195.0	140	140	245.0	283.0	2xM32x1.5	346	357
LA160M/L	909.5	1 028.0	947.5	1 066.0	313.5	227.0	165	165	160.0	198.0	2xM40x1.5	359	371
LA160ZL	957.5	1 076.0	995.5	1 114.0	313.5	227.0	165	165	313.0	351.0	2xM40x1.5	398	410
LG180M/L	969.0	1 091.0	1 007.0	1 129.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	455	467
LG180ZM/ZL	1 020.0	1 142.0	1 058.0	1 180.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	485	497
LG200L	1 025.0	1 151.0	1 063.0	1 189.0	385.0	301.0	260	192	207.0	245.0	2xM50x1.5	535	547
LG225S	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	608	621
LG225M	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	596	609
LG225ZM	1 156.0	1 395.0	1 194.0	1 433.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	654	667
LG250M	1 189.5	1 414.5	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	698	–
LG250ZM	1 259.5	1 485.0	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	801	–
K4-LGI280S	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	929	–
K4-LGI280M	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	941	–
K4-LGI280ZM	1 578.5	1 805.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	1 029	–

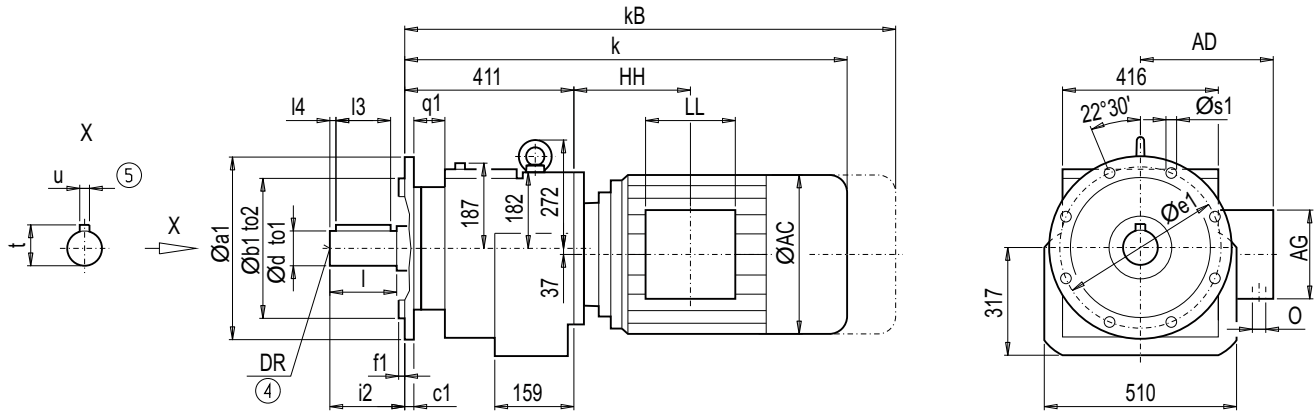
MOTOX Geared Motors

Helical geared motors

Dimensions

Gearbox DF/ZF148 (3- / 2-stage), flange-mounted design (A-type)

DZF011



Flange	a1	b1	to2	c1	e1	f1	q1	s1	d	to1	l	l4	l3	t	u	i2	DR
A450	450	350	h6	22	400	5	68	17.5	90 ^{*)}	m6	170	15	140	95	25	170	M24x50
									100	m6	210	15	180	106	28	210	M24x50
A550	550	450	h6	25	500	5	65	17.5	90 ^{*)}	m6	170	15	140	95	25	170	M24x50
									100	m6	210	15	180	106	28	210	M24x50

^{*)} Preferred series

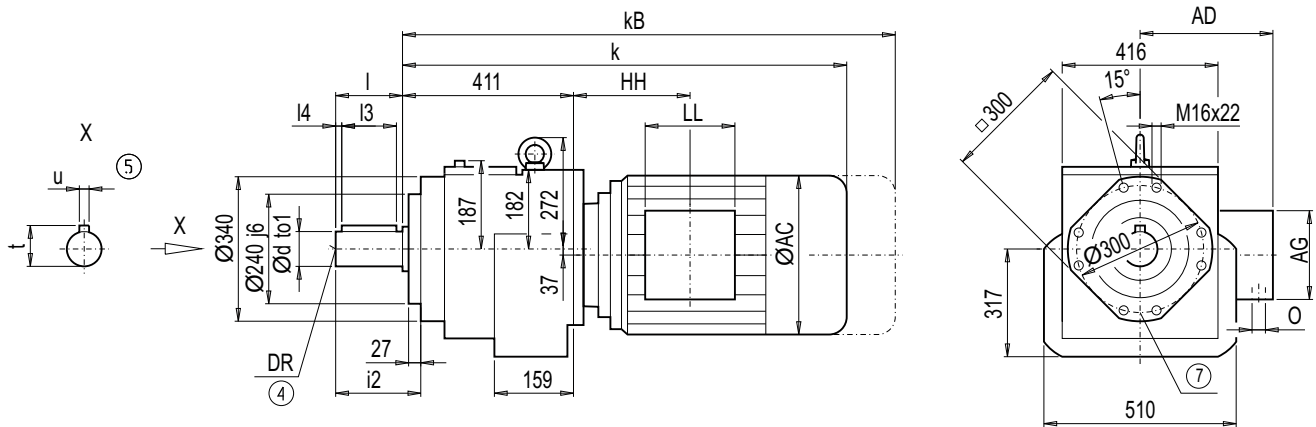
Motor	ZF148		DF148		AC	AD	AG	LL	HH	HH	O	Weight	
	k	kB	k	kB								ZF148	DF148
LA100L	–	–	764.0	845.0	195.0	168.0	120	120	–	135.0	2xM32x1.5	–	307
LA100ZL	–	–	834.0	915.0	195.0	168.0	120	120	–	267.0	2xM32x1.5	–	317
LA112M	–	–	789.5	870.5	219.0	181.0	120	120	–	136.5	2xM32x1.5	–	318
LA112ZM	–	–	817.5	898.5	219.0	181.0	120	120	–	240.5	2xM32x1.5	–	325
LA132S/M	809.5	911.5	847.5	949.5	259.0	195.0	140	140	137.0	175.0	2xM32x1.5	319	330
LA132ZM	855.5	957.5	893.5	995.5	259.0	195.0	140	140	245.0	283.0	2xM32x1.5	340	351
LA160M/L	909.5	1 028.0	947.5	1 066.0	313.5	227.0	165	165	160.0	198.0	2xM40x1.5	353	365
LA160ZL	957.5	1 076.0	995.5	1 114.0	313.5	227.0	165	165	313.0	351.0	2xM40x1.5	392	404
LG180M/L	969.0	1 091.0	1 007.0	1 129.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	449	461
LG180ZM/ZL	1 020.0	1 142.0	1 058.0	1 180.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	479	491
LG200L	1 025.0	1 151.0	1 063.0	1 189.0	385.0	301.0	260	192	207.0	245.0	2xM50x1.5	529	541
LG225S	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	602	615
LG225M	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	590	603
LG225ZM	1 156.0	1 395.0	1 194.0	1 433.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	648	661
LG250M	1 189.5	1 414.5	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	692	–
LG250ZM	1 259.5	1 485.0	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	795	–
K4-LGI280S	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	923	–
K4-LGI280M	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	941	–
K4-LGI280ZM	1 578.5	1 805.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	1 029	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox DZ/ZZ148 (3- / 2-stage), housing-flange-mounted design (C-type)

DZZ011



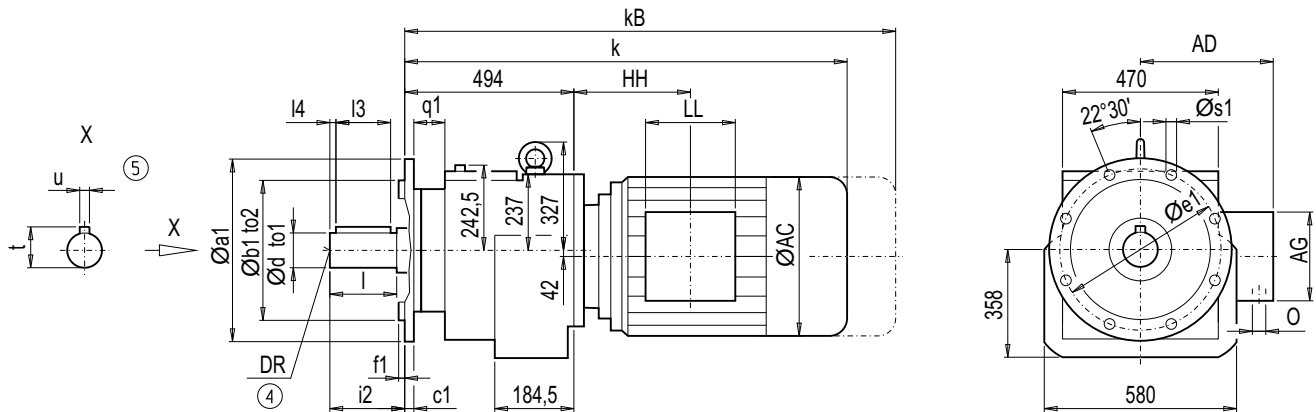
d	to1	l	l4	l3	t	u	i2	DR
90 ^{a)}	m6	170	15	140	95	25	203	M24x50
100	m6	210	15	180	106	28	243	M24x50

*¹) Preferred series

Motor	ZZ148		DZ148						ZZ148	DZ148		Weight	
	k	kB	k	kB	AC	AD	AG	LL	HH	HH	O	ZZ148	DZ148
LA100L	–	–	764.0	845.0	195.0	168.0	120	120	–	135.0	2xM32x1.5	–	283
LA100ZL	–	–	834.0	915.0	195.0	168.0	120	120	–	267.0	2xM32x1.5	–	293
LA112M	–	–	789.5	870.5	219.0	181.0	120	120	–	136.5	2xM32x1.5	–	294
LA112ZM	–	–	817.5	898.5	219.0	181.0	120	120	–	240.5	2xM32x1.5	–	301
LA132S/M	809.5	911.5	847.5	949.5	259.0	195.0	140	140	137.0	175.0	2xM32x1.5	302	306
LA132ZM	855.5	957.5	893.5	995.5	259.0	195.0	140	140	245.0	283.0	2xM32x1.5	323	327
LA160M/L	909.5	1 028.0	947.5	1 066.0	313.5	227.0	165	165	160.0	198.0	2xM40x1.5	336	341
LA160ZL	957.5	1 076.0	995.5	1 114.0	313.5	227.0	165	165	313.0	351.0	2xM40x1.5	375	380
LG180M/L	969.0	1 091.0	1 007.0	1 129.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	432	437
LG180ZM/ZL	1 020.0	1 142.0	1 058.0	1 180.0	348.0	322.5	260	192	177.0	215.0	2xM40x1.5	462	467
LG200L	1 025.0	1 151.0	1 063.0	1 189.0	385.0	301.0	260	192	207.0	245.0	2xM50x1.5	512	517
LG225S	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	585	547
LG225M	1 096.0	1 335.0	1 134.0	1 373.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	573	591
LG225ZM	1 156.0	1 395.0	1 194.0	1 433.0	442.0	325.0	260	192	243.0	281.0	2xM50x1.5	631	637
LG250M	1 189.5	1 414.5	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	675	–
LG250ZM	1 259.5	1 485.0	–	–	495.0	392.0	300	236	278.5	–	2xM63x1.5	778	–
K4-LGI280S	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	906	–
K4-LGI280M	1 468.5	1 695.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	918	–
K4-LGI280ZM	1 578.5	1 805.5	–	–	555.0	432.0	300	236	489.5	–	2xM63x1.5	1 006	–

Gearbox DF/ZF168 (3- / 2-stage), flange-mounted design (A-type)

DZF011



Flange	a1	b1	to2	c1	e1	f1	q1	s1	d	to1	i	l4	l3	t	u	i2	DR
A450	450	350	h6	31	400	5	65	17.5	100 *)	m6	210	15	180	106	28	210	M24x50
									110	m6	210	15	180	116	28	210	M24x50
									120	m6	210	15	180	127	32	210	M24x50
A550	550	450	h6	31	500	5	65	17.5	100 *)	m6	210	15	180	106	28	210	M24x50
									110	m6	210	15	180	116	28	210	M24x50
									120	m6	210	15	180	127	32	210	M24x50
A660	660	550	h6	31	600	5	65	22.0	100 *)	m6	210	15	180	106	28	210	M24x50
									110	m6	210	15	180	116	28	210	M24x50
									120	m6	210	15	180	127	32	210	M24x50

*) Preferred series

Motor	ZF168		DF168						ZF168	DF168		Weight	
	k	kB	k	kB	AC	AD	AG	LL	HH	HH	O	ZF168	DF168
LA132S/M	878.0	980.0	919.0	1 021.0	259.0	195.0	140	140	122.5	163.5	2xM32x1.5	466	484
LA132ZM	924.0	1 026.0	965.0	1 067.0	259.0	195.0	140	140	230.5	271.5	2xM32x1.5	487	505
LA160M/L	978.0	1 096.5	1 019.0	1 137.5	313.5	227.0	165	165	145.5	186.5	2xM40x1.5	500	518
LA160ZL	1 026.0	1 144.5	1 067.0	1 185.5	313.5	227.0	165	165	298.5	339.5	2xM40x1.5	539	557
LG180M/L	1 037.5	1 159.5	1 078.5	1 200.5	348.0	322.5	260	192	162.5	203.5	2xM40x1.5	595	614
LG180ZM/ZL	1 088.5	1 210.5	1 129.5	1 251.5	348.0	322.5	260	192	162.5	203.5	2xM40x1.5	625	644
LG200L	1 093.5	1 219.5	1 134.5	1 260.5	385.0	301.0	260	192	192.5	233.5	2xM50x1.5	675	694
LG225S	1 164.5	1 403.5	1 205.5	1 444.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	747	767
LG225M	1 164.5	1 403.5	1 205.5	1 444.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	735	755
LG225ZM	1 224.5	1 463.5	1 265.5	1 504.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	793	813
LG250M	1 258.0	1 483.0	–	–	495.0	392.0	300	236	264.0	–	2xM63x1.5	837	–
LG250ZM	1 328.0	1 553.5	–	–	495.0	392.0	300	236	264.0	–	2xM63x1.5	940	–
K4-LGI280S	1 537.5	1 764.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	966	–
K4-LGI280M	1 537.5	1 764.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	1 072	–
K4-LGI280ZM	1 647.5	1 874.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	1 160	–

© DIN 332

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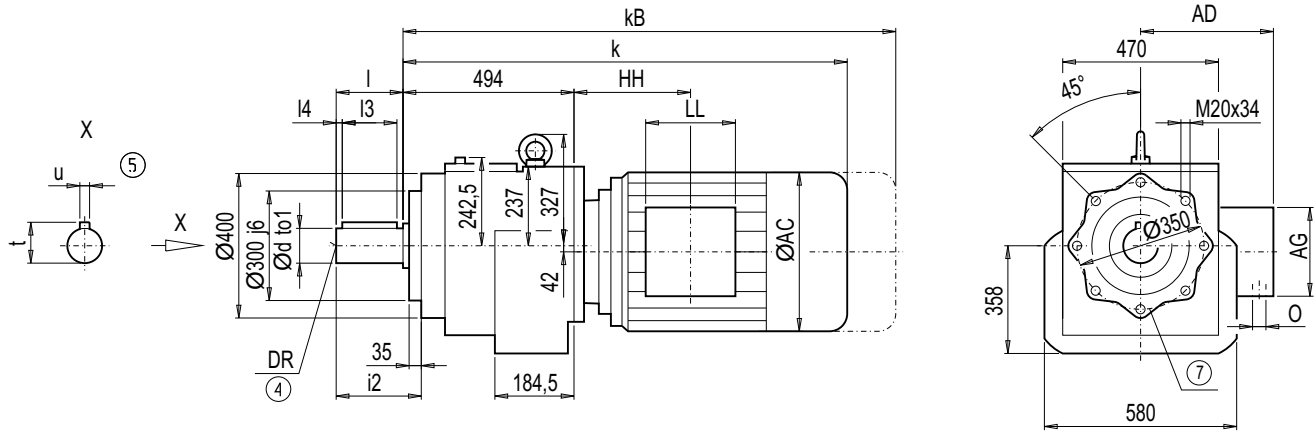
MOTEX Geared Motors

Helical geared motors

Dimensions

Gearbox DZ/ZZ168 (3- / 2-stage), housing-flange-mounted design (C-type)

DZZ011



d	to1	l	l4	l3	t	u	i2	DR
100	m6	210	15	180	106	28	251	M24x50
110	m6	210	15	180	116	28	251	M24x50
120 ^{*)}	m6	210	15	180	127	32	251	M24x50

*) Preferred series

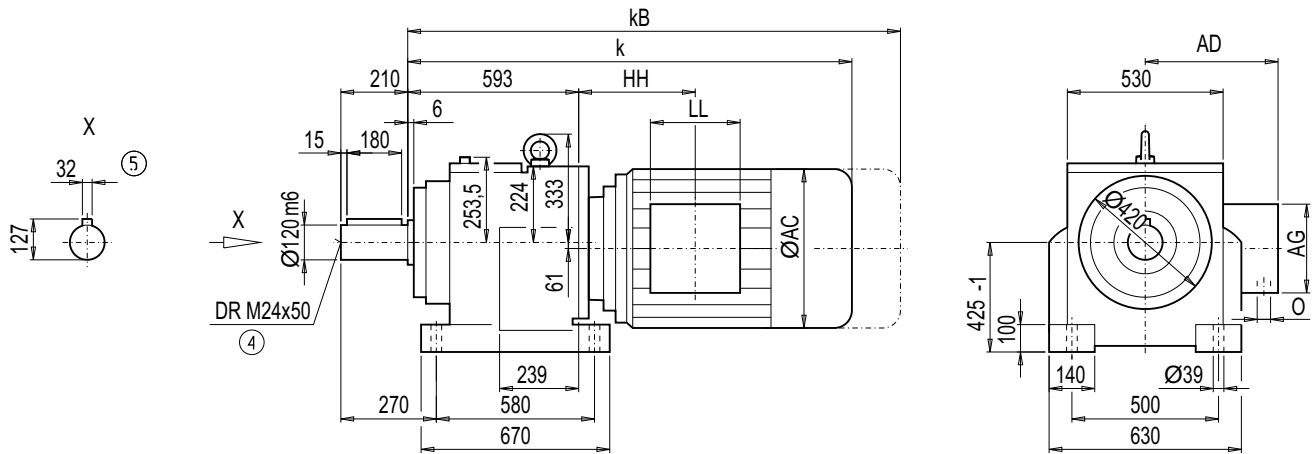
Motor	ZZ168		DZ168						ZZ168	DZ168		Weight	
	k	kB	k	kB	AC	AD	AG	LL	HH	HH	O	ZZ168	DZ168
LA132S/M	878.0	980.0	919.0	1 021.0	259.0	195.0	140	140	122.5	163.5	2xM32x1.5	447	465
LA132ZM	924.0	1 026.0	965.0	1 067.0	259.0	195.0	140	140	230.5	271.5	2xM32x1.5	468	486
LA160M/L	978.0	1 096.5	1 019.0	1 137.5	313.5	227.0	165	165	145.5	186.5	2xM40x1.5	481	499
LA160ZL	1 026.0	1 144.5	1 067.0	1 185.5	313.5	227.0	165	165	298.5	339.5	2xM40x1.5	520	538
LG180M/L	1 037.5	1 159.5	1 078.5	1 200.5	348.0	322.5	260	192	162.5	203.5	2xM40x1.5	576	595
LG180ZM/ZL	1 088.5	1 210.5	1 129.5	1 251.5	348.0	322.5	260	192	162.5	203.5	2xM40x1.5	606	625
LG200L	1 093.5	1 219.5	1 134.5	1 260.5	385.0	301.0	260	192	192.5	233.5	2xM50x1.5	656	675
LG225S	1 164.5	1 403.5	1 205.5	1 444.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	728	748
LG225M	1 164.5	1 403.5	1 205.5	1 444.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	716	736
LG225ZM	1 224.5	1 463.5	1 265.5	1 504.5	442.0	325.0	260	192	228.5	269.5	2xM50x1.5	774	794
LG250M	1 258.0	1 483.0	–	–	495.0	392.0	300	236	264.0	–	2xM63x1.5	818	–
LG250ZM	1 328.0	1 553.5	–	–	495.0	392.0	300	236	264.0	–	2xM63x1.5	921	–
K4-LGI280S	1 537.5	1 764.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	947	–
K4-LGI280M	1 537.5	1 764.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	1 053	–
K4-LGI280ZM	1 647.5	1 874.5	–	–	555.0	432.0	300	236	475.5	–	2xM63x1.5	1 141	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 2/192

Gearbox D/Z188 (3- / 2-stage), foot-mounted design

DZ011

2

Motor	Z188		D188		AC	AD	AG	LL	Z188	D188	O	Weight	
	k	kB	k	kB					HH	HH		Z188	D188
LA132S/M	–	–	977.0	1 079.0	259.0	195.0	140	140	–	122.5	2xM32x1.5	–	652
LA132ZM	–	–	1 023.0	1 125.0	259.0	195.0	140	140	–	230.5	2xM32x1.5	–	673
LA160M/L	1 077.0	1 195.5	1 077.0	1 195.5	313.5	227.0	165	165	145.5	145.5	2xM40x1.5	654	684
LA160ZL	1 125.0	1 243.5	1 125.0	1 243.5	313.5	227.0	165	165	298.5	298.5	2xM40x1.5	693	723
LG180M/L	1 136.5	1 258.5	1 136.5	1 258.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	750	779
LG180ZM/ZL	1 187.5	1 309.5	1 187.5	1 309.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	780	809
LG200L	1 192.5	1 318.5	1 192.5	1 318.5	385.0	301.0	260	192	192.5	192.5	2xM50x1.5	830	859
LG225S	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	903	932
LG225M	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	891	920
LG225ZM	1 323.5	1 562.5	1 323.5	1 562.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	949	978
LG250M	1 357.0	1 582.0	1 357.0	1 582.0	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	993	1022
LG250ZM	1 427.0	1 652.5	1 427.0	1 652.5	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	1 096	1 125
K4-LGI280S	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 121	1 151
K4-LGI280M	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 227	1 256
K4-LGI280ZM	1 746.5	1 973.5	1 746.5	1 973.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 315	1 344
K2-LGI315S/M	1 824.5	2 089.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 356	–
K2-LGI315ZM	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 511	–
K2-LGI315L	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 651	–
K2-LGI315ZL	2 124.5	2 389.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 851	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

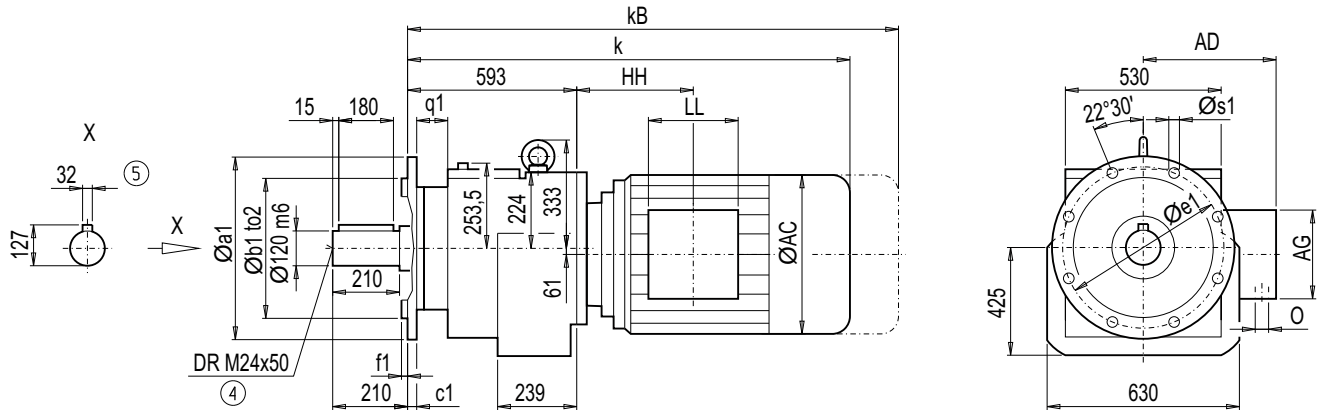
MOTOX Geared Motors

Helical geared motors

Dimensions

Gearbox DF/ZF188 (3- / 2-stage), flange-mounted design (A-type)

DZF011



Flange	a1	b1	to2	c1	e1	f1	q1	s1
A550	550	450	h6	31	500	5	83	17.5
A660	660	550	h6	31	600	6	83	22.0

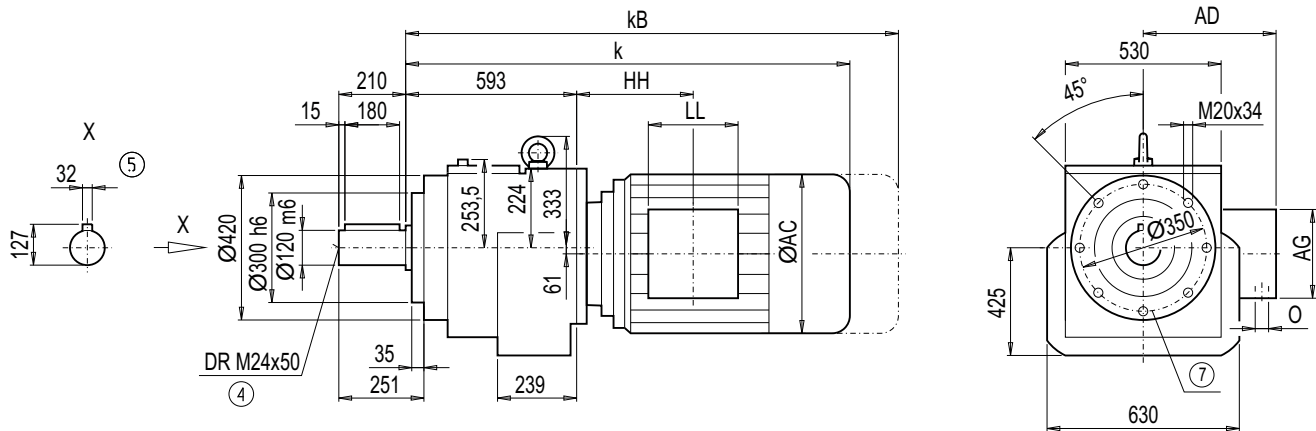
Motor	ZF188		DF188		AC	AD	AG	LL	ZF188	DF188	O	Weight	
	k	kB	k	kB					HH	HH		ZF188	DF188
LA132S/M	–	–	977.0	1 079.0	259.0	195.0	140	140	–	122.5	2xM32x1.5	–	600
LA132ZM	–	–	1 023.0	1 125.0	259.0	195.0	140	140	–	230.5	2xM32x1.5	–	609
LA160M/L	1 077.0	1 195.5	1 077.0	1 195.5	313.5	227.0	165	165	145.5	145.5	2xM40x1.5	602	632
LA160ZL	1 125.0	1 243.5	1 125.0	1 243.5	313.5	227.0	165	165	298.5	298.5	2xM40x1.5	602	632
LG180M/L	1 136.5	1 258.5	1 136.5	1 258.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	698	727
LG180ZM/ZL	1 187.5	1 309.5	1 187.5	1 309.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	728	757
LG200L	1 192.5	1 318.5	1 192.5	1 318.5	385.0	301.0	260	192	192.5	192.5	2xM50x1.5	778	807
LG225S	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	851	880
LG225M	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	839	868
LG225ZM	1 323.5	1 562.5	1 323.5	1 562.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	897	926
LG250M	1 357.0	1 582.0	1 357.0	1 582.0	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	941	970
LG250ZM	1 427.0	1 652.5	1 427.0	1 652.5	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	1 044	1 073
K4-LGI280S	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 069	1 099
K4-LGI280M	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 175	1 204
K4-LGI280ZM	1 746.5	1 973.5	1 746.5	1 973.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 263	1 292
K2-LGI315S/M	1 824.5	2 089.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 304	–
K2-LGI315ZM	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 459	–
K2-LGI315L	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 599	–
K2-LGI315ZL	2 124.5	2 389.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 801	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox DZ/ZZ188 (3- / 2-stage), housing-flange-mounted design (C-type)

DZZ011



Motor	ZZ188		DZ188			ZZ188				DZ188		Weight	
	k	kB	k	kB	AC	AD	AG	LL	HH	HH	O	ZZ188	DZ188
LA132S/M	–	–	977.0	1 079.0	259.0	195.0	140	140	–	122.5	2xM32x1.5	–	580
LA132ZM	–	–	1 023.0	1 125.0	259.0	195.0	140	140	–	230.5	2xM32x1.5	–	589
LA160M/L	1 077.0	1 195.5	1 077.0	1 195.5	313.5	227.0	165	165	145.5	145.5	2xM40x1.5	582	612
LA160ZL	1 125.0	1 243.5	1 125.0	1 243.5	313.5	227.0	165	165	298.5	298.5	2xM40x1.5	582	612
LG180M/L	1 136.5	1 258.5	1 136.5	1 258.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	678	707
LG180ZM/ZL	1 187.5	1 309.5	1 187.5	1 309.5	348.0	322.5	260	192	162.5	162.5	2xM40x1.5	708	737
LG200L	1 192.5	1 318.5	1 192.5	1 318.5	385.0	301.0	260	192	192.5	192.5	2xM50x1.5	758	787
LG225S	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	831	860
LG225M	1 263.5	1 502.5	1 263.5	1 502.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	819	848
LG225ZM	1 323.5	1 562.5	1 323.5	1 562.5	442.0	325.0	260	192	228.5	228.5	2xM50x1.5	877	906
LG250M	1 357.0	1 582.0	1 357.0	1 582.0	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	921	950
LG250ZM	1 427.0	1 652.5	1 427.0	1 652.5	495.0	392.0	300	236	264.0	264.0	2xM63x1.5	1 024	1 053
K4-LGI280S	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 049	1 079
K4-LGI280M	1 636.5	1 863.5	1 636.5	1 863.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 155	1 184
K4-LGI280ZM	1 746.5	1 973.5	1 746.5	1 973.5	555.0	432.0	300	236	475.5	475.5	2xM63x1.5	1 243	1 272
K2-LGI315S/M	1 824.5	2 089.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 284	–
K2-LGI315ZM	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 511	–
K2-LGI315L	1 984.5	2 249.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 651	–
K2-LGI315ZL	2 124.5	2 389.5	–	–	610.0	500.0	380	307	584.5	–	2xM63x1.5	1 851	–

④ DIN 332

⑤ Feather key / keyway DIN 6885

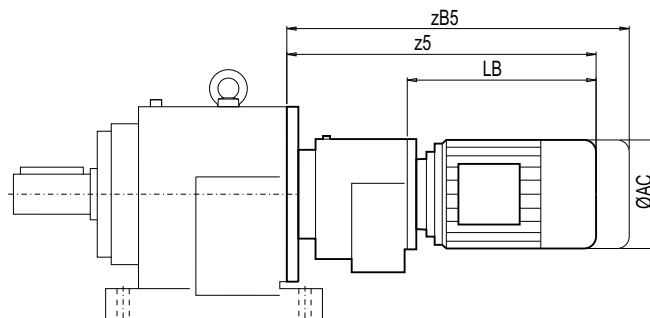
⑦ For note, see page 2/192

MOTOX Geared Motors

Helical geared motors

Dimensions

Helical tandem geared motors

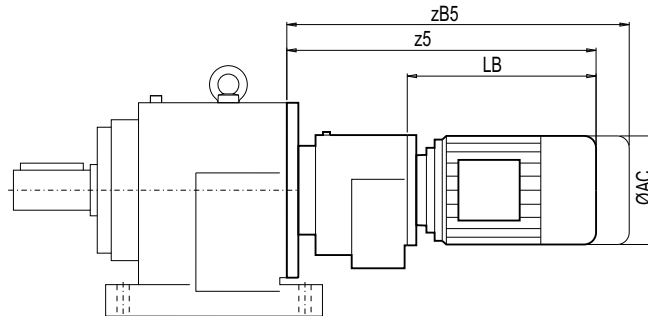


Gearbox	Motor	AC	z5	zB5	LB
Z.38-Z28	LA71	139.0	363.0	418.0	202.5
	LA71Z	139.0	382.0	437.0	221.5
	LA80	156.5	465.0	528.5	304.5
	LA80Z	156.5	487.5	551.0	327.0
	LA90S/L	174.0	460.0	531.0	299.5
	LA90ZL	174.0	505.0	576.0	344.5
	LA100L	195.0	542.0	623.0	381.5
	LA100ZL	195.0	612.0	693.0	451.5
Z.38-D28	LA71	139.0	363.0	418.0	202.5
	LA71Z	139.0	382.0	437.0	221.5
	LA80	156.5	465.0	528.5	304.5
	LA80Z	156.5	487.5	551.0	327.0
	LA90S/L	174.0	460.0	531.0	299.5
	LA90ZL	174.0	505.0	576.0	344.5
D.48-Z28	LA71	139.0	374.5	429.5	202.5
	LA71Z	139.0	393.5	448.5	221.5
	LA80	156.5	476.5	540.0	304.5
	LA80Z	156.5	499.0	562.5	327.0
	LA90S/L	174.0	471.5	542.5	299.5
	LA90ZL	174.0	516.5	587.5	344.5
	LA100L	195.0	553.5	634.5	381.5
	LA100ZL	195.0	623.5	704.5	451.5
D.48-D28	LA71	139.0	374.5	429.5	202.5
	LA71Z	139.0	393.5	448.5	221.5
	LA80	156.5	476.5	540.0	304.5
	LA80Z	156.5	499.0	562.5	327.0
	LA90S/L	174.0	471.5	542.5	299.5
	LA90ZL	174.0	516.5	587.5	344.5
D.68-Z28	LA71	139.0	370.0	425.0	202.5
	LA71Z	139.0	389.0	444.0	221.5
	LA80	156.5	472.0	535.5	304.5
	LA80Z	156.5	494.5	558.0	327.0
	LA90S/L	174.0	467.0	538.0	299.5
	LA90ZL	174.0	512.0	583.0	344.5
	LA100L	195.0	549.0	630.0	381.5
	LA100ZL	195.0	619.0	700.0	451.5

Gearbox	Motor	AC	z5	zB5	LB
D.68-D28	LA71	139.0	370.0	425.0	202.5
	LA71Z	139.0	389.0	444.0	221.5
	LA80	156.5	472.0	535.5	304.5
	LA80Z	156.5	494.5	558.0	327.0
	LA90S/L	174.0	467.0	538.0	299.5
	LA90ZL	174.0	512.0	583.0	344.5
D.88-Z28	LA71	139.0	361.5	416.5	202.5
	LA71Z	139.0	380.5	435.5	221.5
	LA80	156.5	463.5	527.0	304.5
	LA80Z	156.5	486.0	549.5	327.0
	LA90S/L	174.0	458.5	529.5	299.5
	LA90ZL	174.0	503.5	574.5	344.5
	LA100L	195.0	540.5	621.5	381.5
	LA100ZL	195.0	610.5	691.5	451.5
D.88-D28	LA71	139.0	361.5	416.5	202.5
	LA71Z	139.0	380.5	435.5	221.5
	LA80	156.5	463.5	527.0	304.5
	LA80Z	156.5	486.0	549.5	327.0
	LA90S/L	174.0	458.5	529.5	299.5
	LA90ZL	174.0	503.5	574.5	344.5
D.108-Z38	LA71 ¹⁾	139.0	484.5	539.5	258.5
	LA71Z ¹⁾	139.0	503.5	558.5	277.5
	LA80 ¹⁾	156.5	521.5	585.0	295.5
	LA80Z ¹⁾	156.5	544.0	607.5	318.0
	LA90S/L ¹⁾	174.0	552.5	623.5	326.5
	LA90ZL ¹⁾	174.0	597.5	668.5	371.5
	LA100L ¹⁾	195.0	598.5	679.5	372.5
	LA100ZL ¹⁾	195.0	668.5	749.5	442.5
	LA112M ¹⁾	219.0	628.0	709.0	402.0
	LA112ZM ¹⁾	219.0	656.0	737.0	430.0
	LA71 ²⁾	139.0	496.0	551.0	258.5
	LA71Z ²⁾	139.0	515.0	570.0	277.5
	LA80 ²⁾	156.5	533.0	596.5	295.5
	LA80Z ²⁾	156.5	555.5	619.0	318.0
	LA90S/L ²⁾	174.0	564.0	635.0	326.5
	LA90ZL ²⁾	174.0	609.0	680.0	371.5
	LA100L ²⁾	195.0	610.0	691.0	372.5
	LA100ZL ²⁾	195.0	680.0	761.0	442.5
	LA112M ²⁾	219.0	639.5	720.5	402.0
	LA112ZM ²⁾	219.0	667.5	748.5	430.0

1) $i_{tot} \geq 3797$ 2) $i_{tot} < 3797$

Helical tandem geared motors (continued)



2

Gearbox	Motor	AC	z5	zB5	LB
D.108-D38	LA71	139.0	499.5	554.5	273.5
	LA71Z	139.0	518.5	573.5	292.5
	LA80	156.5	536.5	600.0	310.5
	LA80Z	156.5	559.0	622.5	333.0
	LA90S/L	174.0	567.5	638.5	341.5
	LA90ZL	174.0	612.5	683.5	386.5
D.128-Z38	LA71	139.0	488.0	543.0	258.5
	LA71Z	139.0	507.0	562.0	277.5
	LA80	156.5	525.0	588.5	295.5
	LA80Z	156.5	547.5	611.0	318.0
	LA90S/L	174.0	556.0	627.0	326.5
	LA90ZL	174.0	601.0	672.0	371.5
	LA100L	195.0	602.0	683.0	372.5
	LA100ZL	195.0	672.0	753.0	442.5
	LA112M	219.0	631.5	712.5	402.0
	LA112ZM	219.0	659.5	740.5	430.0
D.128-D38	LA71	139.0	503.0	558.0	273.5
	LA71Z	139.0	522.0	577.0	292.5
	LA80	156.5	540.0	603.5	310.5
	LA80Z	156.5	562.5	626.0	333.0
	LA90S/L	174.0	571.0	642.0	341.5
	LA90ZL	174.0	616.0	687.0	386.5
D.128-Z48	LA71	139.0	555.5	610.5	253.0
	LA71Z	139.0	574.5	629.5	272.0
	LA80	156.5	592.5	656.0	290.0
	LA80Z	156.5	615.0	678.5	312.5
	LA90S/L	174.0	623.5	694.5	321.0
	LA90ZL	174.0	668.5	739.5	366.0
	LA100L	195.0	669.5	750.5	367.0
	LA100ZL	195.0	739.5	820.5	437.0
	LA112M	219.0	698.5	779.5	396.0
	LA112ZM	219.0	726.5	807.5	424.0
	LA132S/M	259.0	760.5	862.5	458.0
	LA132ZM	259.0	806.5	908.5	504.0
D.148-Z38	LA71	139.0	485.0	540.0	258.5
	LA71Z	139.0	504.0	559.0	277.5
	LA80	156.5	522.0	585.5	295.5
	LA80Z	156.5	544.5	608.0	318.0

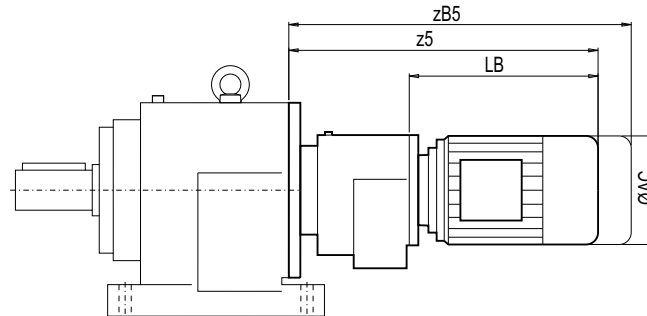
Gearbox	Motor	AC	z5	zB5	LB
D.148-Z38	LA90S/L	174.0	553.0	624.0	326.5
	LA90ZL	174.0	598.0	669.0	371.5
	LA100L	195.0	599.0	680.0	372.5
	LA100ZL	195.0	669.0	750.0	442.5
	LA112M	219.0	628.5	709.5	402.0
	LA112ZM	219.0	656.5	737.5	430.0
D.148-D38	LA71	139.0	500.0	555.0	273.5
	LA71Z	139.0	519.0	574.0	292.5
	LA80	156.5	537.0	600.5	310.5
	LA80Z	156.5	559.5	623.0	333.0
	LA90S/L	174.0	568.0	639.0	341.5
	LA90ZL	174.0	613.0	684.0	386.5
D.148-Z48	LA71	139.0	551.5	606.5	253.0
	LA71Z	139.0	570.5	625.5	272.0
	LA80	156.5	588.5	652.0	290.0
	LA80Z	156.5	611.0	674.5	312.5
	LA90S/L	174.0	619.5	690.5	321.0
	LA90ZL	174.0	664.5	735.5	366.0
	LA100L	195.0	665.5	746.5	367.0
	LA100ZL	195.0	735.5	816.5	437.0
	LA112M	219.0	694.5	775.5	396.0
	LA112ZM	219.0	722.5	803.5	424.0
D.168-Z48	LA132S/M	259.0	756.5	858.5	458.0
	LA132ZM	259.0	802.5	904.5	504.0
	LA71	139.0	540.0	595.0	253.0
D.168-Z38	LA71Z	139.0	559.0	614.0	272.0
	LA80	156.5	577.0	640.5	290.0
	LA80Z	156.5	599.5	663.0	312.5
	LA90S/L	174.0	608.0	679.0	321.0
	LA90ZL	174.0	653.0	724.0	366.0
	LA100L	195.0	654.0	735.0	367.0
	LA100ZL	195.0	724.0	805.0	437.0
	LA112M	219.0	683.0	764.0	396.0
	LA112ZM	219.0	711.0	792.0	424.0
	LA132S/M	259.0	745.0	847.0	458.0
	LA132ZM	259.0	791.0	893.0	504.0

MOTOX Geared Motors

Helical geared motors

Dimensions

Helical tandem geared motors (continued)

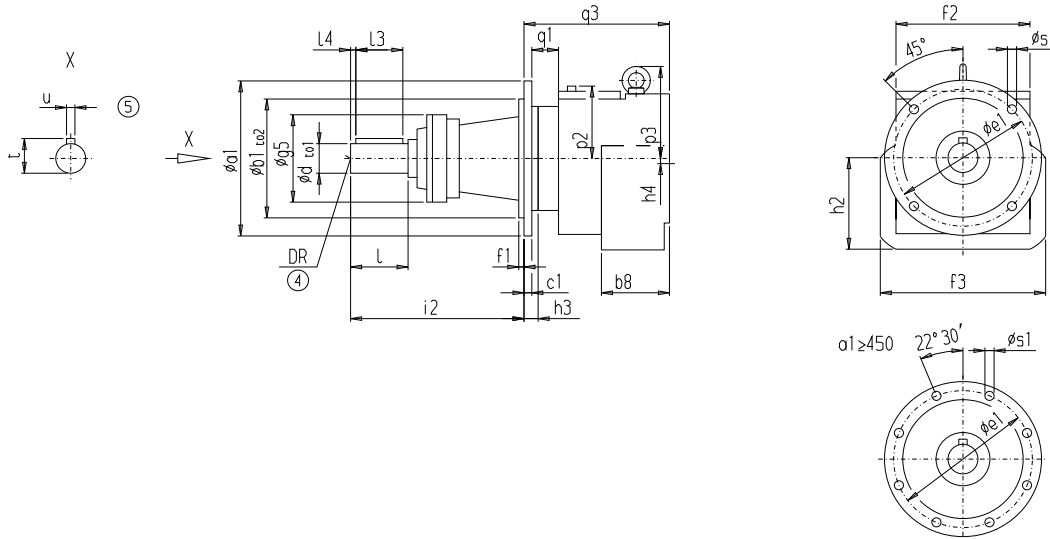


Gearbox	Motor	AC	z5	zB5	LB
D.168-D48	LA71	139.0	557.0	612.0	270.0
	LA71Z	139.0	576.0	631.0	289.0
	LA80	156.5	594.0	657.5	307.0
	LA80Z	156.5	616.5	680.0	329.5
	LA90S	174.0	625.0	696.0	338.0
	LA90L	174.0	625.0	696.0	338.0
	LA90ZL	174.0	670.0	741.0	383.0
	LA100L	195.0	671.0	752.0	384.0
	LA100ZL	195.0	741.0	822.0	454.0
	LA100ZL	195.0	741.0	822.0	454.0
D.168-Z68	LA71	139.0	626.0	681.0	247.0
	LA71Z	139.0	645.0	700.0	266.0
	LA80	156.5	663.0	726.5	284.0
	LA80Z	156.5	685.5	749.0	306.5
	LA90S/L	174.0	694.0	765.0	315.0
	LA90ZL	174.0	739.0	810.0	360.0
	LA100L	195.0	740.0	821.0	361.0
	LA100ZL	195.0	810.0	891.0	431.0
	LA132S/M	259.0	827.0	929.0	448.0
	LA132ZM	259.0	873.0	975.0	494.0
	LA160M/L	313.5	929.5	1 048.0	550.5
	LA160ZL	313.5	977.5	1 096.0	598.5
D.188-Z48	LA71	139.0	499.0	554.0	253.0
	LA71Z	139.0	518.0	573.0	272.0
	LA80	156.5	536.0	599.5	290.0
	LA80Z	156.5	558.5	622.0	312.5
	LA90S/L	174.0	567.0	638.0	321.0
	LA90ZL	174.0	612.0	683.0	366.0
	LA100L	195.0	613.0	694.0	367.0
	LA100ZL	195.0	683.0	764.0	437.0
	LA112M	219.0	642.0	723.0	396.0
	LA112ZM	219.0	670.0	751.0	424.0
	LA132S/M	259.0	704.0	806.0	458.0
	LA132ZM	259.0	750.0	852.0	504.0

Gearbox	Motor	AC	z5	zB5	LB
D.188-D48	LA71	139.0	516.0	571.0	270.0
	LA71Z	139.0	535.0	590.0	289.0
	LA80	156.5	553.0	616.5	307.0
	LA80Z	156.5	575.5	639.0	329.5
	LA90S/L	174.0	584.0	655.0	338.0
	LA90ZL	174.0	629.0	700.0	383.0
	LA100L	195.0	630.0	711.0	384.0
	LA100ZL	195.0	700.0	781.0	454.0
	LA100ZL	195.0	700.0	781.0	454.0
	LA100ZL	195.0	700.0	781.0	454.0
D.188-Z68	LA71	139.0	585.0	640.0	247.0
	LA71Z	139.0	604.0	659.0	266.0
	LA80	156.5	622.0	685.5	284.0
	LA80Z	156.5	644.5	708.0	306.5
	LA90S/L	174.0	653.0	724.0	315.0
	LA90ZL	174.0	698.0	769.0	360.0
	LA100L	195.0	699.0	780.0	361.0
	LA100ZL	195.0	769.0	850.0	431.0
	LA132S/M	259.0	786.0	888.0	448.0
	LA132ZM	259.0	832.0	934.0	494.0
	LA160M/L	313.5	888.5	1 007.0	550.5
	LA160ZL	313.5	936.5	1 055.0	598.5

Gearbox DR/ZR68-168 (3- / 2-stage) with agitator flange

DZZ011



Gearbox	p2	p3	h2	b8	q3	f3	f2	h4	Additional weight ¹⁾
DR/ZR68	109.0	149	144.0	91.5	248	263	206	0	24
DR/ZR88	134.0	181	182.0	129.0	306	332	260	0	46
DR/ZR108	177.0	228	219.5	126.5	355	410	326	0	82
DR/ZR128	194.0	263	250.0	146.0	422	462	364	0	85
DR/ZR148	190.5	270	317.0	160.0	459	510	416	37	94
DR/ZR168	248.0	325	358.0	188.5	539	580	470	42	248

Gearbox	a1	b1	to2	c1	e1	f1	q1	s1	g5	h3	d	to1	l	l4	l3	t	u	DR	i2
DR/ZR68	350	250	h6	18	300	7	79	17.5	165	57	50	k6	100	10.0	80	53.5	14	M16x36	300
DR/ZR88	350	250	h6	18	300	7	92	17.5	185	62	60	m6	120	10.0	100	64.0	18	M20x42	360
DR/ZR108	450	350	h6	22	400	7	78	17.5	210	72	70	m6	140	7.5	125	74.5	20	M20x42	420
DR/ZR128	550	450	h6	25	500	8	101	17.5	252	81	80	m6	170	20.0	125	85.0	22	M20x42	500
DR/ZR148	550	450	h6	25	500	8	113	17.5	252	81	100	m6	210	15.0	180	106	28	M24x50	600
DR/ZR168	660	550	h6	28	600	8	113	22.0	270	86	110	m6	210	15.0	180	116	28	M24x50	660

1) To calculate the overall weight of the drive, add the additional weight to the weight of the DZ/ZZ gearbox, flange-mounted design.
For example: weight of DZ88-M112M (97 kg) + additional weight DR88 (46 kg) = total weight of DR88-M112M (143 kg).

MOTEX Geared Motors

Helical geared motors

Dimensions

Pin holes

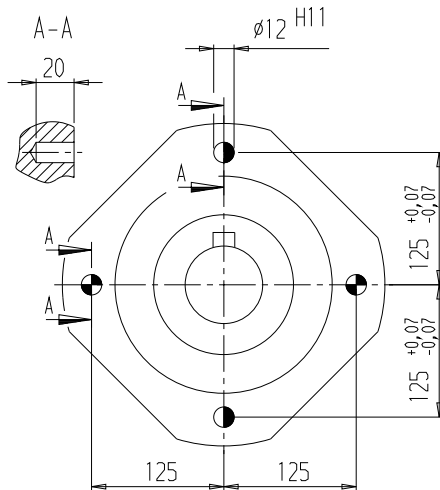
The customer's interface can be pinned to the housing flange (C-type) for sizes EZ128 to EZ148 and DZ/ZZ108 to DZ/ZZ188.

The output flanges have been designed to ensure the reliable transmission of the permissible torques and radial forces by the bolt connections.

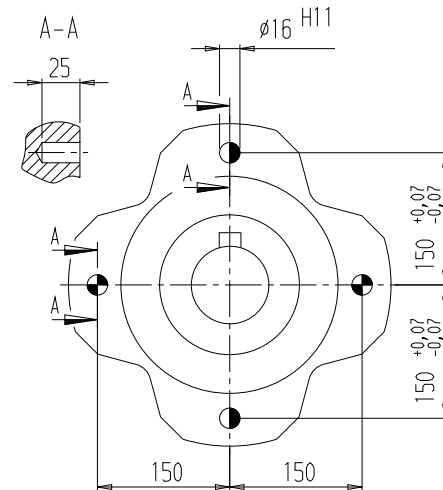
If an additional fuse, e. g. for high shock loads, is required, the existing pin holes can be used.

The gearbox and the machine can be drilled and pinned together. To do so, the provided dimensions must be observed.

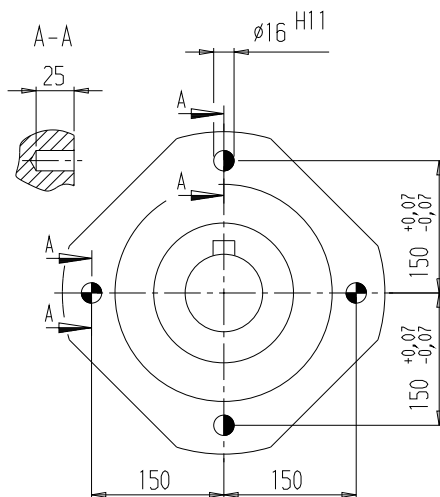
EZ128, DZ/ZZ108



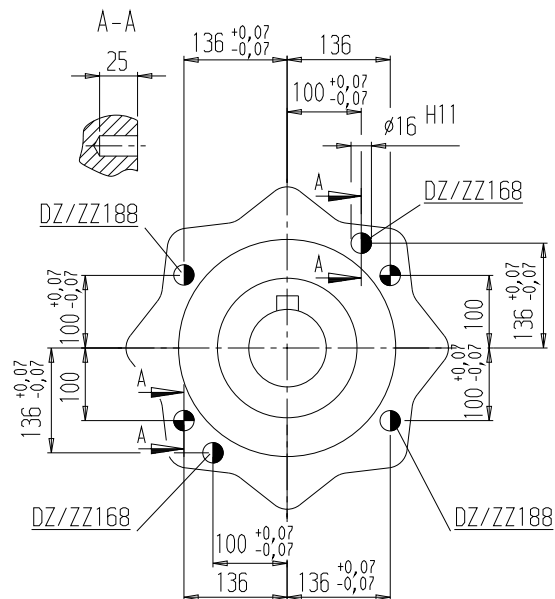
EZ148, DZ/ZZ128



DZ/ZZ148



DZ/ZZ168, DZ/ZZ188



- Spring pins, heavy-duty design, to DIN 1481: Use pin holes provided in the housing flange.
- Grooved cylindrical pins with chamfer to DIN EN 28740 / ISO 8740: Drill connecting component together with housing.

Parallel shaft geared motors



3/2 3/4	Orientation Overview Modular system
3/5	General technical data Permissible radial force
3/6	Geared motors up to 200 kW Selection and ordering data
3/65	Transmission ratios and maximum torques Selection and ordering data
3/86	Mounting types Selection and ordering data
3/89	Shaft designs Selection and ordering data
3/91	Flange-mounted designs Selection and ordering data
3/92	Mounting types and mounting positions Selection and ordering data
3/96 3/96 3/97 3/97 3/98 3/99 3/99 3/100	Special versions Lubricants Oil level control Gearbox ventilation Oil drain Sealing Hollow shaft cover (protection cover) Radially reinforced output shaft bearings Mixer flange in dry-well design
3/101 3/104	Dimensions Dimension drawing overview Dimension drawings

MOTOX Geared Motors

Parallel shaft geared motors

Orientation

Overview



MOTOX parallel shaft gearboxes are part of the MOTOX modular system. With helical, bevel helical, helical worm, or variable speed gearboxes and three-phase AC motors with or without brakes, this system covers all possible drive combinations, right up to electronic variable speed drives.

MOTOX parallel shaft gearboxes are designed for continuous duty. The gearbox housings made of gray cast iron or aluminium are developed in 3D CAD and have an optimized structure in terms of rigidity and vibration absorption. Radial shaft seals with dust-protection lips prevent oil from leaking out of the housing and dust and water from entering it. The tooth flanks are ground or honed so that they are convex and corrected in terms of the profile. Optimum running smoothness is achieved thanks to the gear wheels' helical teeth. The output shaft is parallel to the input shaft on 2-stage and 3-stage gearboxes.

MOTOX parallel shaft gearboxes are available in 2-stage and 3-stage designs. Standard series gearboxes can be supplied for attaching in any position. The gearboxes are available in a solid-shaft or hollow-shaft design with a feather key connection, shrink disk connection, or splined shaft.

MOTOX Geared Motors

Parallel shaft geared motors

Orientation

Overview (continued)

The parallel shaft gearboxes are designated as follows:

Gearbox type:

F Parallel shaft gearbox

Transmission stage **Z** 2-stage
D 3-stage

Type:

Shaft (-) Solid shaft
A Hollow shaft

Mounting (-) Foot-mounted design
F Flange-mounted design (A-type)
Z Housing flange (C-type)
D Torque arm
M Mixer flange
E Extruder flange

Connections (-) Feather key
S Shrink disk
T Hollow shaft with splined shaft

Special features **W** Reduced-backlash version

Type of intermediate gearbox:

(-) Helical gearbox

Transmission stage **Z** 2-stage
D 3-stage

Input unit:

K2 Coupling lantern with flexible coupling for connecting an IEC motor

K2TC Coupling lantern with flexible coupling for connecting a NEMA motor ¹⁾

K4 Short coupling lantern with clamp connection for connecting an IEC motor

K5 Short coupling lantern with clamp connection for connecting a NEMA motor ¹⁾

KQ Lantern for servomotor with feather key and zero-backlash flexible coupling for connecting a servomotor

KQS Lantern for servomotor without feather key and zero-backlash flexible coupling for connecting a servomotor

A Input unit with free input shaft

A5 Input unit with free input shaft (NEMA design) ¹⁾

P Input unit with free input shaft and piggy back for connecting an IEC motor

P5 Input unit with free input shaft and piggy back for connecting a NEMA motor ¹⁾

PS Input unit with free input shaft and piggy back with protection cover

Example:

F Z F 108 B - Z 38 - K4 (100)

Gearbox type	_____
Transmission stages	_____
Type	_____
Size	_____
Revision marks	_____
Type of intermediate gearbox	_____
Size	_____
Input unit (for motor size)	_____

The series currently comprises 10 gearbox sizes.

The basic designs available are 2- and 3-stage gearboxes.

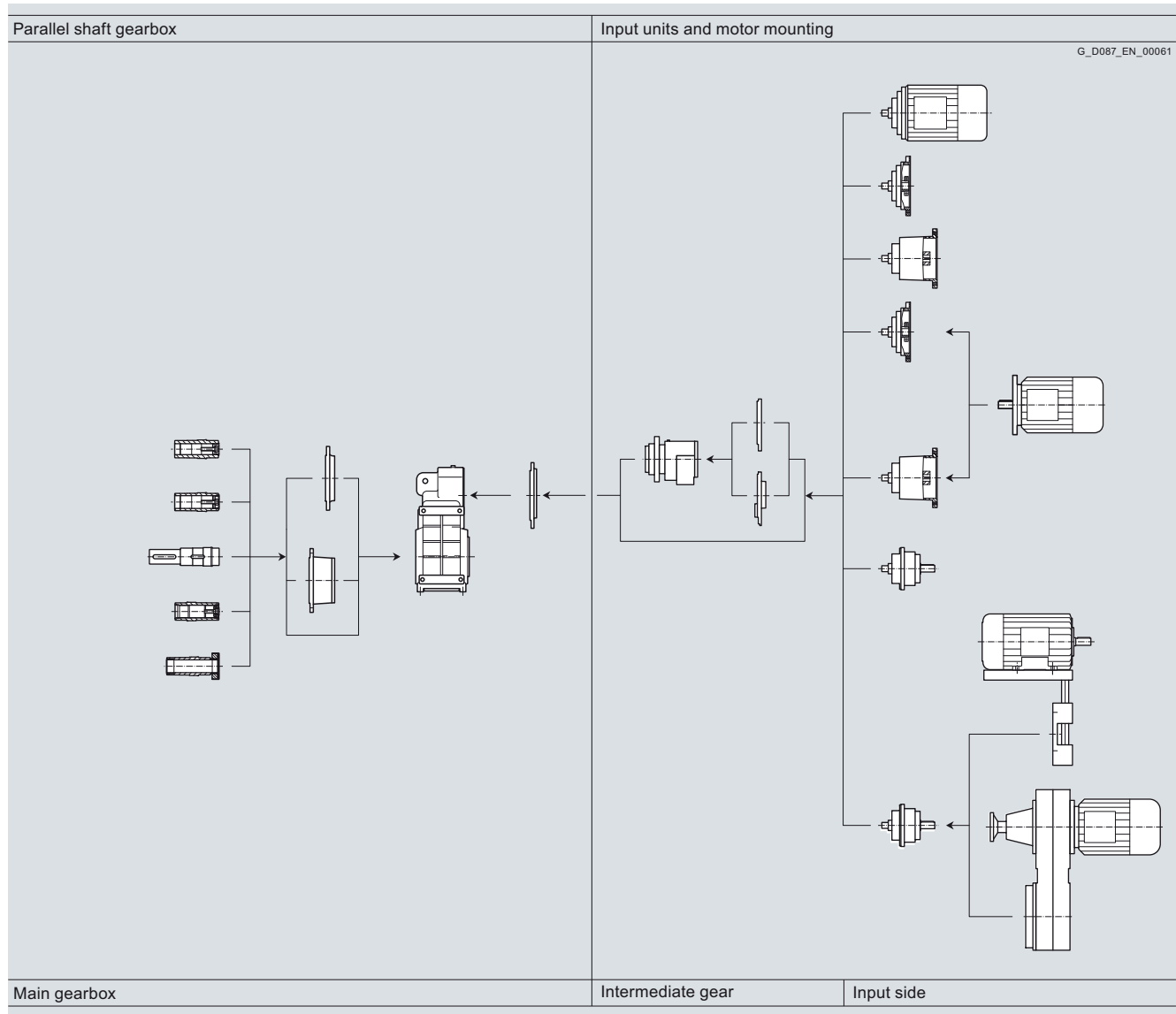
¹⁾ These designs can be selected from our MOTOX Configurator electronic catalog.

MOTOX Geared Motors

Parallel shaft geared motors

Orientation

Modular system



Use

MOTOX parallel shaft gearboxes are the ideal solution when space is at a premium, thanks to their compact, well-shaped structures.

The variety of output shafts – hollow or solid – and the range of mounting options available, which allow the device to be used as a shaft-mounted gearbox with a torque arm, or as a foot- or

flange-mounted design, enable you to achieve exactly the right solution, at the right price.

Parallel shaft gearboxes are extremely efficient. They are very economical, thanks to their low price and low maintenance requirements.

Oil quantities

The oil quantities corresponding to the applicable mounting positions are specified in the operating instructions and on the rating plate.

Permissible radial force F_{Rperm}

2-stage and 3-stage parallel shaft gearbox – standard bearing arrangement

Gearbox type	d mm	l mm	y mm	z mm	a kNmm	F_{Rperm} in N with $x = l/2$ for output speeds n_2 in rpm Direction of rotation when viewing the output shaft	F_{Rperm} in N with $x = l/2$ for output speeds n_2 in rpm							
							≤ 16	≤ 25	≤ 40	≤ 63	≤ 100	≤ 160	≤ 250	≤ 400
F.F28	25	50	128.5	104	115	Left	4 600	4 600	4 150	3 330	2 730	2 350	1 840	1 780
						Right	4 600	4 600	3 950	3 120	2 520	2 160	1 650	1 650
F.F38B	25	50	146.0	121	131	Left	5 246	5 246	4 810	4 020	2 980	2 870	2 590	2 480
						Right	5 246	5 246	4 360	3 610	2 500	2 480	2 450	2 370
F.F48B	30	60	176.0	146	245	Left	8 154	8 060	6 640	5 270	4 840	4 530	4 070	3 770
						Right	8 150	7 500	6 080	4 720	4 400	4 280	3 900	3 650
F.F68B	40	80	213.0	173	357	Left	8 927	7 680	6 160	5 050	3 710	3 930	3 710	3 650
						Right	8 927	6 830	5 310	4 200	2 860	3 290	3 300	3 440
F.F88B	50	100	262.0	212	741	Left	14 825	13 420	10 040	8 310	7 020	6 590	6 320	6 130
						Right	14 340	12 360	8 740	7 010	5 800	5 960	5 920	5 800
F.F108B	60	120	298.0	238	1 100	Left	17 930	13 620	10 750	8 190	6 070	6 610	6 840	7 080
						Right	15 860	11 550	8 680	6 120	4 040	4 960	5 780	6 390
F.F128B	70	140	371.5	302	1 786	Left	25 516	19 950	15 710	10 270	9 120	10 890	10 860	10 360
						Right	23 190	17 570	13 530	7 900	6 740	9 300	9 920	9 810
F.F148B	90	170	434.0	349	2 241	Left	23 390	17 850	13 190	8 530	9 840	11 680	11 800	11 660
						Right	20 390	14 850	10 180	5 620	7 380	10 030	10 530	10 830
F.F168B	110	210	517.5	413	4 814	Left	35 450	27 240	20 850	13 740	12 970	17 210	16 400	16 450
						Right	31 510	23 300	17 200	9 800	9 280	15 230	14 590	15 330
F.F188B	120	210	538.0	433	11 898	Left	113 314	113 314	113 314	106 120	88 810	78 120	76 850	–
						Right	113 314	113 314	113 314	102 690	84 350	75 050	74 100	–
F.F208	160	250	622.0	497	18 750	Left	150 000	150 000	150 000	150 000	143 760	127 130	121 290	–
						Right	150 000	150 000	150 000	150 000	135 990	120 310	114 800	–

2-stage and 3-stage parallel shaft gearbox – reinforced bearing arrangement

Gearbox type	d mm	l mm	y mm	z mm	a kNmm	F_{Rperm} in N with $x = l/2$ for output speeds n_2 in rpm Direction of rotation when viewing the output shaft	F_{Rperm} in N with $x = l/2$ for output speeds n_2 in rpm							
							≤ 16	≤ 25	≤ 40	≤ 63	≤ 100	≤ 160	≤ 250	≤ 400
F.F68B	40	80	213.0	173	546	Left	13 643	13 643	13 643	13 643	13 643	13 260	11 920	10 620
						Right	13 643	13 643	13 643	13 643	13 230	12 690	11 540	10 390
F.F88B	50	100	262.0	212	1 171	Left	23 411	23 411	23 411	23 411	23 411	21 180	19 050	18 130
						Right	23 411	23 411	23 411	23 411	22 960	20 520	18 620	17 790
F.F108B	60	120	298.0	238	1 723	Left	28 718	28 718	28 718	28 718	28 718	26 040	24 150	23 420
						Right	28 718	28 718	28 718	28 718	26 590	24 740	23 300	22 680
F.F128B	70	140	371.5	302	2 514	Left	35 921	35 921	35 921	35 921	35 921	35 921	35 921	34 420
						Right	35 921	35 921	35 921	35 921	35 921	35 921	35 921	33 830
F.F148B	90	170	434.0	349	5 737	Left	67 493	67 493	67 300	55 150	52 240	46 910	44 010	41 380
						Right	67 493	67 493	64 110	52 070	50 180	45 380	42 870	40 510
F.F168B	110	210	517.5	413	9 566	Left	91 102	91 102	91 102	87 720	78 620	71 650	65 350	62 000
						Right	91 102	91 102	91 102	83 520	75 920	69 990	63 850	60 810
F.F188B	120	210	538.0	433	11 898	Left	113 314	113 314	113 314	106 120	88 810	78 120	76 850	–
						Right	113 314	113 314	113 314	102 690	84 350	75 050	74 100	–
F.F208	160	250	622.0	497	18 750	Left	150 000	150 000	150 000	150 000	150 000	150 000	150 000	–
						Right	150 000	150 000	150 000	150 000	150 000	150 000	150 000	–

The values in the table apply to the worst-case scenario.
The output shaft bearing arrangement can be calculated using our MOTOX Configurator electronic catalog.

See Chapter 1 of the configuring guide for more information on calculating the permissible radial force.

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data

The selection tables show the most common variants and combinations. Other combinations can be selected using our MOTOX Configurator or made available on request.

At an identical power rating and output speed, priority is given in the selection tables to 4-pole geared motors.

At the available transmission ratios, they cover the majority of output speeds.

Due to their prevalence, 4-pole geared motors are easily available, with short delivery times and at a low cost. They also feature a favorable size / power ratio.

Power rating P_{Motor} kW (50 Hz)	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight *) kg
0.09	FD.48B-LA71M8						
	2.3	367	1.5	268.8	★ 2KJ1402 - ■CE13 - ■■S1	P02	27
	2.6	326	1.7	238.65	2KJ1402 - ■CE13 - ■■R1	P02	27
	3.0	285	1.9	209.23	★ 2KJ1402 - ■CE13 - ■■Q1	P02	27
	FD.38B-LA71M8						
	2.6	330	0.88	241.91	★ 2KJ1401 - ■CE13 - ■■M1	P02	20
	3.0	284	1.0	207.83	2KJ1401 - ■CE13 - ■■L1	P02	20
	FD.38B-LA71B6						
	3.2	269	1.1	280.41	2KJ1401 - ■CB13 - ■■N1	P01	20
	3.7	232	1.2	241.91	★ 2KJ1401 - ■CB13 - ■■M1	P01	20
	4.3	200	1.5	207.83	2KJ1401 - ■CB13 - ■■L1	P01	20
	4.7	184	1.6	191.34	★ 2KJ1401 - ■CB13 - ■■K1	P01	20
0.12	FD.188B-D48-LA71B4						
	0.05	15 668	1.3	28 045	★ 2KJ1440 - ■CB13 - ■■E1		638
	0.06	12 819	1.6	22 946	★ 2KJ1440 - ■CB13 - ■■C1		638
	0.06	14 134	1.4	25 299	2KJ1440 - ■CB13 - ■■D1		638
	0.07	10 683	1.9	19 122	★ 2KJ1440 - ■CB13 - ■■A1		638
	0.07	11 680	1.7	20 906	2KJ1440 - ■CB13 - ■■B1		638
	FD.188B-Z48-LA71B4						
	0.08	10 013	2.0	17 537	2KJ1438 - ■CB13 - ■■A2		638
	FD.168B-D48-LA71B4						
	0.05	16 202	0.86	29 000	2KJ1436 - ■CB13 - ■■F1		455
	0.06	12 901	1.1	23 093	2KJ1436 - ■CB13 - ■■D1		455
	0.06	14 302	0.98	25 599	★ 2KJ1436 - ■CB13 - ■■E1		455
	0.07	10 661	1.3	19 083	2KJ1436 - ■CB13 - ■■B1		455
	0.07	11 701	1.2	20 944	★ 2KJ1436 - ■CB13 - ■■C1		455
	0.08	9 751	1.4	17 454	★ 2KJ1436 - ■CB13 - ■■A1		455
	FD.168B-Z48-LA71B4						
	0.09	9 139	1.5	16 007	2KJ1435 - ■CB13 - ■■A2		454
	0.10	8 088	1.7	14 165	★ 2KJ1435 - ■CB13 - ■■X1		454
	0.11	7 353	1.9	12 878	2KJ1435 - ■CB13 - ■■W1		454
	FD.148B-D38-LA71B4						
	0.07	10 870	0.83	19 456	2KJ1433 - ■CB13 - ■■B1		288
	0.08	9 891	0.91	17 704	★ 2KJ1433 - ■CB13 - ■■A1		288
	FD.148B-Z38-LA71B4						
	0.09	9 272	0.97	16 239	★ 2KJ1432 - ■CB13 - ■■W1		287
	0.10	8 245	1.1	14 441	2KJ1432 - ■CB13 - ■■V1		287
	0.11	7 152	1.3	12 527	★ 2KJ1432 - ■CB13 - ■■U1		287

★ Preferred transmission ratio

Shaft designs, see page 3/89

1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20

1 to 9

Gearbox housing mounting position, see page 3/92

A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight *) kg
0.12	FD.148B-Z38-LA71B4						
	0.12	6 683	1.3	11 705	2KJ1432 - ■CB13 - ■■T1		287
	0.14	5 878	1.5	10 295	★ 2KJ1432 - ■CB13 - ■■S1		287
	0.16	5 148	1.7	9 016	2KJ1432 - ■CB13 - ■■R1		287
	0.18	4 553	2.0	7 975	★ 2KJ1432 - ■CB13 - ■■Q1		287
	FD.128B-Z38-LA71B4						
	0.12	6 445	0.95	11 289	2KJ1428 - ■CB13 - ■■T1		197
	0.12	6 899	0.88	12 083	★ 2KJ1428 - ■CB13 - ■■U1		197
	0.14	5 669	1.1	9 929	★ 2KJ1428 - ■CB13 - ■■S1		197
	0.16	4 965	1.2	8 696	2KJ1428 - ■CB13 - ■■R1		197
	0.18	4 391	1.4	7 691	★ 2KJ1428 - ■CB13 - ■■Q1		197
	0.20	3 980	1.5	6 971	2KJ1428 - ■CB13 - ■■P1		197
	0.23	3 513	1.7	6 153	★ 2KJ1428 - ■CB13 - ■■N1		197
	0.25	3 169	1.9	5 551	2KJ1428 - ■CB13 - ■■M1		197
	FD.108B-Z38-LA71B4						
	0.19	4 270	0.80	7 479	★ 2KJ1426 - ■CB13 - ■■F2		122
	0.21	3 870	0.88	6 778	2KJ1426 - ■CB13 - ■■E2		122
	0.23	3 416	1.0	5 983	★ 2KJ1426 - ■CB13 - ■■D2		122
	0.26	3 081	1.1	5 397	2KJ1426 - ■CB13 - ■■C2		122
	0.29	2 795	1.2	4 895	★ 2KJ1426 - ■CB13 - ■■B2		122
	0.31	2 546	1.3	4 460	2KJ1426 - ■CB13 - ■■A2		122
	0.34	2 329	1.5	4 079	★ 2KJ1426 - ■CB13 - ■■X1		122
	0.38	2 083	1.6	3 648	2KJ1426 - ■CB13 - ■■W1		122
	0.42	1 912	1.8	3 349	★ 2KJ1426 - ■CB13 - ■■V1		122
	0.46	1 724	2.0	3 019	2KJ1426 - ■CB13 - ■■U1		122
	FD.88B-Z28-LA71B4						
	0.34	2 386	0.8	4 179	2KJ1422 - ■CB13 - ■■W1		73
	0.38	2 118	0.9	3 709	★ 2KJ1422 - ■CB13 - ■■V1		73
	0.43	1 856	1.0	3 251	2KJ1422 - ■CB13 - ■■U1		73
	0.49	1 632	1.2	2 858	★ 2KJ1422 - ■CB13 - ■■T1		73
	0.54	1 474	1.3	2 582	2KJ1422 - ■CB13 - ■■S1		73
	0.62	1 285	1.5	2 250	★ 2KJ1422 - ■CB13 - ■■R1		73
	0.69	1 154	1.6	2 021	2KJ1422 - ■CB13 - ■■Q1		73
	0.77	1 041	1.8	1 824	★ 2KJ1422 - ■CB13 - ■■P1		73
	0.85	944	2.0	1 654	2KJ1422 - ■CB13 - ■■N1		73
	FD.68B-Z28-LA71B4						
	0.69	1 162	0.86	2 035	2KJ1417 - ■CB13 - ■■T1		43
	0.78	1 021	0.98	1 789	★ 2KJ1417 - ■CB13 - ■■S1		43
	0.87	923	1.1	1 616	2KJ1417 - ■CB13 - ■■R1		43
	0.99	804	1.2	1 408	★ 2KJ1417 - ■CB13 - ■■Q1		43
	1.1	722	1.4	1 265	2KJ1417 - ■CB13 - ■■P1		43
	1.2	652	1.5	1 142	★ 2KJ1417 - ■CB13 - ■■N1		43
	1.4	592	1.7	1 036	2KJ1417 - ■CB13 - ■■M1		43
	1.5	538	1.9	942	★ 2KJ1417 - ■CB13 - ■■L1		43

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.12	FD.68B-LA71MB8						
	2.2	526	1.9	296.18	★	2KJ1403 - ■CB13 - ■■S1	P02 43
	FD.48B-Z28-LA71B4						
	1.3	614	0.88	1 076	★	2KJ1413 - ■CB13 - ■■N1	29
	1.4	557	0.97	976		2KJ1413 - ■CB13 - ■■M1	29
	1.6	507	1.1	888	★	2KJ1413 - ■CB13 - ■■L1	29
	1.8	448	1.2	785		2KJ1413 - ■CB13 - ■■K1	29
	1.9	414	1.3	725	★	2KJ1413 - ■CB13 - ■■J1	29
	2.2	356	1.5	624		2KJ1413 - ■CB13 - ■■H1	29
	FD.48B-LA71MB8						
	2.4	478	1.1	268.8	★	2KJ1402 - ■CF13 - ■■S1	P02 27
	2.7	424	1.3	238.65		2KJ1402 - ■CF13 - ■■R1	P02 27
	3.1	372	1.5	209.23	★	2KJ1402 - ■CF13 - ■■Q1	P02 27
	FD.48B-LA71C6						
	3.2	358	1.5	268.8	★	2KJ1402 - ■CC13 - ■■S1	P01 27
	3.6	318	1.7	238.65		2KJ1402 - ■CC13 - ■■R1	P01 27
	4.1	279	1.9	209.23	★	2KJ1402 - ■CC13 - ■■Q1	P01 27
	FZ.38B-Z28-LA71B4						
	2.4	339	0.86	587		2KJ1313 - ■CB13 - ■■G1	22
	FD.38B-LA71C6						
	3.6	322	0.9	241.91	★	2KJ1401 - ■CC13 - ■■M1	P01 20
	4.1	277	1.0	207.83		2KJ1401 - ■CC13 - ■■L1	P01 20
	4.5	255	1.1	191.34	★	2KJ1401 - ■CC13 - ■■K1	P01 20
	FD.38B-LA71B4						
	5.0	230	1.3	280.41		2KJ1401 - ■CB13 - ■■N1	20
	5.8	198	1.5	241.91	★	2KJ1401 - ■CB13 - ■■M1	20
	6.7	170	1.7	207.83		2KJ1401 - ■CB13 - ■■L1	20
	7.3	157	1.9	191.34	★	2KJ1401 - ■CB13 - ■■K1	20
	8.0	142	2.0	173.94		2KJ1401 - ■CB13 - ■■J1	20
	FD.28-LA71B4						
	6.7	170	0.88	207.53		2KJ1400 - ■CB13 - ■■L1	11
	7.3	156	0.96	191.06	★	2KJ1400 - ■CB13 - ■■K1	11
	8.1	142	1.1	173.69		2KJ1400 - ■CB13 - ■■J1	11
	9.1	126	1.2	153.74	★	2KJ1400 - ■CB13 - ■■H1	11
	10.9	105	1.4	128.77		2KJ1400 - ■CB13 - ■■G1	11
	12.8	90	1.7	109.79	★	2KJ1400 - ■CB13 - ■■F1	11
	15.0	76	2.0	93.32	★	2KJ1400 - ■CB13 - ■■E1	11
	17.3	66	2.3	81.1		2KJ1400 - ■CB13 - ■■D1	11
	19.8	58	2.6	70.59	★	2KJ1400 - ■CB13 - ■■C1	11
	22	52	2.9	63.68		2KJ1400 - ■CB13 - ■■B1	11
	25	46	3.3	56.2		2KJ1400 - ■CB13 - ■■A1	11
	FZ.28-LA71B4						
	24	49	3.1	59.65		2KJ1300 - ■CB13 - ■■C2	11
	28	41	3.6	50.3	★	2KJ1300 - ■CB13 - ■■B2	11

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.12	FZ.28-LA71B4						
	31	37	4.1	44.66	2KJ1300 - ■CB13 - ■■A2		11
	36	32	4.7	39.15 ★	2KJ1300 - ■CB13 - ■■X1		11
	40	29	5.2	35.04	2KJ1300 - ■CB13 - ■■W1		11
	45	26	5.9	31.1 ★	2KJ1300 - ■CB13 - ■■V1		11
	51	22	6.7	27.25	2KJ1300 - ■CB13 - ■■U1		11
	58	20	7.6	23.96 ★	2KJ1300 - ■CB13 - ■■T1		11
	65	18	8.5	21.64	2KJ1300 - ■CB13 - ■■S1		11
	74	15	9.7	18.86 ★	2KJ1300 - ■CB13 - ■■R1		11
	83	14	10.8	16.94	2KJ1300 - ■CB13 - ■■Q1		11
	92	12	12.0	15.29 ★	2KJ1300 - ■CB13 - ■■P1		11
	101	11	13.2	13.87	2KJ1300 - ■CB13 - ■■N1		11
	111	10	14.3	12.62 ★	2KJ1300 - ■CB13 - ■■M1		11
0.18	FD.188B-D48-LA71C4						
	0.05	24 072	0.83	25 299	2KJ1440 - ■CC13 - ■■D1		638
	0.06	21 833	0.92	22 946 ★	2KJ1440 - ■CC13 - ■■C1		638
	0.07	18 195	1.1	19 122 ★	2KJ1440 - ■CC13 - ■■A1		638
	0.07	19 892	1.0	20 906	2KJ1440 - ■CC13 - ■■B1		638
	FD.188B-Z48-LA71C4						
	0.08	17 053	1.2	17 537	2KJ1438 - ■CC13 - ■■A2		638
	0.09	15 091	1.3	15 519 ★	2KJ1438 - ■CC13 - ■■X1		638
	0.10	13 719	1.5	14 108	2KJ1438 - ■CC13 - ■■W1		638
	0.11	12 325	1.6	12 674 ★	2KJ1438 - ■CC13 - ■■V1		638
	0.13	10 563	1.9	10 863	2KJ1438 - ■CC13 - ■■U1		638
	FD.168B-D48-LA71C4						
	0.08	16 608	0.84	17 454 ★	2KJ1436 - ■CC13 - ■■A1		455
	FD.168B-Z48-LA71C4						
	0.09	15 566	0.90	16 007	2KJ1435 - ■CC13 - ■■A2		454
	0.10	13 774	1.0	14 165 ★	2KJ1435 - ■CC13 - ■■X1		454
	0.11	12 523	1.1	12 878	2KJ1435 - ■CC13 - ■■W1		454
	0.12	11 249	1.2	11 568 ★	2KJ1435 - ■CC13 - ■■V1		454
	0.14	9 643	1.5	9 916	2KJ1435 - ■CC13 - ■■U1		454
	0.15	8 724	1.6	8 971 ★	2KJ1435 - ■CC13 - ■■T1		454
	0.16	8 053	1.7	8 281	2KJ1435 - ■CC13 - ■■S1		454
	0.19	7 002	2.0	7 201 ★	2KJ1435 - ■CC13 - ■■R1		454
	FD.148B-Z38-LA71C4						
	0.13	10 011	0.9	10 295 ★	2KJ1432 - ■CC13 - ■■S1		287
	0.15	8 767	1.0	9 016	2KJ1432 - ■CC13 - ■■R1		287
	0.17	7 755	1.2	7 975 ★	2KJ1432 - ■CC13 - ■■Q1		287
	0.19	7 028	1.3	7 227	2KJ1432 - ■CC13 - ■■P1		287
	0.22	6 204	1.5	6 380 ★	2KJ1432 - ■CC13 - ■■N1		287
	0.24	5 596	1.6	5 755	2KJ1432 - ■CC13 - ■■M1		287
	0.26	5 076	1.8	5 220 ★	2KJ1432 - ■CC13 - ■■L1		287
	0.29	4 25	1.9	4 756	2KJ1432 - ■CC13 - ■■K1		287

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.18	FD.128B-Z38-LA71C4						
	0.18	7 479	0.82	7 691	★	2KJ1428 - ■ CC13 - ■■ Q1	197
	0.20	6 779	0.9	6 971		2KJ1428 - ■ CC13 - ■■ P1	197
	0.22	5 983	1.0	6 153	★	2KJ1428 - ■ CC13 - ■■ N1	197
	0.25	5 398	1.1	5 551		2KJ1428 - ■ CC13 - ■■ M1	197
	0.27	4 895	1.2	5 034	★	2KJ1428 - ■ CC13 - ■■ L1	197
	0.30	4 461	1.4	4 587		2KJ1428 - ■ CC13 - ■■ K1	197
	0.33	4 079	1.5	4 195	★	2KJ1428 - ■ CC13 - ■■ J1	197
	0.36	3 648	1.7	3 751		2KJ1428 - ■ CC13 - ■■ H1	197
	0.40	3 350	1.8	3 445	★	2KJ1428 - ■ CC13 - ■■ G1	197
	0.44	3 019	2.0	3 105		2KJ1428 - ■ CC13 - ■■ F1	197
	FD.108B-Z38-LA71C4						
	0.34	3 967	0.86	4 079	★	2KJ1426 - ■ CC13 - ■■ X1	122
	0.38	3 547	0.96	3 648		2KJ1426 - ■ CC13 - ■■ W1	122
	0.41	3 257	1.0	3 349	★	2KJ1426 - ■ CC13 - ■■ V1	122
	0.45	2 936	1.2	3 019		2KJ1426 - ■ CC13 - ■■ U1	122
	0.53	2 524	1.3	2 596	★	2KJ1426 - ■ CC13 - ■■ T1	122
	0.59	2 251	1.5	2 315		2KJ1426 - ■ CC13 - ■■ S1	122
	0.64	2 067	1.6	2 126	★	2KJ1426 - ■ CC13 - ■■ R1	122
	0.72	1 863	1.8	1 916		2KJ1426 - ■ CC13 - ■■ Q1	122
	FD.88B-Z28-LA71C4						
	0.61	2 188	0.87	2 250	★	2KJ1422 - ■ CC13 - ■■ R1	73
	0.68	1 965	0.97	2 021		2KJ1422 - ■ CC13 - ■■ Q1	73
	0.75	1 774	1.1	1 824	★	2KJ1422 - ■ CC13 - ■■ P1	73
	0.83	1 608	1.2	1 654		2KJ1422 - ■ CC13 - ■■ N1	73
	0.91	1 464	1.3	1 505	★	2KJ1422 - ■ CC13 - ■■ M1	73
	1.0	1 294	1.5	1 331		2KJ1422 - ■ CC13 - ■■ L1	73
	1.1	1 195	1.6	1 229	★	2KJ1422 - ■ CC13 - ■■ K1	73
	1.3	1 029	1.8	1 058		2KJ1422 - ■ CC13 - ■■ J1	73
	1.4	935	2.0	962	★	2KJ1422 - ■ CC13 - ■■ H1	73
	FD.88B-LA80S8						
	1.7	1 031	1.8	404.92		2KJ1404 - ■ DB13 - ■■ V1 P02	78
	FD.68B-Z28-LA71C4						
	1.1	1 230	0.81	1 265		2KJ1417 - ■ CC13 - ■■ P1	43
	1.2	1 111	0.90	1 142	★	2KJ1417 - ■ CC13 - ■■ N1	43
	1.3	1 007	0.99	1 036		2KJ1417 - ■ CC13 - ■■ M1	43
	1.5	916	1.1	942	★	2KJ1417 - ■ CC13 - ■■ L1	43
	1.6	810	1.2	833		2KJ1417 - ■ CC13 - ■■ K1	43
	1.8	748	1.3	769	★	2KJ1417 - ■ CC13 - ■■ J1	43
	2.1	644	1.6	662		2KJ1417 - ■ CC13 - ■■ H1	43
	FD.68B-LA80S8						
	2.3	754	1.3	296.18	★	2KJ1403 - ■ DB13 - ■■ S1 P02	47
	2.6	671	1.5	263.39		2KJ1403 - ■ DB13 - ■■ R1 P02	47

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg	
0.18	FD.68B-LA71S6							
	2.9	599	1.7	296.18	★	2KJ1403 - ■CD13 - ■■S1	P01	43
	3.2	533	1.9	263.39		2KJ1403 - ■CD13 - ■■R1	P01	43
	FD.48B-Z28-LA71C4							
	2.2	607	0.89	624		2KJ1413 - ■CC13 - ■■H1		29
	FD.48B-LA80S8							
	2.8	608	0.89	238.65		2KJ1402 - ■DB13 - ■■R1	P02	31
	FD.48B-LA71S6							
	3.2	544	0.99	268.8	★	2KJ1402 - ■CD13 - ■■S1	P01	27
	3.6	483	1.1	238.65		2KJ1402 - ■CD13 - ■■R1	P01	27
	4.1	423	1.3	209.23	★	2KJ1402 - ■CD13 - ■■Q1	P01	27
	4.5	379	1.4	187.24		2KJ1402 - ■CD13 - ■■P1	P01	27
	FD.48B-LA71C4							
	5.1	337	1.6	268.8	★	2KJ1402 - ■CC13 - ■■S1		27
	5.7	299	1.8	238.65		2KJ1402 - ■CC13 - ■■R1		27
	6.5	263	2.1	209.23	★	2KJ1402 - ■CC13 - ■■Q1		27
	FD.38B-LA71C4							
	4.9	352	0.82	280.41		2KJ1401 - ■CC13 - ■■N1		20
	5.7	304	0.96	241.91	★	2KJ1401 - ■CC13 - ■■M1		20
	6.6	261	1.1	207.83		2KJ1401 - ■CC13 - ■■L1		20
	7.2	240	1.2	191.34	★	2KJ1401 - ■CC13 - ■■K1		20
	7.9	218	1.3	173.94		2KJ1401 - ■CC13 - ■■J1		20
	8.9	193	1.5	153.96	★	2KJ1401 - ■CC13 - ■■H1		20
	10.6	162	1.8	128.95		2KJ1401 - ■CC13 - ■■G1		20
	12.5	138	2.1	109.95	★	2KJ1401 - ■CC13 - ■■F1		20
	FD.28-LA71C4							
	10.6	162	0.93	128.77		2KJ1400 - ■CC13 - ■■G1		11
	12.5	138	1.1	109.79	★	2KJ1400 - ■CC13 - ■■F1		11
	14.7	117	1.3	93.32	★	2KJ1400 - ■CC13 - ■■E1		11
	16.9	102	1.5	81.1		2KJ1400 - ■CC13 - ■■D1		11
	19.4	89	1.7	70.59	★	2KJ1400 - ■CC13 - ■■C1		11
	22	80	1.9	63.68		2KJ1400 - ■CC13 - ■■B1		11
	24	70	2.1	56.2		2KJ1400 - ■CC13 - ■■A1		11
	FZ.28-LA71C4							
	23	75	2.0	59.65		2KJ1300 - ■CC13 - ■■C2		11
	27	63	2.4	50.3	★	2KJ1300 - ■CC13 - ■■B2		11
	31	56	2.7	44.66		2KJ1300 - ■CC13 - ■■A2		11
	35	49	3.1	39.15	★	2KJ1300 - ■CC13 - ■■X1		11
	39	44	3.4	35.04		2KJ1300 - ■CC13 - ■■W1		11
	44	39	3.8	31.1	★	2KJ1300 - ■CC13 - ■■V1		11
	50	34	4.4	27.25		2KJ1300 - ■CC13 - ■■U1		11
	57	30	5.0	23.96	★	2KJ1300 - ■CC13 - ■■T1		11
	63	27	5.5	21.64		2KJ1300 - ■CC13 - ■■S1		11
	73	24	6.3	18.86	★	2KJ1300 - ■CC13 - ■■R1		11

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.18	FZ.28-LA71C4						
	81	21	7.1	16.94	2KJ1300 - ■ CC13 - ■■ Q1		11
	90	19	7.8	15.29 ★	2KJ1300 - ■ CC13 - ■■ P1		11
	99	17	8.6	13.87	2KJ1300 - ■ CC13 - ■■ N1		11
	109	16	9.3	12.62 ★	2KJ1300 - ■ CC13 - ■■ M1		11
	123	14	10.1	11.16	2KJ1300 - ■ CC13 - ■■ L1		11
	133	13	10.7	10.3 ★	2KJ1300 - ■ CC13 - ■■ K1		11
	154	11	11.8	8.87	2KJ1300 - ■ CC13 - ■■ J1		11
	170	10	12.6	8.06 ★	2KJ1300 - ■ CC13 - ■■ H1		11
	190	9	13.9	7.2 ★	2KJ1300 - ■ CC13 - ■■ G1		11
	210	8.2	14.9	6.53	2KJ1300 - ■ CC13 - ■■ F1		11
0.25	FD.188B-Z48-LA71S4						
	0.09	22 462	0.89	15 519 ★	2KJ1438 - ■ CD13 - ■■ X1		638
	0.10	20 419	0.98	14 108	2KJ1438 - ■ CD13 - ■■ W1		638
	0.11	18 344	1.1	12 674 ★	2KJ1438 - ■ CD13 - ■■ V1		638
	0.12	15 723	1.3	10 863	2KJ1438 - ■ CD13 - ■■ U1		638
	0.14	14 226	1.4	9 829 ★	2KJ1438 - ■ CD13 - ■■ T1		638
	0.15	13 132	1.5	9 073	2KJ1438 - ■ CD13 - ■■ S1		638
	0.17	11 418	1.8	7 889 ★	2KJ1438 - ■ CD13 - ■■ R1		638
	0.19	10 367	1.9	7 163	2KJ1438 - ■ CD13 - ■■ Q1		638
	FD.168B-Z48-LA71S4						
	0.12	16 743	0.84	11 568 ★	2KJ1435 - ■ CD13 - ■■ V1		454
	0.14	14 352	0.98	9 916	2KJ1435 - ■ CD13 - ■■ U1		454
	0.15	12 984	1.1	8 971 ★	2KJ1435 - ■ CD13 - ■■ T1		454
	0.16	11 986	1.2	8 281	2KJ1435 - ■ CD13 - ■■ S1		454
	0.19	10 422	1.3	7 201 ★	2KJ1435 - ■ CD13 - ■■ R1		454
	0.21	9 463	1.5	6 538	2KJ1435 - ■ CD13 - ■■ Q1		454
	0.23	8 641	1.6	5 970 ★	2KJ1435 - ■ CD13 - ■■ P1		454
	0.25	7 927	1.8	5 477	2KJ1435 - ■ CD13 - ■■ N1		454
	0.27	7 303	1.9	5 046 ★	2KJ1435 - ■ CD13 - ■■ M1		454
	FD.148B-Z38-LA71S4						
	0.19	10 460	0.86	7 227	2KJ1432 - ■ CD13 - ■■ P1		287
	0.21	9 234	0.97	6 380 ★	2KJ1432 - ■ CD13 - ■■ N1		287
	0.24	8 330	1.1	5 755	2KJ1432 - ■ CD13 - ■■ M1		287
	0.26	7 555	1.2	5 220 ★	2KJ1432 - ■ CD13 - ■■ L1		287
	0.28	6 884	1.3	4 756	2KJ1432 - ■ CD13 - ■■ K1		287
	0.31	6 296	1.4	4 350 ★	2KJ1432 - ■ CD13 - ■■ J1		287
	0.35	5 629	1.6	3 889	2KJ1432 - ■ CD13 - ■■ H1		287
	0.38	5 169	1.7	3 571 ★	2KJ1432 - ■ CD13 - ■■ G1		287
	0.42	4 659	1.9	3 219	2KJ1432 - ■ CD13 - ■■ F1		287
	FD.128B-Z38-LA71S4						
	0.27	7 286	0.84	5 034 ★	2KJ1428 - ■ CD13 - ■■ L1		197
	0.29	6 639	0.92	4 587	2KJ1428 - ■ CD13 - ■■ K1		197
	0.32	6 072	1.0	4 195 ★	2KJ1428 - ■ CD13 - ■■ J1		197

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.25	FD.128B-Z38-LA71S4						
	0.36	5 429	1.1	3 751	2KJ1428 - ■CD13 - ■■H1		197
	0.39	4 986	1.2	3 445	★ 2KJ1428 - ■CD13 - ■■G1		197
	0.44	4 494	1.4	3 105	2KJ1428 - ■CD13 - ■■F1		197
	0.51	3 864	1.6	2 670	★ 2KJ1428 - ■CD13 - ■■E1		197
	0.57	3 446	1.8	2 381	2KJ1428 - ■CD13 - ■■D1		197
	0.62	3 164	1.9	2 186	★ 2KJ1428 - ■CD13 - ■■C1		197
	FD.108B-Z38-LA71S4						
	0.52	3 757	0.9	2 596	★ 2KJ1426 - ■CD13 - ■■T1		122
	0.58	3 351	1.0	2 315	2KJ1426 - ■CD13 - ■■S1		122
	0.64	3 077	1.1	2 126	★ 2KJ1426 - ■CD13 - ■■R1		122
	0.70	2 773	1.2	1 916	2KJ1426 - ■CD13 - ■■Q1		122
	0.82	2 384	1.4	1 647	★ 2KJ1426 - ■CD13 - ■■P1		122
	0.88	2 209	1.5	1 526	2KJ1426 - ■CD13 - ■■N1		122
	0.98	2 003	1.7	1 384	★ 2KJ1426 - ■CD13 - ■■M1		122
	1.1	1 825	1.9	1 261	2KJ1426 - ■CD13 - ■■L1		122
	1.2	1 669	2.0	1 153	★ 2KJ1426 - ■CD13 - ■■K1		122
	FD.88B-Z28-LA71S4						
	0.9	2 178	0.87	1 505	★ 2KJ1422 - ■CD13 - ■■M1		73
	1.0	1 926	0.99	1 331	2KJ1422 - ■CD13 - ■■L1		73
	1.1	1 779	1.1	1 229	★ 2KJ1422 - ■CD13 - ■■K1		73
	1.3	1 531	1.2	1 058	2KJ1422 - ■CD13 - ■■J1		73
	1.4	1 392	1.4	962	★ 2KJ1422 - ■CD13 - ■■H1		73
	1.5	1 265	1.5	874	★ 2KJ1422 - ■CD13 - ■■G1		73
	FD.88B-LA80M8						
	1.7	1 411	1.3	404.92	2KJ1404 - ■DC13 - ■■V1	P02	78
	1.9	1 249	1.5	358.33	★ 2KJ1404 - ■DC13 - ■■U1	P02	78
	FD.88B-LA71M6						
	2.1	1 124	1.7	404.92	2KJ1404 - ■CE13 - ■■V1	P01	74
	2.4	995	1.9	358.33	★ 2KJ1404 - ■CE13 - ■■U1	P01	74
	FD.68B-Z28-LA71S4						
	1.6	1 206	0.83	833	2KJ1417 - ■CD13 - ■■K1		43
	1.8	1 113	0.9	769	★ 2KJ1417 - ■CD13 - ■■J1		43
	2.0	958	1.0	662	2KJ1417 - ■CD13 - ■■H1		43
	FD.68B-LA80M8						
	2.3	1 032	0.97	296.18	★ 2KJ1403 - ■DC13 - ■■S1	P02	47
	2.6	918	1.1	263.39	2KJ1403 - ■DC13 - ■■R1	P02	47
	FD.68B-LA71M6						
	2.9	822	1.2	296.18	★ 2KJ1403 - ■CE13 - ■■S1	P01	43
	3.3	731	1.4	263.39	2KJ1403 - ■CE13 - ■■R1	P01	43
	3.8	634	1.6	228.48	★ 2KJ1403 - ■CE13 - ■■Q1	P01	43
	4.0	593	1.7	213.48	2KJ1403 - ■CE13 - ■■P1	P01	43

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg	
0.25	FD.68B-LA71S4							
	4.6	524	1.9	296.18	★	2KJ1403 - ■CD13 - ■■S1	43	
	5.1	466	2.1	263.39		2KJ1403 - ■CD13 - ■■R1	43	
	FD.48B-LA71M6							
	3.6	663	0.82	238.65		2KJ1402 - ■CE13 - ■■R1	P01	27
	4.1	581	0.93	209.23	★	2KJ1402 - ■CE13 - ■■Q1	P01	27
	4.6	520	1.0	187.24		2KJ1402 - ■CE13 - ■■P1	P01	27
	FD.48B-LA71S4							
	5.0	475	1.1	268.8	★	2KJ1402 - ■CD13 - ■■S1		27
	5.7	422	1.3	238.65		2KJ1402 - ■CD13 - ■■R1		27
	6.5	370	1.5	209.23	★	2KJ1402 - ■CD13 - ■■Q1		27
	7.2	331	1.6	187.24		2KJ1402 - ■CD13 - ■■P1		27
	8.1	294	1.8	166.19	★	2KJ1402 - ■CD13 - ■■N1		27
	9.3	258	2.1	145.63		2KJ1402 - ■CD13 - ■■M1		27
	FD.38B-LA71S4							
	7.1	338	0.86	191.34	★	2KJ1401 - ■CD13 - ■■K1		20
	7.8	308	0.94	173.94		2KJ1401 - ■CD13 - ■■J1		20
	8.8	272	1.1	153.96	★	2KJ1401 - ■CD13 - ■■H1		20
	10.5	228	1.3	128.95		2KJ1401 - ■CD13 - ■■G1		20
	12.3	194	1.5	109.95	★	2KJ1401 - ■CD13 - ■■F1		20
	14.4	165	1.8	93.46		2KJ1401 - ■CD13 - ■■E1		20
	16.6	144	2.0	81.22	★	2KJ1401 - ■CD13 - ■■D1		20
	19.1	125	2.3	70.7		2KJ1401 - ■CD13 - ■■C1		20
	FZ.38B-LA71S4							
	24	100	2.1	56.72	★	2KJ1301 - ■CD13 - ■■B2		19
	FD.28-LA71S4							
	14.5	165	0.91	93.32	★	2KJ1400 - ■CD13 - ■■E1		11
	16.6	143	1.0	81.1		2KJ1400 - ■CD13 - ■■D1		11
	19.1	125	1.2	70.59	★	2KJ1400 - ■CD13 - ■■C1		11
	21	113	1.3	63.68		2KJ1400 - ■CD13 - ■■B1		11
	24	99	1.5	56.2		2KJ1400 - ■CD13 - ■■A1		11
	FZ.28-LA71S4							
	23	105	1.4	59.65		2KJ1300 - ■CD13 - ■■C2		11
	27	89	1.7	50.3	★	2KJ1300 - ■CD13 - ■■B2		11
	30	79	1.9	44.66		2KJ1300 - ■CD13 - ■■A2		11
	34	69	2.2	39.15	★	2KJ1300 - ■CD13 - ■■X1		11
	38	62	2.4	35.04		2KJ1300 - ■CD13 - ■■W1		11
	43	55	2.7	31.1	★	2KJ1300 - ■CD13 - ■■V1		11
	50	48	3.1	27.25		2KJ1300 - ■CD13 - ■■U1		11
	56	42	3.5	23.96	★	2KJ1300 - ■CD13 - ■■T1		11
	62	38	3.9	21.64		2KJ1300 - ■CD13 - ■■S1		11
	72	33	4.5	18.86	★	2KJ1300 - ■CD13 - ■■R1		11
	80	30	5.0	16.94		2KJ1300 - ■CD13 - ■■Q1		11
	88	27	5.5	15.29	★	2KJ1300 - ■CD13 - ■■P1		11

★ Preferred transmission ratio

Shaft designs, see page 3/89

1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20

1 to 9

Gearbox housing mounting position, see page 3/92

A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.25	FZ.28-LA71S4						
	97	24	6.1	13.87	2KJ1300 - ■CD13 - ■■N1		11
	107	22	6.6	12.62 ★	2KJ1300 - ■CD13 - ■■M1		11
	121	20	7.2	11.16	2KJ1300 - ■CD13 - ■■L1		11
	131	18	7.6	10.3 ★	2KJ1300 - ■CD13 - ■■K1		11
	152	16	8.4	8.87	2KJ1300 - ■CD13 - ■■J1		11
	167	14	8.9	8.06 ★	2KJ1300 - ■CD13 - ■■H1		11
	188	13	9.9	7.2 ★	2KJ1300 - ■CD13 - ■■G1		11
	207	12	10.6	6.53	2KJ1300 - ■CD13 - ■■F1		11
	227	10	11.2	5.94 ★	2KJ1300 - ■CD13 - ■■E1		11
	257	9.3	12.0	5.25	2KJ1300 - ■CD13 - ■■D1		11
	278	8.6	12.8	4.85 ★	2KJ1300 - ■CD13 - ■■C1		11
	323	7.4	13.4	4.18	2KJ1300 - ■CD13 - ■■B1		11
	355	6.7	14.3	3.8 ★	2KJ1300 - ■CD13 - ■■A1		11
0.37	FD.188B-Z48-LA71M4						
	0.13	23 944	0.84	10 863	2KJ1438 - ■CE13 - ■■U1		638
	0.14	21 665	0.92	9 829 ★	2KJ1438 - ■CE13 - ■■T1		638
	0.15	19 998	1.0	9 073	2KJ1438 - ■CE13 - ■■S1		638
	0.17	17 389	1.2	7 889 ★	2KJ1438 - ■CE13 - ■■R1		638
	0.19	15 788	1.3	7 163	2KJ1438 - ■CE13 - ■■Q1		638
	0.21	14 415	1.4	6 540 ★	2KJ1438 - ■CE13 - ■■P1		638
	0.23	13 227	1.5	6 001	2KJ1438 - ■CE13 - ■■N1		638
	0.25	12 187	1.6	5 529 ★	2KJ1438 - ■CE13 - ■■M1		638
	0.27	11 067	1.8	5 021	2KJ1438 - ■CE13 - ■■L1		638
	0.30	10 082	2.0	4 574 ★	2KJ1438 - ■CE13 - ■■K1		638
	FD.168B-Z48-LA71M4						
	0.19	15 872	0.88	7 201 ★	2KJ1435 - ■CE13 - ■■R1		454
	0.21	14 411	0.97	6 538	2KJ1435 - ■CE13 - ■■Q1		454
	0.23	13 159	1.1	5 970 ★	2KJ1435 - ■CE13 - ■■P1		454
	0.25	12 072	1.2	5 477	2KJ1435 - ■CE13 - ■■N1		454
	0.27	11 122	1.3	5 046 ★	2KJ1435 - ■CE13 - ■■M1		454
	0.30	10 102	1.4	4 583	2KJ1435 - ■CE13 - ■■L1		454
	0.33	9 202	1.5	4 175 ★	2KJ1435 - ■CE13 - ■■K1		454
	0.36	8 431	1.7	3 825	2KJ1435 - ■CE13 - ■■J1		454
	0.40	7 523	1.9	3 413 ★	2KJ1435 - ■CE13 - ■■H1		454
	FD.148B-Z38-LA71M4						
	0.29	10 483	0.86	4 756	2KJ1432 - ■CE13 - ■■K1		287
	0.32	9 588	0.94	4 350 ★	2KJ1432 - ■CE13 - ■■J1		287
	0.35	8 572	1.0	3 889	2KJ1432 - ■CE13 - ■■H1		287
	0.38	7 871	1.1	3 571 ★	2KJ1432 - ■CE13 - ■■G1		287
	0.43	7 095	1.3	3 219	2KJ1432 - ■CE13 - ■■F1		287
	0.50	6 101	1.5	2 768 ★	2KJ1432 - ■CE13 - ■■E1		287
	0.56	5 440	1.7	2 468	2KJ1432 - ■CE13 - ■■D1		287
	0.60	4 995	1.8	2 266 ★	2KJ1432 - ■CE13 - ■■C1		287

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.37	FD.148B-Z38-LA71M4						
	0.67	4 503	2.0	2 043	2KJ1432 - ■CE13 - ■■B1		287
	FD.128B-Z38-LA71M4						
	0.40	7 593	0.80	3 445	★	2KJ1428 - ■CE13 - ■■G1	197
	0.44	6 844	0.89	3 105		2KJ1428 - ■CE13 - ■■F1	197
	0.51	5 885	1.0	2 670	★	2KJ1428 - ■CE13 - ■■E1	197
	0.57	5 248	1.2	2 381		2KJ1428 - ■CE13 - ■■D1	197
	0.63	4 818	1.3	2 186	★	2KJ1428 - ■CE13 - ■■C1	197
	0.70	4 342	1.4	1 970		2KJ1428 - ■CE13 - ■■B1	197
	0.81	3 734	1.6	1 694	★	2KJ1428 - ■CE13 - ■■A1	197
	FD.128B-Z48-LA71M4						
	0.91	3 315	1.8	1 504		2KJ1431 - ■CE13 - ■■L1	206
	1.0	3 020	2.0	1 370	★	2KJ1431 - ■CE13 - ■■K1	206
	FD.108B-Z38-LA71M4						
	0.72	4 223	0.81	1 916		2KJ1426 - ■CE13 - ■■Q1	122
	0.83	3 630	0.94	1 647	★	2KJ1426 - ■CE13 - ■■P1	122
	0.90	3 364	1.0	1 526		2KJ1426 - ■CE13 - ■■N1	122
	0.99	3 051	1.1	1 384	★	2KJ1426 - ■CE13 - ■■M1	122
	1.1	2 779	1.2	1 261		2KJ1426 - ■CE13 - ■■L1	122
	1.2	2 541	1.3	1 153	★	2KJ1426 - ■CE13 - ■■K1	122
	1.3	2 272	1.5	1 031		2KJ1426 - ■CE13 - ■■J1	122
	1.4	2 087	1.6	947	★	2KJ1426 - ■CE13 - ■■H1	122
	FD.108B-LA90SA8						
	1.6	2 222	1.5	424.49	★	2KJ1405 - ■EB13 - ■■V1 P02	128
	1.8	2 004	1.7	382.79		2KJ1405 - ■EB13 - ■■U1 P02	128
	2.0	1 807	1.9	345.19	★	2KJ1405 - ■EB13 - ■■T1 P02	128
	FD.88B-Z28-LA71M4						
	1.3	2 332	0.81	1 058		2KJ1422 - ■CE13 - ■■J1	73
	1.4	2 120	0.90	962	★	2KJ1422 - ■CE13 - ■■H1	73
	1.6	1 926	0.99	874	★	2KJ1422 - ■CE13 - ■■G1	73
	FD.88B-LA90SA8						
	1.9	1 876	1.0	358.33	★	2KJ1404 - ■EB13 - ■■U1 P02	81
	FD.88B-LA80S6						
	2.3	1 555	1.2	404.92		2KJ1404 - ■DB13 - ■■V1 P01	78
	2.6	1 376	1.4	358.33	★	2KJ1404 - ■DB13 - ■■U1 P01	78
	2.8	1 251	1.5	325.76		2KJ1404 - ■DB13 - ■■T1 P01	78
	3.1	1 124	1.7	292.64	★	2KJ1404 - ■DB13 - ■■S1 P01	78
	FD.88B-LA71M4						
	3.4	1 044	1.8	404.92		2KJ1404 - ■CE13 - ■■V1	74
	FD.68B-LA80S6						
	3.1	1 138	0.88	296.18	★	2KJ1403 - ■DB13 - ■■S1 P01	47
	3.5	1 012	0.99	263.39		2KJ1403 - ■DB13 - ■■R1 P01	47
	4.0	878	1.1	228.48	★	2KJ1403 - ■DB13 - ■■Q1 P01	47
	4.3	820	1.2	213.48		2KJ1403 - ■DB13 - ■■P1 P01	47

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.37	FD.68B-LA71M4						
	4.6	764	1.3	296.18	★	2KJ1403 - ■CE13 - ■■S1	43
	5.2	679	1.5	263.39		2KJ1403 - ■CE13 - ■■R1	43
	6.0	589	1.7	228.48	★	2KJ1403 - ■CE13 - ■■Q1	43
	6.4	551	1.8	213.48		2KJ1403 - ■CE13 - ■■P1	43
	7.3	484	2.1	187.76	★	2KJ1403 - ■CE13 - ■■N1	43
	FD.48B-LA71M4						
	5.7	616	0.88	238.65		2KJ1402 - ■CE13 - ■■R1	27
	6.5	540	1.0	209.23	★	2KJ1402 - ■CE13 - ■■Q1	27
	7.3	483	1.1	187.24		2KJ1402 - ■CE13 - ■■P1	27
	8.2	429	1.3	166.19	★	2KJ1402 - ■CE13 - ■■N1	27
	9.4	376	1.4	145.63		2KJ1402 - ■CE13 - ■■M1	27
	10.7	330	1.6	128.04	★	2KJ1402 - ■CE13 - ■■L1	27
	11.8	298	1.8	115.68		2KJ1402 - ■CE13 - ■■K1	27
	13.6	260	2.1	100.8	★	2KJ1402 - ■CE13 - ■■J1	27
	FD.38B-LA71M4						
	10.6	333	0.87	128.95		2KJ1401 - ■CE13 - ■■G1	20
	12.5	284	1.0	109.95	★	2KJ1401 - ■CE13 - ■■F1	20
	14.7	241	1.2	93.46	★	2KJ1401 - ■CE13 - ■■E1	20
	16.9	209	1.4	81.22		2KJ1401 - ■CE13 - ■■D1	20
	19.4	182	1.6	70.7	★	2KJ1401 - ■CE13 - ■■C1	20
	22	164	1.8	63.77		2KJ1401 - ■CE13 - ■■B1	20
	24	145	2.0	56.28		2KJ1401 - ■CE13 - ■■A1	20
	FZ.38B-LA71M4						
	24	146	1.4	56.72	★	2KJ1301 - ■CE13 - ■■B2	19
	27	130	1.8	50.44		2KJ1301 - ■CE13 - ■■A2	19
	31	113	2.2	43.75	★	2KJ1301 - ■CE13 - ■■X1	19
	34	105	2.6	40.88		2KJ1301 - ■CE13 - ■■W1	19
	FD.28-LA71M4						
	19.4	182	0.82	70.59	★	2KJ1400 - ■CE13 - ■■C1	11
	22	164	0.91	63.68		2KJ1400 - ■CE13 - ■■B1	11
	24	145	1.0	56.2		2KJ1400 - ■CE13 - ■■A1	11
	FZ.28-LA71M4						
	23	154	0.97	59.65		2KJ1300 - ■CE13 - ■■C2	11
	27	130	1.2	50.3	★	2KJ1300 - ■CE13 - ■■B2	11
	31	115	1.3	44.66		2KJ1300 - ■CE13 - ■■A2	11
	35	101	1.5	39.15	★	2KJ1300 - ■CE13 - ■■X1	11
	39	90	1.7	35.04		2KJ1300 - ■CE13 - ■■W1	11
	44	80	1.9	31.1	★	2KJ1300 - ■CE13 - ■■V1	11
	50	70	2.1	27.25		2KJ1300 - ■CE13 - ■■U1	11
	57	62	2.4	23.96	★	2KJ1300 - ■CE13 - ■■T1	11
	63	56	2.7	21.64		2KJ1300 - ■CE13 - ■■S1	11
	73	49	3.1	18.86	★	2KJ1300 - ■CE13 - ■■R1	11

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.55	FD.188B-Z48-LA71ZMP4						
	0.19	24 147	0.83	7 163	2KJ1438 - ■CG13 - ■■Q1		638
	0.21	22 047	0.91	6 540	★ 2KJ1438 - ■CG13 - ■■P1		638
	0.23	20 230	0.99	6 001	2KJ1438 - ■CG13 - ■■N1		638
	0.25	18 639	1.1	5 529	★ 2KJ1438 - ■CG13 - ■■M1		638
	0.27	16 926	1.2	5 021	2KJ1438 - ■CG13 - ■■L1		638
	0.30	15 419	1.3	4 574	★ 2KJ1438 - ■CG13 - ■■K1		638
	0.33	14 125	1.4	4 190	2KJ1438 - ■CG13 - ■■J1		638
	0.37	12 604	1.6	3 739	★ 2KJ1438 - ■CG13 - ■■H1		638
	FD.168B-Z48-LA71ZMP4						
	0.27	17 010	0.82	5 046	★ 2KJ1435 - ■CG13 - ■■M1		454
	0.30	15 450	0.91	4 583	2KJ1435 - ■CG13 - ■■L1		454
	0.33	14 074	0.99	4 175	★ 2KJ1435 - ■CG13 - ■■K1		454
	0.36	12 894	1.1	3 825	2KJ1435 - ■CG13 - ■■J1		454
	0.40	11 505	1.2	3 413	★ 2KJ1435 - ■CG13 - ■■H1		454
	0.65	7 143	2.0	2 119	★ 2KJ1435 - ■CG13 - ■■D1		454
	FD.148B-Z48-LA71ZMP4						
	0.84	5 508	1.6	1 634	2KJ1434 - ■CG13 - ■■K1		296
	0.92	5 020	1.8	1 489	★ 2KJ1434 - ■CG13 - ■■J1		296
	1.0	4 598	2.0	1 364	2KJ1434 - ■CG13 - ■■H1		296
	FD.148B-Z38-LA71ZMP4						
	0.43	10 852	0.83	3 219	2KJ1432 - ■CG13 - ■■F1		287
	0.50	9 331	0.96	2 768	★ 2KJ1432 - ■CG13 - ■■E1		287
	0.56	8 320	1.1	2 468	2KJ1432 - ■CG13 - ■■D1		287
	0.60	7 639	1.2	2 266	★ 2KJ1432 - ■CG13 - ■■C1		287
	0.67	6 887	1.3	2 043	2KJ1432 - ■CG13 - ■■B1		287
	0.78	5 923	1.5	1 757	★ 2KJ1432 - ■CG13 - ■■A1		287
	FD.128B-Z48-LA71ZMP4						
	0.91	5 070	1.2	1 504	2KJ1431 - ■CG13 - ■■L1		206
	1.0	4 618	1.3	1 370	★ 2KJ1431 - ■CG13 - ■■K1		206
	1.1	4 231	1.4	1 255	2KJ1431 - ■CG13 - ■■J1		206
	1.2	3 776	1.6	1 120	★ 2KJ1431 - ■CG13 - ■■H1		206
	FD.128B-Z38-LA71ZMP4						
	0.63	7 369	0.83	2 186	★ 2KJ1428 - ■CG13 - ■■C1		197
	0.70	6 641	0.92	1 970	2KJ1428 - ■CG13 - ■■B1		197
	0.81	5 711	1.1	1 694	★ 2KJ1428 - ■CG13 - ■■A1		197
	FD.128B-LA90LA8						
	1.5	3 486	1.7	447.96	2KJ1406 - ■EE13 - ■■V1	P02	212
	1.7	3 155	1.9	405.47	★ 2KJ1406 - ■EE13 - ■■U1	P02	212
	FD.108B-Z38-LA71ZMP4						
	1.1	4 251	0.80	1 261	2KJ1426 - ■CG13 - ■■L1		122
	1.2	3 887	0.87	1 153	★ 2KJ1426 - ■CG13 - ■■K1		122
	1.3	3 476	0.98	1 031	2KJ1426 - ■CG13 - ■■J1		122
	1.4	3 192	1.1	947	★ 2KJ1426 - ■CG13 - ■■H1		122

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.55	FD.108B-LA90LA8						
	1.6	3 303	1.0	424.49 ★	2KJ1405 - ■EE13 - ■■V1	P02	131
	1.8	2 979	1.1	382.79	2KJ1405 - ■EE13 - ■■U1	P02	131
	2.0	2 686	1.3	345.19 ★	2KJ1405 - ■EE13 - ■■T1	P02	131
	FD.108B-LA80M6						
	2.1	2 450	1.4	424.49 ★	2KJ1405 - ■DC13 - ■■V1	P01	125
	2.4	2 209	1.5	382.79	2KJ1405 - ■DC13 - ■■U1	P01	125
	2.6	1 992	1.7	345.19 ★	2KJ1405 - ■DC13 - ■■T1	P01	125
	3.0	1 742	2.0	301.88	2KJ1405 - ■DC13 - ■■S1	P01	125
	FD.88B-LA80M6						
	2.2	2 337	0.81	404.92	2KJ1404 - ■DC13 - ■■V1	P01	78
	2.5	2 068	0.92	358.33 ★	2KJ1404 - ■DC13 - ■■U1	P01	78
	2.8	1 880	1.0	325.76	2KJ1404 - ■DC13 - ■■T1	P01	78
	3.1	1 689	1.1	292.64 ★	2KJ1404 - ■DC13 - ■■S1	P01	78
	FD.88B-LA71ZMP4						
	3.4	1 552	1.2	404.92	2KJ1404 - ■CG13 - ■■V1		74
	3.8	1 374	1.4	358.33 ★	2KJ1404 - ■CG13 - ■■U1		74
	4.2	1 249	1.5	325.76	2KJ1404 - ■CG13 - ■■T1		74
	4.7	1 122	1.7	292.64 ★	2KJ1404 - ■CG13 - ■■S1		74
	5.5	962	2.0	250.83	2KJ1404 - ■CG13 - ■■R1		74
	FD.68B-LA80M6						
	4.3	1 232	0.81	213.48	2KJ1403 - ■DC13 - ■■P1	P01	47
	FD.68B-LA71ZMP4						
	4.6	1 136	0.88	296.18 ★	2KJ1403 - ■CG13 - ■■S1		43
	5.2	1 010	0.99	263.39	2KJ1403 - ■CG13 - ■■R1		43
	6.0	876	1.1	228.48 ★	2KJ1403 - ■CG13 - ■■Q1		43
	6.4	818	1.2	213.48	2KJ1403 - ■CG13 - ■■P1		43
	7.3	720	1.4	187.76 ★	2KJ1403 - ■CG13 - ■■N1		43
	8.3	630	1.6	164.44	2KJ1403 - ■CG13 - ■■M1		43
	9.4	558	1.8	145.44 ★	2KJ1403 - ■CG13 - ■■L1		43
	10.4	505	2.0	131.82	2KJ1403 - ■CG13 - ■■K1		43
	11.8	446	2.2	116.36 ★	2KJ1403 - ■CG13 - ■■J1		43
	FD.48B-LA71ZMP4						
	8.2	637	0.85	166.19 ★	2KJ1402 - ■CG13 - ■■N1		27
	9.4	558	0.97	145.63	2KJ1402 - ■CG13 - ■■M1		27
	10.7	491	1.1	128.04 ★	2KJ1402 - ■CG13 - ■■L1		27
	11.8	444	1.2	115.68	2KJ1402 - ■CG13 - ■■K1		27
	13.6	386	1.4	100.8 ★	2KJ1402 - ■CG13 - ■■J1		27
	15.1	347	1.6	90.53	2KJ1402 - ■CG13 - ■■H1		27
	16.8	313	1.7	81.73 ★	2KJ1402 - ■CG13 - ■■G1		27
	18.5	284	1.9	74.1	2KJ1402 - ■CG13 - ■■F1		27
	20	259	2.1	67.43 ★	2KJ1402 - ■CG13 - ■■E1		27
	23	229	2.4	59.62	2KJ1402 - ■CG13 - ■■D1		27

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.55	FZ.48B-LA71ZMP4						
	23	233	1.7	60.71	★	2KJ1302 - ■CG13 - ■■B2	27
	25	212	2.4	55.19		2KJ1302 - ■CG13 - ■■A2	27
	FD.38B-LA71ZMP4						
	14.7	358	0.81	93.46	★	2KJ1401 - ■CG13 - ■■E1	20
	16.9	311	0.93	81.22		2KJ1401 - ■CG13 - ■■D1	20
	19.4	271	1.1	70.7	★	2KJ1401 - ■CG13 - ■■C1	20
	22	244	1.2	63.77		2KJ1401 - ■CG13 - ■■B1	20
	24	216	1.3	56.28		2KJ1401 - ■CG13 - ■■A1	20
	FZ.38B-LA71ZMP4						
	24	217	0.97	56.72	★	2KJ1301 - ■CG13 - ■■B2	19
	27	193	1.2	50.44		2KJ1301 - ■CG13 - ■■A2	19
	31	168	1.5	43.75	★	2KJ1301 - ■CG13 - ■■X1	19
	34	157	1.8	40.88		2KJ1301 - ■CG13 - ■■W1	19
	38	138	2.1	35.96	★	2KJ1301 - ■CG13 - ■■V1	19
	44	121	2.4	31.49		2KJ1301 - ■CG13 - ■■U1	19
	49	107	2.7	27.85	★	2KJ1301 - ■CG13 - ■■T1	19
	FZ.28-LA71ZMP4						
	31	171	0.88	44.66		2KJ1300 - ■CG13 - ■■A2	11
	35	150	1.0	39.15	★	2KJ1300 - ■CG13 - ■■X1	11
	39	134	1.1	35.04		2KJ1300 - ■CG13 - ■■W1	11
	44	119	1.3	31.1	★	2KJ1300 - ■CG13 - ■■V1	11
	50	104	1.4	27.25		2KJ1300 - ■CG13 - ■■U1	11
	57	92	1.6	23.96	★	2KJ1300 - ■CG13 - ■■T1	11
	63	83	1.8	21.64		2KJ1300 - ■CG13 - ■■S1	11
	73	72	2.1	18.86	★	2KJ1300 - ■CG13 - ■■R1	11
	81	65	2.3	16.94		2KJ1300 - ■CG13 - ■■Q1	11
	90	59	2.6	15.29	★	2KJ1300 - ■CG13 - ■■P1	11
	99	53	2.8	13.87		2KJ1300 - ■CG13 - ■■N1	11
	109	48	3.1	12.62	★	2KJ1300 - ■CG13 - ■■M1	11
	123	43	3.3	11.16		2KJ1300 - ■CG13 - ■■L1	11
	133	40	3.5	10.3	★	2KJ1300 - ■CG13 - ■■K1	11
	154	34	3.9	8.87		2KJ1300 - ■CG13 - ■■J1	11
	170	31	4.1	8.06	★	2KJ1300 - ■CG13 - ■■H1	11
0.75	FD.188B-Z48-LA80ZMB4E						
	0.28	22 934	0.87	5 021		2KJ1438 - ■DE13 - ■■L1	642
	0.31	20 892	0.96	4 574	★	2KJ1438 - ■DE13 - ■■K1	642
	0.33	19 138	1.0	4 190		2KJ1438 - ■DE13 - ■■J1	642
	0.37	17 078	1.2	3 739	★	2KJ1438 - ■DE13 - ■■H1	642
	0.42	15 242	1.3	3 337		2KJ1438 - ■DE13 - ■■G1	642
	0.49	12 945	1.5	2 834		2KJ1438 - ■DE13 - ■■F1	642
	0.59	10 775	1.9	2 359	★	2KJ1438 - ■DE13 - ■■E1	642
	0.60	10 606	1.9	2 322	★	2KJ1438 - ■DE13 - ■■D1	642

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.75	FD.168B-Z48-LA80ZMB4E						
	0.37	17 471	0.80	3 825	2KJ1435 - ■ DE13 - ■ ■ J1		458
	0.41	15 589	0.90	3 413	★ 2KJ1435 - ■ DE13 - ■ ■ H1		458
	0.46	13 913	1.0	3 046	2KJ1435 - ■ DE13 - ■ ■ G1		458
	0.54	11 816	1.2	2 587	2KJ1435 - ■ DE13 - ■ ■ F1		458
	0.65	9 834	1.4	2 153	★ 2KJ1435 - ■ DE13 - ■ ■ E1		458
	0.66	9 679	1.4	2 119	★ 2KJ1435 - ■ DE13 - ■ ■ D1		458
	0.74	8 637	1.6	1 891	2KJ1435 - ■ DE13 - ■ ■ C1		458
	0.87	7 336	1.9	1 606	2KJ1435 - ■ DE13 - ■ ■ B1		458
	FD.148B-Z38-LA80ZMB4E						
	0.57	11 273	0.80	2 468	2KJ1432 - ■ DE13 - ■ ■ D1		291
	0.62	10 350	0.87	2 266	★ 2KJ1432 - ■ DE13 - ■ ■ C1		291
	0.68	9 332	0.96	2 043	2KJ1432 - ■ DE13 - ■ ■ B1		291
	0.80	8 025	1.1	1 757	★ 2KJ1432 - ■ DE13 - ■ ■ A1		291
	FD.148B-Z48-LA80ZMB4E						
	0.86	7 464	1.2	1 634	2KJ1434 - ■ DE13 - ■ ■ K1		300
	0.94	6 801	1.3	1 489	★ 2KJ1434 - ■ DE13 - ■ ■ J1		300
	1.0	6 230	1.4	1 364	2KJ1434 - ■ DE13 - ■ ■ H1		300
	1.2	5 559	1.6	1 217	★ 2KJ1434 - ■ DE13 - ■ ■ G1		300
	1.3	4 960	1.8	1 086	2KJ1434 - ■ DE13 - ■ ■ F1		300
	FD.148B-LA100LA8						
	1.5	4 732	1.9	449.21	★ 2KJ1407 - ■ FB13 - ■ ■ U1	P02	316
	FD.128B-Z48-LA80ZMB4E						
	0.93	6 870	0.89	1 504	2KJ1431 - ■ DE13 - ■ ■ L1		210
	1.0	6 258	0.97	1 370	★ 2KJ1431 - ■ DE13 - ■ ■ K1		210
	1.1	5 732	1.1	1 255	2KJ1431 - ■ DE13 - ■ ■ J1		210
	1.2	5 116	1.2	1 120	★ 2KJ1431 - ■ DE13 - ■ ■ H1		210
	1.4	4 563	1.3	999	2KJ1431 - ■ DE13 - ■ ■ G1		210
	FD.128B-LA100LA8						
	1.9	3 739	1.6	354.99	2KJ1406 - ■ FB13 - ■ ■ T1	P02	220
	FD.128B-LA90SB6E						
	2.1	3 469	1.8	447.96	2KJ1406 - ■ ED13 - ■ ■ V1	P01	209
	2.3	3 140	1.9	405.47	★ 2KJ1406 - ■ ED13 - ■ ■ U1	P01	209
	FD.108B-LA90SB6E						
	2.2	3 287	1.0	424.49	★ 2KJ1405 - ■ ED13 - ■ ■ V1	P01	128
	2.4	2 964	1.1	382.79	2KJ1405 - ■ ED13 - ■ ■ U1	P01	128
	2.7	2 673	1.3	345.19	★ 2KJ1405 - ■ ED13 - ■ ■ T1	P01	128
	3.1	2 338	1.5	301.88	2KJ1405 - ■ ED13 - ■ ■ S1	P01	128
	FD.108B-LA80ZMB4E						
	3.3	2 172	1.6	424.49	★ 2KJ1405 - ■ DE13 - ■ ■ V1		125
	3.7	1 958	1.7	382.79	2KJ1405 - ■ DE13 - ■ ■ U1		125
	4.1	1 766	1.9	345.19	★ 2KJ1405 - ■ DE13 - ■ ■ T1		125
	FD.88B-LA90SB6E						
	3.2	2 266	0.84	292.64	★ 2KJ1404 - ■ ED13 - ■ ■ S1	P01	81

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.75	FD.88B-LA80ZMB4E						
	3.5	2 072	0.92	404.92	2KJ1404 - ■ DE13 - ■ ■ V1		78
	3.9	1 833	1.0	358.33 ★	2KJ1404 - ■ DE13 - ■ ■ U1		78
	4.3	1 667	1.1	325.76	2KJ1404 - ■ DE13 - ■ ■ T1		78
	4.8	1 497	1.3	292.64 ★	2KJ1404 - ■ DE13 - ■ ■ S1		78
	5.6	1 283	1.5	250.83	2KJ1404 - ■ DE13 - ■ ■ R1		78
	6.2	1 161	1.6	226.94 ★	2KJ1404 - ■ DE13 - ■ ■ P1		78
	6.7	1 072	1.8	209.49	2KJ1404 - ■ DE13 - ■ ■ N1		78
	7.7	932	2.0	182.15 ★	2KJ1404 - ■ DE13 - ■ ■ M1		78
	FD.68B-LA80ZMB4E						
	6.1	1 169	0.86	228.48 ★	2KJ1403 - ■ DE13 - ■ ■ Q1		47
	6.6	1 092	0.92	213.48	2KJ1403 - ■ DE13 - ■ ■ P1		47
	7.5	961	1.0	187.76 ★	2KJ1403 - ■ DE13 - ■ ■ N1		47
	8.5	841	1.2	164.44	2KJ1403 - ■ DE13 - ■ ■ M1		47
	9.6	744	1.3	145.44 ★	2KJ1403 - ■ DE13 - ■ ■ L1		47
	10.6	674	1.5	131.82	2KJ1403 - ■ DE13 - ■ ■ K1		47
	12.0	595	1.7	116.36 ★	2KJ1403 - ■ DE13 - ■ ■ J1		47
	13.3	537	1.9	104.96	2KJ1403 - ■ DE13 - ■ ■ H1		47
	14.7	487	2.1	95.2 ★	2KJ1403 - ■ DE13 - ■ ■ G1		47
	16.1	444	2.3	86.74	2KJ1403 - ■ DE13 - ■ ■ F1		47
	FD.48B-LA80ZMB4E						
	10.9	655	0.82	128.04 ★	2KJ1402 - ■ DE13 - ■ ■ L1		31
	12.1	592	0.91	115.68	2KJ1402 - ■ DE13 - ■ ■ K1		31
	13.9	516	1.0	100.8 ★	2KJ1402 - ■ DE13 - ■ ■ J1		31
	15.5	463	1.2	90.53	2KJ1402 - ■ DE13 - ■ ■ H1		31
	17.1	418	1.3	81.73 ★	2KJ1402 - ■ DE13 - ■ ■ G1		31
	18.9	379	1.4	74.1	2KJ1402 - ■ DE13 - ■ ■ F1		31
	21	345	1.6	67.43 ★	2KJ1402 - ■ DE13 - ■ ■ E1		31
	24	305	1.8	59.62	2KJ1402 - ■ DE13 - ■ ■ D1		31
	25	282	1.9	55.06 ★	2KJ1402 - ■ DE13 - ■ ■ C1		31
	30	243	2.2	47.4	2KJ1402 - ■ DE13 - ■ ■ B1		31
	32	220	2.4	43.09 ★	2KJ1402 - ■ DE13 - ■ ■ A1		31
	FZ.48B-LA80ZMB4E						
	23	311	1.3	60.71 ★	2KJ1302 - ■ DE13 - ■ ■ B2		31
	25	282	1.8	55.19	2KJ1302 - ■ DE13 - ■ ■ A2		31
	28	254	2.1	49.58 ★	2KJ1302 - ■ DE13 - ■ ■ X1		31
	33	217	2.5	42.5	2KJ1302 - ■ DE13 - ■ ■ W1		31
	FD.38B-LA80ZMB4E						
	19.8	362	0.80	70.7 ★	2KJ1401 - ■ DE13 - ■ ■ C1		24
	22	326	0.89	63.77	2KJ1401 - ■ DE13 - ■ ■ B1		24
	25	288	1.0	56.28	2KJ1401 - ■ DE13 - ■ ■ A1		24
	FZ.38B-LA80ZMB4E						
	28	258	0.89	50.44	2KJ1301 - ■ DE13 - ■ ■ A2		23
	32	224	1.1	43.75 ★	2KJ1301 - ■ DE13 - ■ ■ X1		23

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
0.75	FZ.38B-LA80ZMB4E						
	34	209	1.3	40.88	2KJ1301 - ■DE13 - ■■W1		23
	39	184	1.6	35.96 ★	2KJ1301 - ■DE13 - ■■V1		23
	44	161	1.8	31.49	2KJ1301 - ■DE13 - ■■U1		23
	50	142	2.0	27.85 ★	2KJ1301 - ■DE13 - ■■T1		23
	56	129	2.2	25.24	2KJ1301 - ■DE13 - ■■S1		23
	63	114	2.5	22.28 ★	2KJ1301 - ■DE13 - ■■R1		23
	70	103	2.8	20.1	2KJ1301 - ■DE13 - ■■Q1		23
	77	93	3.1	18.23 ★	2KJ1301 - ■DE13 - ■■P1		23
	FZ.28-LA80ZMB4E						
	40	179	0.84	35.04	2KJ1300 - ■DE13 - ■■W1		15
	45	159	0.94	31.1 ★	2KJ1300 - ■DE13 - ■■V1		15
	51	139	1.1	27.25	2KJ1300 - ■DE13 - ■■U1		15
	58	123	1.2	23.96 ★	2KJ1300 - ■DE13 - ■■T1		15
	65	111	1.4	21.64	2KJ1300 - ■DE13 - ■■S1		15
	74	96	1.6	18.86 ★	2KJ1300 - ■DE13 - ■■R1		15
	83	87	1.7	16.94	2KJ1300 - ■DE13 - ■■Q1		15
	92	78	1.9	15.29 ★	2KJ1300 - ■DE13 - ■■P1		15
	101	71	2.1	13.87	2KJ1300 - ■DE13 - ■■N1		15
	111	65	2.3	12.62 ★	2KJ1300 - ■DE13 - ■■M1		15
	125	57	2.5	11.16	2KJ1300 - ■DE13 - ■■L1		15
	136	53	2.6	10.3 ★	2KJ1300 - ■DE13 - ■■K1		15
	158	45	2.9	8.87	2KJ1300 - ■DE13 - ■■J1		15
	174	41	3.1	8.06 ★	2KJ1300 - ■DE13 - ■■H1		15
	194	37	3.4	7.2 ★	2KJ1300 - ■DE13 - ■■G1		15
	214	33	3.7	6.53	2KJ1300 - ■DE13 - ■■F1		15
	236	30	3.9	5.94 ★	2KJ1300 - ■DE13 - ■■E1		15
	267	27	4.1	5.25	2KJ1300 - ■DE13 - ■■D1		15
	289	25	4.4	4.85 ★	2KJ1300 - ■DE13 - ■■C1		15
	335	21	4.6	4.18	2KJ1300 - ■DE13 - ■■B1		15
	368	19	4.9	3.8 ★	2KJ1300 - ■DE13 - ■■A1		15
1.1	FD.188B-Z48-LA90SB4E						
	0.38	24 675	0.81	3 739 ★	2KJ1438 - ■EM13 - ■■H1		645
	0.43	22 022	0.91	3 337	2KJ1438 - ■EM13 - ■■G1		645
	0.51	18 703	1.1	2 834	2KJ1438 - ■EM13 - ■■F1		645
	0.61	15 568	1.3	2 359 ★	2KJ1438 - ■EM13 - ■■E1		645
	0.62	15 324	1.3	2 322 ★	2KJ1438 - ■EM13 - ■■D1		645
	0.70	13 674	1.5	2 072	2KJ1438 - ■EM13 - ■■C1		645
	0.82	11 615	1.7	1 760	2KJ1438 - ■EM13 - ■■B1		645
	FD.168B-Z48-LA90SB4E						
	0.56	17 073	0.82	2 587	2KJ1435 - ■EM13 - ■■F1		461
	0.67	14 209	0.99	2 153 ★	2KJ1435 - ■EM13 - ■■E1		461
	0.68	13 984	1.0	2 119 ★	2KJ1435 - ■EM13 - ■■D1		461
	0.76	12 480	1.1	1 891	2KJ1435 - ■EM13 - ■■C1		461

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.1	FD.168B-Z48-LA90SB4E						
	0.9	10 599	1.3	1 606	2KJ1435 - ■EM13 - ■■B1		461
	1.1	8 823	1.6	1 337 ★	2KJ1435 - ■EM13 - ■■A1		461
	FD.168B-Z68-LA90SB4E						
	1.1	8 566	1.6	1 298	2KJ1437 - ■EM13 - ■■H1		478
	1.3	7 312	1.9	1 108 ★	2KJ1437 - ■EM13 - ■■G1		478
	FD.148B-Z48-LA90SB4E						
	0.88	10 783	0.83	1 634	2KJ1434 - ■EM13 - ■■K1		303
	0.97	9 827	0.92	1 489 ★	2KJ1434 - ■EM13 - ■■J1		303
	1.1	9 002	1.0	1 364	2KJ1434 - ■EM13 - ■■H1		303
	1.2	8 032	1.1	1 217 ★	2KJ1434 - ■EM13 - ■■G1		303
	1.3	7 167	1.3	1 086	2KJ1434 - ■EM13 - ■■F1		303
	FD.148B-LA100L8						
	1.5	6 940	1.3	449.21 ★	2KJ1407 - ■FL13 - ■■U1	P02	316
	1.7	6 364	1.4	411.98	2KJ1407 - ■FL13 - ■■T1	P02	316
	1.8	5 686	1.6	368.06 ★	2KJ1407 - ■FL13 - ■■S1	P02	316
	2.0	5 207	1.7	337.07	2KJ1407 - ■FL13 - ■■R1	P02	316
	FD.128B-Z48-LA90SB4E						
	1.3	7 391	0.83	1 120 ★	2KJ1431 - ■EM13 - ■■H1		213
	1.4	6 593	0.93	999	2KJ1431 - ■EM13 - ■■G1		213
	FD.128B-LA100L8						
	1.9	5 484	1.1	354.99	2KJ1406 - ■FL13 - ■■T1	P02	220
	FD.128B-LA90ZLD6E						
	2.1	5 006	1.2	447.96	2KJ1406 - ■EQ13 - ■■V1	P01	212
	2.3	4 531	1.3	405.47 ★	2KJ1406 - ■EQ13 - ■■U1	P01	212
	2.6	3 967	1.5	354.99	2KJ1406 - ■EQ13 - ■■T1	P01	212
	2.9	3 579	1.7	320.24 ★	2KJ1406 - ■EQ13 - ■■S1	P01	212
	FD.128B-LA90SB4E						
	3.2	3 268	1.9	447.96	2KJ1406 - ■EM13 - ■■V1		209
	FD.108B-LA90ZLD6E						
	2.7	3 858	0.88	345.19 ★	2KJ1405 - ■EQ13 - ■■T1	P01	131
	3.1	3 374	1.0	301.88	2KJ1405 - ■EQ13 - ■■S1	P01	131
	FD.108B-LA90SB4E						
	3.4	3 097	1.1	424.49 ★	2KJ1405 - ■EM13 - ■■V1		128
	3.8	2 793	1.2	382.79	2KJ1405 - ■EM13 - ■■U1		128
	4.2	2 518	1.4	345.19 ★	2KJ1405 - ■EM13 - ■■T1		128
	4.8	2 202	1.5	301.88	2KJ1405 - ■EM13 - ■■S1		128
	5.3	1 977	1.7	271.01 ★	2KJ1405 - ■EM13 - ■■R1		128
	5.8	1 806	1.9	247.53	2KJ1405 - ■EM13 - ■■Q1		128
	6.6	1 602	2.1	219.66 ★	2KJ1405 - ■EM13 - ■■P1		128
	FD.88B-LA90SB4E						
	4.4	2 376	0.8	325.76	2KJ1404 - ■EM13 - ■■T1		81
	4.9	2 135	0.89	292.64 ★	2KJ1404 - ■EM13 - ■■S1		81
	5.7	1 830	1.0	250.83	2KJ1404 - ■EM13 - ■■R1		81

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.1	FD.88B-LA90SB4E						
	6.3	1 656	1.1	226.94	★	2KJ1404 - ■EM13 - ■■P1	81
	6.9	1 528	1.2	209.49		2KJ1404 - ■EM13 - ■■N1	81
	7.9	1 329	1.4	182.15	★	2KJ1404 - ■EM13 - ■■M1	81
	8.7	1 206	1.6	165.38		2KJ1404 - ■EM13 - ■■L1	81
	9.5	1 102	1.7	151.01	★	2KJ1404 - ■EM13 - ■■K1	81
	10.4	1 011	1.9	138.56		2KJ1404 - ■EM13 - ■■J1	81
	11.3	931	2.0	127.66	★	2KJ1404 - ■EM13 - ■■H1	81
	12.4	846	2.2	115.93		2KJ1404 - ■EM13 - ■■G1	81
	FD.68B-LA90SB4E						
	8.8	1 200	0.83	164.44		2KJ1403 - ■EM13 - ■■M1	50
	9.9	1 061	0.94	145.44	★	2KJ1403 - ■EM13 - ■■L1	50
	10.9	962	1.0	131.82		2KJ1403 - ■EM13 - ■■K1	50
	12.4	849	1.2	116.36	★	2KJ1403 - ■EM13 - ■■J1	50
	13.7	766	1.3	104.96		2KJ1403 - ■EM13 - ■■H1	50
	15.1	694	1.4	95.2	★	2KJ1403 - ■EM13 - ■■G1	50
	16.6	633	1.6	86.74		2KJ1403 - ■EM13 - ■■F1	50
	18.2	579	1.7	79.33	★	2KJ1403 - ■EM13 - ■■E1	50
	20	517	1.9	70.93		2KJ1403 - ■EM13 - ■■D1	50
	22	475	2.1	65.14	★	2KJ1403 - ■EM13 - ■■C1	50
	24	428	2.3	58.71		2KJ1403 - ■EM13 - ■■B1	50
	FZ.68B-LA90SB4E						
	24	446	1.9	61.17	★	2KJ1303 - ■EM13 - ■■B2	49
	FD.48B-LA90SB4E						
	15.9	660	0.82	90.53		2KJ1402 - ■EM13 - ■■H1	34
	17.6	596	0.91	81.73	★	2KJ1402 - ■EM13 - ■■G1	34
	19.4	541	1.0	74.1		2KJ1402 - ■EM13 - ■■F1	34
	21	492	1.1	67.43	★	2KJ1402 - ■EM13 - ■■E1	34
	24	435	1.2	59.62		2KJ1402 - ■EM13 - ■■D1	34
	26	402	1.3	55.06	★	2KJ1402 - ■EM13 - ■■C1	34
	30	346	1.6	47.4		2KJ1402 - ■EM13 - ■■B1	34
	33	314	1.7	43.09	★	2KJ1402 - ■EM13 - ■■A1	34
	FZ.48B-LA90SB4E						
	24	443	0.90	60.71	★	2KJ1302 - ■EM13 - ■■B2	34
	26	403	1.2	55.19		2KJ1302 - ■EM13 - ■■A2	34
	29	362	1.5	49.58	★	2KJ1302 - ■EM13 - ■■X1	34
	34	310	1.7	42.5		2KJ1302 - ■EM13 - ■■W1	34
	38	280	1.9	38.45	★	2KJ1302 - ■EM13 - ■■V1	34
	41	259	2.1	35.49		2KJ1302 - ■EM13 - ■■U1	34
	47	225	2.4	30.86	★	2KJ1302 - ■EM13 - ■■T1	34
	51	204	2.6	28.02		2KJ1302 - ■EM13 - ■■S1	34
	56	187	2.9	25.59	★	2KJ1302 - ■EM13 - ■■R1	34
	FZ.38B-LA90SB4E						
	35	298	0.92	40.88		2KJ1301 - ■EM13 - ■■W1	26

★ Preferred transmission ratio

Shaft designs, see page 3/89 ————— 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 ————— 1 to 9

Gearbox housing mounting position, see page 3/92 ————— A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.1	FZ.38B-LA90SB4E						
	40	262	1.1	35.96 ★	2KJ1301 - ■EM13 - ■■V1		26
	46	230	1.3	31.49	2KJ1301 - ■EM13 - ■■U1		26
	52	203	1.4	27.85 ★	2KJ1301 - ■EM13 - ■■T1		26
	57	184	1.6	25.24	2KJ1301 - ■EM13 - ■■S1		26
	65	163	1.8	22.28 ★	2KJ1301 - ■EM13 - ■■R1		26
	72	147	2.0	20.1	2KJ1301 - ■EM13 - ■■Q1		26
	79	133	2.2	18.23 ★	2KJ1301 - ■EM13 - ■■P1		26
	87	121	2.4	16.61	2KJ1301 - ■EM13 - ■■N1		26
	95	111	2.6	15.19 ★	2KJ1301 - ■EM13 - ■■M1		26
	106	99	2.9	13.58	2KJ1301 - ■EM13 - ■■L1		26
	115	91	3.2	12.47 ★	2KJ1301 - ■EM13 - ■■K1		26
	128	82	3.5	11.24	2KJ1301 - ■EM13 - ■■J1		26
	FZ.28-LA90SB4E						
	60	175	0.86	23.96 ★	2KJ1300 - ■EM13 - ■■T1		18
	66	158	0.95	21.64	2KJ1300 - ■EM13 - ■■S1		18
	76	138	1.1	18.86 ★	2KJ1300 - ■EM13 - ■■R1		18
	85	124	1.2	16.94	2KJ1300 - ■EM13 - ■■Q1		18
	94	112	1.3	15.29 ★	2KJ1300 - ■EM13 - ■■P1		18
	104	101	1.5	13.87	2KJ1300 - ■EM13 - ■■N1		18
	114	92	1.6	12.62 ★	2KJ1300 - ■EM13 - ■■M1		18
	129	81	1.7	11.16	2KJ1300 - ■EM13 - ■■L1		18
	140	75	1.8	10.3 ★	2KJ1300 - ■EM13 - ■■K1		18
	162	65	2.0	8.87	2KJ1300 - ■EM13 - ■■J1		18
	179	59	2.2	8.06 ★	2KJ1300 - ■EM13 - ■■H1		18
	200	52	2.4	7.2 ★	2KJ1300 - ■EM13 - ■■G1		18
	221	48	2.6	6.53	2KJ1300 - ■EM13 - ■■F1		18
	242	43	2.7	5.94 ★	2KJ1300 - ■EM13 - ■■E1		18
	274	38	2.9	5.25	2KJ1300 - ■EM13 - ■■D1		18
	297	35	3.1	4.85 ★	2KJ1300 - ■EM13 - ■■C1		18
	344	30	3.2	4.18	2KJ1300 - ■EM13 - ■■B1		18
	379	28	3.5	3.8 ★	2KJ1300 - ■EM13 - ■■A1		18
1.5	FD.188B-Z48-LA90ZLB4E						
	0.61	21 388	0.94	2 359 ★	2KJ1438 - ■EQ13 - ■■E1		648
	0.62	21 052	0.95	2 322 ★	2KJ1438 - ■EQ13 - ■■D1		648
	0.70	18 786	1.1	2 072	2KJ1438 - ■EQ13 - ■■C1		648
	0.82	15 957	1.3	1 760	2KJ1438 - ■EQ13 - ■■B1		648
	0.98	13 282	1.5	1 465 ★	2KJ1438 - ■EQ13 - ■■A1		648
	FD.188B-Z68-LA90ZLB4E						
	0.99	13 137	1.5	1 449	2KJ1441 - ■EQ13 - ■■H1		665
	1.2	11 206	1.8	1 236 ★	2KJ1441 - ■EQ13 - ■■G1		665
	FD.168B-Z48-LA90ZLB4E						
	0.76	17 145	0.82	1 891	2KJ1435 - ■EQ13 - ■■C1		464
	0.90	14 561	0.96	1 606	2KJ1435 - ■EQ13 - ■■B1		464

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.5	FD.168B-Z48-LA90ZLB4E						
	1.1	12 122	1.2	1 337	★	2KJ1435 - ■EQ13 - ■A1	464
	FD.168B-Z68-LA90ZLB4E						
	1.1	11 768	1.2	1 298		2KJ1437 - ■EQ13 - ■H1	481
	1.3	10 046	1.4	1 108	★	2KJ1437 - ■EQ13 - ■G1	481
	FD.148B-Z48-LA90ZLB4E						
	1.2	11 034	0.82	1 217	★	2KJ1434 - ■EQ13 - ■G1	306
	1.3	9 846	0.91	1 086		2KJ1434 - ■EQ13 - ■F1	306
	FD.148B-LA112M8						
	1.6	9 128	0.99	449.21	★	2KJ1407 - ■GG13 - ■U1 P02	323
	1.7	8 371	1.1	411.98		2KJ1407 - ■GG13 - ■T1 P02	323
	1.9	7 479	1.2	368.06	★	2KJ1407 - ■GG13 - ■S1 P02	323
	FD.148B-LA100ZLP6E						
	2.1	6 882	1.3	449.21	★	2KJ1407 - ■FM13 - ■U1 P01	316
	2.3	6 312	1.4	411.98		2KJ1407 - ■FM13 - ■T1 P01	316
	2.5	5 639	1.6	368.06	★	2KJ1407 - ■FM13 - ■S1 P01	316
	2.8	5 164	1.7	337.07		2KJ1407 - ■FM13 - ■R1 P01	316
	3.0	4 757	1.9	310.51	★	2KJ1407 - ■FM13 - ■Q1 P01	316
	FD.128B-LA100ZLP6E						
	2.6	5 439	1.1	354.99		2KJ1406 - ■FM13 - ■T1 P01	220
	2.9	4 906	1.2	320.24	★	2KJ1406 - ■FM13 - ■S1 P01	220
	FD.128B-LA90ZLB4E						
	3.2	4 456	1.4	447.96		2KJ1406 - ■EQ13 - ■V1	212
	3.6	4 034	1.5	405.47	★	2KJ1406 - ■EQ13 - ■U1	212
	4.1	3 531	1.7	354.99		2KJ1406 - ■EQ13 - ■T1	212
	4.5	3 186	1.9	320.24	★	2KJ1406 - ■EQ13 - ■S1	212
	FD.108B-LA90ZLB4E						
	3.4	4 223	0.81	424.49	★	2KJ1405 - ■EQ13 - ■V1	131
	3.8	3 808	0.89	382.79		2KJ1405 - ■EQ13 - ■U1	131
	4.2	3 434	0.99	345.19	★	2KJ1405 - ■EQ13 - ■T1	131
	4.8	3 003	1.1	301.88		2KJ1405 - ■EQ13 - ■S1	131
	5.3	2 696	1.3	271.01	★	2KJ1405 - ■EQ13 - ■R1	131
	5.8	2 462	1.4	247.53		2KJ1405 - ■EQ13 - ■Q1	131
	6.6	2 185	1.6	219.66	★	2KJ1405 - ■EQ13 - ■P1	131
	7.1	2 017	1.7	202.77		2KJ1405 - ■EQ13 - ■N1	131
	7.9	1 824	1.9	183.39	★	2KJ1405 - ■EQ13 - ■M1	131
	8.5	1 680	2.0	168.88		2KJ1405 - ■EQ13 - ■L1	131
	FD.88B-LA90ZLB4E						
	6.3	2 258	0.84	226.94	★	2KJ1404 - ■EQ13 - ■P1	84
	6.9	2 084	0.91	209.49		2KJ1404 - ■EQ13 - ■N1	84
	7.9	1 812	1.0	182.15	★	2KJ1404 - ■EQ13 - ■M1	84
	8.7	1 645	1.2	165.38		2KJ1404 - ■EQ13 - ■L1	84
	9.5	1 502	1.3	151.01	★	2KJ1404 - ■EQ13 - ■K1	84
	10.4	1 378	1.4	138.56		2KJ1404 - ■EQ13 - ■J1	84

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.5	FD.88B-LA90ZLB4E						
	11.3	1 270	1.5	127.66	★	2KJ1404 - ■EQ13 - ■■H1	84
	12.4	1 153	1.6	115.93		2KJ1404 - ■EQ13 - ■■G1	84
	13.6	1 051	1.8	105.61	★	2KJ1404 - ■EQ13 - ■■F1	84
	14.9	962	2.0	96.75		2KJ1404 - ■EQ13 - ■■E1	84
	16.7	859	2.2	86.33	★	2KJ1404 - ■EQ13 - ■■D1	84
	FD.68B-LA90ZLB4E						
	12.4	1 158	0.86	116.36	★	2KJ1403 - ■EQ13 - ■■J1	53
	13.7	1 044	0.96	104.96		2KJ1403 - ■EQ13 - ■■H1	53
	15.1	947	1.1	95.2	★	2KJ1403 - ■EQ13 - ■■G1	53
	16.6	863	1.2	86.74		2KJ1403 - ■EQ13 - ■■F1	53
	18.2	789	1.3	79.33	★	2KJ1403 - ■EQ13 - ■■E1	53
	20	706	1.4	70.93		2KJ1403 - ■EQ13 - ■■D1	53
	22	648	1.5	65.14	★	2KJ1403 - ■EQ13 - ■■C1	53
	24	584	1.7	58.71		2KJ1403 - ■EQ13 - ■■B1	53
	28	502	2.0	50.48	★	2KJ1403 - ■EQ13 - ■■A1	53
	FZ.68B-LA90ZLB4E						
	24	609	1.4	61.17	★	2KJ1303 - ■EQ13 - ■■B2	52
	27	532	1.9	53.5		2KJ1303 - ■EQ13 - ■■A2	52
	30	478	2.1	48.03	★	2KJ1303 - ■EQ13 - ■■X1	52
	33	436	2.3	43.87		2KJ1303 - ■EQ13 - ■■V1	52
	37	387	2.6	38.93	★	2KJ1303 - ■EQ13 - ■■U1	52
	FD.48B-LA90ZLB4E						
	21	671	0.81	67.43	★	2KJ1402 - ■EQ13 - ■■E1	37
	24	593	0.91	59.62		2KJ1402 - ■EQ13 - ■■D1	37
	26	548	0.99	55.06	★	2KJ1402 - ■EQ13 - ■■C1	37
	30	472	1.1	47.4		2KJ1402 - ■EQ13 - ■■B1	37
	33	429	1.3	43.09	★	2KJ1402 - ■EQ13 - ■■A1	37
	FZ.48B-LA90ZLB4E						
	26	549	0.91	55.19		2KJ1302 - ■EQ13 - ■■A2	37
	29	493	1.1	49.58	★	2KJ1302 - ■EQ13 - ■■X1	37
	34	423	1.3	42.5		2KJ1302 - ■EQ13 - ■■W1	37
	38	382	1.4	38.45	★	2KJ1302 - ■EQ13 - ■■V1	37
	41	353	1.5	35.49		2KJ1302 - ■EQ13 - ■■U1	37
	47	307	1.8	30.86	★	2KJ1302 - ■EQ13 - ■■T1	37
	51	279	1.9	28.02		2KJ1302 - ■EQ13 - ■■S1	37
	56	255	2.1	25.59	★	2KJ1302 - ■EQ13 - ■■R1	37
	61	234	2.3	23.48		2KJ1302 - ■EQ13 - ■■Q1	37
	67	215	2.5	21.63	★	2KJ1302 - ■EQ13 - ■■P1	37
	73	195	2.8	19.64		2KJ1302 - ■EQ13 - ■■N1	37
	80	178	3.0	17.89	★	2KJ1302 - ■EQ13 - ■■M1	37
	88	163	3.3	16.39		2KJ1302 - ■EQ13 - ■■L1	37

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
1.5	FZ.38B-LA90ZLB4E						
	40	358	0.81	35.96	★	2KJ1301 - ■EQ13 - ■■V1	29
	46	313	0.93	31.49		2KJ1301 - ■EQ13 - ■■U1	29
	52	277	1.0	27.85	★	2KJ1301 - ■EQ13 - ■■T1	29
	57	251	1.2	25.24		2KJ1301 - ■EQ13 - ■■S1	29
	65	222	1.3	22.28	★	2KJ1301 - ■EQ13 - ■■R1	29
	72	200	1.5	20.1		2KJ1301 - ■EQ13 - ■■Q1	29
	79	181	1.6	18.23	★	2KJ1301 - ■EQ13 - ■■P1	29
	87	165	1.8	16.61		2KJ1301 - ■EQ13 - ■■N1	29
	95	151	1.9	15.19	★	2KJ1301 - ■EQ13 - ■■M1	29
	106	135	2.1	13.58		2KJ1301 - ■EQ13 - ■■L1	29
	115	124	2.3	12.47	★	2KJ1301 - ■EQ13 - ■■K1	29
	128	112	2.6	11.24		2KJ1301 - ■EQ13 - ■■J1	29
	149	96	3.0	9.67	★	2KJ1301 - ■EQ13 - ■■H1	29
	169	85	3.4	8.52	★	2KJ1301 - ■EQ13 - ■■G1	29
	186	77	3.8	7.76		2KJ1301 - ■EQ13 - ■■F1	29
	203	71	4.1	7.1	★	2KJ1301 - ■EQ13 - ■■E1	29
	227	63	4.4	6.35		2KJ1301 - ■EQ13 - ■■D1	29
	247	58	4.7	5.83	★	2KJ1301 - ■EQ13 - ■■C1	29
	274	52	4.8	5.25		2KJ1301 - ■EQ13 - ■■B1	29
	319	45	5.1	4.52	★	2KJ1301 - ■EQ13 - ■■A1	29
	FZ.38B-LA90SB2E						
	159	90	3.2	18.23	★	2KJ1301 - ■EM13 - ■■P1 P00	26
	174	82	3.5	16.61		2KJ1301 - ■EM13 - ■■N1 P00	26
	190	75	3.9	15.19	★	2KJ1301 - ■EM13 - ■■M1 P00	26
	213	67	4.3	13.58		2KJ1301 - ■EM13 - ■■L1 P00	26
	FZ.28-LA90ZLB4E						
	76	188	0.80	18.86	★	2KJ1300 - ■EQ13 - ■■R1	21
	85	169	0.89	16.94		2KJ1300 - ■EQ13 - ■■Q1	21
	94	152	0.99	15.29	★	2KJ1300 - ■EQ13 - ■■P1	21
	104	138	1.1	13.87		2KJ1300 - ■EQ13 - ■■N1	21
	114	126	1.2	12.62	★	2KJ1300 - ■EQ13 - ■■M1	21
	129	111	1.3	11.16		2KJ1300 - ■EQ13 - ■■L1	21
	140	102	1.3	10.3	★	2KJ1300 - ■EQ13 - ■■K1	21
	162	88	1.5	8.87		2KJ1300 - ■EQ13 - ■■J1	21
	179	80	1.6	8.06	★	2KJ1300 - ■EQ13 - ■■H1	21
	200	72	1.8	7.2	★	2KJ1300 - ■EQ13 - ■■G1	21
	221	65	1.9	6.53		2KJ1300 - ■EQ13 - ■■F1	21
	242	59	2.0	5.94	★	2KJ1300 - ■EQ13 - ■■E1	21
	274	52	2.1	5.25		2KJ1300 - ■EQ13 - ■■D1	21
	297	48	2.3	4.85	★	2KJ1300 - ■EQ13 - ■■C1	21
	344	42	2.4	4.18		2KJ1300 - ■EQ13 - ■■B1	21
	379	38	2.5	3.8	★	2KJ1300 - ■EQ13 - ■■A1	21

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
2.2	FD.188B-Z48-LA100ZLP4E						
	0.82	23 638	0.85	1 760	2KJ1438 - ■ FM13 - ■■ B1		656
	0.98	19 676	1.0	1 465	★ 2KJ1438 - ■ FM13 - ■■ A1		656
	FD.188B-Z68-LA100ZLP4E						
	0.99	19 461	1.0	1 449	2KJ1441 - ■ FM13 - ■■ H1		673
	1.2	16 600	1.2	1 236	★ 2KJ1441 - ■ FM13 - ■■ G1		673
	1.4	13 833	1.4	1 030	2KJ1441 - ■ FM13 - ■■ F1		673
	FD.188B-LA132S8						
	1.7	12 122	1.6	403.86	★ 2KJ1410 - ■ HE13 - ■■ U1	P02	676
	1.9	11 121	1.8	370.52	2KJ1410 - ■ HE13 - ■■ T1	P02	676
	2.0	10 263	1.9	341.94	★ 2KJ1410 - ■ HE13 - ■■ S1	P02	676
	FD.168B-Z68-LA100ZLP4E						
	1.1	17 433	0.80	1 298	2KJ1437 - ■ FM13 - ■■ H1		489
	1.3	14 881	0.94	1 108	★ 2KJ1437 - ■ FM13 - ■■ G1		489
	1.6	12 396	1.1	923	2KJ1437 - ■ FM13 - ■■ F1		489
	FD.168B-LA132S8						
	1.9	11 083	1.3	369.26	★ 2KJ1408 - ■ HE13 - ■■ V1	P02	495
	2.1	10 160	1.4	338.49	2KJ1408 - ■ HE13 - ■■ U1	P02	495
	2.2	9 368	1.5	312.12	★ 2KJ1408 - ■ HE13 - ■■ T1	P02	495
	2.4	8 682	1.6	289.26	2KJ1408 - ■ HE13 - ■■ S1	P02	495
	2.5	8 255	1.7	275.03	★ 2KJ1408 - ■ HE13 - ■■ R1	P02	495
	FD.148B-LA132S8						
	1.9	11 047	0.81	368.06	★ 2KJ1407 - ■ HE13 - ■■ S1	P02	333
	FD.148B-LA112ZMP6E						
	2.1	9 883	0.91	449.21	★ 2KJ1407 - ■ GJ13 - ■■ U1	P01	323
	2.3	9 064	0.99	411.98	2KJ1407 - ■ GJ13 - ■■ T1	P01	323
	2.6	8 097	1.1	368.06	★ 2KJ1407 - ■ GJ13 - ■■ S1	P01	323
	2.8	7 416	1.2	337.07	2KJ1407 - ■ GJ13 - ■■ R1	P01	323
	3.1	6 831	1.3	310.51	★ 2KJ1407 - ■ GJ13 - ■■ Q1	P01	323
	FD.148B-LA100ZLP4E						
	3.2	6 577	1.4	449.21	★ 2KJ1407 - ■ FM13 - ■■ U1		316
	3.5	6 032	1.5	411.98	2KJ1407 - ■ FM13 - ■■ T1		316
	3.9	5 389	1.7	368.06	★ 2KJ1407 - ■ FM13 - ■■ S1		316
	4.3	4 935	1.8	337.07	2KJ1407 - ■ FM13 - ■■ R1		316
	4.6	4 546	2.0	310.51	★ 2KJ1407 - ■ FM13 - ■■ Q1		316
	5.0	4 209	2.1	287.49	2KJ1407 - ■ FM13 - ■■ P1		316
	FD.128B-LA112ZMP6E						
	3.0	7 045	0.87	320.24	★ 2KJ1406 - ■ GJ13 - ■■ S1	P01	227
	FD.128B-LA100ZLP4E						
	4.0	5 197	1.2	354.99	2KJ1406 - ■ FM13 - ■■ T1		220
	4.5	4 689	1.3	320.24	★ 2KJ1406 - ■ FM13 - ■■ S1		220
	4.9	4 293	1.4	293.22	2KJ1406 - ■ FM13 - ■■ R1		220
	5.5	3 819	1.6	260.84	★ 2KJ1406 - ■ FM13 - ■■ Q1		220
	6	3 490	1.7	238.39	2KJ1406 - ■ FM13 - ■■ P1		220

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
2.2	FD.128B-LA100ZLP4E						
	6.5	3 209	1.9	219.15	★	2KJ1406 - ■ FM13 - ■■ N1	220
	7.1	2 965	2.1	202.48		2KJ1406 - ■ FM13 - ■■ M1	220
	FD.108B-LA100ZLP4E						
	5.3	3 968	0.86	271.01	★	2KJ1405 - ■ FM13 - ■■ R1	139
	5.8	3 624	0.94	247.53		2KJ1405 - ■ FM13 - ■■ Q1	139
	6.5	3 216	1.1	219.66	★	2KJ1405 - ■ FM13 - ■■ P1	139
	7.1	2 969	1.1	202.77		2KJ1405 - ■ FM13 - ■■ N1	139
	7.8	2 685	1.3	183.39	★	2KJ1405 - ■ FM13 - ■■ M1	139
	8.5	2 473	1.4	168.88		2KJ1405 - ■ FM13 - ■■ L1	139
	9.2	2 287	1.5	156.19	★	2KJ1405 - ■ FM13 - ■■ K1	139
	9.9	2 123	1.6	144.99		2KJ1405 - ■ FM13 - ■■ J1	139
	11.2	1 873	1.8	127.92	★	2KJ1405 - ■ FM13 - ■■ H1	139
	12.1	1 729	2.0	118.11		2KJ1405 - ■ FM13 - ■■ G1	139
	13.6	1 549	2.2	105.81	★	2KJ1405 - ■ FM13 - ■■ F1	139
	FD.88B-LA100ZLP4E						
	9.5	2 211	0.86	151.01	★	2KJ1404 - ■ FM13 - ■■ K1	92
	10.4	2 029	0.94	138.56		2KJ1404 - ■ FM13 - ■■ J1	92
	11.2	1 869	1.0	127.66	★	2KJ1404 - ■ FM13 - ■■ H1	92
	12.4	1 697	1.1	115.93		2KJ1404 - ■ FM13 - ■■ G1	92
	13.6	1 546	1.2	105.61	★	2KJ1404 - ■ FM13 - ■■ F1	92
	14.8	1 417	1.3	96.75		2KJ1404 - ■ FM13 - ■■ E1	92
	16.6	1 264	1.5	86.33	★	2KJ1404 - ■ FM13 - ■■ D1	92
	18.6	1 128	1.7	77.04		2KJ1404 - ■ FM13 - ■■ C1	92
	22	958	2.0	65.43		2KJ1404 - ■ FM13 - ■■ B1	92
	26	798	2.4	54.47	★	2KJ1404 - ■ FM13 - ■■ A1	92
	FZ.88B-LA100ZLP4E						
	22	946	2.0	64.58	★	2KJ1304 - ■ FM13 - ■■ X1	91
	24	866	2.2	59.13		2KJ1304 - ■ FM13 - ■■ W1	91
	27	770	2.5	52.6	★	2KJ1304 - ■ FM13 - ■■ V1	91
	FD.68B-LA100ZLP4E						
	18.1	1 161	0.86	79.33	★	2KJ1403 - ■ FM13 - ■■ E1	61
	20	1 038	0.96	70.93		2KJ1403 - ■ FM13 - ■■ D1	61
	22	954	1.0	65.14	★	2KJ1403 - ■ FM13 - ■■ C1	61
	24	860	1.2	58.71		2KJ1403 - ■ FM13 - ■■ B1	61
	28	739	1.4	50.48	★	2KJ1403 - ■ FM13 - ■■ A1	61
	FZ.68B-LA100ZLP4E						
	27	783	1.3	53.5		2KJ1303 - ■ FM13 - ■■ A2	60
	30	703	1.4	48.03	★	2KJ1303 - ■ FM13 - ■■ X1	60
	33	642	1.6	43.87		2KJ1303 - ■ FM13 - ■■ V1	60
	37	570	1.8	38.93	★	2KJ1303 - ■ FM13 - ■■ U1	60
	40	526	1.9	35.93		2KJ1303 - ■ FM13 - ■■ T1	60
	44	476	2.1	32.5	★	2KJ1303 - ■ FM13 - ■■ S1	60
	48	438	2.3	29.93		2KJ1303 - ■ FM13 - ■■ R1	60

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
2.2	FZ.68B-LA100ZLP4E						
	52	405	2.5	27.68	★	2KJ1303 - ■ FM13 - ■■ Q1	60
	56	376	2.7	25.69		2KJ1303 - ■ FM13 - ■■ P1	60
	63	332	3.0	22.67	★	2KJ1303 - ■ FM13 - ■■ N1	60
	FD.48B-LA100ZLP4E						
	33	631	0.86	43.09	★	2KJ1402 - ■ FM13 - ■■ A1	45
	FZ.48B-LA100ZLP4E						
	34	622	0.87	42.5		2KJ1302 - ■ FM13 - ■■ W1	45
	37	563	0.96	38.45	★	2KJ1302 - ■ FM13 - ■■ V1	45
	40	520	1.0	35.49		2KJ1302 - ■ FM13 - ■■ U1	45
	46	452	1.2	30.86	★	2KJ1302 - ■ FM13 - ■■ T1	45
	51	410	1.3	28.02		2KJ1302 - ■ FM13 - ■■ S1	45
	56	375	1.4	25.59	★	2KJ1302 - ■ FM13 - ■■ R1	45
	61	344	1.6	23.48		2KJ1302 - ■ FM13 - ■■ Q1	45
	66	317	1.7	21.63	★	2KJ1302 - ■ FM13 - ■■ P1	45
	73	288	1.9	19.64		2KJ1302 - ■ FM13 - ■■ N1	45
	80	262	2.1	17.89	★	2KJ1302 - ■ FM13 - ■■ M1	45
	88	240	2.3	16.39		2KJ1302 - ■ FM13 - ■■ L1	45
	98	214	2.5	14.63	★	2KJ1302 - ■ FM13 - ■■ K1	45
	110	191	2.8	13.05		2KJ1302 - ■ FM13 - ■■ J1	45
	129	162	3.3	11.09		2KJ1302 - ■ FM13 - ■■ H1	45
	155	135	3.9	9.23	★	2KJ1302 - ■ FM13 - ■■ G1	45
	171	123	4.2	8.39	★	2KJ1302 - ■ FM13 - ■■ F1	45
	187	112	4.2	7.68		2KJ1302 - ■ FM13 - ■■ E1	45
	209	100	4.4	6.86	★	2KJ1302 - ■ FM13 - ■■ D1	45
	234	90	4.5	6.12		2KJ1302 - ■ FM13 - ■■ C1	45
	331	63	5.1	4.33	★	2KJ1302 - ■ FM13 - ■■ A1	45
	FZ.38B-LA100ZLP4E						
	64	326	0.89	22.28	★	2KJ1301 - ■ FM13 - ■■ R1	37
	71	294	0.99	20.1		2KJ1301 - ■ FM13 - ■■ Q1	37
	79	267	1.1	18.23	★	2KJ1301 - ■ FM13 - ■■ P1	37
	86	243	1.2	16.61		2KJ1301 - ■ FM13 - ■■ N1	37
	94	222	1.3	15.19	★	2KJ1301 - ■ FM13 - ■■ M1	37
	106	199	1.5	13.58		2KJ1301 - ■ FM13 - ■■ L1	37
	115	183	1.6	12.47	★	2KJ1301 - ■ FM13 - ■■ K1	37
	128	165	1.8	11.24		2KJ1301 - ■ FM13 - ■■ J1	37
	148	142	2.0	9.67	★	2KJ1301 - ■ FM13 - ■■ H1	37
	168	125	2.3	8.52	★	2KJ1301 - ■ FM13 - ■■ G1	37
	185	114	2.6	7.76		2KJ1301 - ■ FM13 - ■■ F1	37
	202	104	2.8	7.1	★	2KJ1301 - ■ FM13 - ■■ E1	37
	226	93	3.0	6.35		2KJ1301 - ■ FM13 - ■■ D1	37
	246	85	3.2	5.83	★	2KJ1301 - ■ FM13 - ■■ C1	37
	273	77	3.3	5.25		2KJ1301 - ■ FM13 - ■■ B1	37
	317	66	3.4	4.52	★	2KJ1301 - ■ FM13 - ■■ A1	37

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
2.2	FZ.38B-LA90ZLB2E						
	159	133	2.2	18.23 ★	2KJ1301 - ■EQ13 - ■■P1	P00	29
	174	121	2.4	16.61	2KJ1301 - ■EQ13 - ■■N1	P00	29
	190	110	2.6	15.19 ★	2KJ1301 - ■EQ13 - ■■M1	P00	29
	213	99	2.9	13.58	2KJ1301 - ■EQ13 - ■■L1	P00	29
	232	91	3.2	12.47 ★	2KJ1301 - ■EQ13 - ■■K1	P00	29
	257	82	3.5	11.24	2KJ1301 - ■EQ13 - ■■J1	P00	29
	299	70	4.1	9.67 ★	2KJ1301 - ■EQ13 - ■■H1	P00	29
	339	62	4.7	8.52 ★	2KJ1301 - ■EQ13 - ■■G1	P00	29
	372	56	5.1	7.76	2KJ1301 - ■EQ13 - ■■F1	P00	29
3	FD.188B-Z68-LA100ZLD4E						
	1.2	22 720	0.88	1236 ★	2KJ1441 - ■FP13 - ■■G1		673
	1.4	18 933	1.1	1030	2KJ1441 - ■FP13 - ■■F1		673
	FD.188B-LA132MA8						
	1.7	16 529	1.2	403.86 ★	2KJ1410 - ■HG13 - ■■U1	P02	684
	1.9	15 165	1.3	370.52	2KJ1410 - ■HG13 - ■■T1	P02	684
	2.0	13 995	1.4	341.94 ★	2KJ1410 - ■HG13 - ■■S1	P02	684
	2.2	12 982	1.5	317.18	2KJ1410 - ■HG13 - ■■R1	P02	684
	2.3	12 246	1.6	299.2 ★	2KJ1410 - ■HG13 - ■■Q1	P02	684
	FD.188B-LA132SB6E						
	2.4	12 180	1.6	403.86 ★	2KJ1410 - ■HF13 - ■■U1	P01	684
	2.6	11 174	1.8	370.52	2KJ1410 - ■HF13 - ■■T1	P01	684
	2.8	10 312	1.9	341.94 ★	2KJ1410 - ■HF13 - ■■S1	P01	684
	FD.168B-Z68-LA100ZLD4E						
	1.6	16 967	0.83	923	2KJ1437 - ■FP13 - ■■F1		489
	FD.168B-LA132MA8						
	1.9	15 113	0.93	369.26 ★	2KJ1408 - ■HG13 - ■■V1	P02	503
	2.1	13 854	1.0	338.49	2KJ1408 - ■HG13 - ■■U1	P02	503
	2.2	12 775	1.1	312.12 ★	2KJ1408 - ■HG13 - ■■T1	P02	503
	2.4	11 839	1.2	289.26	2KJ1408 - ■HG13 - ■■S1	P02	503
	2.5	11 257	1.2	275.03 ★	2KJ1408 - ■HG13 - ■■R1	P02	503
	FD.168B-LA132SB6E						
	2.6	11 136	1.3	369.26 ★	2KJ1408 - ■HF13 - ■■V1	P01	503
	2.8	10 208	1.4	338.49	2KJ1408 - ■HF13 - ■■U1	P01	503
	3.0	9 413	1.5	312.12 ★	2KJ1408 - ■HF13 - ■■T1	P01	503
	3.3	8 723	1.6	289.26	2KJ1408 - ■HF13 - ■■S1	P01	503
	3.5	8 294	1.7	275.03 ★	2KJ1408 - ■HF13 - ■■R1	P01	503
	3.7	7 752	1.8	257.04	2KJ1408 - ■HF13 - ■■Q1	P01	503
	FD.148B-LA132SB6E						
	2.6	11 100	0.81	368.06 ★	2KJ1407 - ■HF13 - ■■S1	P01	341
	2.8	10 165	0.89	337.07	2KJ1407 - ■HF13 - ■■R1	P01	341
	3.1	9 364	0.96	310.51 ★	2KJ1407 - ■HF13 - ■■Q1	P01	341
	FD.148B-LA100ZLD4E						
	3.2	8 969	1.0	449.21 ★	2KJ1407 - ■FP13 - ■■U1		316

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
3	FD.148B-LA100ZLD4E						
	3.5	8 225	1.1	411.98	2KJ1407 - ■FP13 - ■■T1		316
	3.9	7 348	1.2	368.06	★ 2KJ1407 - ■FP13 - ■■S1		316
	4.3	6 730	1.3	337.07	2KJ1407 - ■FP13 - ■■R1		316
	4.6	6 199	1.5	310.51	★ 2KJ1407 - ■FP13 - ■■Q1		316
	5.0	5 740	1.6	287.49	2KJ1407 - ■FP13 - ■■P1		316
	5.4	5 338	1.7	267.35	★ 2KJ1407 - ■FP13 - ■■N1		316
	5.7	4 983	1.8	249.58	2KJ1407 - ■FP13 - ■■M1		316
	6.4	4 458	2.0	223.31	★ 2KJ1407 - ■FP13 - ■■L1		316
	FD.128B-LA100ZLD4E						
	4.0	7 087	0.86	354.99	2KJ1406 - ■FP13 - ■■T1		220
	4.5	6 394	0.95	320.24	★ 2KJ1406 - ■FP13 - ■■S1		220
	4.9	5 854	1.0	293.22	2KJ1406 - ■FP13 - ■■R1		220
	5.5	5 208	1.2	260.84	★ 2KJ1406 - ■FP13 - ■■Q1		220
	6.0	4 759	1.3	238.39	2KJ1406 - ■FP13 - ■■P1		220
	6.5	4 375	1.4	219.15	★ 2KJ1406 - ■FP13 - ■■N1		220
	7.1	4 043	1.5	202.48	2KJ1406 - ■FP13 - ■■M1		220
	7.6	3 751	1.6	187.88	★ 2KJ1406 - ■FP13 - ■■L1		220
	8.2	3 494	1.7	175.01	2KJ1406 - ■FP13 - ■■K1		220
	9.1	3 159	1.9	158.22	★ 2KJ1406 - ■FP13 - ■■J1		220
	9.9	2 908	2.1	145.66	2KJ1406 - ■FP13 - ■■H1		220
	FD.108B-LA100ZLD4E						
	7.1	4 048	0.84	202.77	2KJ1405 - ■FP13 - ■■N1		139
	7.8	3 661	0.93	183.39	★ 2KJ1405 - ■FP13 - ■■M1		139
	8.5	3 372	1.0	168.88	2KJ1405 - ■FP13 - ■■L1		139
	9.2	3 118	1.1	156.19	★ 2KJ1405 - ■FP13 - ■■K1		139
	9.9	2 895	1.2	144.99	2KJ1405 - ■FP13 - ■■J1		139
	11.2	2 554	1.3	127.92	★ 2KJ1405 - ■FP13 - ■■H1		139
	12.1	2 358	1.4	118.11	2KJ1405 - ■FP13 - ■■G1		139
	13.6	2 113	1.6	105.81	★ 2KJ1405 - ■FP13 - ■■F1		139
	14.7	1 948	1.7	97.57	2KJ1405 - ■FP13 - ■■E1		139
	17.5	1 634	2.1	81.86	2KJ1405 - ■FP13 - ■■D1		139
	FZ.108B-LA100ZLD4E						
	22	1 282	2.3	64.21	★ 2KJ1305 - ■FP13 - ■■A2		138
	FD.88B-LA100ZLD4E						
	12.4	2 315	0.82	115.93	2KJ1404 - ■FP13 - ■■G1		92
	13.6	2 109	0.90	105.61	★ 2KJ1404 - ■FP13 - ■■F1		92
	14.8	1 932	0.98	96.75	2KJ1404 - ■FP13 - ■■E1		92
	16.6	1 724	1.1	86.33	★ 2KJ1404 - ■FP13 - ■■D1		92
	18.6	1 538	1.2	77.04	2KJ1404 - ■FP13 - ■■C1		92
	22	1 306	1.5	65.43	2KJ1404 - ■FP13 - ■■B1		92
	26	1 088	1.7	54.47	★ 2KJ1404 - ■FP13 - ■■A1		92
	FZ.88B-LA100ZLD4E						
	22	1289	1.5	64.58	★ 2KJ1304 - ■FP13 - ■■X1		91

★ Preferred transmission ratio

Shaft designs, see page 3/89 ————— 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 ————— 1 to 9

Gearbox housing mounting position, see page 3/92 ————— A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
3	FZ.88B-LA100ZLD4E						
	24	1 181	1.6	59.13	2KJ1304 - ■FP13 - ■■W1		91
	27	1 050	1.8	52.6 ★	2KJ1304 - ■FP13 - ■■V1		91
	30	959	2.0	48.03	2KJ1304 - ■FP13 - ■■U1		91
	32	882	2.2	44.2 ★	2KJ1304 - ■FP13 - ■■T1		91
	35	815	2.3	40.83	2KJ1304 - ■FP13 - ■■S1		91
	38	756	2.5	37.89 ★	2KJ1304 - ■FP13 - ■■R1		91
	41	705	2.7	35.29	2KJ1304 - ■FP13 - ■■Q1		91
	FD.68B-LA100ZLD4E						
	24	1 172	0.85	58.71	2KJ1403 - ■FP13 - ■■B1		61
	28	1 008	0.99	50.48 ★	2KJ1403 - ■FP13 - ■■A1		61
	FZ.68B-LA100ZLD4E						
	27	1 068	0.94	53.5	2KJ1303 - ■FP13 - ■■A2		60
	30	959	1.0	48.03 ★	2KJ1303 - ■FP13 - ■■X1		60
	33	876	1.1	43.87	2KJ1303 - ■FP13 - ■■V1		60
	37	777	1.3	38.93 ★	2KJ1303 - ■FP13 - ■■U1		60
	40	717	1.4	35.93	2KJ1303 - ■FP13 - ■■T1		60
	44	649	1.5	32.5 ★	2KJ1303 - ■FP13 - ■■S1		60
	48	598	1.7	29.93	2KJ1303 - ■FP13 - ■■R1		60
	52	553	1.8	27.68 ★	2KJ1303 - ■FP13 - ■■Q1		60
	56	513	1.9	25.69	2KJ1303 - ■FP13 - ■■P1		60
	63	453	2.2	22.67 ★	2KJ1303 - ■FP13 - ■■N1		60
	69	418	2.4	20.93	2KJ1303 - ■FP13 - ■■M1		60
	76	374	2.7	18.75 ★	2KJ1303 - ■FP13 - ■■L1		60
	83	345	2.9	17.29	2KJ1303 - ■FP13 - ■■K1		60
	99	290	3.5	14.51	2KJ1303 - ■FP13 - ■■J1		60
	FZ.48B-LA100ZLD4E						
	46	616	0.88	30.86 ★	2KJ1302 - ■FP13 - ■■T1		45
	51	559	0.97	28.02	2KJ1302 - ■FP13 - ■■S1		45
	56	511	1.1	25.59 ★	2KJ1302 - ■FP13 - ■■R1		45
	61	469	1.2	23.48	2KJ1302 - ■FP13 - ■■Q1		45
	66	432	1.3	21.63 ★	2KJ1302 - ■FP13 - ■■P1		45
	73	392	1.4	19.64	2KJ1302 - ■FP13 - ■■N1		45
	80	357	1.5	17.89 ★	2KJ1302 - ■FP13 - ■■M1		45
	88	327	1.7	16.39	2KJ1302 - ■FP13 - ■■L1		45
	98	292	1.8	14.63 ★	2KJ1302 - ■FP13 - ■■K1		45
	110	261	2.1	13.05	2KJ1302 - ■FP13 - ■■J1		45
	129	221	2.4	11.09	2KJ1302 - ■FP13 - ■■H1		45
	155	184	2.9	9.23 ★	2KJ1302 - ■FP13 - ■■G1		45
	171	168	3.0	8.39 ★	2KJ1302 - ■FP13 - ■■F1		45
	187	153	3.0	7.68	2KJ1302 - ■FP13 - ■■E1		45
	209	137	3.2	6.86 ★	2KJ1302 - ■FP13 - ■■D1		45
	234	122	3.3	6.12	2KJ1302 - ■FP13 - ■■C1		45
	276	104	3.6	5.2	2KJ1302 - ■FP13 - ■■B1		45
	331	86	3.8	4.33 ★	2KJ1302 - ■FP13 - ■■A1		45

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg	
3	FZ.48B-LA100ZLB2E							
	162	177	3.0	17.89	★	2KJ1302 - ■ FM13 - ■ ■ M1	P00	45
	176	162	3.3	16.39		2KJ1302 - ■ FM13 - ■ ■ L1	P00	45
	198	145	3.7	14.63	★	2KJ1302 - ■ FM13 - ■ ■ K1	P00	45
	221	129	4.2	13.05		2KJ1302 - ■ FM13 - ■ ■ J1	P00	45
	FZ.38B-LA100ZLD4E							
	79	364	0.80	18.23	★	2KJ1301 - ■ FP13 - ■ ■ P1		37
	86	332	0.87	16.61		2KJ1301 - ■ FP13 - ■ ■ N1		37
	94	303	0.96	15.19	★	2KJ1301 - ■ FP13 - ■ ■ M1		37
	106	271	1.1	13.58		2KJ1301 - ■ FP13 - ■ ■ L1		37
	115	249	1.2	12.47	★	2KJ1301 - ■ FP13 - ■ ■ K1		37
	128	224	1.3	11.24		2KJ1301 - ■ FP13 - ■ ■ J1		37
	148	193	1.5	9.67	★	2KJ1301 - ■ FP13 - ■ ■ H1		37
	168	170	1.7	8.52	★	2KJ1301 - ■ FP13 - ■ ■ G1		37
	185	155	1.9	7.76		2KJ1301 - ■ FP13 - ■ ■ F1		37
	202	142	2.0	7.1	★	2KJ1301 - ■ FP13 - ■ ■ E1		37
	226	127	2.2	6.35		2KJ1301 - ■ FP13 - ■ ■ D1		37
	246	116	2.4	5.83	★	2KJ1301 - ■ FP13 - ■ ■ C1		37
	273	105	2.4	5.25		2KJ1301 - ■ FP13 - ■ ■ B1		37
	317	90	2.5	4.52	★	2KJ1301 - ■ FP13 - ■ ■ A1		37
	FZ.38B-LA100ZLB2E							
	159	181	1.6	18.23	★	2KJ1301 - ■ FM13 - ■ ■ P1	P00	37
	174	165	1.8	16.61		2KJ1301 - ■ FM13 - ■ ■ N1	P00	37
	190	151	1.9	15.19	★	2KJ1301 - ■ FM13 - ■ ■ M1	P00	37
	213	135	2.2	13.58		2KJ1301 - ■ FM13 - ■ ■ L1	P00	37
	232	124	2.3	12.47	★	2KJ1301 - ■ FM13 - ■ ■ K1	P00	37
	257	111	2.6	11.24		2KJ1301 - ■ FM13 - ■ ■ J1	P00	37
	299	96	3.0	9.67	★	2KJ1301 - ■ FM13 - ■ ■ H1	P00	37
	339	84	3.4	8.52	★	2KJ1301 - ■ FM13 - ■ ■ G1	P00	37
	372	77	3.8	7.76		2KJ1301 - ■ FM13 - ■ ■ F1	P00	37
	407	70	4.1	7.1	★	2KJ1301 - ■ FM13 - ■ ■ E1	P00	37
	455	63	4.4	6.35		2KJ1301 - ■ FM13 - ■ ■ D1	P00	37
	496	58	4.8	5.83	★	2KJ1301 - ■ FM13 - ■ ■ C1	P00	37
	550	52	4.9	5.25		2KJ1301 - ■ FM13 - ■ ■ B1	P00	37
	639	45	5.1	4.52	★	2KJ1301 - ■ FM13 - ■ ■ A1	P00	37
	FZ.28-LA100ZLD4E							
	199	144	0.88	7.2	★	2KJ1300 - ■ FP13 - ■ ■ G1		29
	220	130	0.94	6.53		2KJ1300 - ■ FP13 - ■ ■ F1		29
	242	119	0.99	5.94	★	2KJ1300 - ■ FP13 - ■ ■ E1		29
	273	105	1.1	5.25		2KJ1300 - ■ FP13 - ■ ■ D1		29
	296	97	1.1	4.85	★	2KJ1300 - ■ FP13 - ■ ■ C1		29
	343	84	1.2	4.18		2KJ1300 - ■ FP13 - ■ ■ B1		29
	378	76	1.3	3.8	★	2KJ1300 - ■ FP13 - ■ ■ A1		29

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
4	FD.188B-Z68-LA112ZMP4						
	1.4	24 905	0.80	1 030	2KJ1441 - GJ13 - F1		680
	FD.188B-LA132ZMB6E						
	2.4	16 239	1.2	403.86 ★	2KJ1410 - HJ13 - U1	P01	684
	2.6	14 899	1.3	370.52	2KJ1410 - HJ13 - T1	P01	684
	2.8	13 750	1.5	341.94 ★	2KJ1410 - HJ13 - S1	P01	684
	3.0	12 754	1.6	317.18	2KJ1410 - HJ13 - R1	P01	684
	3.2	12 031	1.7	299.2 ★	2KJ1410 - HJ13 - Q1	P01	684
	3.4	11 253	1.8	279.86	2KJ1410 - HJ13 - P1	P01	684
	FD.168B-LA132ZMB6E						
	2.6	14 848	0.94	369.26 ★	2KJ1408 - HJ13 - V1	P01	503
	2.8	13 611	1.0	338.49	2KJ1408 - HJ13 - U1	P01	503
	3.0	12 551	1.1	312.12 ★	2KJ1408 - HJ13 - T1	P01	503
	3.3	11 631	1.2	289.26	2KJ1408 - HJ13 - S1	P01	503
	3.5	11 059	1.3	275.03 ★	2KJ1408 - HJ13 - R1	P01	503
	3.7	10 336	1.4	257.04	2KJ1408 - HJ13 - Q1	P01	503
	FD.148B-LA112ZMP4E						
	3.5	10 929	0.82	411.98	2KJ1407 - GJ13 - T1		323
	3.9	9 764	0.92	368.06 ★	2KJ1407 - GJ13 - S1		323
	4.3	8 942	1.0	337.07	2KJ1407 - GJ13 - R1		323
	4.6	8 237	1.1	310.51 ★	2KJ1407 - GJ13 - Q1		323
	5.0	7 626	1.2	287.49	2KJ1407 - GJ13 - P1		323
	5.4	7 092	1.3	267.35 ★	2KJ1407 - GJ13 - N1		323
	5.8	6 621	1.4	249.58	2KJ1407 - GJ13 - M1		323
	6.4	5 924	1.5	223.31 ★	2KJ1407 - GJ13 - L1		323
	7.0	5 489	1.6	206.93	2KJ1407 - GJ13 - K1		323
	7.6	5 032	1.8	189.69 ★	2KJ1407 - GJ13 - J1		323
	8.3	4 613	2.0	173.89	2KJ1407 - GJ13 - H1		323
	FD.128B-LA112ZMP4E						
	5.5	6 920	0.88	260.84 ★	2KJ1406 - GJ13 - Q1		227
	6.0	6 324	0.96	238.39	2KJ1406 - GJ13 - P1		227
	6.6	5 814	1.0	219.15 ★	2KJ1406 - GJ13 - N1		227
	7.1	5 371	1.1	202.48	2KJ1406 - GJ13 - M1		227
	7.7	4 984	1.2	187.88 ★	2KJ1406 - GJ13 - L1		227
	8.2	4 643	1.3	175.01	2KJ1406 - GJ13 - K1		227
	9.1	4 197	1.5	158.22 ★	2KJ1406 - GJ13 - J1		227
	9.9	3 864	1.6	145.66	2KJ1406 - GJ13 - H1		227
	11.0	3 475	1.8	131.01 ★	2KJ1406 - GJ13 - G1		227
	11.9	3 206	1.9	120.87	2KJ1406 - GJ13 - F1		227
	14.1	2 717	2.2	102.41	2KJ1406 - GJ13 - E1		227

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
4	FD.108B-LA112ZMP4E						
	9.2	4 143	0.82	156.19	★	2KJ1405 - ■GJ13 - ■■K1	146
	9.9	3 846	0.88	144.99		2KJ1405 - ■GJ13 - ■■J1	146
	11.3	3 393	1.0	127.92	★	2KJ1405 - ■GJ13 - ■■H1	146
	12.2	3 133	1.1	118.11		2KJ1405 - ■GJ13 - ■■G1	146
	13.6	2 807	1.2	105.81	★	2KJ1405 - ■GJ13 - ■■F1	146
	14.8	2 588	1.3	97.57		2KJ1405 - ■GJ13 - ■■E1	146
	17.6	2 172	1.6	81.86		2KJ1405 - ■GJ13 - ■■D1	146
	21	1 853	1.8	69.84	★	2KJ1405 - ■GJ13 - ■■C1	146
	25	1 544	2.2	58.2		2KJ1405 - ■GJ13 - ■■B1	146
	FZ.108B-LA112ZMP4E						
	22	1 703	1.8	64.21	★	2KJ1305 - ■GJ13 - ■■A2	145
	24	1 560	1.9	58.8		2KJ1305 - ■GJ13 - ■■X1	145
	27	1 437	2.4	54.17	★	2KJ1305 - ■GJ13 - ■■W1	145
	FD.88B-LA112ZMP4E						
	16.7	2 290	0.83	86.33	★	2KJ1404 - ■GJ13 - ■■D1	99
	18.7	2 044	0.93	77.04		2KJ1404 - ■GJ13 - ■■C1	99
	22	1 736	1.1	65.43		2KJ1404 - ■GJ13 - ■■B1	99
	26	1 445	1.3	54.47	★	2KJ1404 - ■GJ13 - ■■A1	99
	FZ.88B-LA112ZMP4E						
	22	1 713	1.1	64.58	★	2KJ1304 - ■GJ13 - ■■X1	98
	24	1 569	1.2	59.13		2KJ1304 - ■GJ13 - ■■W1	98
	27	1 395	1.4	52.6	★	2KJ1304 - ■GJ13 - ■■V1	98
	30	1 274	1.5	48.03		2KJ1304 - ■GJ13 - ■■U1	98
	33	1 173	1.6	44.2	★	2KJ1304 - ■GJ13 - ■■T1	98
	35	1 083	1.8	40.83		2KJ1304 - ■GJ13 - ■■S1	98
	38	1 005	1.9	37.89	★	2KJ1304 - ■GJ13 - ■■R1	98
	41	936	2.0	35.29		2KJ1304 - ■GJ13 - ■■Q1	98
	45	847	2.2	31.91	★	2KJ1304 - ■GJ13 - ■■P1	98
	49	779	2.4	29.38		2KJ1304 - ■GJ13 - ■■N1	98
	54	701	2.7	26.42	★	2KJ1304 - ■GJ13 - ■■M1	98
	59	647	2.9	24.38		2KJ1304 - ■GJ13 - ■■L1	98
	FZ.68B-LA112ZMP4E						
	33	1 164	0.86	43.87		2KJ1303 - ■GJ13 - ■■V1	67
	37	1 033	0.97	38.93	★	2KJ1303 - ■GJ13 - ■■U1	67
	40	953	1.0	35.93		2KJ1303 - ■GJ13 - ■■T1	67
	44	862	1.2	32.5	★	2KJ1303 - ■GJ13 - ■■S1	67
	48	794	1.3	29.93		2KJ1303 - ■GJ13 - ■■R1	67
	52	734	1.4	27.68	★	2KJ1303 - ■GJ13 - ■■Q1	67
	56	681	1.5	25.69		2KJ1303 - ■GJ13 - ■■P1	67
	64	601	1.7	22.67	★	2KJ1303 - ■GJ13 - ■■N1	67
	69	555	1.8	20.93		2KJ1303 - ■GJ13 - ■■M1	67
	77	497	2.0	18.75	★	2KJ1303 - ■GJ13 - ■■L1	67
	83	459	2.2	17.29		2KJ1303 - ■GJ13 - ■■K1	67

★ Preferred transmission ratio

Shaft designs, see page 3/89 ————— 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 ————— 1 to 9

Gearbox housing mounting position, see page 3/92 ————— A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
4	FZ.68B-LA112ZMP4E						
	99	385	2.6	14.51	2KJ1303 - ■GJ13 - ■■J1		67
	116	328	3.0	12.38 ★	2KJ1303 - ■GJ13 - ■■H1		67
	140	274	3.7	10.31	2KJ1303 - ■GJ13 - ■■G1		67
	179	213	4.2	8.03	2KJ1303 - ■GJ13 - ■■E1		67
	FZ.48B-LA112ZMP4E						
	56	679	0.8	25.59 ★	2KJ1302 - ■GJ13 - ■■R1		52
	61	623	0.87	23.48	2KJ1302 - ■GJ13 - ■■Q1		52
	67	574	0.94	21.63 ★	2KJ1302 - ■GJ13 - ■■P1		52
	73	521	1.0	19.64	2KJ1302 - ■GJ13 - ■■N1		52
	80	475	1.1	17.89 ★	2KJ1302 - ■GJ13 - ■■M1		52
	88	435	1.2	16.39	2KJ1302 - ■GJ13 - ■■L1		52
	98	388	1.4	14.63 ★	2KJ1302 - ■GJ13 - ■■K1		52
	110	346	1.6	13.05	2KJ1302 - ■GJ13 - ■■J1		52
	130	294	1.8	11.09	2KJ1302 - ■GJ13 - ■■H1		52
	156	245	2.1	9.23 ★	2KJ1302 - ■GJ13 - ■■G1		52
	172	223	2.3	8.39 ★	2KJ1302 - ■GJ13 - ■■F1		52
	188	204	2.3	7.68	2KJ1302 - ■GJ13 - ■■E1		52
	210	182	2.4	6.86 ★	2KJ1302 - ■GJ13 - ■■D1		52
	235	162	2.5	6.12	2KJ1302 - ■GJ13 - ■■C1		52
	277	138	2.7	5.2	2KJ1302 - ■GJ13 - ■■B1		52
	333	115	2.8	4.33 ★	2KJ1302 - ■GJ13 - ■■A1		52
	FZ.48B-LA112ZMP2E						
	162	235	2.3	17.89 ★	2KJ1302 - ■GJ13 - ■■M1	P00	52
	177	216	2.5	16.39	2KJ1302 - ■GJ13 - ■■L1	P00	52
	199	192	2.8	14.63 ★	2KJ1302 - ■GJ13 - ■■K1	P00	52
	223	172	3.1	13.05	2KJ1302 - ■GJ13 - ■■J1	P00	52
	262	146	3.7	11.09	2KJ1302 - ■GJ13 - ■■H1	P00	52
	315	121	4.3	9.23 ★	2KJ1302 - ■GJ13 - ■■G1	P00	52
	346	110	4.6	8.39 ★	2KJ1302 - ■GJ13 - ■■F1	P00	52
	378	101	4.6	7.68	2KJ1302 - ■GJ13 - ■■E1	P00	52
	423	90	4.9	6.86 ★	2KJ1302 - ■GJ13 - ■■D1	P00	52
	475	80	5.0	6.12	2KJ1302 - ■GJ13 - ■■C1	P00	52
	559	68	5.5	5.2	2KJ1302 - ■GJ13 - ■■B1	P00	52
	671	57	5.7	4.33 ★	2KJ1302 - ■GJ13 - ■■A1	P00	52
5.5	FD.208-LA160MB8						
	2.9	17 904	1.9	242.01	2KJ1411 - ■JF13 - ■■T1	P02	1123
	FD.188B-LA132ZMD6E						
	2.4	22 097	0.91	403.86 ★	2KJ1410 - ■HK13 - ■■U1	P01	684
	2.6	20 272	0.99	370.52	2KJ1410 - ■HK13 - ■■T1	P01	684
	2.8	18 709	1.1	341.94 ★	2KJ1410 - ■HK13 - ■■S1	P01	684
	3.0	17 354	1.2	317.18	2KJ1410 - ■HK13 - ■■R1	P01	684
	3.2	16 370	1.2	299.2 ★	2KJ1410 - ■HK13 - ■■Q1	P01	684
	3.4	15 312	1.3	279.86	2KJ1410 - ■HK13 - ■■P1	P01	684

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
5.5	FD.188B-LA132SP4E						
	3.6	14 680	1.4	403.86	★	2KJ1410 - ■ HG13 - ■■ U1	684
	3.9	13 468	1.5	370.52		2KJ1410 - ■ HG13 - ■■ T1	684
	4.2	12 429	1.6	341.94	★	2KJ1410 - ■ HG13 - ■■ S1	684
	4.6	11 529	1.7	317.18		2KJ1410 - ■ HG13 - ■■ R1	684
	4.8	10 876	1.8	299.2	★	2KJ1410 - ■ HG13 - ■■ Q1	684
	5.2	10 173	2.0	279.86		2KJ1410 - ■ HG13 - ■■ P1	684
	FD.168B-LA132ZMD6E						
	3.1	17 077	0.82	312.12	★	2KJ1408 - ■ HK13 - ■■ T1 P01	503
	3.3	15 826	0.88	289.26		2KJ1408 - ■ HK13 - ■■ S1 P01	503
	3.5	15 048	0.93	275.03	★	2KJ1408 - ■ HK13 - ■■ R1 P01	503
	3.7	14 064	1.0	257.04		2KJ1408 - ■ HK13 - ■■ Q1 P01	503
	FD.168B-LA132SP4E						
	3.9	13 422	1.0	369.26	★	2KJ1408 - ■ HG13 - ■■ V1	503
	4.3	12 304	1.1	338.49		2KJ1408 - ■ HG13 - ■■ U1	503
	4.6	11 345	1.2	312.12	★	2KJ1408 - ■ HG13 - ■■ T1	503
	5.0	10 514	1.3	289.26		2KJ1408 - ■ HG13 - ■■ S1	503
	5.3	9 997	1.4	275.03	★	2KJ1408 - ■ HG13 - ■■ R1	503
	5.6	9 343	1.5	257.04		2KJ1408 - ■ HG13 - ■■ Q1	503
	6.4	8 242	1.7	226.74	★	2KJ1408 - ■ HG13 - ■■ P1	503
	6.8	7 774	1.8	213.87		2KJ1408 - ■ HG13 - ■■ N1	503
	7.5	6 966	2.0	191.63	★	2KJ1408 - ■ HG13 - ■■ M1	503
	FD.148B-LA132SP4E						
	4.7	11 287	0.80	310.51	★	2KJ1407 - ■ HG13 - ■■ Q1	341
	5.0	10 450	0.86	287.49		2KJ1407 - ■ HG13 - ■■ P1	341
	5.4	9 718	0.93	267.35	★	2KJ1407 - ■ HG13 - ■■ N1	341
	5.8	9 072	0.99	249.58		2KJ1407 - ■ HG13 - ■■ M1	341
	6.5	8 117	1.1	223.31	★	2KJ1407 - ■ HG13 - ■■ L1	341
	7.0	7 522	1.2	206.93		2KJ1407 - ■ HG13 - ■■ K1	341
	7.6	6 895	1.3	189.69	★	2KJ1407 - ■ HG13 - ■■ J1	341
	8.3	6 321	1.4	173.89		2KJ1407 - ■ HG13 - ■■ H1	341
	9.8	5 386	1.7	148.18		2KJ1407 - ■ HG13 - ■■ G1	341
	11.1	4 753	1.9	130.76	★	2KJ1407 - ■ HG13 - ■■ F1	341
	13.0	4 045	2.2	111.29		2KJ1407 - ■ HG13 - ■■ E1	341
	FZ.148B-LA132SP4E						
	21	2 480	2.3	68.23		2KJ1307 - ■ HG13 - ■■ V1	333
	FD.128B-LA132SP4E						
	7.1	7 360	0.83	202.48		2KJ1406 - ■ HG13 - ■■ M1	245
	7.7	6 829	0.89	187.88	★	2KJ1406 - ■ HG13 - ■■ L1	245
	8.3	6 362	0.96	175.01		2KJ1406 - ■ HG13 - ■■ K1	245
	9.1	5 751	1.1	158.22	★	2KJ1406 - ■ HG13 - ■■ J1	245
	9.9	5 295	1.2	145.66		2KJ1406 - ■ HG13 - ■■ H1	245
	11.0	4 762	1.3	131.01	★	2KJ1406 - ■ HG13 - ■■ G1	245
	12.0	4 394	1.4	120.87		2KJ1406 - ■ HG13 - ■■ F1	245

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
5.5	FD.128B-LA132SP4E						
	14.1	3 723	1.6	102.41	2KJ1406 - ■ HG13 - ■ ■ E1		245
	16.2	3 244	1.9	89.25 ★	2KJ1406 - ■ HG13 - ■ ■ D1		245
	19.0	2 760	2.2	75.93	2KJ1406 - ■ HG13 - ■ ■ C1		245
	FZ.128B-LA132SP4E						
	26	2 051	2.1	56.42 ★	2KJ1306 - ■ HG13 - ■ ■ A2		241
	28	1 901	2.4	52.29	2KJ1306 - ■ HG13 - ■ ■ X1		241
	FD.108B-LA132SP4E						
	13.7	3 846	0.88	105.81 ★	2KJ1405 - ■ HG13 - ■ ■ F1		164
	14.8	3 547	0.96	97.57	2KJ1405 - ■ HG13 - ■ ■ E1		164
	17.7	2 976	1.1	81.86	2KJ1405 - ■ HG13 - ■ ■ D1		164
	21	2 539	1.3	69.84 ★	2KJ1405 - ■ HG13 - ■ ■ C1		164
	25	2 116	1.6	58.2	2KJ1405 - ■ HG13 - ■ ■ B1		164
	30	1 753	1.9	48.24 ★	2KJ1405 - ■ HG13 - ■ ■ A1		164
	FZ.108B-LA132SP4E						
	22	2 334	1.3	64.21 ★	2KJ1305 - ■ HG13 - ■ ■ A2		163
	25	2 137	1.4	58.8	2KJ1305 - ■ HG13 - ■ ■ X1		163
	27	1 969	1.7	54.17 ★	2KJ1305 - ■ HG13 - ■ ■ W1		163
	29	1 823	1.9	50.15	2KJ1305 - ■ HG13 - ■ ■ V1		163
	31	1 695	2.0	46.64 ★	2KJ1305 - ■ HG13 - ■ ■ U1		163
	33	1 583	2.1	43.54	2KJ1305 - ■ HG13 - ■ ■ T1		163
	37	1 416	2.4	38.95 ★	2KJ1305 - ■ HG13 - ■ ■ S1		163
	40	1 312	2.6	36.1	2KJ1305 - ■ HG13 - ■ ■ R1		163
	FD.88B-LA132SP4E						
	22	2 378	0.80	65.43	2KJ1404 - ■ HG13 - ■ ■ B1		117
	26	1 980	0.96	54.47 ★	2KJ1404 - ■ HG13 - ■ ■ A1		117
	FZ.88B-LA132SP4E						
	28	1 912	0.99	52.6 ★	2KJ1304 - ■ HG13 - ■ ■ V1		116
	30	1 746	1.1	48.03	2KJ1304 - ■ HG13 - ■ ■ U1		116
	33	1 607	1.2	44.2 ★	2KJ1304 - ■ HG13 - ■ ■ T1		116
	35	1 484	1.3	40.83	2KJ1304 - ■ HG13 - ■ ■ S1		116
	38	1 377	1.4	37.89 ★	2KJ1304 - ■ HG13 - ■ ■ R1		116
	41	1 283	1.5	35.29	2KJ1304 - ■ HG13 - ■ ■ Q1		116
	45	1 160	1.6	31.91 ★	2KJ1304 - ■ HG13 - ■ ■ P1		116
	49	1 068	1.8	29.38	2KJ1304 - ■ HG13 - ■ ■ N1		116
	55	960	2.0	26.42 ★	2KJ1304 - ■ HG13 - ■ ■ M1		116
	59	886	2.1	24.38	2KJ1304 - ■ HG13 - ■ ■ L1		116
	70	751	2.5	20.65	2KJ1304 - ■ HG13 - ■ ■ K1		116
	80	654	2.9	18 ★	2KJ1304 - ■ HG13 - ■ ■ J1		116
	94	557	3.4	15.31	2KJ1304 - ■ HG13 - ■ ■ H1		116
	FZ.68B-LA132SP4E						
	44	1 181	0.85	32.5 ★	2KJ1303 - ■ HG13 - ■ ■ S1		85
	48	1 088	0.92	29.93	2KJ1303 - ■ HG13 - ■ ■ R1		85
	52	1 006	0.99	27.68 ★	2KJ1303 - ■ HG13 - ■ ■ Q1		85

★ Preferred transmission ratio

Shaft designs, see page 3/89 ————— 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 ————— 1 to 9

Gearbox housing mounting position, see page 3/92 ————— A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
5.5	FZ.68B-LA132SP4E						
	56	934	1.1	25.69	2KJ1303 - ■ HG13 - ■ ■ P1		85
	64	824	1.2	22.67 ★	2KJ1303 - ■ HG13 - ■ ■ N1		85
	69	761	1.3	20.93	2KJ1303 - ■ HG13 - ■ ■ M1		85
	77	682	1.5	18.75 ★	2KJ1303 - ■ HG13 - ■ ■ L1		85
	84	628	1.6	17.29	2KJ1303 - ■ HG13 - ■ ■ K1		85
	100	527	1.9	14.51	2KJ1303 - ■ HG13 - ■ ■ J1		85
	117	450	2.2	12.38 ★	2KJ1303 - ■ HG13 - ■ ■ H1		85
	140	375	2.7	10.31	2KJ1303 - ■ HG13 - ■ ■ G1		85
	169	311	3.2	8.55 ★	2KJ1303 - ■ HG13 - ■ ■ F1		85
	180	292	3.1	8.03	2KJ1303 - ■ HG13 - ■ ■ E1		85
	214	245	3.4	6.74	2KJ1303 - ■ HG13 - ■ ■ D1		85
	251	209	3.6	5.75 ★	2KJ1303 - ■ HG13 - ■ ■ C1		85
	302	174	3.9	4.79	2KJ1303 - ■ HG13 - ■ ■ B1		85
	364	144	4.1	3.97 ★	2KJ1303 - ■ HG13 - ■ ■ A1		85
	FZ.68B-LA132SB2E						
	156	336	3.0	18.75 ★	2KJ1303 - ■ HF13 - ■ ■ L1	P00	77
	169	310	3.2	17.29	2KJ1303 - ■ HF13 - ■ ■ K1	P00	77
	202	260	3.8	14.51	2KJ1303 - ■ HF13 - ■ ■ J1	P00	77
	237	222	4.5	12.38 ★	2KJ1303 - ■ HF13 - ■ ■ H1	P00	77
7.5	FD.208-LA160LB8						
	3.0	24 243	1.4	242.01	2KJ1411 - ■ JJ13 - ■ ■ T1	P02	1 135
	3.3	21 892	1.6	218.54	2KJ1411 - ■ JJ13 - ■ ■ S1	P02	1 135
	3.5	20 517	1.7	204.81	2KJ1411 - ■ JJ13 - ■ ■ R1	P02	1 135
	FD.208-LA160MD6E						
	4.0	17 963	1.9	242.01	2KJ1411 - ■ JJ13 - ■ ■ T1	P01	1 135
	FD.188B-LA160MD6E						
	3.2	22 207	0.90	299.2 ★	2KJ1410 - ■ JJ13 - ■ ■ Q1	P01	720
	3.4	20 772	0.96	279.86	2KJ1410 - ■ JJ13 - ■ ■ P1	P01	720
	FD.188B-LA132ZMP4E						
	3.6	19 881	1.0	403.86 ★	2KJ1410 - ■ HK13 - ■ ■ U1		684
	3.9	18 240	1.1	370.52	2KJ1410 - ■ HK13 - ■ ■ T1		684
	4.3	16 833	1.2	341.94 ★	2KJ1410 - ■ HK13 - ■ ■ S1		684
	4.6	15 614	1.3	317.18	2KJ1410 - ■ HK13 - ■ ■ R1		684
	4.9	14 729	1.4	299.2 ★	2KJ1410 - ■ HK13 - ■ ■ Q1		684
	5.2	13 777	1.5	279.86	2KJ1410 - ■ HK13 - ■ ■ P1		684
	5.8	12 250	1.6	248.85 ★	2KJ1410 - ■ HK13 - ■ ■ N1		684
	6.2	11 565	1.7	234.93	2KJ1410 - ■ HK13 - ■ ■ M1		684
	6.9	10 381	1.9	210.89 ★	2KJ1410 - ■ HK13 - ■ ■ L1		684
	7.5	9 528	2.1	193.56	2KJ1410 - ■ HK13 - ■ ■ K1		684
	FD.168B-LA132ZMP4E						
	4.3	16 663	0.84	338.49	2KJ1408 - ■ HK13 - ■ ■ U1		503
	4.7	15 365	0.91	312.12 ★	2KJ1408 - ■ HK13 - ■ ■ T1		503
	5.0	14 239	0.98	289.26	2KJ1408 - ■ HK13 - ■ ■ S1		503

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
7.5	FD.168B-LA132ZMP4E						
	5.3	13 539	1.0	275.03	★	2KJ1408 - ■ HK13 - ■ ■ R1	503
	5.7	12 653	1.1	257.04		2KJ1408 - ■ HK13 - ■ ■ Q1	503
	6.4	11 162	1.3	226.74	★	2KJ1408 - ■ HK13 - ■ ■ P1	503
	6.8	10 528	1.3	213.87		2KJ1408 - ■ HK13 - ■ ■ N1	503
	7.6	9 433	1.5	191.63	★	2KJ1408 - ■ HK13 - ■ ■ M1	503
	8.2	8 710	1.6	176.94		2KJ1408 - ■ HK13 - ■ ■ L1	503
	9.6	7 442	1.9	151.18		2KJ1408 - ■ HK13 - ■ ■ K1	503
	10.6	6 726	2.1	136.63	★	2KJ1408 - ■ HK13 - ■ ■ J1	503
	11.1	6 480	2.2	131.64		2KJ1408 - ■ HK13 - ■ ■ H1	503
	FD.148B-LA132ZMP4E						
	6.5	10 993	0.82	223.31	★	2KJ1407 - ■ HK13 - ■ ■ L1	341
	7.0	10 187	0.88	206.93		2KJ1407 - ■ HK13 - ■ ■ K1	341
	7.7	9 338	0.96	189.69	★	2KJ1407 - ■ HK13 - ■ ■ J1	341
	8.4	8 560	1.1	173.89		2KJ1407 - ■ HK13 - ■ ■ H1	341
	9.8	7 294	1.2	148.18		2KJ1407 - ■ HK13 - ■ ■ G1	341
	11.1	6 437	1.4	130.76	★	2KJ1407 - ■ HK13 - ■ ■ F1	341
	13.1	5 478	1.6	111.29		2KJ1407 - ■ HK13 - ■ ■ E1	341
	15.1	4 747	1.9	96.43	★	2KJ1407 - ■ HK13 - ■ ■ D1	341
	17.9	3 995	2.3	81.15	★	2KJ1407 - ■ HK13 - ■ ■ C1	341
	FZ.148B-LA132ZMP4E						
	21	3 359	1.7	68.23		2KJ1307 - ■ HK13 - ■ ■ V1	333
	23	3 169	2.1	64.37	★	2KJ1307 - ■ HK13 - ■ ■ U1	333
	24	2 964	2.4	60.21		2KJ1307 - ■ HK13 - ■ ■ T1	333
	FD.128B-LA132ZMP4E						
	10.0	7 170	0.85	145.66		2KJ1406 - ■ HK13 - ■ ■ H1	245
	11.1	6 449	0.95	131.01	★	2KJ1406 - ■ HK13 - ■ ■ G1	245
	12.0	5 950	1.0	120.87		2KJ1406 - ■ HK13 - ■ ■ F1	245
	14.2	5 041	1.2	102.41		2KJ1406 - ■ HK13 - ■ ■ E1	245
	16.3	4 393	1.4	89.25	★	2KJ1406 - ■ HK13 - ■ ■ D1	245
	19.2	3 738	1.6	75.93		2KJ1406 - ■ HK13 - ■ ■ C1	245
	22	3 190	1.9	64.8	★	2KJ1406 - ■ HK13 - ■ ■ B1	245
	27	2 615	2.3	53.13	★	2KJ1406 - ■ HK13 - ■ ■ A1	245
	FZ.128B-LA132ZMP4E						
	26	2 777	1.5	56.42	★	2KJ1306 - ■ HK13 - ■ ■ A2	241
	28	2 574	1.8	52.29		2KJ1306 - ■ HK13 - ■ ■ X1	241
	29	2 447	2.0	49.71	★	2KJ1306 - ■ HK13 - ■ ■ W1	241
	31	2 287	2.3	46.46		2KJ1306 - ■ HK13 - ■ ■ V1	241
	FD.108B-LA132ZMP4E						
	17.8	4 030	0.84	81.86		2KJ1405 - ■ HK13 - ■ ■ D1	164
	21	3 438	0.99	69.84	★	2KJ1405 - ■ HK13 - ■ ■ C1	164
	25	2 865	1.2	58.2		2KJ1405 - ■ HK13 - ■ ■ B1	164
	30	2 375	1.4	48.24	★	2KJ1405 - ■ HK13 - ■ ■ A1	164

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
7.5	FZ.108B-LA132ZMP4E						
	23	3 161	0.95	64.21	★	2KJ1305 - ■ HK13 - ■■ A2	163
	25	2 895	1.0	58.8		2KJ1305 - ■ HK13 - ■■ X1	163
	27	2 667	1.3	54.17	★	2KJ1305 - ■ HK13 - ■■ W1	163
	29	2 469	1.4	50.15		2KJ1305 - ■ HK13 - ■■ V1	163
	31	2 296	1.5	46.64	★	2KJ1305 - ■ HK13 - ■■ U1	163
	33	2 143	1.6	43.54		2KJ1305 - ■ HK13 - ■■ T1	163
	37	1 917	1.8	38.95	★	2KJ1305 - ■ HK13 - ■■ S1	163
	40	1 777	1.9	36.1		2KJ1305 - ■ HK13 - ■■ R1	163
	44	1 629	2.1	33.09	★	2KJ1305 - ■ HK13 - ■■ Q1	163
	48	1 493	2.3	30.33		2KJ1305 - ■ HK13 - ■■ P1	163
	56	1 273	2.7	25.85		2KJ1305 - ■ HK13 - ■■ N1	163
	64	1 123	3.0	22.81	★	2KJ1305 - ■ HK13 - ■■ M1	163
	FZ.88B-LA132ZMP4E						
	30	2 364	0.80	48.03		2KJ1304 - ■ HK13 - ■■ U1	116
	33	2 176	0.87	44.2	★	2KJ1304 - ■ HK13 - ■■ T1	116
	36	2 010	0.95	40.83		2KJ1304 - ■ HK13 - ■■ S1	116
	38	1 865	1.0	37.89	★	2KJ1304 - ■ HK13 - ■■ R1	116
	41	1 737	1.1	35.29		2KJ1304 - ■ HK13 - ■■ Q1	116
	46	1 571	1.2	31.91	★	2KJ1304 - ■ HK13 - ■■ P1	116
	50	1 446	1.3	29.38		2KJ1304 - ■ HK13 - ■■ N1	116
	55	1 301	1.5	26.42	★	2KJ1304 - ■ HK13 - ■■ M1	116
	60	1 200	1.6	24.38		2KJ1304 - ■ HK13 - ■■ L1	116
	70	1 017	1.9	20.65		2KJ1304 - ■ HK13 - ■■ K1	116
	81	886	2.1	18	★	2KJ1304 - ■ HK13 - ■■ J1	116
	95	754	2.5	15.31		2KJ1304 - ■ HK13 - ■■ H1	116
	111	643	3.0	13.07	★	2KJ1304 - ■ HK13 - ■■ G1	116
	136	527	3.6	10.71	★	2KJ1304 - ■ HK13 - ■■ F1	116
	158	452	3.7	9.19		2KJ1304 - ■ HK13 - ■■ E1	116
	182	394	3.9	8.01	★	2KJ1304 - ■ HK13 - ■■ D1	116
	213	336	4.3	6.82		2KJ1304 - ■ HK13 - ■■ C1	116
	250	286	4.7	5.82	★	2KJ1304 - ■ HK13 - ■■ B1	116
	FZ.68B-LA132ZMP4E						
	64	1 116	0.90	22.67	★	2KJ1303 - ■ HK13 - ■■ N1	85
	70	1 030	0.97	20.93		2KJ1303 - ■ HK13 - ■■ M1	85
	78	923	1.1	18.75	★	2KJ1303 - ■ HK13 - ■■ L1	85
	84	851	1.2	17.29		2KJ1303 - ■ HK13 - ■■ K1	85
	100	714	1.4	14.51		2KJ1303 - ■ HK13 - ■■ J1	85
	118	609	1.6	12.38	★	2KJ1303 - ■ HK13 - ■■ H1	85
	141	508	2.0	10.31		2KJ1303 - ■ HK13 - ■■ G1	85
	170	421	2.4	8.55	★	2KJ1303 - ■ HK13 - ■■ F1	85
	181	395	2.3	8.03		2KJ1303 - ■ HK13 - ■■ E1	85
	216	332	2.5	6.74		2KJ1303 - ■ HK13 - ■■ D1	85
	253	283	2.7	5.75	★	2KJ1303 - ■ HK13 - ■■ C1	85

★ Preferred transmission ratio

Shaft designs, see page 3/89

1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20

1 to 9

Gearbox housing mounting position, see page 3/92

A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
7.5	FZ.68B-LA132ZMP4E						
	304	236	2.9	4.79	2KJ1303 - ■ HK13 - ■ ■ B1		85
	366	195	3.0	3.97 ★	2KJ1303 - ■ HK13 - ■ ■ A1		85
	FZ.68B-LA132ZSD2E						
	156	458	2.2	18.75 ★	2KJ1303 - ■ HJ13 - ■ ■ L1	P00	85
	169	423	2.4	17.29	2KJ1303 - ■ HJ13 - ■ ■ K1	P00	85
	202	355	2.8	14.51	2KJ1303 - ■ HJ13 - ■ ■ J1	P00	85
	237	303	3.3	12.38 ★	2KJ1303 - ■ HJ13 - ■ ■ H1	P00	85
	284	252	4.0	10.31	2KJ1303 - ■ HJ13 - ■ ■ G1	P00	85
	343	209	4.8	8.55 ★	2KJ1303 - ■ HJ13 - ■ ■ F1	P00	85
	365	196	4.6	8.03	2KJ1303 - ■ HJ13 - ■ ■ E1	P00	85
	435	165	5.1	6.74	2KJ1303 - ■ HJ13 - ■ ■ D1	P00	85
	510	141	5.4	5.75 ★	2KJ1303 - ■ HJ13 - ■ ■ C1	P00	85
	612	117	5.8	4.79	2KJ1303 - ■ HJ13 - ■ ■ B1	P00	85
	738	97	6.1	3.97 ★	2KJ1303 - ■ HJ13 - ■ ■ A1	P00	85
9.2	FD.188B-LA160MB4E						
	4.9	18 067	1.1	299.2 ★	2KJ1410 - ■ JP13 - ■ ■ Q1		708
	5.2	16 899	1.2	279.86	2KJ1410 - ■ JP13 - ■ ■ P1		708
	5.8	15 027	1.3	248.85 ★	2KJ1410 - ■ JP13 - ■ ■ N1		708
	6.2	14 186	1.4	234.93	2KJ1410 - ■ JP13 - ■ ■ M1		708
	6.9	12 735	1.6	210.89 ★	2KJ1410 - ■ JP13 - ■ ■ L1		708
	7.5	11 688	1.7	193.56	2KJ1410 - ■ JP13 - ■ ■ K1		708
	8.7	10 086	2.0	167.03	2KJ1410 - ■ JP13 - ■ ■ J1		708
	FD.168B-LA160MB4E						
	5.3	16 608	0.84	275.03 ★	2KJ1408 - ■ JP13 - ■ ■ R1		527
	5.7	15 521	0.9	257.04	2KJ1408 - ■ JP13 - ■ ■ Q1		527
	6.4	13 692	1.0	226.74 ★	2KJ1408 - ■ JP13 - ■ ■ P1		527
	6.8	12 915	1.1	213.87	2KJ1408 - ■ JP13 - ■ ■ N1		527
	7.6	11 572	1.2	191.63 ★	2KJ1408 - ■ JP13 - ■ ■ M1		527
	8.2	10 685	1.3	176.94	2KJ1408 - ■ JP13 - ■ ■ L1		527
	9.6	9 129	1.5	151.18	2KJ1408 - ■ JP13 - ■ ■ K1		527
	10.6	8 250	1.7	136.63 ★	2KJ1408 - ■ JP13 - ■ ■ J1		527
	11.1	7 949	1.8	131.64	2KJ1408 - ■ JP13 - ■ ■ H1		527
	12.8	6 875	2.0	113.86	2KJ1408 - ■ JP13 - ■ ■ G1		527
	FD.148B-LA160MB4E						
	8.4	10 500	0.86	173.89	2KJ1407 - ■ JP13 - ■ ■ H1		365
	9.8	8 948	1.0	148.18	2KJ1407 - ■ JP13 - ■ ■ G1		365
	11.1	7 896	1.1	130.76 ★	2KJ1407 - ■ JP13 - ■ ■ F1		365
	13.1	6 720	1.3	111.29	2KJ1407 - ■ JP13 - ■ ■ E1		365
	15.1	5 823	1.5	96.43 ★	2KJ1407 - ■ JP13 - ■ ■ D1		365
	17.9	4 900	1.8	81.15 ★	2KJ1407 - ■ JP13 - ■ ■ C1		365
	19.9	4 421	2.0	73.22	2KJ1407 - ■ JP13 - ■ ■ B1		365
	23	3 800	2.4	62.93 ★	2KJ1407 - ■ JP13 - ■ ■ A1		365

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
9.2	FZ.148B-LA160MB4E						
	23	3 887	1.7	64.37	★	2KJ1307 - ■JP13 - ■■U1	357
	24	3 636	1.9	60.21		2KJ1307 - ■JP13 - ■■T1	357
	27	3 232	2.5	53.53	★	2KJ1307 - ■JP13 - ■■S1	357
	FD.128B-LA160MB4E						
	12.0	7 299	0.84	120.87		2KJ1406 - ■JP13 - ■■F1	269
	14.2	6 184	0.99	102.41		2KJ1406 - ■JP13 - ■■E1	269
	16.3	5 389	1.1	89.25	★	2KJ1406 - ■JP13 - ■■D1	269
	19.2	4 585	1.3	75.93		2KJ1406 - ■JP13 - ■■C1	269
	22	3 913	1.6	64.8	★	2KJ1406 - ■JP13 - ■■B1	269
	27	3 208	1.9	53.13	★	2KJ1406 - ■JP13 - ■■A1	269
	FZ.128B-LA160MB4E						
	29	3 002	1.6	49.71	★	2KJ1306 - ■JP13 - ■■W1	265
	31	2 805	1.8	46.46		2KJ1306 - ■JP13 - ■■V1	265
	36	2 475	2.3	40.99	★	2KJ1306 - ■JP13 - ■■U1	265
	38	2 334	2.6	38.66		2KJ1306 - ■JP13 - ■■T1	265
	FD.108B-LA160MB4E						
	21	4 217	0.81	69.84	★	2KJ1405 - ■JP13 - ■■C1	188
	25	3 514	0.97	58.2		2KJ1405 - ■JP13 - ■■B1	188
	30	2 913	1.2	48.24	★	2KJ1405 - ■JP13 - ■■A1	188
	FZ.108B-LA160MB4E						
	31	2 816	1.2	46.64	★	2KJ1305 - ■JP13 - ■■U1	187
	33	2 629	1.3	43.54		2KJ1305 - ■JP13 - ■■T1	187
	37	2 352	1.4	38.95	★	2KJ1305 - ■JP13 - ■■S1	187
	40	2 180	1.6	36.1		2KJ1305 - ■JP13 - ■■R1	187
	44	1 998	1.7	33.09	★	2KJ1305 - ■JP13 - ■■Q1	187
	48	1 831	1.9	30.33		2KJ1305 - ■JP13 - ■■P1	187
	56	1 561	2.2	25.85		2KJ1305 - ■JP13 - ■■N1	187
	64	1 377	2.5	22.81	★	2KJ1305 - ■JP13 - ■■M1	187
	75	1 172	2.9	19.41		2KJ1305 - ■JP13 - ■■L1	187
	86	1 016	3.3	16.82	★	2KJ1305 - ■JP13 - ■■K1	187
	FZ.88B-LA160MB4E						
	38	2 288	0.83	37.89	★	2KJ1304 - ■JP13 - ■■R1	140
	41	2 131	0.89	35.29		2KJ1304 - ■JP13 - ■■Q1	140
	46	1 927	0.99	31.91	★	2KJ1304 - ■JP13 - ■■P1	140
	50	1 774	1.1	29.38		2KJ1304 - ■JP13 - ■■N1	140
	55	1 595	1.2	26.42	★	2KJ1304 - ■JP13 - ■■M1	140
	60	1 472	1.3	24.38		2KJ1304 - ■JP13 - ■■L1	140
	70	1 247	1.5	20.65		2KJ1304 - ■JP13 - ■■K1	140
	81	1 087	1.7	18	★	2KJ1304 - ■JP13 - ■■J1	140
	95	924	2.1	15.31		2KJ1304 - ■JP13 - ■■H1	140
	111	789	2.4	13.07	★	2KJ1304 - ■JP13 - ■■G1	140
	136	647	2.9	10.71	★	2KJ1304 - ■JP13 - ■■F1	140
	158	555	3.0	9.19		2KJ1304 - ■JP13 - ■■E1	140

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg	
9.2	FZ.88B-LA160MB4E							
	182	484	3.2	8.01	★	2KJ1304 - ■JP13 - ■■D1	140	
	213	412	3.5	6.82		2KJ1304 - ■JP13 - ■■C1	140	
	250	351	3.8	5.82	★	2KJ1304 - ■JP13 - ■■B1	140	
	305	288	4.2	4.77	★	2KJ1304 - ■JP13 - ■■A1	140	
11	FD.208-LG180LA8							
	3.0	35 066	0.97	242.01		2KJ1411 - ■KM13 - ■■T1	P02	1 205
	3.3	31 666	1.1	218.54		2KJ1411 - ■KM13 - ■■S1	P02	1 205
	3.5	29 676	1.1	204.81		2KJ1411 - ■KM13 - ■■R1	P02	1 205
	FD.208-LA160ZLP6E							
	4.0	26 482	1.3	242.01		2KJ1411 - ■JT13 - ■■T1	P01	1 135
	4.4	23 914	1.4	218.54		2KJ1411 - ■JT13 - ■■S1	P01	1 135
	4.7	22 412	1.5	204.81		2KJ1411 - ■JT13 - ■■R1	P01	1 135
	5.5	19 250	1.8	175.92	★	2KJ1411 - ■JT13 - ■■Q1	P01	1 135
	FD.208-LA160MP4E							
	6.0	17 413	2.0	242.01		2KJ1411 - ■JQ13 - ■■T1		1 123
	FD.188B-LA160MP4E							
	4.9	21 528	0.93	299.2	★	2KJ1410 - ■JQ13 - ■■Q1		708
	5.2	20 137	0.99	279.86		2KJ1410 - ■JQ13 - ■■P1		708
	5.9	17 905	1.1	248.85	★	2KJ1410 - ■JQ13 - ■■N1		708
	6.2	16 904	1.2	234.93		2KJ1410 - ■JQ13 - ■■M1		708
	6.9	15 174	1.3	210.89	★	2KJ1410 - ■JQ13 - ■■L1		708
	7.5	13 927	1.4	193.56		2KJ1410 - ■JQ13 - ■■K1		708
	8.7	12 018	1.7	167.03		2KJ1410 - ■JQ13 - ■■J1		708
	10.0	10 513	1.9	146.11		2KJ1410 - ■JQ13 - ■■H1		708
	11.5	9 143	2.2	127.07		2KJ1410 - ■JQ13 - ■■G1		708
	FD.168B-LA160MP4E							
	6.4	16 314	0.86	226.74	★	2KJ1408 - ■JQ13 - ■■P1		527
	6.8	15 388	0.91	213.87		2KJ1408 - ■JQ13 - ■■N1		527
	7.6	13 788	1.0	191.63	★	2KJ1408 - ■JQ13 - ■■M1		527
	8.3	12 731	1.1	176.94		2KJ1408 - ■JQ13 - ■■L1		527
	9.7	10 878	1.3	151.18		2KJ1408 - ■JQ13 - ■■K1		527
	10.7	9 831	1.4	136.63	★	2KJ1408 - ■JQ13 - ■■J1		527
	11.1	9 472	1.5	131.64		2KJ1408 - ■JQ13 - ■■H1		527
	12.8	8 192	1.7	113.86		2KJ1408 - ■JQ13 - ■■G1		527
	14.7	7 146	2.0	99.31	★	2KJ1408 - ■JQ13 - ■■F1		527
	17.2	6 115	2.3	84.99	★	2KJ1408 - ■JQ13 - ■■E1		527
	FZ.168B-LA160MP4E							
	27	3 848	2.3	53.48		2KJ1308 - ■JQ13 - ■■R1		510
	FD.148B-LA160MP4E							
	9.9	10 662	0.84	148.18		2KJ1407 - ■JQ13 - ■■G1		365
	11.2	9 408	0.96	130.76	★	2KJ1407 - ■JQ13 - ■■F1		365
	13.1	8 008	1.1	111.29		2KJ1407 - ■JQ13 - ■■E1		365
	15.1	6 938	1.3	96.43	★	2KJ1407 - ■JQ13 - ■■D1		365

★ Preferred transmission ratio

Shaft designs, see page 3/89 ————— 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 ————— 1 to 9

Gearbox housing mounting position, see page 3/92 ————— A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
11	FD.148B-LA160MP4E						
	18.0	5 839	1.5	81.15 ★	2KJ1407 - ■ JQ13 - ■■ C1		365
	19.9	5 268	1.7	73.22	2KJ1407 - ■ JQ13 - ■■ B1		365
	23	4 528	2.0	62.93 ★	2KJ1407 - ■ JQ13 - ■■ A1		365
	FZ.148B-LA160MP4E						
	23	4 632	1.4	64.37 ★	2KJ1307 - ■ JQ13 - ■■ U1		357
	24	4 332	1.6	60.21	2KJ1307 - ■ JQ13 - ■■ T1		357
	27	3 852	2.1	53.53 ★	2KJ1307 - ■ JQ13 - ■■ S1		357
	29	3 636	2.2	50.54	2KJ1307 - ■ JQ13 - ■■ R1		357
	FD.128B-LA160MP4E						
	14.3	7 369	0.83	102.41	2KJ1406 - ■ JQ13 - ■■ E1		269
	16.4	6 422	0.95	89.25 ★	2KJ1406 - ■ JQ13 - ■■ D1		269
	19.2	5 463	1.1	75.93	2KJ1406 - ■ JQ13 - ■■ C1		269
	22	4 662	1.3	64.8 ★	2KJ1406 - ■ JQ13 - ■■ B1		269
	28	3 823	1.6	53.13 ★	2KJ1406 - ■ JQ13 - ■■ A1		269
	FZ.128B-LA160MP4E						
	29	3 577	1.4	49.71 ★	2KJ1306 - ■ JQ13 - ■■ W1		265
	31	3 343	1.5	46.46	2KJ1306 - ■ JQ13 - ■■ V1		265
	36	2 949	1.9	40.99 ★	2KJ1306 - ■ JQ13 - ■■ U1		265
	38	2 782	2.2	38.66	2KJ1306 - ■ JQ13 - ■■ T1		265
	42	2 492	2.4	34.64 ★	2KJ1306 - ■ JQ13 - ■■ S1		265
	46	2 301	2.7	31.98	2KJ1306 - ■ JQ13 - ■■ R1		265
	FD.108B-LA160MP4E						
	25	4 188	0.81	58.2	2KJ1405 - ■ JQ13 - ■■ B1		188
	30	3 471	0.98	48.24 ★	2KJ1405 - ■ JQ13 - ■■ A1		188
	FZ.108B-LA160MP4E						
	31	3 356	1.0	46.64 ★	2KJ1305 - ■ JQ13 - ■■ U1		187
	34	3 133	1.1	43.54	2KJ1305 - ■ JQ13 - ■■ T1		187
	38	2 803	1.2	38.95 ★	2KJ1305 - ■ JQ13 - ■■ S1		187
	40	2 597	1.3	36.1	2KJ1305 - ■ JQ13 - ■■ R1		187
	44	2 381	1.4	33.09 ★	2KJ1305 - ■ JQ13 - ■■ Q1		187
	48	2 182	1.6	30.33	2KJ1305 - ■ JQ13 - ■■ P1		187
	56	1 860	1.8	25.85	2KJ1305 - ■ JQ13 - ■■ N1		187
	64	1 641	2.1	22.81 ★	2KJ1305 - ■ JQ13 - ■■ M1		187
	75	1 397	2.4	19.41	2KJ1305 - ■ JQ13 - ■■ L1		187
	87	1 210	2.8	16.82 ★	2KJ1305 - ■ JQ13 - ■■ K1		187
	103	1 019	3.2	14.16 ★	2KJ1305 - ■ JQ13 - ■■ J1		187
	FZ.88B-LA160MP4E						
	46	2 296	0.83	31.91 ★	2KJ1304 - ■ JQ13 - ■■ P1		140
	50	2 114	0.90	29.38	2KJ1304 - ■ JQ13 - ■■ N1		140
	55	1 901	1.0	26.42 ★	2KJ1304 - ■ JQ13 - ■■ M1		140
	60	1 754	1.1	24.38	2KJ1304 - ■ JQ13 - ■■ L1		140
	71	1 486	1.3	20.65	2KJ1304 - ■ JQ13 - ■■ K1		140

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
11	FZ.88B-LA160MP4E						
	81	1 295	1.5	18 ★	2KJ1304 - ■ JQ13 - ■ ■ J1		140
	95	1 102	1.7	15.31	2KJ1304 - ■ JQ13 - ■ ■ H1		140
	112	940	2.0	13.07 ★	2KJ1304 - ■ JQ13 - ■ ■ G1		140
	136	771	2.5	10.71 ★	2KJ1304 - ■ JQ13 - ■ ■ F1		140
	159	661	2.5	9.19	2KJ1304 - ■ JQ13 - ■ ■ E1		140
	182	576	2.7	8.01 ★	2KJ1304 - ■ JQ13 - ■ ■ D1		140
	214	491	3.0	6.82	2KJ1304 - ■ JQ13 - ■ ■ C1		140
	251	419	3.2	5.82 ★	2KJ1304 - ■ JQ13 - ■ ■ B1		140
	306	343	3.5	4.77 ★	2KJ1304 - ■ JQ13 - ■ ■ A1		140
15	FD.208-LG200L8						
	3.5	40 468	0.84	204.81	2KJ1411 - ■ LL13 - ■ ■ R1	P02	1 255
	FD.208-LG180ZLB6E						
	4.0	35 557	0.96	242.01	2KJ1411 - ■ KP13 - ■ ■ T1	P01	1 205
	4.5	32 109	1.1	218.54	2KJ1411 - ■ KP13 - ■ ■ S1	P01	1 205
	4.8	30 091	1.1	204.81	2KJ1411 - ■ KP13 - ■ ■ R1	P01	1 205
	5.5	25 847	1.3	175.92 ★	2KJ1411 - ■ KP13 - ■ ■ Q1	P01	1 205
	FD.208-LA160ZLP4E						
	6.0	23 745	1.4	242.01	2KJ1411 - ■ JT13 - ■ ■ T1		1 135
	6.7	21 442	1.6	218.54	2KJ1411 - ■ JT13 - ■ ■ S1		1 135
	7.1	20 095	1.7	204.81	2KJ1411 - ■ JT13 - ■ ■ R1		1 135
	8.3	17 261	2.0	175.92 ★	2KJ1411 - ■ JT13 - ■ ■ Q1		1 135
	FD.188B-LA160ZLP4E						
	5.9	24 416	0.82	248.85 ★	2KJ1410 - ■ JT13 - ■ ■ N1		720
	6.2	23 050	0.87	234.93	2KJ1410 - ■ JT13 - ■ ■ M1		720
	6.9	20 692	0.97	210.89 ★	2KJ1410 - ■ JT13 - ■ ■ L1		720
	7.5	18 991	1.1	193.56	2KJ1410 - ■ JT13 - ■ ■ K1		720
	8.7	16 388	1.2	167.03	2KJ1410 - ■ JT13 - ■ ■ J1		720
	10.0	14 336	1.4	146.11	2KJ1410 - ■ JT13 - ■ ■ H1		720
	11.5	12 468	1.6	127.07	2KJ1410 - ■ JT13 - ■ ■ G1		720
	13.1	10 939	1.8	111.49 ★	2KJ1410 - ■ JT13 - ■ ■ F1		720
	15.5	9 250	2.2	94.28 ★	2KJ1410 - ■ JT13 - ■ ■ E1		720
	FD.168B-LA160ZLP4E						
	8.3	17 361	0.81	176.94	2KJ1408 - ■ JT13 - ■ ■ L1		539
	9.7	14 833	0.94	151.18	2KJ1408 - ■ JT13 - ■ ■ K1		539
	10.7	13 406	1.0	136.63 ★	2KJ1408 - ■ JT13 - ■ ■ J1		539
	11.1	12 916	1.1	131.64	2KJ1408 - ■ JT13 - ■ ■ H1		539
	12.8	11 172	1.3	113.86	2KJ1408 - ■ JT13 - ■ ■ G1		539
	14.7	9 744	1.4	99.31 ★	2KJ1408 - ■ JT13 - ■ ■ F1		539
	17.2	8 339	1.7	84.99 ★	2KJ1408 - ■ JT13 - ■ ■ E1		539
	19.2	7 469	1.9	76.12	2KJ1408 - ■ JT13 - ■ ■ D1		539
	23	6 326	2.2	64.47 ★	2KJ1408 - ■ JT13 - ■ ■ C1		539

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
15	FZ.168B-LA160ZLP4E						
	27	5 247	1.7	53.48	2KJ1308 - ■ JT13 - ■■ R1		522
	30	4 738	2.2	48.29	2KJ1308 - ■ JT13 - ■■ Q1		522
	FD.148B-LA160ZLP4E						
	13.1	10 919	0.82	111.29	2KJ1407 - ■ JT13 - ■■ E1		377
	15.1	9 461	0.95	96.43	★ 2KJ1407 - ■ JT13 - ■■ D1		377
	18.0	7 962	1.1	81.15	★ 2KJ1407 - ■ JT13 - ■■ C1		377
	19.9	7 184	1.3	73.22	2KJ1407 - ■ JT13 - ■■ B1		377
	23	6 174	1.5	62.93	★ 2KJ1407 - ■ JT13 - ■■ A1		377
	FZ.148B-LA160ZLP4E						
	23	6 316	1.0	64.37	★ 2KJ1307 - ■ JT13 - ■■ U1		369
	24	5 908	1.2	60.21	2KJ1307 - ■ JT13 - ■■ T1		369
	27	5 252	1.5	53.53	★ 2KJ1307 - ■ JT13 - ■■ S1		369
	29	4 959	1.6	50.54	2KJ1307 - ■ JT13 - ■■ R1		369
	32	4 452	2.0	45.37	★ 2KJ1307 - ■ JT13 - ■■ Q1		369
	35	4 086	2.2	41.64	2KJ1307 - ■ JT13 - ■■ P1		369
	41	3 525	2.6	35.93	2KJ1307 - ■ JT13 - ■■ N1		369
	FD.128B-LA160ZLP4E						
	19.2	7 450	0.82	75.93	2KJ1406 - ■ JT13 - ■■ C1		281
	22	6 358	0.96	64.8	★ 2KJ1406 - ■ JT13 - ■■ B1		281
	28	5 213	1.2	53.13	★ 2KJ1406 - ■ JT13 - ■■ A1		281
	FZ.128B-LA160ZLP4E						
	29	4 877	1.0	49.71	★ 2KJ1306 - ■ JT13 - ■■ W1		277
	31	4 558	1.1	46.46	2KJ1306 - ■ JT13 - ■■ V1		277
	36	4 022	1.4	40.99	★ 2KJ1306 - ■ JT13 - ■■ U1		277
	38	3 793	1.6	38.66	2KJ1306 - ■ JT13 - ■■ T1		277
	42	3 399	1.8	34.64	★ 2KJ1306 - ■ JT13 - ■■ S1		277
	46	3 138	1.9	31.98	2KJ1306 - ■ JT13 - ■■ R1		277
	53	2 682	2.3	27.33	2KJ1306 - ■ JT13 - ■■ Q1		277
	59	2 423	2.5	24.7	★ 2KJ1306 - ■ JT13 - ■■ P1		277
	61	2 335	2.6	23.8	2KJ1306 - ■ JT13 - ■■ N1		277
	71	2 019	3.0	20.58	2KJ1306 - ■ JT13 - ■■ L1		277
	FZ.108B-LA160ZLP4E						
	34	4 272	0.80	43.54	2KJ1305 - ■ JT13 - ■■ T1		199
	38	3 822	0.89	38.95	★ 2KJ1305 - ■ JT13 - ■■ S1		199
	40	3 542	0.96	36.1	2KJ1305 - ■ JT13 - ■■ R1		199
	44	3 247	1.0	33.09	★ 2KJ1305 - ■ JT13 - ■■ Q1		199
	48	2 976	1.1	30.33	2KJ1305 - ■ JT13 - ■■ P1		199
	56	2 536	1.3	25.85	2KJ1305 - ■ JT13 - ■■ N1		199
	64	2 238	1.5	22.81	★ 2KJ1305 - ■ JT13 - ■■ M1		199
	75	1 904	1.8	19.41	2KJ1305 - ■ JT13 - ■■ L1		199
	87	1 650	2.1	16.82	★ 2KJ1305 - ■ JT13 - ■■ K1		199
	103	1 389	2.4	14.16	★ 2KJ1305 - ■ JT13 - ■■ J1		199
	114	1 253	2.6	12.77	2KJ1305 - ■ JT13 - ■■ H1		199

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
15	FZ.108B-LA160ZLP4E						
	133	1 077	2.9	10.98	★	2KJ1305 - ■JT13 - ■■G1	199
	145	985	3.4	10.04		2KJ1305 - ■JT13 - ■■F1	199
	168	854	3.6	8.7	★	2KJ1305 - ■JT13 - ■■E1	199
	199	718	4.0	7.32	★	2KJ1305 - ■JT13 - ■■D1	199
	221	648	4.1	6.6		2KJ1305 - ■JT13 - ■■C1	199
	257	557	4.3	5.68	★	2KJ1305 - ■JT13 - ■■B1	199
	FZ.88B-LA160ZLP4E						
	71	2 026	0.94	20.65		2KJ1304 - ■JT13 - ■■K1	152
	81	1 766	1.1	18	★	2KJ1304 - ■JT13 - ■■J1	152
	95	1 502	1.3	15.31		2KJ1304 - ■JT13 - ■■H1	152
	112	1 282	1.5	13.07	★	2KJ1304 - ■JT13 - ■■G1	152
	136	1 051	1.8	10.71	★	2KJ1304 - ■JT13 - ■■F1	152
	159	902	1.8	9.19		2KJ1304 - ■JT13 - ■■E1	152
	182	786	2.0	8.01	★	2KJ1304 - ■JT13 - ■■D1	152
	214	669	2.2	6.82		2KJ1304 - ■JT13 - ■■C1	152
	251	571	2.4	5.82	★	2KJ1304 - ■JT13 - ■■B1	152
	306	468	2.6	4.77	★	2KJ1304 - ■JT13 - ■■A1	152
18.5	FD.208-LG200L6E						
	4.5	39 601	0.86	218.54		2KJ1411 - ■LL13 - ■■S1 P01	1 255
	4.8	37 113	0.92	204.81		2KJ1411 - ■LL13 - ■■R1 P01	1 255
	5.5	31 878	1.1	175.92	★	2KJ1411 - ■LL13 - ■■Q1 P01	1 255
	FD.208-LG180ZMB4E						
	6.1	29 086	1.2	242.01		2KJ1411 - ■KL13 - ■■T1	1 190
	6.7	26 266	1.3	218.54		2KJ1411 - ■KL13 - ■■S1	1 190
	7.2	24 616	1.4	204.81		2KJ1411 - ■KL13 - ■■R1	1 190
	8.4	21 143	1.6	175.92	★	2KJ1411 - ■KL13 - ■■Q1	1 190
	9.7	18 267	1.9	151.99		2KJ1411 - ■KL13 - ■■P1	1 190
	11.0	16 124	2.1	134.16		2KJ1411 - ■KL13 - ■■N1	1 190
	FD.188B-LG180ZMB4E						
	7.6	23 263	0.86	193.56		2KJ1410 - ■KL13 - ■■K1	775
	8.8	20 075	1.0	167.03		2KJ1410 - ■KL13 - ■■J1	775
	10.1	17 561	1.1	146.11		2KJ1410 - ■KL13 - ■■H1	775
	11.6	15 272	1.3	127.07		2KJ1410 - ■KL13 - ■■G1	775
	13.2	13 400	1.5	111.49	★	2KJ1410 - ■KL13 - ■■F1	775
	15.6	11 331	1.8	94.28	★	2KJ1410 - ■KL13 - ■■E1	775
	17.2	10 281	1.9	85.54		2KJ1410 - ■KL13 - ■■D1	775
	19.7	8 964	2.2	74.58	★	2KJ1410 - ■KL13 - ■■C1	775
	FD.168B-LG180ZMB4E						
	10.8	16 421	0.85	136.63	★	2KJ1408 - ■KL13 - ■■J1	594
	11.2	15 821	0.88	131.64		2KJ1408 - ■KL13 - ■■H1	594
	12.9	13 685	1.0	113.86		2KJ1408 - ■KL13 - ■■G1	594
	14.8	11 936	1.2	99.31	★	2KJ1408 - ■KL13 - ■■F1	594
	17.3	10 215	1.4	84.99	★	2KJ1408 - ■KL13 - ■■E1	594

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
18.5	FD.168B-LG180ZMB4E						
	19.3	9 149	1.5	76.12	2KJ1408 - ■ KL13 - ■■ D1		594
	23	7 748	1.8	64.47 ★	2KJ1408 - ■ KL13 - ■■ C1		594
	26	6 692	2.1	55.68	2KJ1408 - ■ KL13 - ■■ B1		594
	FZ.168B-LG180ZMB4E						
	28	6 428	1.4	53.48	2KJ1308 - ■ KL13 - ■■ R1		577
	30	5 804	1.8	48.29	2KJ1308 - ■ KL13 - ■■ Q1		577
	32	5 438	2.1	45.25	2KJ1308 - ■ KL13 - ■■ P1		577
	FD.148B-LG180ZMB4E						
	18.1	9 753	0.92	81.15 ★	2KJ1407 - ■ KL13 - ■■ C1		432
	20	8 800	1.0	73.22	2KJ1407 - ■ KL13 - ■■ B1		432
	23	7 563	1.2	62.93 ★	2KJ1407 - ■ KL13 - ■■ A1		432
	FZ.148B-LG180ZMB4E						
	28	6 434	1.2	53.53 ★	2KJ1307 - ■ KL13 - ■■ S1		424
	29	6 074	1.3	50.54	2KJ1307 - ■ KL13 - ■■ R1		424
	32	5 453	1.6	45.37 ★	2KJ1307 - ■ KL13 - ■■ Q1		424
	35	5 005	1.8	41.64	2KJ1307 - ■ KL13 - ■■ P1		424
	41	4 318	2.1	35.93	2KJ1307 - ■ KL13 - ■■ N1		424
	47	3 777	2.4	31.43	2KJ1307 - ■ KL13 - ■■ M1		424
	54	3 286	2.7	27.34	2KJ1307 - ■ KL13 - ■■ L1		424
	FD.128B-LG180ZMB4E						
	28	6 386	0.96	53.13 ★	2KJ1406 - ■ KL13 - ■■ A1		336
	FZ.128B-LG180ZMB4E						
	36	4 926	1.2	40.99 ★	2KJ1306 - ■ KL13 - ■■ U1		332
	38	4 646	1.3	38.66	2KJ1306 - ■ KL13 - ■■ T1		332
	42	4 163	1.5	34.64 ★	2KJ1306 - ■ KL13 - ■■ S1		332
	46	3 844	1.6	31.98	2KJ1306 - ■ KL13 - ■■ R1		332
	54	3 285	1.9	27.33	2KJ1306 - ■ KL13 - ■■ Q1		332
	60	2 969	2.1	24.7 ★	2KJ1306 - ■ KL13 - ■■ P1		332
	62	2 860	2.1	23.8	2KJ1306 - ■ KL13 - ■■ N1		332
	71	2 473	2.5	20.58	2KJ1306 - ■ KL13 - ■■ L1		332
	82	2 157	2.8	17.95 ★	2KJ1306 - ■ KL13 - ■■ K1		332
	96	1 846	3.2	15.36 ★	2KJ1306 - ■ KL13 - ■■ J1		332
	107	1 654	3.4	13.76	2KJ1306 - ■ KL13 - ■■ H1		332
	126	1 400	3.8	11.65 ★	2KJ1306 - ■ KL13 - ■■ G1		332
	213	830	4.3	6.91	2KJ1306 - ■ KL13 - ■■ D1		332
	251	703	4.7	5.85 ★	2KJ1306 - ■ KL13 - ■■ C1		332
	FZ.108B-LG180ZMB4E						
	44	3 977	0.85	33.09 ★	2KJ1305 - ■ KL13 - ■■ Q1		254
	48	3 645	0.93	30.33	2KJ1305 - ■ KL13 - ■■ P1		254
	57	3 107	1.1	25.85	2KJ1305 - ■ KL13 - ■■ N1		254
	64	2 741	1.2	22.81 ★	2KJ1305 - ■ KL13 - ■■ M1		254
	76	2 333	1.5	19.41	2KJ1305 - ■ KL13 - ■■ L1		254
	87	2 022	1.7	16.82 ★	2KJ1305 - ■ KL13 - ■■ K1		254

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
18.5	FZ.108B-LG180ZMB4E						
	104	1 702	1.9	14.16 ★	2KJ1305 - ■ KL13 - ■ J1		254
	115	1 535	2.1	12.77	2KJ1305 - ■ KL13 - ■ H1		254
	134	1 320	2.4	10.98 ★	2KJ1305 - ■ KL13 - ■ G1		254
	146	1 207	2.8	10.04	2KJ1305 - ■ KL13 - ■ F1		254
	169	1 046	3.0	8.7 ★	2KJ1305 - ■ KL13 - ■ E1		254
	201	880	3.2	7.32 ★	2KJ1305 - ■ KL13 - ■ D1		254
	223	793	3.3	6.6	2KJ1305 - ■ KL13 - ■ C1		254
	259	683	3.5	5.68 ★	2KJ1305 - ■ KL13 - ■ B1		254
22	FD.208-LG200ZLB6E						
	5.5	37 909	0.90	175.92 ★	2KJ1411 - ■ LM13 - ■ Q1	P01	1 255
	FD.208-LG180ZLB4E						
	6.1	34 707	0.98	242.01	2KJ1411 - ■ KP13 - ■ T1		1 205
	6.7	31 341	1.1	218.54	2KJ1411 - ■ KP13 - ■ S1		1 205
	7.2	29 372	1.2	204.81	2KJ1411 - ■ KP13 - ■ R1		1 205
	8.3	25 229	1.3	175.92 ★	2KJ1411 - ■ KP13 - ■ Q1		1 205
	9.6	21 797	1.6	151.99	2KJ1411 - ■ KP13 - ■ P1		1 205
	10.9	19 240	1.8	134.16	2KJ1411 - ■ KP13 - ■ N1		1 205
	12.1	17 320	2.0	120.77 ★	2KJ1411 - ■ KP13 - ■ M1		1 205
	FD.188B-LG180ZLB4E						
	8.8	23 954	0.83	167.03	2KJ1410 - ■ KP13 - ■ J1		790
	10.0	20 954	0.95	146.11	2KJ1410 - ■ KP13 - ■ H1		790
	11.5	18 223	1.1	127.07	2KJ1410 - ■ KP13 - ■ G1		790
	13.1	15 989	1.3	111.49 ★	2KJ1410 - ■ KP13 - ■ F1		790
	15.5	13 521	1.5	94.28 ★	2KJ1410 - ■ KP13 - ■ E1		790
	17.1	12 268	1.6	85.54	2KJ1410 - ■ KP13 - ■ D1		790
	19.6	10 696	1.9	74.58 ★	2KJ1410 - ■ KP13 - ■ C1		790
	23	9 081	2.2	63.32	2KJ1410 - ■ KP13 - ■ B1		790
	FZ.188B-LG180ZLB4E						
	28	7 548	2.2	52.63	2KJ1310 - ■ KP13 - ■ P1		767
	30	6 951	2.4	48.47	2KJ1310 - ■ KP13 - ■ N1		767
	FD.168B-LG180ZLB4E						
	12.9	16 329	0.86	113.86	2KJ1408 - ■ KP13 - ■ G1		609
	14.8	14 242	0.98	99.31 ★	2KJ1408 - ■ KP13 - ■ F1		609
	17.2	12 189	1.1	84.99 ★	2KJ1408 - ■ KP13 - ■ E1		609
	19.2	10 917	1.3	76.12	2KJ1408 - ■ KP13 - ■ D1		609
	23	9 246	1.5	64.47 ★	2KJ1408 - ■ KP13 - ■ C1		609
	26	7 985	1.8	55.68	2KJ1408 - ■ KP13 - ■ B1		609
	35	6 002	2.3	41.85 ★	2KJ1408 - ■ KP13 - ■ A1		609
	FZ.168B-LG180ZLB4E						
	27	7 670	1.2	53.48	2KJ1308 - ■ KP13 - ■ R1		592
	30	6 925	1.5	48.29	2KJ1308 - ■ KP13 - ■ Q1		592
	32	6 489	1.8	45.25	2KJ1308 - ■ KP13 - ■ P1		592
	38	5 574	2.3	38.87 ★	2KJ1308 - ■ KP13 - ■ N1		592
	44	4 816	2.7	33.58	2KJ1308 - ■ KP13 - ■ M1		592

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
22	FD.148B-LG180ZLB4E						
	20	10 501	0.86	73.22	2KJ1407 - ■ KP13 - ■■ B1		447
	23	9 025	1.0	62.93 ★	2KJ1407 - ■ KP13 - ■■ A1		447
	FZ.148B-LG180ZLB4E						
	27	7 677	1.0	53.53 ★	2KJ1307 - ■ KP13 - ■■ S1		439
	29	7 248	1.1	50.54	2KJ1307 - ■ KP13 - ■■ R1		439
	32	6 507	1.3	45.37 ★	2KJ1307 - ■ KP13 - ■■ Q1		439
	35	5 972	1.5	41.64	2KJ1307 - ■ KP13 - ■■ P1		439
	41	5 153	1.7	35.93	2KJ1307 - ■ KP13 - ■■ N1		439
	47	4 507	2.0	31.43	2KJ1307 - ■ KP13 - ■■ M1		439
	54	3 921	2.3	27.34	2KJ1307 - ■ KP13 - ■■ L1		439
	61	3 439	2.6	23.98 ★	2KJ1307 - ■ KP13 - ■■ K1		439
	72	2 908	3.1	20.28 ★	2KJ1307 - ■ KP13 - ■■ J1		439
	FD.128B-LG180ZLB4E						
	28	7 620	0.80	53.13 ★	2KJ1406 - ■ KP13 - ■■ A1		351
	FZ.128B-LG180ZLB4E						
	36	5 878	0.97	40.99 ★	2KJ1306 - ■ KP13 - ■■ U1		347
	38	5 544	1.1	38.66	2KJ1306 - ■ KP13 - ■■ T1		347
	42	4 968	1.2	34.64 ★	2KJ1306 - ■ KP13 - ■■ S1		347
	46	4 586	1.3	31.98	2KJ1306 - ■ KP13 - ■■ R1		347
	54	3 919	1.6	27.33	2KJ1306 - ■ KP13 - ■■ Q1		347
	59	3 542	1.7	24.7 ★	2KJ1306 - ■ KP13 - ■■ P1		347
	62	3 413	1.8	23.8	2KJ1306 - ■ KP13 - ■■ N1		347
	71	2 951	2.1	20.58	2KJ1306 - ■ KP13 - ■■ L1		347
	82	2 574	2.4	17.95 ★	2KJ1306 - ■ KP13 - ■■ K1		347
	95	2 203	2.7	15.36 ★	2KJ1306 - ■ KP13 - ■■ J1		347
	106	1 973	2.9	13.76	2KJ1306 - ■ KP13 - ■■ H1		347
	126	1 671	3.2	11.65 ★	2KJ1306 - ■ KP13 - ■■ G1		347
	145	1 444	3.5	10.07	2KJ1306 - ■ KP13 - ■■ F1		347
	194	1 086	4.2	7.57 ★	2KJ1306 - ■ KP13 - ■■ E1		347
	212	991	3.6	6.91	2KJ1306 - ■ KP13 - ■■ D1		347
	250	839	3.9	5.85 ★	2KJ1306 - ■ KP13 - ■■ C1		347
	290	724	4.3	5.05	2KJ1306 - ■ KP13 - ■■ B1		347
	386	545	5.0	3.8 ★	2KJ1306 - ■ KP13 - ■■ A1		347
	FZ.108B-LG180ZLB4E						
	57	3 707	0.92	25.85	2KJ1305 - ■ KP13 - ■■ N1		269
	64	3 271	1.0	22.81 ★	2KJ1305 - ■ KP13 - ■■ M1		269
	76	2 784	1.2	19.41	2KJ1305 - ■ KP13 - ■■ L1		269
	87	2 412	1.4	16.82 ★	2KJ1305 - ■ KP13 - ■■ K1		269
	103	2 031	1.6	14.16 ★	2KJ1305 - ■ KP13 - ■■ J1		269
	115	1 831	1.8	12.77	2KJ1305 - ■ KP13 - ■■ H1		269
	133	1 575	2.0	10.98 ★	2KJ1305 - ■ KP13 - ■■ G1		269
	146	1 440	2.3	10.04	2KJ1305 - ■ KP13 - ■■ F1		269
	168	1 248	2.5	8.7 ★	2KJ1305 - ■ KP13 - ■■ E1		269

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
22	FZ.108B-LG180ZLB4E						
	200	1 050	2.7	7.32 ★	2KJ1305 - ■ KP13 - ■■ D1		269
	222	947	2.8	6.6	2KJ1305 - ■ KP13 - ■■ C1		269
	258	815	3.0	5.68 ★	2KJ1305 - ■ KP13 - ■■ B1		269
30	FD.208-LG200LB4E						
	6.7	42 449	0.80	218.54	2KJ1411 - ■ LM13 - ■■ S1		1 255
	7.2	39 782	0.85	204.81	2KJ1411 - ■ LM13 - ■■ R1		1 255
	8.4	34 170	1.0	175.92 ★	2KJ1411 - ■ LM13 - ■■ Q1		1 255
	9.7	29 522	1.2	151.99	2KJ1411 - ■ LM13 - ■■ P1		1 255
	11.0	26 059	1.3	134.16	2KJ1411 - ■ LM13 - ■■ N1		1 255
	12.2	23 458	1.4	120.77 ★	2KJ1411 - ■ LM13 - ■■ M1		1 255
	14.7	19 465	1.7	100.21 ★	2KJ1411 - ■ LM13 - ■■ L1		1 255
	16.1	17 749	1.9	91.38	2KJ1411 - ■ LM13 - ■■ K1		1 255
	18.4	15 572	2.2	80.17 ★	2KJ1411 - ■ LM13 - ■■ J1		1 255
	FD.188B-LG200LB4E						
	11.6	24 682	0.81	127.07	2KJ1410 - ■ LM13 - ■■ G1		840
	13.2	21 656	0.92	111.49 ★	2KJ1410 - ■ LM13 - ■■ F1		840
	15.6	18 313	1.1	94.28 ★	2KJ1410 - ■ LM13 - ■■ E1		840
	17.2	16 615	1.2	85.54	2KJ1410 - ■ LM13 - ■■ D1		840
	19.8	14 486	1.4	74.58 ★	2KJ1410 - ■ LM13 - ■■ C1		840
	23	12 299	1.6	63.32	2KJ1410 - ■ LM13 - ■■ B1		840
	30	9 413	2.1	48.46 ★	2KJ1410 - ■ LM13 - ■■ A1		840
	FZ.188B-LG200LB4E						
	28	10 223	1.6	52.63	2KJ1310 - ■ LM13 - ■■ P1		817
	30	9 415	1.8	48.47	2KJ1310 - ■ LM13 - ■■ N1		817
	35	8 172	2.1	42.07 ★	2KJ1310 - ■ LM13 - ■■ M1		817
	40	7 202	2.4	37.08	2KJ1310 - ■ LM13 - ■■ L1		817
	FD.168B-LG200LB4E						
	17.4	16 508	0.85	84.99 ★	2KJ1408 - ■ LM13 - ■■ E1		659
	19.4	14 785	0.95	76.12	2KJ1408 - ■ LM13 - ■■ D1		659
	23	12 522	1.1	64.47 ★	2KJ1408 - ■ LM13 - ■■ C1		659
	26	10 815	1.3	55.68	2KJ1408 - ■ LM13 - ■■ B1		659
	35	8 129	1.7	41.85 ★	2KJ1408 - ■ LM13 - ■■ A1		659
	FZ.168B-LG200LB4E						
	28	10 388	0.87	53.48	2KJ1308 - ■ LM13 - ■■ R1		642
	30	9 380	1.1	48.29	2KJ1308 - ■ LM13 - ■■ Q1		642
	33	8 789	1.3	45.25	2KJ1308 - ■ LM13 - ■■ P1		642
	38	7 550	1.7	38.87 ★	2KJ1308 - ■ LM13 - ■■ N1		642
	44	6 522	2.0	33.58	2KJ1308 - ■ LM13 - ■■ M1		642
	50	5 757	2.4	29.64	2KJ1308 - ■ LM13 - ■■ L1		642
	55	5 182	2.7	26.68 ★	2KJ1308 - ■ LM13 - ■■ K1		642
	FZ.148B-LG200LB4E						
	29	9 817	0.81	50.54	2KJ1307 - ■ LM13 - ■■ R1		489
	32	8 813	0.99	45.37 ★	2KJ1307 - ■ LM13 - ■■ Q1		489

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
30	FZ.148B-LG200LB4E						
	35	8 088	1.1	41.64	2KJ1307 - ■ LM13 - ■■ P1		489
	41	6 979	1.3	35.93	2KJ1307 - ■ LM13 - ■■ N1		489
	47	6 105	1.5	31.43	2KJ1307 - ■ LM13 - ■■ M1		489
	54	5 310	1.7	27.34	2KJ1307 - ■ LM13 - ■■ L1		489
	62	4 658	1.9	23.98	★ 2KJ1307 - ■ LM13 - ■■ K1		489
	73	3 939	2.3	20.28	★ 2KJ1307 - ■ LM13 - ■■ J1		489
	80	3 574	2.5	18.4	2KJ1307 - ■ LM13 - ■■ H1		489
	92	3 116	2.9	16.04	★ 2KJ1307 - ■ LM13 - ■■ G1		489
	108	2 646	3.2	13.62	2KJ1307 - ■ LM13 - ■■ F1		489
	141	2 026	3.9	10.43	★ 2KJ1307 - ■ LM13 - ■■ E1		489
	155	1 847	3.6	9.51	2KJ1307 - ■ LM13 - ■■ D1		489
	178	1 610	3.9	8.29	★ 2KJ1307 - ■ LM13 - ■■ C1		489
	210	1 367	4.3	7.04	2KJ1307 - ■ LM13 - ■■ B1		489
	274	1 047	4.9	5.39	★ 2KJ1307 - ■ LM13 - ■■ A1		489
	FZ.128B-LG200LB4E						
	38	7 509	0.80	38.66	2KJ1306 - ■ LM13 - ■■ T1		397
	43	6 728	0.91	34.64	★ 2KJ1306 - ■ LM13 - ■■ S1		397
	46	6 212	0.98	31.98	2KJ1306 - ■ LM13 - ■■ R1		397
	54	5 309	1.1	27.33	2KJ1306 - ■ LM13 - ■■ Q1		397
	60	4 798	1.3	24.7	★ 2KJ1306 - ■ LM13 - ■■ P1		397
	62	4 623	1.3	23.8	2KJ1306 - ■ LM13 - ■■ N1		397
	72	3 997	1.5	20.58	2KJ1306 - ■ LM13 - ■■ L1		397
	82	3 487	1.7	17.95	★ 2KJ1306 - ■ LM13 - ■■ K1		397
	96	2 983	2.0	15.36	★ 2KJ1306 - ■ LM13 - ■■ J1		397
	107	2 673	2.1	13.76	2KJ1306 - ■ LM13 - ■■ H1		397
	127	2 263	2.4	11.65	★ 2KJ1306 - ■ LM13 - ■■ G1		397
	146	1 956	2.6	10.07	2KJ1306 - ■ LM13 - ■■ F1		397
	195	1 470	3.1	7.57	★ 2KJ1306 - ■ LM13 - ■■ E1		397
	213	1 342	2.7	6.91	2KJ1306 - ■ LM13 - ■■ D1		397
	252	1 136	2.9	5.85	★ 2KJ1306 - ■ LM13 - ■■ C1		397
	292	981	3.2	5.05	2KJ1306 - ■ LM13 - ■■ B1		397
	388	738	3.7	3.8	★ 2KJ1306 - ■ LM13 - ■■ A1		397
37	FD.208-LG225S4E						
	8.4	42 287	0.80	175.92	★ 2KJ1411 - ■ ME13 - ■■ Q1		1 335
	9.7	36 534	0.93	151.99	2KJ1411 - ■ ME13 - ■■ P1		1 335
	11.0	32 249	1.1	134.16	2KJ1411 - ■ ME13 - ■■ N1		1 335
	12.2	29 030	1.2	120.77	★ 2KJ1411 - ■ ME13 - ■■ M1		1 335
	14.7	24 088	1.4	100.21	★ 2KJ1411 - ■ ME13 - ■■ L1		1 335
	16.1	21 965	1.5	91.38	2KJ1411 - ■ ME13 - ■■ K1		1 335
	18.3	19 271	1.8	80.17	★ 2KJ1411 - ■ ME13 - ■■ J1		1 335
	21	16 562	2.1	68.9	2KJ1411 - ■ ME13 - ■■ H1		1 335
	FD.188B-LG225S4E						
	15.6	22 662	0.88	94.28	★ 2KJ1410 - ■ ME13 - ■■ E1		920

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
37	FD.188B-LG225S4E						
	17.2	20 562	0.97	85.54	2KJ1410 - ■ME13 - ■■D1		920
	19.7	17 927	1.1	74.58 ★	2KJ1410 - ■ME13 - ■■C1		920
	23	15 220	1.3	63.32	2KJ1410 - ■ME13 - ■■B1		920
	30	11 649	1.7	48.46 ★	2KJ1410 - ■ME13 - ■■A1		920
	FZ.188B-LG225S4E						
	28	12 651	1.3	52.63	2KJ1310 - ■ME13 - ■■P1		897
	30	11 651	1.4	48.47	2KJ1310 - ■ME13 - ■■N1		897
	35	10 113	1.7	42.07 ★	2KJ1310 - ■ME13 - ■■M1		897
	40	8 913	2.0	37.08	2KJ1310 - ■ME13 - ■■L1		897
	45	7 822	2.4	32.54	2KJ1310 - ■ME13 - ■■K1		897
	50	7 050	2.8	29.33 ★	2KJ1310 - ■ME13 - ■■J1		897
	FD.168B-LG225S4E						
	23	15 497	0.90	64.47 ★	2KJ1408 - ■ME13 - ■■C1		739
	26	13 384	1.0	55.68	2KJ1408 - ■ME13 - ■■B1		739
	35	10 060	1.4	41.85 ★	2KJ1408 - ■ME13 - ■■A1		739
	FZ.168B-LG225S4E						
	30	11 608	0.90	48.29	2KJ1308 - ■ME13 - ■■Q1		722
	32	10 877	1.1	45.25	2KJ1308 - ■ME13 - ■■P1		722
	38	9 343	1.4	38.87 ★	2KJ1308 - ■ME13 - ■■N1		722
	44	8 072	1.6	33.58	2KJ1308 - ■ME13 - ■■M1		722
	50	7 125	2.0	29.64	2KJ1308 - ■ME13 - ■■L1		722
	55	6 413	2.2	26.68 ★	2KJ1308 - ■ME13 - ■■K1		722
	66	5 322	2.6	22.14 ★	2KJ1308 - ■ME13 - ■■J1		722
	73	4 853	2.9	20.19	2KJ1308 - ■ME13 - ■■H1		722
	83	4 257	3.3	17.71 ★	2KJ1308 - ■ME13 - ■■G1		722
	FZ.148B-LG225S4E						
	32	10 906	0.80	45.37 ★	2KJ1307 - ■ME13 - ■■Q1		569
	35	10 009	0.90	41.64	2KJ1307 - ■ME13 - ■■P1		569
	41	8 637	1.0	35.93	2KJ1307 - ■ME13 - ■■N1		569
	47	7 555	1.2	31.43	2KJ1307 - ■ME13 - ■■M1		569
	54	6 572	1.4	27.34	2KJ1307 - ■ME13 - ■■L1		569
	61	5 764	1.6	23.98 ★	2KJ1307 - ■ME13 - ■■K1		569
	72	4 875	1.8	20.28 ★	2KJ1307 - ■ME13 - ■■J1		569
	80	4 423	2.0	18.4	2KJ1307 - ■ME13 - ■■H1		569
	92	3 856	2.3	16.04 ★	2KJ1307 - ■ME13 - ■■G1		569
	108	3 274	2.6	13.62	2KJ1307 - ■ME13 - ■■F1		569
	141	2 507	3.1	10.43 ★	2KJ1307 - ■ME13 - ■■E1		569
	155	2 286	2.9	9.51	2KJ1307 - ■ME13 - ■■D1		569
	177	1 993	3.1	8.29 ★	2KJ1307 - ■ME13 - ■■C1		569
	209	1 692	3.4	7.04	2KJ1307 - ■ME13 - ■■B1		569
	273	1 296	4.0	5.39 ★	2KJ1307 - ■ME13 - ■■A1		569
	FZ.128B-K4-LGI225S4E						
	54	6 569	0.93	27.33	2KJ1306 - ■ME13 - ■■Q1		477

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
37	FZ.128B-K4-LGI225S4E						
	60	5 937	1.0	24.7 ★	2KJ1306 - ■ME13 - ■■P1		477
	62	5 721	1.1	23.8	2KJ1306 - ■ME13 - ■■N1		477
	71	4 947	1.2	20.58	2KJ1306 - ■ME13 - ■■L1		477
	82	4 315	1.4	17.95 ★	2KJ1306 - ■ME13 - ■■K1		477
	96	3 692	1.6	15.36 ★	2KJ1306 - ■ME13 - ■■J1		477
	107	3 308	1.7	13.76	2KJ1306 - ■ME13 - ■■H1		477
	126	2 800	1.9	11.65 ★	2KJ1306 - ■ME13 - ■■G1		477
	146	2 421	2.1	10.07	2KJ1306 - ■ME13 - ■■F1		477
	194	1 820	2.5	7.57 ★	2KJ1306 - ■ME13 - ■■E1		477
	213	1 661	2.2	6.91	2KJ1306 - ■ME13 - ■■D1		477
	251	1 406	2.3	5.85 ★	2KJ1306 - ■ME13 - ■■C1		477
	291	1 214	2.6	5.05	2KJ1306 - ■ME13 - ■■B1		477
	387	913	3.0	3.8 ★	2KJ1306 - ■ME13 - ■■A1		477
45	FD.208-LG225ZM4E						
	11.0	39 088	0.87	134.16	2KJ1411 - ■MU13 - ■■N1		1 335
	12.2	35 187	0.97	120.77 ★	2KJ1411 - ■MU13 - ■■M1		1 335
	14.7	29 197	1.2	100.21 ★	2KJ1411 - ■MU13 - ■■L1		1 335
	16.1	26 624	1.3	91.38	2KJ1411 - ■MU13 - ■■K1		1 335
	18.4	23 358	1.5	80.17 ★	2KJ1411 - ■MU13 - ■■J1		1 335
	21	20 074	1.7	68.9	2KJ1411 - ■MU13 - ■■H1		1 335
	28	15 634	2.2	53.66	2KJ1411 - ■MU13 - ■■G1		1 335
	FD.188B-LG225ZM4E						
	17.2	24 923	0.80	85.54	2KJ1410 - ■MU13 - ■■D1		920
	19.8	21 729	0.92	74.58 ★	2KJ1410 - ■MU13 - ■■C1		920
	23	18 449	1.1	63.32	2KJ1410 - ■MU13 - ■■B1		920
	30	14 119	1.4	48.46 ★	2KJ1410 - ■MU13 - ■■A1		920
	FZ.188B-LG225ZM4E						
	28	15 334	1.1	52.63	2KJ1310 - ■MU13 - ■■P1		897
	30	14 122	1.2	48.47	2KJ1310 - ■MU13 - ■■N1		897
	35	12 257	1.4	42.07 ★	2KJ1310 - ■MU13 - ■■M1		897
	40	10 803	1.6	37.08	2KJ1310 - ■MU13 - ■■L1		897
	45	9 481	2.0	32.54	2KJ1310 - ■MU13 - ■■K1		897
	50	8 545	2.3	29.33 ★	2KJ1310 - ■MU13 - ■■J1		897
	59	7 255	2.8	24.9 ★	2KJ1310 - ■MU13 - ■■H1		897
	64	6 739	3.0	23.13	2KJ1310 - ■MU13 - ■■G1		897
	FD.168B-LG225ZM4E						
	26	16 223	0.86	55.68	2KJ1408 - ■MU13 - ■■B1		739
	35	12 193	1.1	41.85 ★	2KJ1408 - ■MU13 - ■■A1		739
	FZ.168B-LG225ZM4E						
	33	13 184	0.87	45.25	2KJ1308 - ■MU13 - ■■P1		722
	38	11 325	1.1	38.87 ★	2KJ1308 - ■MU13 - ■■N1		722
	44	9 784	1.3	33.58	2KJ1308 - ■MU13 - ■■M1		722
	50	8 636	1.6	29.64	2KJ1308 - ■MU13 - ■■L1		722

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
45	FZ.168B-LG225ZM4E						
	55	7 773	1.8	26.68 ★	2KJ1308 - ■MU13 - ■■K1		722
	67	6 451	2.2	22.14 ★	2KJ1308 - ■MU13 - ■■J1		722
	73	5 882	2.4	20.19	2KJ1308 - ■MU13 - ■■H1		722
	83	5 160	2.7	17.71 ★	2KJ1308 - ■MU13 - ■■G1		722
	97	4 434	3.2	15.22	2KJ1308 - ■MU13 - ■■F1		722
	FZ.148B-LG225ZM4E						
	41	10 468	0.86	35.93	2KJ1307 - ■MU13 - ■■N1		569
	47	9 157	0.98	31.43	2KJ1307 - ■MU13 - ■■M1		569
	54	7 966	1.1	27.34	2KJ1307 - ■MU13 - ■■L1		569
	62	6 987	1.3	23.98 ★	2KJ1307 - ■MU13 - ■■K1		569
	73	5 909	1.5	20.28 ★	2KJ1307 - ■MU13 - ■■J1		569
	80	5 361	1.7	18.4	2KJ1307 - ■MU13 - ■■H1		569
	92	4 673	1.9	16.04 ★	2KJ1307 - ■MU13 - ■■G1		569
	108	3 968	2.1	13.62	2KJ1307 - ■MU13 - ■■F1		569
	141	3 039	2.6	10.43 ★	2KJ1307 - ■MU13 - ■■E1		569
	155	2 771	2.4	9.51	2KJ1307 - ■MU13 - ■■D1		569
	178	2 415	2.6	8.29 ★	2KJ1307 - ■MU13 - ■■C1		569
	210	2 051	2.8	7.04	2KJ1307 - ■MU13 - ■■B1		569
	274	1 570	3.3	5.39 ★	2KJ1307 - ■MU13 - ■■A1		569
	FZ.128B-K4-LGI225ZM4E						
	60	7 188	0.85	24.7 ★	2KJ1306 - ■MU13 - ■■P1		477
	62	6 926	0.88	23.8	2KJ1306 - ■MU13 - ■■N1		477
	72	5 989	1.0	20.58	2KJ1306 - ■MU13 - ■■L1		477
	82	5 223	1.2	17.95 ★	2KJ1306 - ■MU13 - ■■K1		477
	96	4 470	1.3	15.36 ★	2KJ1306 - ■MU13 - ■■J1		477
	107	4 004	1.4	13.76	2KJ1306 - ■MU13 - ■■H1		477
	127	3 390	1.6	11.65 ★	2KJ1306 - ■MU13 - ■■G1		477
	146	2 930	1.7	10.07	2KJ1306 - ■MU13 - ■■F1		477
	195	2 203	2.1	7.57 ★	2KJ1306 - ■MU13 - ■■E1		477
	213	2 011	1.8	6.91	2KJ1306 - ■MU13 - ■■D1		477
	252	1 702	1.9	5.85 ★	2KJ1306 - ■MU13 - ■■C1		477
	292	1 470	2.1	5.05	2KJ1306 - ■MU13 - ■■B1		477
	388	1 106	2.4	3.8 ★	2KJ1306 - ■MU13 - ■■A1		477
55	FD.208-LG250ZM4E						
	14.8	35 564	0.96	100.21 ★	2KJ1411 - ■NN13 - ■■L1		1 425
	16.2	32 431	1.0	91.38	2KJ1411 - ■NN13 - ■■K1		1 425
	18.5	28 452	1.2	80.17 ★	2KJ1411 - ■NN13 - ■■J1		1 425
	22	24 453	1.4	68.9	2KJ1411 - ■NN13 - ■■H1		1 425
	28	19 044	1.8	53.66	2KJ1411 - ■NN13 - ■■G1		1 425
	35	15 129	2.2	42.63 ★	2KJ1411 - ■NN13 - ■■F1		1 425
	38	13 784	2.5	38.84	2KJ1411 - ■NN13 - ■■E1		1 425

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
55	FD.188B-LG250ZM4E						
	23	22 472	0.89	63.32	2KJ1410 - ■ NN13 - ■ ■ B1		1 010
	30	17 198	1.2	48.46 ★	2KJ1410 - ■ NN13 - ■ ■ A1		1 010
	FZ.188B-LG250ZM4E						
	30	17 202	0.98	48.47	2KJ1310 - ■ NN13 - ■ ■ N1		987
	35	14 931	1.2	42.07 ★	2KJ1310 - ■ NN13 - ■ ■ M1		987
	40	13 160	1.3	37.08	2KJ1310 - ■ NN13 - ■ ■ L1		987
	46	11 548	1.6	32.54	2KJ1310 - ■ NN13 - ■ ■ K1		987
	50	10 409	1.9	29.33 ★	2KJ1310 - ■ NN13 - ■ ■ J1		987
	59	8 837	2.3	24.9 ★	2KJ1310 - ■ NN13 - ■ ■ H1		987
	64	8 209	2.4	23.13	2KJ1310 - ■ NN13 - ■ ■ G1		987
	74	7 052	2.8	19.87 ★	2KJ1310 - ■ NN13 - ■ ■ F1		987
	87	6 016	3.1	16.95	2KJ1310 - ■ NN13 - ■ ■ E1		987
	FD.168B-LG250ZM4E						
	35	14 853	0.94	41.85 ★	2KJ1408 - ■ NN13 - ■ ■ A1		829
	FZ.168B-LG250ZM4E						
	38	13 795	0.94	38.87 ★	2KJ1308 - ■ NN13 - ■ ■ N1		812
	44	11 917	1.1	33.58	2KJ1308 - ■ NN13 - ■ ■ M1		812
	50	10 519	1.3	29.64	2KJ1308 - ■ NN13 - ■ ■ L1		812
	56	9 469	1.5	26.68 ★	2KJ1308 - ■ NN13 - ■ ■ K1		812
	67	7 857	1.8	22.14 ★	2KJ1308 - ■ NN13 - ■ ■ J1		812
	73	7 165	2.0	20.19	2KJ1308 - ■ NN13 - ■ ■ H1		812
	84	6 285	2.2	17.71 ★	2KJ1308 - ■ NN13 - ■ ■ G1		812
	97	5 402	2.6	15.22	2KJ1308 - ■ NN13 - ■ ■ F1		812
	125	4 209	3.1	11.86 ★	2KJ1308 - ■ NN13 - ■ ■ E1		812
	157	3 343	3.6	9.42 ★	2KJ1308 - ■ NN13 - ■ ■ D1		812
	173	3 031	3.7	8.54	2KJ1308 - ■ NN13 - ■ ■ C1		812
	223	2 360	4.2	6.65 ★	2KJ1308 - ■ NN13 - ■ ■ B1		812
	280	1 874	4.6	5.28 ★	2KJ1308 - ■ NN13 - ■ ■ A1		812
	FZ.148B-K4-LGI250ZM4E						
	47	11 154	0.81	31.43	2KJ1307 - ■ NN13 - ■ ■ M1		659
	54	9 703	0.93	27.34	2KJ1307 - ■ NN13 - ■ ■ L1		659
	62	8 510	1.1	23.98 ★	2KJ1307 - ■ NN13 - ■ ■ K1		659
	73	7 197	1.3	20.28 ★	2KJ1307 - ■ NN13 - ■ ■ J1		659
	80	6 530	1.4	18.4	2KJ1307 - ■ NN13 - ■ ■ H1		659
	92	5 693	1.6	16.04 ★	2KJ1307 - ■ NN13 - ■ ■ G1		659
	109	4 834	1.8	13.62	2KJ1307 - ■ NN13 - ■ ■ F1		659
	142	3 702	2.1	10.43 ★	2KJ1307 - ■ NN13 - ■ ■ E1		659
	156	3 375	1.9	9.51	2KJ1307 - ■ NN13 - ■ ■ D1		659
	179	2 942	2.1	8.29 ★	2KJ1307 - ■ NN13 - ■ ■ C1		659
	210	2 498	2.3	7.04	2KJ1307 - ■ NN13 - ■ ■ B1		659
	275	1 913	2.7	5.39 ★	2KJ1307 - ■ NN13 - ■ ■ A1		659

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
75	FD.208-K4-LGI280S4E						
	18.5	38 668	0.88	80.17 ★	2KJ1411 - PG13 - J1		1 550
	22	33 232	1.0	68.9	2KJ1411 - PG13 - H1		1 550
	28	25 881	1.3	53.66	2KJ1411 - PG13 - G1		1 550
	35	20 561	1.7	42.63 ★	2KJ1411 - PG13 - F1		1 550
	38	18 733	1.8	38.84	2KJ1411 - PG13 - E1		1 550
	49	14 590	2.2	30.25 ★	2KJ1411 - PG13 - D1		1 550
	62	11 590	2.6	24.03 ★	2KJ1411 - PG13 - C1		1 550
	FD.188B-K4-LGI280S4E						
	31	23 373	0.86	48.46 ★	2KJ1410 - PG13 - A1		1 135
	FZ.188B-K4-LGI280S4E						
	40	17 885	0.98	37.08	2KJ1310 - PG13 - L1		1 112
	46	15 695	1.2	32.54	2KJ1310 - PG13 - K1		1 112
	51	14 147	1.4	29.33 ★	2KJ1310 - PG13 - J1		1 112
	60	12 010	1.7	24.9 ★	2KJ1310 - PG13 - H1		1 112
	64	11 156	1.8	23.13	2KJ1310 - PG13 - G1		1 112
	75	9 584	2.1	19.87 ★	2KJ1310 - PG13 - F1		1 112
	88	8 175	2.3	16.95	2KJ1310 - PG13 - E1		1 112
	111	6 439	2.7	13.35 ★	2KJ1310 - PG13 - D1		1 112
	138	5 180	3.1	10.74 ★	2KJ1310 - PG13 - C1		1 112
	159	4 505	3.3	9.34	2KJ1310 - PG13 - B1		1 112
	178	4 023	3.5	8.34	2KJ1310 - PG13 - A1		1 112
	FZ.168B-K4-LGI280S4E						
	44	16 196	0.80	33.58	2KJ1308 - PG13 - M1		937
	50	14 296	0.98	29.64	2KJ1308 - PG13 - L1		937
	56	12 868	1.1	26.68 ★	2KJ1308 - PG13 - K1		937
	67	10 679	1.3	22.14 ★	2KJ1308 - PG13 - J1		937
	74	9 738	1.4	20.19	2KJ1308 - PG13 - H1		937
	84	8 542	1.6	17.71 ★	2KJ1308 - PG13 - G1		937
	98	7 341	1.9	15.22	2KJ1308 - PG13 - F1		937
	125	5 720	2.3	11.86 ★	2KJ1308 - PG13 - E1		937
	158	4 543	2.7	9.42 ★	2KJ1308 - PG13 - D1		937
	174	4 119	2.7	8.54	2KJ1308 - PG13 - C1		937
	223	3 207	3.1	6.65 ★	2KJ1308 - PG13 - B1		937
	281	2 547	3.4	5.28 ★	2KJ1308 - PG13 - A1		937
90	FD.208-K4-LGI280ZM4E						
	22	39 878	0.85	68.9	2KJ1411 - PW13 - H1		1 590
	28	31 058	1.1	53.66	2KJ1411 - PW13 - G1		1 590
	35	24 674	1.4	42.63 ★	2KJ1411 - PW13 - F1		1 590
	38	22 480	1.5	38.84	2KJ1411 - PW13 - E1		1 590
	49	17 508	1.8	30.25 ★	2KJ1411 - PW13 - D1		1 590
	62	13 908	2.1	24.03 ★	2KJ1411 - PW13 - C1		1 590

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
90	FZ.208-K4-LGI280ZM4E						
	74	11 610	2.8	20.06	2KJ1311 - ■PW13 - ■■H1		1 570
	93	9 272	3.3	16.02 ★	2KJ1311 - ■PW13 - ■■G1		1 570
	FZ.188B-K4-LGI280ZM4E						
	40	21 461	0.82	37.08	2KJ1310 - ■PW13 - ■■L1		1 152
	46	18 834	0.98	32.54	2KJ1310 - ■PW13 - ■■K1		1 152
	51	16 976	1.2	29.33 ★	2KJ1310 - ■PW13 - ■■J1		1 152
	60	14 412	1.4	24.9 ★	2KJ1310 - ■PW13 - ■■H1		1 152
	64	13 387	1.5	23.13	2KJ1310 - ■PW13 - ■■G1		1 152
	75	11 501	1.7	19.87 ★	2KJ1310 - ■PW13 - ■■F1		1 152
	88	9 810	1.9	16.95	2KJ1310 - ■PW13 - ■■E1		1 152
	111	7 727	2.3	13.35 ★	2KJ1310 - ■PW13 - ■■D1		1 152
	138	6 216	2.6	10.74 ★	2KJ1310 - ■PW13 - ■■C1		1 152
	159	5 406	2.8	9.34	2KJ1310 - ■PW13 - ■■B1		1 152
	178	4 827	2.9	8.34	2KJ1310 - ■PW13 - ■■A1		1 152
	FZ.168B-K4-LGI280ZM4E						
	50	17 155	0.82	29.64	2KJ1308 - ■PW13 - ■■L1		977
	56	15 442	0.91	26.68 ★	2KJ1308 - ■PW13 - ■■K1		977
	67	12 814	1.1	22.14 ★	2KJ1308 - ■PW13 - ■■J1		977
	74	11 686	1.2	20.19	2KJ1308 - ■PW13 - ■■H1		977
	84	10 250	1.4	17.71 ★	2KJ1308 - ■PW13 - ■■G1		977
	98	8 809	1.6	15.22	2KJ1308 - ■PW13 - ■■F1		977
	125	6 864	1.9	11.86 ★	2KJ1308 - ■PW13 - ■■E1		977
	158	5 452	2.2	9.42 ★	2KJ1308 - ■PW13 - ■■D1		977
	174	4 943	2.3	8.54	2KJ1308 - ■PW13 - ■■C1		977
	223	3 849	2.6	6.65 ★	2KJ1308 - ■PW13 - ■■B1		977
	281	3 056	2.8	5.28 ★	2KJ1308 - ■PW13 - ■■A1		977
110	FD.208-K2-LGI315S4E						
	28	37 832	0.90	53.66	2KJ1411 - ■QQ13 - ■■G1		1 770
	35	30 056	1.1	42.63 ★	2KJ1411 - ■QQ13 - ■■F1		1 770
	38	27 384	1.2	38.84	2KJ1411 - ■QQ13 - ■■E1		1 770
	49	21 327	1.5	30.25 ★	2KJ1411 - ■QQ13 - ■■D1		1 770
	62	16 942	1.8	24.03 ★	2KJ1411 - ■QQ13 - ■■C1		1 770
	FZ.208-K2-LGI315S4E						
	74	14 143	2.3	20.06	2KJ1311 - ■QQ13 - ■■H1		1 750
	93	11 295	2.7	16.02 ★	2KJ1311 - ■QQ13 - ■■G1		1 750
	114	9 229	3.1	13.09 ★	2KJ1311 - ■QQ13 - ■■F1		1 750
	129	8 115	3.4	11.51	2KJ1311 - ■QQ13 - ■■E1		1 750
	141	7 452	3.6	10.57	2KJ1311 - ■QQ13 - ■■D1		1 750
	165	6 352	4.0	9.01	2KJ1311 - ■QQ13 - ■■C1		1 750
	FZ.188B-K2-LGI315S4E						
	88	11 950	1.6	16.95	2KJ1310 - ■QQ13 - ■■E1		1 332
	112	9 412	1.9	13.35 ★	2KJ1310 - ■QQ13 - ■■D1		1 332
	139	7 572	2.1	10.74 ★	2KJ1310 - ■QQ13 - ■■C1		1 332

★ Preferred transmission ratio

Shaft designs, see page 3/89 — 1, 2, 3, 5, 6 or 9

Frequency and voltage, see page 8/20 — 1 to 9

Gearbox housing mounting position, see page 3/92 — A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_B	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
110	FZ.188B-K2-LGI315S4E						
	160	6 585	2.3	9.34	2KJ1310 - ■■QQ13 - ■■B1	1 332	
	179	5 880	2.4	8.34	2KJ1310 - ■■QQ13 - ■■A1	1 332	
132	FD.208-K2-LGI315ZM4E						
	35	36 188	0.94	42.63 ★	2KJ1411 - ■■QS13 - ■■F1	1 825	
	38	32 971	1.0	38.84	2KJ1411 - ■■QS13 - ■■E1	1 825	
	49	25 679	1.2	30.25 ★	2KJ1411 - ■■QS13 - ■■D1	1 825	
	62	20 399	1.5	24.03 ★	2KJ1411 - ■■QS13 - ■■C1	1 825	
	FZ.208-K2-LGI315ZM4E						
	74	17 029	1.9	20.06	2KJ1311 - ■■QS13 - ■■H1	1 805	
	93	13 599	2.2	16.02 ★	2KJ1311 - ■■QS13 - ■■G1	1 805	
	113	11 112	2.6	13.09 ★	2KJ1311 - ■■QS13 - ■■F1	1 805	
	129	9 771	2.8	11.51	2KJ1311 - ■■QS13 - ■■E1	1 805	
	140	8 973	3.0	10.57	2KJ1311 - ■■QS13 - ■■D1	1 805	
	165	7 648	3.3	9.01	2KJ1311 - ■■QS13 - ■■C1	1 805	
	FZ.188B2-K2-LGI315ZM4E						
	88	14 389	1.3	16.95	2KJ1310 - ■■QS13 - ■■E1	1 387	
	111	11 333	1.5	13.35 ★	2KJ1310 - ■■QS13 - ■■D1	1 387	
	138	9 117	1.8	10.74 ★	2KJ1310 - ■■QS13 - ■■C1	1 387	
	159	7 929	1.9	9.34	2KJ1310 - ■■QS13 - ■■B1	1 387	
	178	7 080	2.0	8.34	2KJ1310 - ■■QS13 - ■■A1	1 387	
160	FD.208-K2-LGI315L4E						
	38	39 965	0.85	38.84	2KJ1411 - ■■QU13 - ■■E1	1 950	
	49	31 126	1.0	30.25 ★	2KJ1411 - ■■QU13 - ■■D1	1 950	
	62	24 726	1.2	24.03 ★	2KJ1411 - ■■QU13 - ■■C1	1 950	
	FZ.208-K2-LGI315L4E						
	74	20 641	1.6	20.06	2KJ1311 - ■■QU13 - ■■H1	1 930	
	93	16 484	1.8	16.02 ★	2KJ1311 - ■■QU13 - ■■G1	1 930	
	113	13 469	2.1	13.09 ★	2KJ1311 - ■■QU13 - ■■F1	1 930	
	129	11 843	2.3	11.51	2KJ1311 - ■■QU13 - ■■E1	1 930	
	140	10 876	2.5	10.57	2KJ1311 - ■■QU13 - ■■D1	1 930	
	165	9 271	2.7	9.01	2KJ1311 - ■■QU13 - ■■C1	1 930	
	FZ.188B-K2-LGI315L4E						
	88	17 441	1.1	16.95	2KJ1310 - ■■QU13 - ■■E1	1 512	
	111	13 737	1.3	13.35 ★	2KJ1310 - ■■QU13 - ■■D1	1 512	
	138	11 051	1.5	10.74 ★	2KJ1310 - ■■QU13 - ■■C1	1 512	
	159	9 610	1.6	9.34	2KJ1310 - ■■QU13 - ■■B1	1 512	
	178	8 581	1.7	8.34	2KJ1310 - ■■QU13 - ■■A1	1 512	
200	FD.208-K2-LGI315ZLB4E						
	49	38 777	0.83	30.25 ★	2KJ1411 - ■■QV13 - ■■D1	2 065	
	62	30 804	0.97	24.03 ★	2KJ1411 - ■■QV13 - ■■C1	2 065	
	FZ.208-K2-LGI315ZLB4E						
	74	25 714	1.3	20.06	2KJ1311 - ■■QV13 - ■■H1	2 045	
93	20 536	1.5	16.02 ★	2KJ1311 - ■■QV13 - ■■G1	2 045		

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

*) For mounting type B5-01

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

MOTOX Geared Motors

Parallel shaft geared motors

Geared motors up to 200 kW

Selection and ordering data (continued)

Power rating P_{Motor} kW	Output speed n_2 (50 Hz) rpm	Output torque T_2 Nm	Service factor f_{B}	Gearbox ratio i_{tot}	Order No.	Order code (No. of poles)	Weight ^{*)} kg
200	FZ.208-K2-LGI315ZLB4E						
	114	16 780	1.7	13.09 ★	2KJ1311 - ■ QV13 - ■■ F1		2 045
	129	14 754	1.9	11.51	2KJ1311 - ■ QV13 - ■■ E1		2 045
	141	13 549	2.0	10.57	2KJ1311 - ■ QV13 - ■■ D1		2 045
	165	11 550	2.2	9.01	2KJ1311 - ■ QV13 - ■■ C1		2 045
	FZ.188B-K2-LGI315ZLB4E						
	88	21 728	0.86	16.95	2KJ1310 - ■ QV13 - ■■ E1		1 627
	112	17 113	1.0	13.35 ★	2KJ1310 - ■ QV13 - ■■ D1		1 627
	139	13 767	1.2	10.74 ★	2KJ1310 - ■ QV13 - ■■ C1		1 627
	160	11 973	1.3	9.34	2KJ1310 - ■ QV13 - ■■ B1		1 627
	179	10 691	1.3	8.34	2KJ1310 - ■ QV13 - ■■ A1		1 627

★ Preferred transmission ratio

Shaft designs, see page 3/89

Frequency and voltage, see page 8/20

Gearbox housing mounting position, see page 3/92

1, 2, 3, 5, 6 or 9

1 to 9

A, D, E, F, H or M

*) For mounting type B5-01

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ¹⁾	Nominal torque	Permissible input torque T_1 [Nm]																	
						2.5x the value is permissible for a brief period (e.g. motor starting torque)																	
						Motor size																	
Max. gearbox torque	Order No	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	3	3	5	10	20	26	61	98	198	198	291	356	580	1290				
Nm	15th and 16th position		rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315				
FD.28 150	N1	280.00	5.2	21	150		•																
	M1	241.56 ★	6.0	21	150		•	•															
	L1	207.53	7.0	21	150		•	•															
	K1	191.06 ★	7.6	21	150		•	•	•														
	J1	173.69	8.3	21	150		•	•	•														
	H1	153.74 ★	9.4	21	150		•	•	•														
	G1	128.77	11.3	21	150		•	•	•														
	F1	109.79 ★	13.2	21	150		•	•	•														
	E1	93.32 ★	15.5	21	150		•	•	•														
	D1	81.10	17.9	22	150		•	•	•														
	C1	70.59 ★	21.0	22	150		•	•	•														
	B1	63.68	23.0	22	150		•	•	•														
	A1	56.20	26.0	22	150		•	•	•														
FZ.28 96 ... 150	C2	59.65	24	20	150		•																
	B2	50.30 ★	29	20	150		•	•															
	A2	44.66	32	20	150		•	•															
	X1	39.15 ★	37	20	150		•	•	•														
	W1	35.04	41	20	150		•	•	•														
	V1	31.10 ★	47	20	150		•	•	•														
	U1	27.25	53	20	150		•	•	•														
	T1	23.96 ★	61	20	150		•	•	•														
	S1	21.64	67	20	150		•	•	•	•													
	R1	18.86 ★	77	20	150		•	•	•														
	Q1	16.94	86	20	150		•	•	•														
	P1	15.29 ★	95	21	150		•	•	•	•													
	N1	13.87	105	21	150		•	•	•	•													
	M1	12.62 ★	115	21	148		•	•	•	•													
	L1	11.16	130	21	142		•	•	•	•													
	K1	10.30 ★	141	21	138		•	•	•	•													
	J1	8.87	163	22	131		•	•	•	•													
	H1	8.06 ★	180	22	127		•	•	•	•													
	G1	7.20 ★	201	27	126		•	•	•	•													
	F1	6.53	222	28	122		•	•	•	•													
	E1	5.94 ★	244	28	118		•	•	•	•													
	D1	5.25	276	28	111		•	•	•	•													
	C1	4.85 ★	299	28	110		•	•	•	•													
	B1	4.18	347	29	99		•	•	•	•													

★ Preferred transmission ratio

¹⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input units:

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
Max. gearbox torque	Order No.					2.5x the value is permissible for a brief period (e.g. motor starting torque)															
	15th and 16th position	i_{tot}	n_2 (50 Hz)	φ	T_{2N} ($f_B=1$)	3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Nm			rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FZ.38B-D28 290	M1	7 591	0.18	–	290		•														
	L1	6 548	★	0.21	290		•	•													
	K1	5 626		0.25	290		•	•													
	J1	5 179	★	0.27	290		•	•	•												
	H1	4 709		0.30	290		•	•	•												
	G1	4 168	★	0.34	290		•	•	•												
	F1	3 491		0.40	290		•	•	•												
	E1	2 976	★	0.47	290		•	•	•												
	D1	2 530	★	0.55	290		•	•	•												
	C1	2 199		0.64	290		•	•	•												
	B1	1 914	★	0.73	290		•	•	•												
	A1	1 726		0.81	290		•	•	•												
FZ.38B-Z28 290	Q1	1 617		0.87	290		•														
	P1	1 364	★	1.00	290		•	•													
	N1	1 211		1.20	290		•	•													
	M1	1 061	★	1.30	290		•	•	•												
	L1	950		1.50	290		•	•	•												
	K1	843	★	1.70	290		•	•	•												
	J1	739		1.90	290		•	•	•												
	H1	650	★	2.20	290		•	•	•												
	G1	587		2.40	290		•	•	•	•											
	F1	511	★	2.70	290		•	•	•												
	E1	459		3.10	290		•	•	•												
	D1	415	★	3.40	290		•	•	•	•											
	C1	376		3.70	290		•	•	•	•											
	B1	342	★	4.10	290		•	•	•	•											
	A1	303		4.60	290		•	•	•	•											

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmis- sion ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
Max. gearbox torque	Order No.					2.5x the value is permissible for a brief period (e.g. motor starting torque)															
	15th and 16th position	i_{tot}	n_2 (50 Hz)	φ	T_{2N} ($f_B=1$)	3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
	Nm		rpm	arcmin	Nm	Motor size															
						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.38B 290	N1	280.41	5.2	20	290	•	•														
	M1	241.91 ★	6.0	20	290	•	•	•													
	L1	207.83	7.0	20	290	•	•	•													
	K1	191.34 ★	7.6	20	290	•	•	•	•												
	J1	173.94	8.3	20	290	•	•	•	•												
	H1	153.96 ★	9.4	20	290	•	•	•	•												
	G1	128.95	11.2	20	290	•	•	•	•												
	F1	109.95 ★	13.2	20	290	•	•	•	•												
	E1	93.46 ★	15.5	20	290	•	•	•	•												
	D1	81.22	17.9	20	290	•	•	•	•												
	C1	70.70 ★	21.0	20	290	•	•	•	•												
	B1	63.77	23.0	20	290	•	•	•	•												
A1	56.28	26.0	20	290	•	•	•	•													
FZ.38B 210 ... 290	B2	56.72 ★	26	19	210	•	•	•													
	A2	50.44	29	19	230	•	•	•													
	X1	43.75 ★	33	19	250	•	•	•	•												
	W1	40.88	35	19	275	•	•	•	•												
	V1	35.96 ★	40	19	290	•	•	•	•												
	U1	31.49	46	19	290	•	•	•	•	•											
	T1	27.85 ★	52	19	290	•	•	•	•	•											
	S1	25.24	57	19	290	•	•	•	•	•											
	R1	22.28 ★	65	19	290	•	•	•	•	•											
	Q1	20.10	72	20	290	•	•	•	•	•											
	P1	18.23 ★	80	20	290	•	•	•	•	•											
	N1	16.61	87	20	290	•	•	•	•	•											
	M1	15.19 ★	95	20	290	•	•	•	•	•											
	L1	13.58	107	20	290	•	•	•	•	•											
	K1	12.47 ★	116	20	290	•	•	•	•	•											
	J1	11.24	129	20	290	•	•	•	•	•											
	H1	9.67 ★	150	20	290	•	•	•	•	•											
	G1	8.52 ★	170	20	290	•	•	•	•	•											
	F1	7.76	187	20	290	•	•	•	•	•											
	E1	7.10 ★	204	20	290	•	•	•	•	•											
	D1	6.35	228	20	275	•	•	•	•	•											
	C1	5.83 ★	249	20	275	•	•	•	•	•											
	B1	5.25	276	20	253	•	•	•	•	•											
	A1	4.52 ★	321	21	228	•	•	•	•	•											

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.48B-D28 540	M1	19 701		0.07	—	540	•														
	L1	16 996	★	0.08	—	540	•	•													
	K1	14 602		0.10	—	540	•	•													
	J1	13 443	★	0.10	—	540	•	•	•												
	H1	12 221		0.11	—	540	•	•	•												
	G1	10 817	★	0.13	—	540	•	•	•												
	F1	9 060		0.15	—	540	•	•	•												
	E1	7 725	★	0.18	—	540	•	•	•												
	D1	6 566	★	0.21	—	540	•	•	•												
	C1	5 706		0.25	—	540	•	•	•												
	B1	4 967	★	0.28	—	540	•	•	•												
	A1	4 480		0.31	—	540	•	•	•												
FD.48B-Z28 540	B2	4 197		0.33	—	540	•														
	A2	3 539	★	0.40	—	540	•	•													
	X1	3 142		0.45	—	540	•	•													
	W1	2 755	★	0.51	—	540	•	•	•												
	V1	2 465		0.57	—	540	•	•	•												
	U1	2 188	★	0.64	—	540	•	•	•												
	T1	1 918		0.73	—	540	•	•	•												
	S1	1 686	★	0.83	—	540	•	•	•												
	R1	1 523		0.92	—	540	•	•	•	•											
	Q1	1 327	★	1.10	—	540	•	•	•												
	P1	1 192		1.20	—	540	•	•	•												
	N1	1 076	★	1.30	—	540	•	•	•	•											
	M1	976		1.40	—	540	•	•	•	•											
	L1	888	★	1.60	—	540	•	•	•	•											
	K1	785		1.80	—	540	•	•	•	•											
	J1	725	★	1.90	—	540	•	•	•	•											
	H1	624		2.20	—	540	•	•	•	•											
	G1	567	★	2.50	—	540	•	•	•	•											
	F1	516	★	2.70	—	540	•	•	•	•											
	E1	468		3.00	—	540	•	•	•	•											
	D1	426	★	3.30	—	540	•	•	•	•											
	C1	376		3.70	—	540	•	•	•	•											
	B1	347	★	4.00	—	540	•	•	•	•											
	A1	299		4.70	—	540	•	•	•	•											

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]													
						2.5x the value is permissible for a brief period (e.g. motor starting torque)													
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size													
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315
FD.48B 540	S1	268.80 ★	5.4	12	540	•	•	•											
	R1	238.65	6.1	12	540	•	•	•											
	Q1	209.23 ★	6.9	12	540	•	•	•	•										
	P1	187.24	7.7	12	540	•	•	•	•										
	N1	166.19 ★	8.7	13	540	•	•	•	•										
	M1	145.63	10.0	13	540	•	•	•	•										
	L1	128.04 ★	11.3	13	540	•	•	•	•										
	K1	115.68	12.5	13	540	•	•	•	•										
	J1	100.80 ★	14.4	13	540	•	•	•	•										
	H1	90.53	16.0	13	540	•	•	•	•										
	G1	81.73 ★	17.7	13	540	•	•	•	•	•									
	F1	74.10	19.6	13	540	•	•	•	•	•									
	E1	67.43 ★	22.0	13	540	•	•	•	•	•									
	D1	59.62	24.0	13	540	•	•	•	•	•									
	C1	55.06 ★	26.0	13	540	•	•	•	•	•									
	B1	47.40	31.0	13	540	•	•	•	•	•									
	A1	43.09 ★	34.0	13	540	•	•	•	•	•									
FZ.48B 325 ... 540	B2	60.71 ★	24	12	400	•	•	•	•										
	A2	55.19	26	12	500	•	•	•	•										
	X1	49.58 ★	29	12	540	•	•	•	•										
	W1	42.50	34	12	540	•	•	•	•	•									
	V1	38.45 ★	38	12	540	•	•	•	•	•	• ¹⁾								
	U1	35.49	41	12	540	•	•	•	•	•	• ¹⁾								
	T1	30.86 ★	47	12	540	•	•	•	•	•	• ¹⁾								
	S1	28.02	52	12	540	•	•	•	•	•	• ¹⁾								
	R1	25.59 ★	57	12	540	•	•	•	•	•	• ¹⁾								
	Q1	23.48	62	12	540	•	•	•	•	•	• ¹⁾								
	P1	21.63 ★	67	12	540	•	•	•	•	•	• ¹⁾								
	N1	19.64	74	12	540	•	•	•	•	•	• ¹⁾								
	M1	17.89 ★	81	12	540	•	•	•	•	•	• ¹⁾								
	L1	16.39	88	12	540	•	•	•	•	•	• ¹⁾								
	K1	14.63 ★	99	12	540	•	•	•	•	•	• ¹⁾								
	J1	13.05	111	13	540			•	•	•	• ¹⁾								
	H1	11.09	131	13	535			•	•	•	• ¹⁾								
	G1	9.23 ★	157	13	526			•	•	•	• ¹⁾								
	F1	8.39 ★	173	16	510	•	•	•	•	•	• ¹⁾								
	E1	7.68	189	16	467	•	•	•	•	•	• ¹⁾								
	D1	6.86 ★	211	17	443	•	•	•	•	•	• ¹⁾								
	C1	6.12	237	17	406			•	•	•	• ¹⁾								
	B1	5.20	279	18	378			•	•	•	• ¹⁾								
	A1	4.33 ★	335	19	325			•	•	•	• ¹⁾								

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]																
						2.5x the value is permissible for a brief period (e.g. motor starting torque)																
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290	Motor size		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	63	71	80	90	100	112	132	160	180	200	225	250	280	315			
Nm	15th and 16th position		rpm	arcmin	Nm																	
FD.68B-D28 1 000	T1	39 638	0.04	—	1 000		•															
	S1	34 196	★	0.04	1 000		•	•														
	R1	29 378		0.05	1 000		•	•														
	Q1	27 047	★	0.05	1 000		•	•	•													
	P1	24 588		0.06	1 000		•	•	•													
	N1	21 763	★	0.06	1 000		•	•	•													
	M1	20 908		0.07	1 000		•															
	L1	18 038	★	0.08	1 000		•	•														
	K1	15 497	★	0.09	1 000		•	•														
	J1	14 267		0.10	1 000		•	•	•													
	H1	12 970	★	0.11	1 000		•	•	•													
	G1	11 480		0.12	1 000		•	•	•													
	F1	9 615		0.15	1 000		•	•	•													
	E1	8 198		0.17	1 000		•	•	•													
	D1	6 969		0.20	1 000		•	•	•													
	C1	6 056		0.23	1 000		•	•	•													
	B1	5 271		0.27	1 000		•	•	•													
	A1	4 755		0.29	1 000		•	•	•													
FD.68B-Z28 1 000	B2	4 454		0.31	1 000		•															
	A2	3 756	★	0.37	1 000		•	•														
	X1	3 335		0.42	1 000		•	•														
	W1	2 924	★	0.48	1 000		•	•	•													
	V1	2 916		0.54	1 000		•	•	•													
	U1	2 322	★	0.60	1 000		•	•	•													
	T1	2 035		0.69	1 000		•	•	•													
	S1	1 789	★	0.78	1 000		•	•	•													
	R1	1 616		0.87	1 000		•	•	•	•												
	Q1	1 408	★	0.99	1 000		•	•	•													
	P1	1 265		1.10	1 000		•	•	•													
	N1	1 142	★	1.20	1 000		•	•	•	•												
	M1	1 036		1.40	1 000		•	•	•	•												
	L1	942	★	1.50	1 000		•	•	•	•	•											
	K1	833		1.70	1 000		•	•	•	•												
	J1	769	★	1.80	1 000		•	•	•	•												
	H1	662		2.10	1 000		•	•	•	•												
	G1	602	★	2.30	1 000		•	•	•	•												
	F1	547	★	2.60	1 000		•	•	•	•												
	E1	496		2.80	1 000		•	•	•	•												
	D1	452	★	3.10	1 000		•	•	•	•												
	C1	399		3.50	1 000		•	•	•	•												
	B1	369	★	3.80	1 000		•	•	•	•												
	A1	317		4.40	1 000		•	•	•	•												

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	2.5x the value is permissible for a brief period (e.g. motor starting torque)															
	3					3	5	10	20	26	61	98	198	198	291	356	580	1290			
	Motor size																				
Nm	15th and 16th position		rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.68B 1 000	S1	296.18 ★	4.9	10.	1 000	•	•	•													
	R1	263.39	5.5	10	1 000	•	•	•													
	Q1	228.48 ★	6.3	10	1 000	•	•	•	•												
	P1	213.48	6.8	10	1 000	•	•	•	•												
	N1	187.76 ★	7.7	11	1 000	•	•	•	•												
	M1	164.44	8.8	11	1 000	•	•	•	•	•											
	L1	145.44 ★	10.0	11	1 000	•	•	•	•	•											
	K1	131.82	11.0	11	1 000	•	•	•	•	•											
	J1	116.36 ★	12.5	11	1 000	•	•	•	•	•											
	H1	104.96	13.8	11	1 000	•	•	•	•	•											
	G1	95.20 ★	15.2	11	1 000	•	•	•	•	•											
	F1	86.74	16.7	11	1 000	•	•	•	•	•											
	E1	79.33 ★	18.3	11	1 000	•	•	•	•	•											
	D1	70.93	20.0	11	1 000	•	•	•	•	•											
	C1	65.14 ★	22.0	11	1 000	•	•	•	•	•											
	B1	58.71	25.0	11	1 000	•	•	•	•	•											
	A1	50.48 ★	29.0	11	1 000	•	•	•	•	•											
FZ.68B 589 ... 1 000	B2	61.17 ★	24	10	850			•	•												
	A2	53.50	27	10	1 000			•	•	•											
	X1	48.03 ★	30	10	1 000			•	•	•	•										
	V1	43.87	33	10	1 000			•	•	•	•										
	U1	38.93 ★	37	10	1 000			•	•	•	•	• ¹⁾									
	T1	35.93	40	10	1 000			•	•	•	•	• ¹⁾									
	S1	32.50 ★	45	10	1 000			•	•	•	•	• ¹⁾									
	R1	29.93	48	10	1 000			•	•	•	•	• ¹⁾									
	Q1	27.68 ★	52	10	1 000			•	•	•	•	• ¹⁾									
	P1	25.69	56	10	1 000			•	•	•	•	• ¹⁾									
	N1	22.67 ★	64	10	1 000			•	•	•	•	• ¹⁾									
	M1	20.93	69	10	1 000			•	•	•	•	• ¹⁾									
	L1	18.75 ★	77	10	1 000			•	•	•	•	• ¹⁾									
	K1	17.29	84	10	1 000			•	•	•	•	• ¹⁾									
	J1	14.51	100	10	1 000			•	•	•	•	• ¹⁾									
	H1	12.38 ★	117	11	1 000			•	•	•	•	• ¹⁾									
	G1	10.31	141	11	1 000					•	•	• ¹⁾									
	F1	8.55 ★	170	11	1 000					•	•	• ¹⁾									
	E1	8.03	181	15	897			•	•	•	•	• ¹⁾									
	D1	6.74	215	15	835			•	•	•	•	• ¹⁾									
	C1	5.75 ★	252	16	755			•	•	•	•	• ¹⁾									
	B1	4.79	303	16	682					•	•	• ¹⁾									
	A1	3.97 ★	365	17	589					•	•	• ¹⁾									

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

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Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.88B-D28 1 900	T1	54 705	0.03	—	1 900		•														
	S1	47 195	★	0.03	1 900		•	•													
	R1	40 546		0.03	1 900		•	•													
	Q1	37 328	★	0.04	1 900		•	•	•												
	P1	33 935		0.04	1 900		•	•	•												
	N1	30 036	★	0.05	1 900		•	•	•												
	M1	28 814		0.05	1 900		•	•													
	L1	24 755	★	0.06	1 900		•	•													
	K1	22 790	★	0.06	1 900		•	•	•												
	J1	20 718		0.07	1 900		•	•	•												
	H1	18 338	★	0.08	1 900		•	•	•												
	G1	15 360		0.09	1 900		•	•	•												
	F1	13 096	★	0.11	1 900		•	•	•												
	E1	11 132	★	0.13	1 900		•	•	•												
	D1	9 674		0.14	1 900		•	•	•												
	C1	8 420	★	0.17	1 900		•	•	•												
	B1	7 595		0.18	1 900		•	•	•												
	A1	6 703		0.21	1 900		•	•	•												
FD.88B-Z28 1 900	B2	6 000	★	0.23	1 900		•	•													
	A2	5 327		0.26	1 900		•	•													
	X1	4 670	★	0.30	1 900		•	•	•												
	W1	4 179		0.33	1 900		•	•	•												
	V1	3 709	★	0.38	1 900		•	•	•												
	U1	3 251		0.43	1 900		•	•	•												
	T1	2 858	★	0.49	1 900		•	•	•												
	S1	2 582		0.54	1 900		•	•	•	•											
	R1	2 250	★	0.62	1 900		•	•	•												
	Q1	2 021		0.69	1 900		•	•	•												
	P1	1 824	★	0.77	1 900		•	•	•	•											
	N1	1 654		0.85	1 900		•	•	•	•											
	M1	1 505	★	0.93	1 900		•	•	•	•											
	L1	1 331		1.10	1 900		•	•	•	•	•										
	K1	1 229	★	1.10	1 900		•	•	•	•											
	J1	1 058		1.30	1 900		•	•	•	•											
	H1	962	★	1.50	1 900		•	•	•	•											
	G1	874	★	1.60	1 900		•	•	•	•											
	F1	793		1.80	1 900		•	•	•	•											
	E1	721	★	1.90	1 900		•	•	•	•											
	D1	638		2.20	1 900		•	•	•	•											
	C1	589	★	2.40	1 900		•	•	•	•											
	B1	507		2.80	1 900		•	•	•	•											
	A1	461	★	3.00	1 900		•	•	•	•											

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.88B 1 900	V1	404.92	3.6	9	1 900	•	•	•													
	U1	358.33 ★	4.0	9	1 900	•	•	•	•												
	T1	325.76	4.5	9	1 900	•	•	•	•												
	S1	292.64 ★	5.0	9	1 900	•	•	•	•												
	R1	250.83	5.8	9	1 900	•	•	•	•	•											
	P1	226.94 ★	6.4	9	1 900	•	•	•	•	•	•										
	N1	209.49	6.9	9	1 900	•	•	•	•	•	•										
	M1	182.15 ★	8.0	9	1 900	•	•	•	•	•	•	•									
	L1	165.38	8.8	9	1 900	•	•	•	•	•	•	•	•								
	K1	151.01 ★	9.6	9	1 900	•	•	•	•	•	•	•	•	•							
	J1	138.56	10.5	9	1 900	•	•	•	•	•	•	•	•	•	•						
	H1	127.66 ★	11.4	9	1 900	•	•	•	•	•	•	•	•	•	•	•					
	G1	115.93	12.5	9	1 900	•	•	•	•	•	•	•	•	•	•	•	•				
	F1	105.61 ★	13.7	9	1 900	•	•	•	•	•	•	•	•	•	•	•	•	•			
	E1	96.75	15.0	9	1 900	•	•	•	•	•	•	•	•	•	•	•	•	•	•		
	D1	86.33 ★	16.8	9	1 900	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
	C1	77.04	18.8	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	
	B1	65.43	22.0	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	
	A1	54.47 ★	27.0	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	
FZ.88B 1 199 ... 1 900	X1	64.58 ★	22	8	1 900			•	•	•	•										
	W1	59.13	25	8	1 900			•	•	•	•										
	V1	52.60 ★	28	8	1 900			•	•	•	•	•									
	U1	48.08	30	8	1 900			•	•	•	•	•	•								
	T1	44.20 ★	33	8	1 900			•	•	•	•	•	•	•							
	S1	40.83	36	8	1 900			•	•	•	•	•	•	•	•						
	R1	37.89 ★	38	8	1 900			•	•	•	•	•	•	•	•	•	1)				
	Q1	35.29	41	8	1 900			•	•	•	•	•	•	•	•	•	•	•	1)		
	P1	31.91 ★	45	8	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	N1	29.38	49	8	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	M1	26.42 ★	55	8	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	L1	24.38	59	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	K1	20.65	70	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	J1	18.00 ★	81	9	1 900			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	H1	15.31	95	9	1 900				•	•	•	•	•	•	•	•	•	•	•	•	1)
	G1	13.07 ★	111	9	1 900				•	•	•	•	•	•	•	•	•	•	•	•	1)
	F1	10.71 ★	135	9	1 900				•	•	•	•	•	•	•	•	•	•	•	•	1)
	E1	9.19	158	12	1 658			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	D1	8.01 ★	181	12	1 548			•	•	•	•	•	•	•	•	•	•	•	•	•	1)
	C1	6.82	213	13	1 454				•	•	•	•	•	•	•	•	•	•	•	•	1)
	B1	5.82 ★	249	13	1 348				•	•	•	•	•	•	•	•	•	•	•	•	1)
	A1	4.77 ★	304	13	1 199				•	•	•	•	•	•	•	•	•	•	•	•	1)

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
	15th and 16th position					rpm	arcmin	Nm	Motor size												
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.108B-D38 3 400	N1	66 190	★	0.02	–	3 400	•	•	•												
	M1	58 766		0.02	–	3 400	•	•	•												
	L1	51 521	★	0.03	–	3 400	•	•	•	•											
	K1	46 105		0.03	–	3 400	•	•	•	•											
	J1	40 922	★	0.04	–	3 400	•	•	•	•											
	H1	35 860		0.04	–	3 400	•	•	•	•											
	G1	31 530	★	0.05	–	3 400	•	•	•	•											
	F1	28 485		0.05	–	3 400	•	•	•	•											
	E1	24 821	★	0.06	–	3 400	•	•	•	•											
	D1	22 293		0.07	–	3 400	•	•	•	•											
	C1	20 125	★	0.07	–	3 400	•	•	•	•											
	B1	18 247		0.08	–	3 400	•	•	•	•											
	A1	16 603	★	0.09	–	3 400	•	•	•	•											
FD.108B-Z38 3 400	M2	15 230	★	0.10	–	3 400	•	•	•												
	L2	13 544		0.11	–	3 400	•	•	•												
	K2	11 749	★	0.12	–	3 400	•	•	•	•											
	J2	10 977		0.13	–	3 400	•	•	•	•											
	H2	9 655	★	0.15	–	3 400	•	•	•	•											
	G2	8 456		0.17	–	3 400	•	•	•	•	•										
	F2	7 479	★	0.19	–	3 400	•	•	•	•	•										
	E2	6 778		0.21	–	3 400	•	•	•	•	•										
	D2	5 983	★	0.24	–	3 400	•	•	•	•	•										
	C2	5 397		0.27	–	3 400	•	•	•	•	•										
	B2	4 895	★	0.30	–	3 400	•	•	•	•	•										
	A2	4 460		0.33	–	3 400	•	•	•	•	•										
	X1	4 079	★	0.36	–	3 400	•	•	•	•	•										
	W1	3 648		0.40	–	3 400	•	•	•	•	•										
	V1	3 349	★	0.43	–	3 400	•	•	•	•	•										
	U1	3 019		0.48	–	3 400	•	•	•	•	•										
	T1	2 596	★	0.56	–	3 400	•	•	•	•	•										
	S1	2 315		0.63	–	3 400	•	•	•	•	•										
	R1	2 126	★	0.68	–	3 400	•	•	•	•	•										
	Q1	1 916		0.76	–	3 400	•	•	•	•	•										
	P1	1 647	★	0.88	–	3 400	•	•	•	•	•										
	N1	1 526		0.95	–	3 400	•	•	•	•	•										
	M1	1 384	★	1.00	–	3 400	•	•	•	•	•										
	L1	1 261		1.10	–	3 400	•	•	•	•	•										
	K1	1 153	★	1.30	–	3 400	•	•	•	•	•										
	J1	1 031		1.40	–	3 400	•	•	•	•	•										
	H1	947	★	1.50	–	3 400	•	•	•	•	•										
	G1	853		1.70	–	3 400	•	•	•	•	•										
	F1	734	★	2.00	–	3 400	•	•	•	•	•										
	E1	732	★	2.00	–	3 400	•	•	•	•	•										
	D1	654		2.20	–	3 400	•	•	•	•	•										
	C1	601	★	2.40	–	3 400	•	•	•	•	•										
	B1	541		2.70	–	3 400	•	•	•	•	•										
	A1	466	★	3.10	–	3 400	•	•	•	•	•										

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}; \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size															
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.108B 3 400	V1	424.49 ★	3.4	7	3 400			•	•												
	U1	382.79	3.8	7	3 400			•	•												
	T1	345.19 ★	4.2	7	3 400			•	•												
	S1	301.88	4.8	7	3 400			•	•	•											
	R1	271.01 ★	5.4	7	3 400			•	•	•	•										
	Q1	247.53	5.9	7	3 400			•	•	•	•										
	P1	219.66 ★	6.6	7	3 400			•	•	•	•	•									
	N1	202.77	7.2	7	3 400			•	•	•	•	•									
	M1	183.39 ★	7.9	7	3 400			•	•	•	•	•									
	L1	168.88	8.6	8	3 400			•	•	•	•	•									
	K1	156.19 ★	9.3	8	3 400			•	•	•	•	•	•								
	J1	144.99	10.0	8	3 400			•	•	•	•	•	•								
	H1	127.92 ★	11.3	8	3 400			•	•	•	•	•	•								
	G1	118.11	12.3	8	3 400			•	•	•	•	•	•								
	F1	105.81 ★	13.7	8	3 400			•	•	•	•	•	•								
	E1	97.57	14.9	8	3 400			•	•	•	•	•	•								
	D1	81.86	17.7	8	3 400			•	•	•	•	•	•								
	C1	69.84 ★	21.0	8	3 400			•	•	•	•	•	•								
	B1	58.20	25.0	8	3 400					•	•	•	•								
	A1	48.24 ★	30.0	8	3 400					•	•	•	•								
FZ.108B 2 422 ... 3 400	A2	64.21 ★	23	7	3 000					•	•	•									
	X1	58.80	25	7	3 000					•	•	•									
	W1	54.17 ★	27	7	3 400					•	•	•									
	V1	50.15	29	7	3 400					•	•	•									
	U1	46.64 ★	31	7	3 400					•	•	•	•								
	T1	43.54	33	7	3 400					•	•	•	•								
	S1	38.95 ★	37	7	3 400					•	•	•	•	•	• ¹⁾						
	R1	36.10	40	7	3 400					•	•	•	•	•	• ¹⁾						
	Q1	33.09 ★	44	7	3 400					•	•	•	•	•	• ¹⁾						
	P1	30.33	48	7	3 400					•	•	•	•	•	• ¹⁾						
	N1	25.85	56	7	3 400					•	•	•	•	•	• ¹⁾						
	M1	22.81 ★	64	7	3 400					•	•	•	•	•	• ¹⁾						
	L1	19.41	75	7	3 400					•	•	•	•	•	• ¹⁾						
	K1	16.82 ★	86	7	3 400					•	•	•	•	•	• ¹⁾						
	J1	14.16 ★	102	7	3 304					•	•	•	•	•	• ¹⁾						
	H1	12.77	114	7	3 249							•	•	•	• ¹⁾						
	G1	10.98 ★	132	8	3 153							•	•	•	• ¹⁾						
	F1	10.04	144	10	3 374					•	•	•	•	•	• ¹⁾						
	E1	8.70 ★	167	10	3 102					•	•	•	•	•	• ¹⁾						
	D1	7.32 ★	198	10	2 853					•	•	•	•	•	• ¹⁾						
	C1	6.60	220	10	2 651							•	•	•	• ¹⁾						
	B1	5.68 ★	255	10	2 422							•	•	•	• ¹⁾						

★ Preferred transmission ratio

1) Only possible with integrated motor.

2) Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	Motor size															
Nm	15th and 16th position		rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.128B-D38 6 100	N1	68 070	★	0.02	—	6 100	•	•	•												
	M1	60 435		0.02	—	6 100	•	•	•												
	L1	52 984	★	0.03	—	6 100	•	•	•	•											
	K1	47 415		0.03	—	6 100	•	•	•	•											
	J1	42 084	★	0.03	—	6 100	•	•	•	•											
	H1	36 878		0.04	—	6 100	•	•	•	•											
	G1	32 425	★	0.04	—	6 100	•	•	•	•											
	F1	29 294		0.05	—	6 100	•	•	•	•											
	E1	25 526	★	0.06	—	6 100	•	•	•	•											
	D1	22 926		0.06	—	6 100	•	•	•	•											
	C1	20 697	★	0.07	—	6 100	•	•	•	•											
	B1	18 765		0.08	—	6 100	•	•	•	•											
	A1	17 075	★	0.08	—	6 100	•	•	•	•											
FD.128B-Z38 6 100	W1	15 663	★	0.09	—	6 100	•	•	•												
	V1	13 928		0.10	—	6 100	•	•	•												
	U1	12 083	★	0.12	—	6 100	•	•	•	•											
	T1	11 289		0.13	—	6 100	•	•	•	•											
	S1	9 929	★	0.15	—	6 100	•	•	•	•											
	R1	8 696		0.17	—	6 100	•	•	•	•	•										
	Q1	7 691	★	0.19	—	6 100	•	•	•	•	•										
	P1	6 971		0.21	—	6 100	•	•	•	•	•										
	N1	6 153	★	0.24	—	6 100	•	•	•	•	•										
	M1	5 551		0.26	—	6 100	•	•	•	•	•										
	L1	5 034	★	0.29	—	6 100	•	•	•	•	•										
	K1	4 587		0.32	—	6 100	•	•	•	•	•										
	J1	4 195	★	0.35	—	6 100	•	•	•	•	•										
	H1	3 751		0.39	—	6 100	•	•	•	•	•										
	G1	3 445	★	0.42	—	6 100	•	•	•	•	•										
	F1	3 105		0.47	—	6 100	•	•	•	•	•										
	E1	2 670	★	0.54	—	6 100	•	•	•	•	•										
	D1	2 381		0.61	—	6 100	•	•	•	•	•										
	C1	2 186	★	0.66	—	6 100	•	•	•	•	•										
	B1	1 970		0.74	—	6 100	•	•	•	•	•										
	A1	1 694	★	0.86	—	6 100	•	•	•	•	•										
FD.128B-Z48 6 100	L1	1 504		0.96	—	6 100	•	•	•	•	•	•									
	K1	1 370	★	1.10	—	6 100	•	•	•	•	•	•									
	J1	1 255		1.20	—	6 100	•	•	•	•	•	•									
	H1	1 120	★	1.30	—	6 100	•	•	•	•	•	•									
	G1	999		1.50	—	6 100			•	•	•	•									
	F1	849		1.70	—	6 100			•	•	•	•									
	E1	706	★	2.10	—	6 100			•	•	•	•									
	D1	695	★	2.10	—	6 100	•	•	•	•	•	•									
	C1	620		2.30	—	6 100			•	•	•	•									
	B1	527		2.80	—	6 100			•	•	•	•									
	A1	439	★	3.30	—	6 100			•	•	•	•									

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot} \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290	Motor size	
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.128B 6 100	V1	447.96	3.2	7	6 100				•												
	U1	405.47 ★	3.6	7	6 100				•												
	T1	354.99	4.1	7	6 100				•	•											
	S1	320.24 ★	4.5	7	6 100				•	•	•										
	R1	293.22	4.9	7	6 100				•	•	•										
	Q1	260.84 ★	5.6	7	6 100				•	•	•	•									
	P1	238.39	6.1	7	6 100				•	•	•	•									
	N1	219.15 ★	6.6	7	6 100				•	•	•	•									
	M1	202.48	7.2	7	6 100				•	•	•	•									
	L1	187.88 ★	7.7	7	6 100				•	•	•	•	•								
	K1	175.01	8.3	7	6 100				•	•	•	•	•								
	J1	158.22 ★	9.2	7	6 100				•	•	•	•	•	•	•						
	H1	145.66	10.0	7	6 100				•	•	•	•	•	•	•						
	G1	131.01 ★	11.1	7	6 100				•	•	•	•	•	•	•						
	F1	120.87	12.0	7	6 100				•	•	•	•	•	•	•						
	E1	102.41	14.2	7	6 100				•	•	•	•	•	•	•						
	D1	89.25 ★	16.2	7	6 100				•	•	•	•	•	•	•						
	C1	75.93	19.1	7	6 100				•	•	•	•	•	•	•						
	B1	64.80 ★	22.0	7	6 100				•	•	•	•	•	•	•						
	A1	53.13 ★	27.0	7	6 100				•	•	•	•	•	•	•						
FZ.128B 2 703 ... 6 100	A2	56.42 ★	26	6	4 300				•	•											
	X1	52.29	28	6	4 600				•	•											
	W1	49.71 ★	29	6	4 900				•	•	•										
	V1	46.46	31	6	5 150				•	•	•										
	U1	40.99 ★	35	6	5 700				•	•	•	•	•								
	T1	38.66	38	6	6 000				•	•	•	•	•	•							
	S1	34.64 ★	42	6	6 100				•	•	•	•	•	•	•	•	1)				
	R1	31.98	45	7	6 100				•	•	•	•	•	•	•	•	•	1)			
	Q1	27.33	53	7	6 100				•	•	•	•	•	•	•	•	•	•	1)		
	P1	24.70 ★	59	7	6 100				•	•	•	•	•	•	•	•	•	•	•	1)	
	N1	23.80	61	7	6 100				•	•	•	•	•	•	•	•	•	•	•	1)	
	L1	20.58	70	7	6 100				•	•	•	•	•	•	•	•	•	•	•	1)	
	K1	17.95 ★	81	7	6 100				•	•	•	•	•	•	•	•	•	•	•	1)	
	J1	15.36 ★	94	7	5 847				•	•	•	•	•	•	•	•	•	•	•	1)	
	H1	13.76	105	7	5 640				•	•	•	•	•	•	•	•	•	•	•	1)	
	G1	11.65 ★	124	7	5 347				•	•	•	•	•	•	•	•	•	•	•	1)	
	F1	10.07	144	7	5 113				•	•	•	•	•	•	•	•	•	•	•	1)	
	E1	7.57 ★	192	7	4 565				•	•	•	•	•	•	•	•	•	•	•	1)	
	D1	6.91	210	9	3 592				•	•	•	•	•	•	•	•	•	•	•	1)	
	C1	5.85 ★	248	10	3 301				•	•	•	•	•	•	•	•	•	•	•	1)	
	B1	5.05	287	10	3 137				•	•	•	•	•	•	•	•	•	•	•	1)	
	A1	3.80 ★	382	10	2 708				•	•	•	•	•	•	•	•	•	•	•	1)	

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No.					Motor size															
Nm	15th and 16th position	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)																
			rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.148B-D38 9 000	N1	70 576	★	0.02	–	9 000	•	•	•												
	M1	62 660		0.02	–	9 000	•	•	•												
	L1	54 935	★	0.03	–	9 000	•	•	•	•											
	K1	49 161		0.03	–	9 000	•	•	•	•											
	J1	43 633	★	0.03	–	9 000	•	•	•	•											
	H1	38 236		0.04	–	9 000	•	•	•	•											
	G1	33 619	★	0.04	–	9 000	•	•	•	•											
	F1	30 373		0.05	–	9 000	•	•	•	•											
	E1	26 466	★	0.05	–	9 000	•	•	•	•											
	D1	23 770		0.06	–	9 000	•	•	•	•											
	C1	21 459	★	0.07	–	9 000	•	•	•	•											
	B1	19 456		0.07	–	9 000	•	•	•	•											
A1	17 704	★	0.08	–	9 000	•	•	•	•												
FD.148B-Z38 9 000	W1	16 239	★	0.09	–	9 000	•	•	•												
	V1	14 441		0.10	–	9 000	•	•	•												
	U1	12 527	★	0.12	–	9 000	•	•	•	•											
	T1	11 705		0.12	–	9 000	•	•	•	•											
	S1	10 295	★	0.14	–	9 000	•	•	•	•											
	R1	9 016		0.16	–	9 000	•	•	•	•	•										
	Q1	7 975	★	0.18	–	9 000	•	•	•	•	•										
	P1	7 227		0.20	–	9 000	•	•	•	•	•										
	N1	6 380	★	0.23	–	9 000	•	•	•	•	•										
	M1	5 755		0.25	–	9 000	•	•	•	•	•										
	L1	5 220	★	0.28	–	9 000	•	•	•	•	•										
	K1	4 756		0.30	–	9 000	•	•	•	•	•										
	J1	4 350	★	0.33	–	9 000	•	•	•	•	•										
	H1	3 889		0.37	–	9 000	•	•	•	•	•										
	G1	3 571	★	0.41	–	9 000	•	•	•	•	•										
	F1	3 219		0.45	–	9 000	•	•	•	•	•										
	E1	2 768	★	0.52	–	9 000	•	•	•	•	•										
	D1	2 468		0.59	–	9 000	•	•	•	•	•										
	C1	2 266	★	0.64	–	9 000	•	•	•	•	•										
	B1	2 043		0.71	–	9 000	•	•	•	•	•										
	A1	1 757	★	0.83	–	9 000	•	•	•	•	•										
FD.148B-Z48 9 000	K1	1 634		0.89	–	9 000	•	•	•	•	•	•									
	J1	1 489	★	0.97	–	9 000	•	•	•	•	•	•									
	H1	1 364		1.10	–	9 000	•	•	•	•	•	•									
	G1	1 217	★	1.20	–	9 000	•	•	•	•	•	•									
	F1	1 086		1.30	–	9 000			•	•	•	•	•								
	E1	922		1.60	–	9 000			•	•	•	•	•								
	D1	768	★	1.90	–	9 000	•	•	•	•	•	•									
	C1	674		2.20	–	9 000			•	•	•	•									
	B1	573		2.50	–	9 000			•	•	•	•									
A1	477	★	3.00	–	9 000			•	•	•	•										

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	Motor size															
Nm	15th and 16th position					rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280
FD.148B 9 000	U1	449.21 ★	3.2	6	9 000					•	•										
	T1	411.98	3.5	6	9 000					•	•										
	S1	368.06 ★	3.9	6	9 000					•	•	•									
	R1	337.07	4.3	6	9 000					•	•	•									
	Q1	310.51 ★	4.7	6	9 000					•	•	•									
	P1	287.49	5.0	6	9 000					•	•	•									
	N1	267.35 ★	5.4	6	9 000					•	•	•	•								
	M1	249.58	5.8	6	9 000					•	•	•	•								
	L1	223.31 ★	6.5	6	9 000					•	•	•	•	•	•						
	K1	206.93	7.0	6	9 000					•	•	•	•	•	•						
	J1	189.69 ★	7.6	6	9 000					•	•	•	•	•	•	•					
	H1	173.89	8.3	6	9 000					•	•	•	•	•	•	•	•				
	G1	148.18	9.8	6	9 000					•	•	•	•	•	•	•	•				
	F1	130.76 ★	11.1	6	9 000					•	•	•	•	•	•	•	•				
	E1	111.29	13.0	6	9 000					•	•	•	•	•	•	•	•				
	D1	96.43 ★	15.0	6	9 000					•	•	•	•	•	•	•	•				
	C1	81.15 ★	17.9	6	9 000					•	•	•	•	•	•	•	•				
	B1	73.22	19.8	6	9 000							•	•	•	•	•	•				
	A1	62.93 ★	23.0	6	9 000							•	•	•	•	•	•				
FZ.148B 5 124 ... 9 000	V1	68.23	21	5	5 600							•									
	U1	64.37 ★	23	5	6 500							•	•								
	T1	60.21	24	5	7 000							•	•								
	S1	53.53 ★	27	6	8 000							•	•	•	•						
	R1	50.54	29	6	8 000							•	•	•	•						
	Q1	45.37 ★	32	6	8 700							•	•	•	•	•					
	P1	41.64	35	6	9 000							•	•	•	•	•					
	N1	35.93	40	6	9 000							•	•	•	•	•	•	• ¹⁾			
	M1	31.43	46	6	9 000							•	•	•	•	•	•	• ¹⁾			
	L1	27.34	53	6	9 000							•	•	•	•	•	•	• ¹⁾			
	K1	23.98 ★	60	6	9 000							•	•	•	•	•	•	• ¹⁾			
	J1	20.28 ★	71	6	9 000							•	•	•	•	•	•	• ¹⁾			
	H1	18.40	79	6	9 000							•	•	•	•	•	•	• ¹⁾			
	G1	16.04 ★	90	6	9 000							•	•	•	•	•	•	• ¹⁾			
	F1	13.62	106	6	8 519								•	•	•	•	•	• ¹⁾			
	E1	10.43 ★	139	6	7 822								•	•	•	•	•	• ¹⁾			
	D1	9.51	152	8	6 581							•	•	•	•	•	•	• ¹⁾			
	C1	8.29 ★	175	9	6 204							•	•	•	•	•	•	• ¹⁾			
	B1	7.04	206	9	5 820								•	•	•	•	•	• ¹⁾			
	A1	5.39 ★	269	9	5 124									•	•	•	•	• ¹⁾			

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No. 15th and 16th position	i_{tot}	n_2 (50 Hz) rpm	ϕ arcmin	T_{2N} ($f_B=1$) Nm	Motor size															
Nm						63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.168B-D48 14 000	N1	65 160	★	0.02	—	14 000	•	•	•												
	M1	57 946		0.03	—	14 000	•	•	•												
	L1	50 267	★	0.03	—	14 000	•	•	•	•											
	K1	46 966		0.03	—	14 000	•	•	•	•											
	J1	41 307	★	0.04	—	14 000	•	•	•	•											
	H1	36 177		0.04	—	14 000	•	•	•	•	•										
	G1	31 998	★	0.05	—	14 000	•	•	•	•	•										
	F1	29 000		0.05	—	14 000	•	•	•	•	•										
	E1	25 599	★	0.06	—	14 000	•	•	•	•	•										
	D1	23 093		0.06	—	14 000	•	•	•	•	•										
	C1	20 944	★	0.07	—	14 000	•	•	•	•	•										
	B1	19 083		0.08	—	14 000	•	•	•	•	•										
A1	17 454	★	0.08	—	14 000	•	•	•	•	•											
FD.168B-Z48 14 000	A2	16 007		0.09	—	14 000	•	•	•												
	X1	14 165	★	0.10	—	14 000	•	•	•	•											
	W1	12 878		0.11	—	14 000	•	•	•	•											
	V1	11 568	★	0.13	—	14 000	•	•	•	•											
	U1	9 916		0.15	—	14 000	•	•	•	•	•										
	T1	8 971	★	0.16	—	14 000	•	•	•	•	•	•									
	S1	8 281		0.18	—	14 000	•	•	•	•	•	•									
	R1	7 201	★	0.20	—	14 000	•	•	•	•	•	•									
	Q1	6 538		0.22	—	14 000	•	•	•	•	•	•									
	P1	5 970	★	0.24	—	14 000	•	•	•	•	•	•									
	N1	5 477		0.26	—	14 000	•	•	•	•	•	•									
	M1	5 046	★	0.29	—	14 000	•	•	•	•	•	•									
	L1	4 583		0.32	—	14 000	•	•	•	•	•	•									
	K1	4 175	★	0.35	—	14 000	•	•	•	•	•	•									
	J1	3 825		0.38	—	14 000	•	•	•	•	•	•									
	H1	3 413	★	0.42	—	14 000	•	•	•	•	•	•									
	G1	3 046		0.48	—	14 000			•	•	•	•									
	F1	2 587		0.56	—	14 000			•	•	•	•									
	E1	2 153	★	0.67	—	14 000			•	•	•	•									
	D1	2 119	★	0.68	—	14 000	•	•	•	•	•	•									
	C1	1 891		0.77	—	14 000			•	•	•	•									
	B1	1 606		0.90	—	14 000			•	•	•	•									
	A1	1 337	★	1.10	—	14 000			•	•	•	•									
FD.168B-Z68 14 000	H1	1 298		1.1	—	14 000			•	•	•	•	•								
	G1	1 108	★	1.3	—	14 000			•	•	•	•	•								
	F1	923		1.6	—	14 000				•	•	•	•								
	E1	765	★	1.9	—	14 000				•	•	•	•								
	D1	675		2.1	—	14 000			•	•	•	•	•								
	C1	576	★	2.5	—	14 000			•	•	•	•	•								
	B1	480		3.0	—	14 000				•	•	•	•								
A1	398	★	3.6	—	14 000				•	•	•	•									

★ Preferred transmission ratio

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In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
Max. gearbox torque	Order No.					2.5x the value is permissible for a brief period (e.g. motor starting torque)															
	15th and 16th position	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	Motor size															
	Nm		rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.168B 14 000	V1	369.26 ★	3.9	6	14 000							•									
	U1	338.49	4.3	6	14 000							•									
	T1	312.12 ★	4.6	6	14 000							•									
	S1	289.26	5.0	6	14 000							•									
	R1	275.03 ★	5.3	6	14 000							•	•								
	Q1	257.04	5.6	6	14 000							•	•								
	P1	226.74 ★	6.4	6	14 000							•	•	•	•						
	N1	213.87	6.8	6	14 000							•	•	•	•						
	M1	191.63 ★	7.6	6	14 000							•	•	•	•	•					
	L1	176.94	8.2	6	14 000							•	•	•	•	•					
	K1	151.18	9.6	6	14 000							•	•	•	•	•					
	J1	136.63 ★	10.6	6	14 000							•	•	•	•	•					
	H1	131.64	11.0	6	14 000							•	•	•	•	•	•				
	G1	113.86	12.7	6	14 000							•	•	•	•	•	•				
	F1	99.31 ★	14.6	6	14 000							•	•	•	•	•	•				
	E1	84.99 ★	17.1	6	14 000							•	•	•	•	•	•				
	D1	76.12	19.0	6	14 000							•	•	•	•	•	•				
	C1	64.47 ★	22.0	6	14 000							•	•	•	•	•	•				
	B1	55.68	26.0	6	14 000								•	•	•	•	•				
	A1	41.85 ★	35.0	6	14 000								•	•	•	•	•				
FZ.168B 8 683 ... 14 000	R1	53.48	27	5	9 000							•	•	•	•						
	Q1	48.29	30	6	10 500							•	•	•	•	•					
	P1	45.25	32	6	11 500							•	•	•	•	•	•				
	N1	38.87 ★	37	6	13 000							•	•	•	•	•	•				
	M1	33.58	43	6	13 000							•	•	•	•	•	•	•			
	L1	29.64	49	6	14 000							•	•	•	•	•	•	•			
	K1	26.68 ★	54	6	14 000							•	•	•	•	•	•	•			
	J1	22.14 ★	65	6	14 000							•	•	•	•	•	•	•			
	H1	20.19	72	6	14 000							•	•	•	•	•	•	•			
	G1	17.71 ★	82	6	14 000							•	•	•	•	•	•	•			
	F1	15.22	95	6	14 000								•	•	•	•	•	•			
	E1	11.86 ★	122	6	13 076								•	•	•	•	•	•			
	D1	9.42 ★	154	6	12 147									•	•	•	•	•			
	C1	8.54	170	7	11 257									•	•	•	•	•			
	B1	6.65 ★	218	8	10 011									•	•	•	•	•			
	A1	5.28 ★	275	8	8 682										•	•	•	•			

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	2.5x the value is permissible for a brief period (e.g. motor starting torque)															
	15th and 16th position					rpm	arcmin	Nm	3	3	5	10	20	26	61	98	198	198	291	356	580
Nm							63	71	80	90	100	112	132	160	180	200	225	250	280	315	
FD.188B-D48 20 000	N1	71 388	★	0.02	–	20 000	•	•	•												
	M1	63 484		0.02	–	20 000	•	•	•												
	L1	55 070	★	0.03	–	20 000	•	•	•	•											
	K1	51 455		0.03	–	20 000	•	•	•	•											
	J1	45 255	★	0.03	–	20 000	•	•	•	•											
	H1	39 634		0.04	–	20 000	•	•	•	•	•										
	G1	35 056	★	0.04	–	20 000	•	•	•	•	•										
	F1	31 771		0.05	–	20 000	•	•	•	•	•										
	E1	28 045	★	0.05	–	20 000	•	•	•	•	•										
	D1	25 299		0.06	–	20 000	•	•	•	•	•										
	C1	22 946	★	0.06	–	20 000	•	•	•	•	•										
	B1	20 906		0.07	–	20 000	•	•	•	•	•										
A1	19 122	★	0.08	–	20 000	•	•	•	•	•											
FD.188B-Z48 20 000	A2	17 537		0.08	–	20 000	•	•	•												
	X1	15 519	★	0.09	–	20 000	•	•	•	•											
	W1	14 108		0.10	–	20 000	•	•	•	•											
	V1	12 674	★	0.11	–	20 000	•	•	•	•											
	U1	10 863		0.13	–	20 000	•	•	•	•	•										
	T1	9 829	★	0.15	–	20 000	•	•	•	•	•	•									
	S1	9 073		0.16	–	20 000	•	•	•	•	•	•									
	R1	7 889	★	0.18	–	20 000	•	•	•	•	•	•									
	Q1	7 163		0.20	–	20 000	•	•	•	•	•	•									
	P1	6 540	★	0.22	–	20 000	•	•	•	•	•	•									
	N1	6 001		0.24	–	20 000	•	•	•	•	•	•									
	M1	5 529	★	0.26	–	20 000	•	•	•	•	•	•									
	L1	5 021		0.29	–	20 000	•	•	•	•	•	•									
	K1	4 574	★	0.32	–	20 000	•	•	•	•	•	•									
	J1	4 190		0.35	–	20 000	•	•	•	•	•	•									
	H1	3 739	★	0.39	–	20 000	•	•	•	•	•	•									
	G1	3 337		0.43	–	20 000			•	•	•	•									
	F1	2 834		0.51	–	20 000			•	•	•	•									
	E1	2 359	★	0.61	–	20 000			•	•	•	•									
	D1	2 322	★	0.62	–	20 000	•	•	•	•	•	•									
	C1	2 072		0.70	–	20 000			•	•	•	•									
	B1	1 760		0.82	–	20 000			•	•	•	•									
	A1	1 465	★	0.99	–	20 000			•	•	•	•									
FD.188B-Z68 20 000	H1	1 449		1.0	–	20 000			•	•	•	•	•								
	G1	1 236	★	1.2	–	20 000			•	•	•	•	•								
	F1	1 030		1.4	–	20 000				•	•	•	•								
	E1	854	★	1.7	–	20 000				•	•	•	•								
	D1	754		1.9	–	20 000			•	•	•	•	•								
	C1	643	★	2.3	–	20 000			•	•	•	•	•								
	B1	536		2.7	–	20 000				•	•	•	•								
A1	444	★	3.3	–	20 000				•	•	•	•									

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	φ	T_{2N} ($f_B=1$)	Motor size															
Nm	15th and 16th position					rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280
FD.188B 20 000	U1	403.86 ★	3.6	6	20 000							•									
	T1	370.52	3.9	6	20 000							•									
	S1	341.94 ★	4.2	6	20 000							•									
	R1	317.18	4.6	6	20 000							•									
	Q1	299.20 ★	4.8	6	20 000							•	•								
	P1	279.86	5.2	6	20 000							•	•								
	N1	248.85 ★	5.8	6	20 000							•	•	•	•						
	M1	234.93	6.2	6	20 000							•	•	•	•						
	L1	210.89 ★	6.9	6	20 000							•	•	•	•	•					
	K1	193.56	7.5	6	20 000							•	•	•	•	•					
	J1	167.03	8.7	6	20 000							•	•	•	•	•	•				
	H1	146.11	9.9	6	20 000							•	•	•	•	•	•				
	G1	127.07	11.4	6	20 000							•	•	•	•	•	•	•			
	F1	111.49 ★	13.0	6	20 000							•	•	•	•	•	•	•			
	E1	94.28 ★	15.4	6	20 000							•	•	•	•	•	•	•	•		
	D1	85.54	17.0	6	20 000							•	•	•	•	•	•	•	•		
	C1	74.58 ★	19.4	6	20 000							•	•	•	•	•	•	•	•		
	B1	63.32	23.0	6	20 000								•	•	•	•	•	•	•		
A1	48.46 ★	30.0	6	20 000								•	•	•	•	•	•	•			
FZ.188B 14 190 ... 20 000	P1	52.63	28	5	16 580								•	•	•	•					
	N1	48.47	30	5	16 870								•	•	•	•	•				
	M1	42.07 ★	34	5	17 500								•	•	•	•	•				
	L1	37.08	39	5	17 510								•	•	•	•	•	•			
	K1	32.54	45	5	18 550								•	•	•	•	•	•			
	J1	29.33 ★	49	5	20 000								•	•	•	•	•	•			
	H1	24.90 ★	58	5	20 000								•	•	•	•	•	•			
	G1	23.13	63	5	20 000								•	•	•	•	•	•			
	F1	19.87 ★	73	5	19 790								•	•	•	•	•	•			
	E1	16.95	86	5	18 870								•	•	•	•	•	•	•		
	D1	13.35 ★	109	6	17 560								•	•	•	•	•	•	•		
	C1	10.74 ★	135	6	16 070									•	•	•	•	•	•		
	B1	9.34	155	6	14 990										•	•	•	•	•		
	A1	8.34	174	6	14 190												•	•	•		

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]															
						2.5x the value is permissible for a brief period (e.g. motor starting torque)															
						3	3	5	10	20	26	61	98	198	198	291	356	580	1290		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$) <th colspan="16">Motor size</th>	Motor size															
Nm	15th and 16th position		rpm	arcmin	Nm	63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.208-D68 34 000	V1	61 412	0.02	–	34 000		•	•													
	U1	54 347	0.03	–	34 000		•	•	•												
	T1	49 406	0.03	–	34 000		•	•	•												
	S1	44 383	0.03	–	34 000		•	•	•												
	R1	38 043	0.04	–	34 000		•	•	•	•											
	Q1	34 420	0.04	–	34 000		•	•	•	•											
	P1	31 772	0.04	–	34 000		•	•	•	•											
	N1	27 626	0.05	–	34 000		•	•	•	•											
	M1	25 083	0.06	–	34 000		•	•	•	•											
	L1	22 903	0.06	–	34 000		•	•	•	•											
	K1	21 014	0.07	–	34 000		•	•	•	•											
	J1	19 361	0.07	–	34 000		•	•	•	•											
	H1	17 583	0.08	–	34 000		•	•	•	•											
	G1	16 018	0.09	–	34 000		•	•	•	•											
	F1	14 674	0.10	–	34 000		•	•	•	•											
	E1	13 093	0.11	–	34 000		•	•	•	•											
	D1	11 685	0.12	–	34 000			•	•	•											
	C1	9 924	0.14	–	34 000			•	•	•											
FD.208-Z68 34 000	X1	8 251	0.17	–	34 000			•	•	•	•										
	W1	7 536	0.19	–	34 000			•	•	•	•										
	V1	6 688	0.21	–	34 000			•	•	•	•	•									
	U1	6 173	0.23	–	34 000			•	•	•	•	•									
	T1	5 584	0.25	–	34 000			•	•	•	•	•									
	S1	5 142	0.27	–	34 000			•	•	•	•	•									
	R1	4 755	0.29	–	34 000			•	•	•	•	•	•								
	Q1	4 414	0.32	–	34 000			•	•	•	•	•	•								
	P1	3 895	0.36	–	34 000			•	•	•	•	•	•								
	N1	3 596	0.39	–	34 000			•	•	•	•	•	•								
	M1	3 222	0.43	–	34 000			•	•	•	•	•	•								
	L1	2 970	0.47	–	34 000				•	•	•	•	•								
	K1	2 492	0.56	–	34 000				•	•	•	•	•								
	J1	2 126	0.66	–	34 000				•	•	•	•	•								
	H1	1 772	0.79	–	34 000					•	•	•	•								
	G1	1 469	0.95	–	34 000					•	•	•	•								
	F1	1 296	1.08	–	34 000				•	•	•	•	•								
	E1	1 106	1.27	–	34 000				•	•	•	•	•								
	D1	921	1.52	–	34 000					•	•	•	•								
	C1	764	1.83	–	34 000					•	•	•	•								
FD.208-Z88 34 000	J1	694	2.02	–	34 000					•	•	•	•	•	•						
	H1	636	2.20	–	34 000					•	•	•	•	•	•						
	G1	543	2.58	–	34 000					•	•	•	•	•	•						
	F1	445	3.15	–	34 000					•	•	•	•	•	•						
	E1	406	3.45	–	34 000					•	•	•	•	•	•						
	D1	347	4.04	–	34 000					•	•	•	•	•	•						
	C1	284	4.92	–	34 000					•	•	•	•	•	•						

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit. $T_{2max} = T_1 \times i_{tot}$, if $T_{2max} \leq T_{2N}$ If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Transmission ratios and maximum torques

Selection and ordering data (continued)

Gearbox size	Ratio code	Transmission ratio	Output speed	Twisting angle ²⁾	Nominal torque	Permissible input torque T_1 [Nm]																		
Max. gearbox torque	Order No.	i_{tot}	n_2 (50 Hz)	ϕ	T_{2N} ($f_B=1$)	2.5x the value is permissible for a brief period (e.g. motor starting torque)																		
	15th and 16th position					rpm	arcmin	Nm	Motor size															
									63	71	80	90	100	112	132	160	180	200	225	250	280	315		
FD.208	T1	242.01	6.1	5	34 000								•	•	•	•								
29 901 ... 34 000	S1	218.54	6.8	5	34 000								•	•	•	•	•							
	R1	204.81	7.2	5	34 000								•	•	•	•	•	•	•					
	Q1	175.92	8.4	5	34 000								•	•	•	•	•	•	•					
	P1	151.99	9.7	5	34 000								•	•	•	•	•	•	•					
	N1	134.16	11	5	34 000								•	•	•	•	•	•	•					
	M1	120.77	12	5	34 000								•	•	•	•	•	•	•					
	L1	100.21	15	5	34 000								•	•	•	•	•	•	•					
	K1	91.38	16	5	34 000								•	•	•	•	•	•	•					
	J1	80.17	18	5	34 000								•	•	•	•	•	•	•					
	H1	68.90	21	5	34 000									•	•	•	•	•	•	•				
	G1	53.66	28	5	34 000									•	•	•	•	•	•	•				
	F1	42.63	35	5	34 000										•	•	•	•	•	•				
	E1	38.84	38	5	34 000									•	•	•	•	•	•	•				
	D1	30.25	49	6	32 038									•	•	•	•	•	•	•				
	C1	24.03	62	6	29 901										•	•	•	•	•	•				
FZ.208	H1	20.06	74	5	32 681								•	•	•	•	•	•	•	•				
	G1	16.02	92	5	30 487								•	•	•	•	•	•	•	•				
	F1	13.09	113	5	28 634									•	•	•	•	•	•	•				
	E1	11.51	129	5	27 499											•	•	•	•	•				
	D1	10.57	140	5	26 785												•	•	•	•				
	C1	9.01	164	5	25 469													•	•	•				

★ Preferred transmission ratio

¹⁾ Only possible with integrated motor.²⁾ Twisting angle applies to reduced-backlash gearboxes.

In the case of gearboxes of size 28, only possible with integrated motor or input unit KQ and KQS.

Calculation of maximum output torque T_{2max} for gearboxes with input unit.

$$T_{2max} = T_1 \times i_{tot}, \text{ if } T_{2max} \leq T_{2N}$$

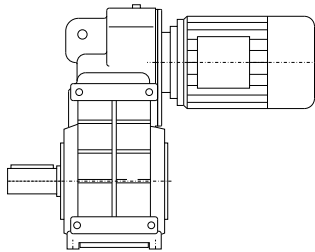
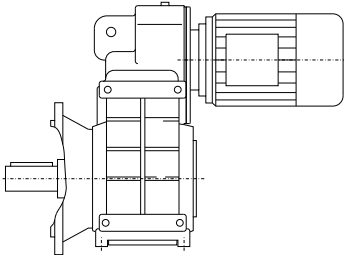
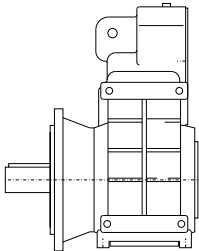
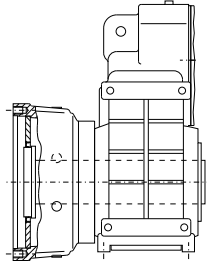
If $T_{2max} \geq T_{2N}$ the max. output torque T_{2N} of the unit is the decisive factor.

MOTOX Geared Motors

Parallel shaft geared motors

Mounting types

Selection and ordering data

Mounting type	Order No. 14th position	Code in type designation 3rd position for solid shaft, 4th position for hollow shaft	Representation
Foot-mounted design	A	—	
Housing flange (C-type)	H	Z	
Design with torque arm	D	D	
Flange-mounted design (A-type)	F	F	
Mixer flange	M	M	
Extruder flange	E	E	

Selection and ordering data (continued)

Parallel shaft gearbox with torque arm

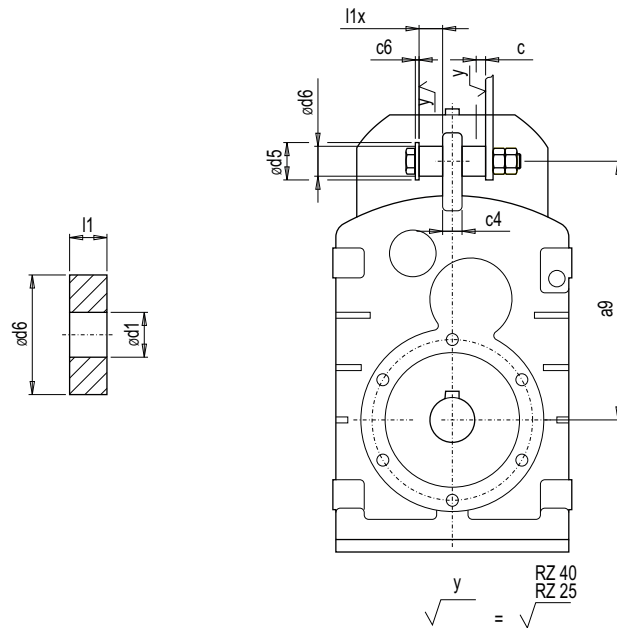
The rubber buffers (supplied loose) are used to flexibly support the gearbox on the housing plate provided. The rubber buffers are suitable for all mounting positions and can withstand temperatures of between -40 °C and $+80\text{ °C}$.

The rubber buffer must be stretched to the dimension $l1$ during installation.

Material: Natural rubber, hardness 70 ± 5 , Shore A.

Order No.: **D** in **14th position**

The shafts, mounting positions, and dimensions correspond to the design featuring a housing flange.



Gearbox type	a9	l1	l1x	d6	d1	d5	c6 _{min}	c4	c ^{*)}
F.28	140	15	14.0	30	10.5 + 0.5	40	2.0	10	1.8
F.38B	140	15	13.1	30	10.5 + 0.5	40	2.5	12	3.8
F.48B	185	20	18.2	40	12.5 + 0.5	50	3.0	12	3.7
F.68B	218	20	17.0	40	12.5 + 0.5	50	3.0	16	5.6
F.88B	278	30	27.2	60	21.0 + 0.5	75	4.0	20	5.0
F.108B	346	30	26.0	60	21.0 + 0.5	75	4.0	26	7.3
F.128B	395	40	35.8	80	25.0 + 0.5	100	6.0	30	8.0
F.148B	485	40	34.8	80	25.0 + 0.5	100	6.0	36	9.4
F.168B	550	50	46.2	120	31.0 + 0.5	140	8.0	50	6.2
F.188B	620	50	45.1	120	31.0 + 0.5	140	8.0	50	8.3

^{*)} Spring compression at max. torque

MOTOX Geared Motors

Parallel shaft geared motors

Mounting types

Selection and ordering data (continued)

Parallel shaft gearbox with mixer flange, sizes 88 to 168

Heavy-duty design

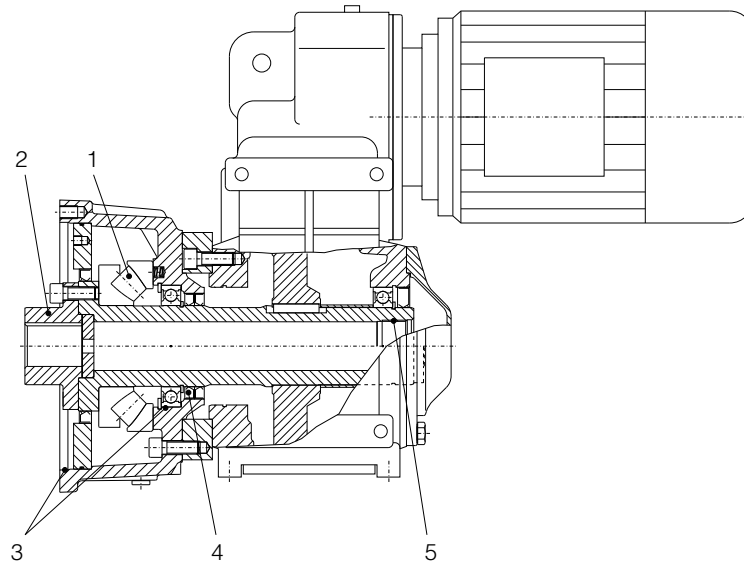
The mixer flange is fitted with a heavy-duty output bearing with a sizable bearing span for absorbing large radial and axial forces.

The optimized design ensures that no axial forces are transferred to the gearbox housing.

Bearing life can be calculated on request or using the MOTOX Configurator calculation program.

Parallel shaft gearbox with extruder flange, sizes 68 to 168

Gearboxes with an extruder flange are ideal for use in the extrusion industry, particularly in the low to medium performance range.



1. Large axial spherical roller bearing

294 series spherical roller bearing for heavy axial loads.

2. Simple, low-cost design

Flange hub supplied by customer, no grinding processes. Standard shaft-hub connection with feather key in acc. with DIN 6885/1.

3. Good radial eccentricity

Radial bearing hole and center hole created in one clamping operation and direction.

Area of application

4. Optimum lubrication

Extruder oil chamber separate from gearbox oil chamber.

5. Standard connection

Metric thread for supporting the extruder worm (worm pulled out from rear).

Parallel shaft gearbox		F.AE 68B	F.AE 88B	F.AE 108B	F.AE 128B	F.AE 148B	F.AE 168B
Max. power	[kW]	9.2	15	30	45	55	90
Transmission ratio min./max.	[2-stage]	3.97 / 61.17	4.77 / 64.58	5.60 / 64.21	3.8 / 56.42	5.39 / 68.23	5.28 / 53.48
Max. torque	[Nm]	1 000	1 900	3 400	6 100	9 000	14 000
Max. axial forces	[kN]	65	105	180	260	400	580
Spherical roller bearing	[.]	29414E	29417E	29420E	29424E	29426E	29432E

MOTOX Geared Motors

Parallel shaft geared motors

Shaft designs

Selection and ordering data

Shaft design	Order No. 8th position	Order No. suffix	Shaft dimensions					
Parallel shaft gearbox FZ, 2-stage and FD, 3-stage, foot-mounted design								
Size			F.28	F.38B	F.48B	F.68B	F.88B	
Hollow shaft	5		H25 x 104 ^{*)}	H30 x 120 ^{*)}	H35 x 150 ^{*)}	H40 x 180 ^{*)}	H50 x 210 ^{*)}	
	6				H40 x 150	H45 x 180	H60 x 210	
Hollow shaft with shrink disk	9	H3A	H25 x 126 ^{*)}	H30 x 146 ^{*)}	H40 x 177	H50 x 209	H60 x 241	
	9	H3B		H30/31 x 146	H40/41 x 177	H50/51 x 209	H60/61 x 241	
	9	H3C			H35 x 177 ^{*)}	H40 x 209 ^{*)}	H50 x 241 ^{*)}	
	9	H3D				H40/42 x 209	H50/52 x 241	
Hollow shaft with splined shaft	9	H4A	N25x1.25x30x18x9H x 104	N35x1.25x30x26x9H x 120	N40x2x30x18x9H x 150	N50x2x30x24x9H x 180	N60x2x30x28x9H x 210	
Size			F.108B	F.128B	F.148B	F.168B	F.188B	F.208
Hollow shaft	5		H60 x 240 ^{*)}	H70 x 300 ^{*)}	H80 x 350	H100 x 410 ^{*)}	H120 x 500 ^{*)}	
	6		H70 x 240	H80 x 300	H90 x 350 ^{*)}	H110 x 410		
Hollow shaft with shrink disk	9	H3A	H70 x 280	H80 x 345	H95 x 404 ^{*)}	H105 x 483 ^{*)}	H125 x 580 ^{*)}	
	9	H3B	H70/71 x 280	H80/81 x 345	H95/96 x 404	H105/106 x483		H145/146 x 728
	9	H3C	H65 x 280 ^{*)}	H75 x 345 ^{*)}				
	9	H3D	H65/66 x 280	H75/76 x 345				
Hollow shaft with splined shaft	9	H4A	N70x2x30x34x9H x240	N80x3x30x25x9H x300	N90x3x30x28x9H x350	N110x3x30x35x9Hx410	N130x5x30x24x9H x 500	
Parallel shaft gearbox FZ.Z, 2-stage and FD.Z, 3-stage with housing flange								
Size			F..Z28	F..Z38B	F..Z48B	F..Z68B	F..Z88B	
Solid shaft with feather key	1		V25 x 50 ^{*)}	V25 x 50 ^{*)}	V30 x 60 ^{*)}	V40 x 80 ^{*)}	V50 x 100 ^{*)}	
	3			V35 x 70	V40 x 80	V50 x 100	V70 x 140	
	4					V35 x 70		
Hollow shaft	5		H25 x 104 ^{*)}	H30 x 120 ^{*)}	H35 x 150 ^{*)}	H40 x 180 ^{*)}	H50 x 210 ^{*)}	
	6				H40 x 150	H45 x 180	H60 x 210	
Hollow shaft with shrink disk	9	H3A	H25 x 126 ^{*)}	H30 x 146 ^{*)}	H40 x 177	H50 x 209	H60 x 241	
	9	H3B		H30/31 x 146	H40/41 x 177	H50/51 x 209	H60/61 x 241	
	9	H3C			H35 x 177 ^{*)}	H40 x 209 ^{*)}	H50 x 241 ^{*)}	
	9	H3D				H40/42 x 209	H50/52 x 241	
Hollow shaft with splined shaft	9	H4A	N25x1.25x30x18x9H x 104	N35x1.25x30x26x9H x 120	N40x2x30x18x9H x 150	N50x2x30x24x9H x 180	N60x2x30x28x9H x 210	
Size			F..Z108B	F..Z128B	F..Z148B	F..Z168B	F..Z188B	F.208
Solid shaft with feather key	1		V60 x 120 ^{*)}	V70 x 140 ^{*)}	V90 x 170 ^{*)}	V110 x 210 ^{*)}	V120 x 210 ^{*)}	V160 x 250 ^{*)}
	3		V80 x 170	V90 x 170	V100 x 210	V120 x 210	V140 x 250	
Hollow shaft	5		H60 x 240 ^{*)}	H70 x 300 ^{*)}	H80 x 350	H100 x 410	H120 x 500 ^{*)}	
	6		H70 x 240	H80 x 300	H90 x 350 ^{*)}	H110 x 410 ^{*)}		
Hollow shaft with shrink disk	9	H3A	H70 x 280	H80 x 345	H95 x 404 ^{*)}	H105 x 483 ^{*)}	H125 x 580 ^{*)}	
	9	H3B	H70/71 x 280	H80/81 x 345	H95/96 x 404	H105/106 x483		H145/146 x 728
	9	H3C	H65 x 280 ^{*)}	H75 x 345 ^{*)}				
	9	H3D	H65/66 x 280	H75/76 x 345				
Hollow shaft with splined shaft	9	H4A	N70x2x30x34x9H x 240	N80x3x30x25x9H x 300	N90x3x30x28x9H x 350	N110x3x30x35x9H x 410	N130x5x30x24x9H x 500	

*) Preferred series

MOTOX Geared Motors

Parallel shaft geared motors

Shaft designs

Selection and ordering data (continued)

Shaft design	Order No. 8th position	Order No. suffix	Shaft dimensions				
Parallel shaft gearbox FZ.F, 2-stage and FD.F, 3-stage, flange-mounted design (A-type)							
Size			F..F28	F..F38B	F..F48B	F..F68B	F..F88B
Solid shaft with feather key	2		V25 x 50 (i2=l) ^{*)}	V25 x 50 (i2=l) ^{*)}	V30 x 60 (i2=l) ^{*)}	V40 x 80 (i2=l) ^{*)}	V50 x 100 (i2=l) ^{*)}
Hollow shaft	5		H25 x 104 ^{*)}	H30 x 120 ^{*)}	H35 x 150 ^{*)}	H40 x 180 ^{*)}	H50 x 210 ^{*)}
	6				H40 x 150	H45 x 180	H60 x 210
Hollow shaft with shrink disk	9	H3A	H25 x 126 ^{*)}	H30 x 146 ^{*)}	H40 x 177	H50 x 209	H60 x 241
	9	H3B		H30/31 x 146	H40/41 x 177	H50/51 x 209	H60/61 x 241
	9	H3C			H35 x 177 ^{*)}	H40 x 209 ^{*)}	H50 x 241 ^{*)}
	9	H3D				H40/42 x 209	H50/52 x 241
Hollow shaft with splined shaft	9	H4A	N25x1.25x30x18 x9H x 104	N35x1.25x30x26 x9H x 120	N40x2x30x18x9H x 150	N50x2x30x24x9H x 180	N60x2x30x28x9H x 210
Size			F..F108B	F..F128B	F..F148B	F..F168B	F..F188B
Solid shaft with feather key	2		V60 x 120 (i2=l) ^{*)}	V70 x 140 (i2=l) ^{*)}	V90 x 170 (i2=l) ^{*)}	V110 x 210 (i2=l) ^{*)}	V120 x 210 (i2=l) ^{*)}
Hollow shaft	5		H60 x 240 ^{*)}	H70 x 300 ^{*)}	H80 x 350	H100 x 410 ^{*)}	H120 x 500 ^{*)}
	6		H70 x 240	H80 x 300	H90 x 350 ^{*)}	H110 x 410	
Hollow shaft with shrink disk	9	H3A	H70 x 280	H80 x 345	H95 x 404 ^{*)}	H105 x 483 ^{*)}	H125 x 580 ^{*)}
	9	H3B	H70/71 x 280	H80/81 x 345	H95/96 x 404	H105/106 x 483	
	9	H3C	H65 x 280 ^{*)}	H75 x 345 ^{*)}			
	9	H3D	H65/66 x 280	H75/76 x 345			
Hollow shaft with splined shaft	9	H4A	N70x2x30x34x9H x 240	N80x3x30x25x9H x 300	N90x3x30x28x9H x 350	N110x3x30x35x9H x 410	N130x5x30x24x9H x 500

^{*)} Preferred series

Shaft designs for parallel shaft gearbox with mixer flange

Shaft design	Order No. 8th position	Order No. suffix	Shaft dimensions				
Parallel shaft gearbox F..M, 2-stage and 3-stage							
Size			F..M88B	F..M108B	F..M128B	F..M148B	F.M168B
Solid shaft with feather key	3		V70 x 140	V80 x 170	V90 x 170	V100 x 210	V120 x 210
Hollow shaft	9	H2F	H60 x 321	H70 x 366	H80 x 456	H90 x 524	H110 x 609

Shaft designs for parallel shaft gearbox with extruder flange

Shaft design	Order No. 8th position	Order No. suffix	Shaft dimensions					
Parallel shaft gearbox F..E, 2-stage and 3-stage								
Size			F.AE68	F.AE88	F.AE108	F.AE128	F.AE148	F.AE168
Hollow shaft	9	H2A	H20 x 48	H30 x 58	H40 x 71	H45 x 87	H60 x 95	H70 x 105
	9	H2B	H25 x 48	H35 x 58	H45 x 71	H50 x 87	H70 x 95	H80 x 105
	9	H2C	H30 x 48 ^{*)}	H40 x 58 ^{*)}	H50 x 71 ^{*)}	H60 x 87 ^{*)}	H75 x 95 ^{*)}	H90 x 105 ^{*)}

^{*)} Preferred series

MOTOX Geared Motors

Parallel shaft geared motors

Flange-mounted designs (A-type)

Selection and ordering data

Order code	Flange diameter									
Parallel shaft gearbox FZ.F, 2-stage										
Size	FZ.F28	FZ.F38B	FZ.F48B	FZ.F68B	FZ.F88B	FZ.F108B	FZ.F128B	FZ.F148B	FZ.F168B	FZ.F188B
H02	120	160	200	250	300	350		450		660
H03	160						450		550	
Parallel shaft gearbox FD.F, 3-stage										
Size	FD.F28	FD.F38B	FD.F48B	FD.F68B	FD.F88B	FD.F108B	FD.F128B	FD.F148B	FD.F168B	FD.F188B
H02	120	160	200	250	300	350		450		660
H03	160						450		550	

MOTOX Geared Motors

Parallel shaft geared motors

Mounting types and mounting positions

Selection and ordering data

The mounting type / mounting position must be specified when you place your order to ensure that the gearbox is supplied with the correct quantity of oil.

Please contact customer service to discuss the oil quantity if you wish to use a mounting position which is not shown here.

Position of the terminal box

The terminal box of the motor can be mounted in four different positions. See Chapter 8 for an accurate representation of the terminal box position and the corresponding order codes.

2-stage and 3-stage parallel shaft gearbox, foot-mounted design, flange-mounted design, and with housing flange

Oil control valves:

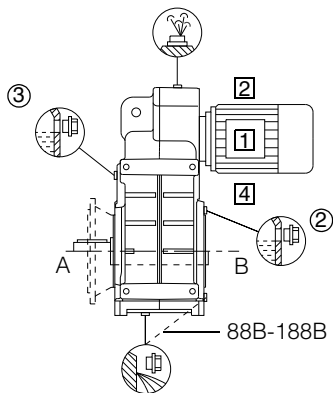
- Size 28: These types are lubricated for life. No ventilation, oil level, or drain plugs are present.
- Size 38B: V Oil inlet
- From size 48B up:  Oil level  Ventilation  Oil drain  Oil dipstick - - - alternative
- ② 2-stage gearbox ③ 3-stage gearbox * On opposite side A,B position of the customer's solid/plug-in shaft

① ... ④ Position of the terminal box, see Chapter 8.

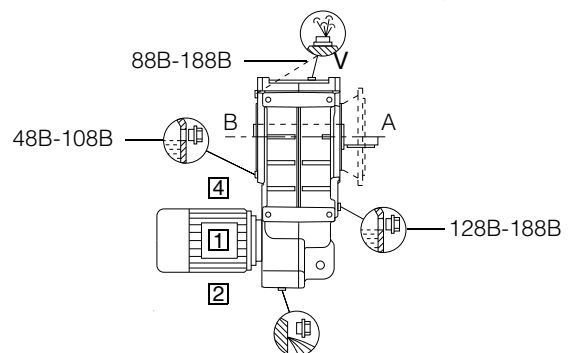
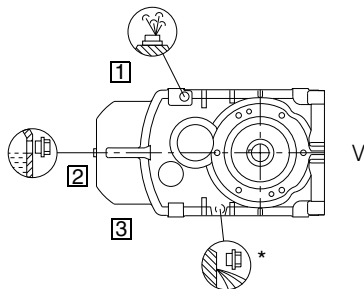
F.Z, F.F: B5-01 (IM B5-01) ¹⁾
Order code (output side A): **D22**
F.AZ, F.AF: H-01 ¹⁾
Order code (output side A): **D76**

1) Standard mounting type

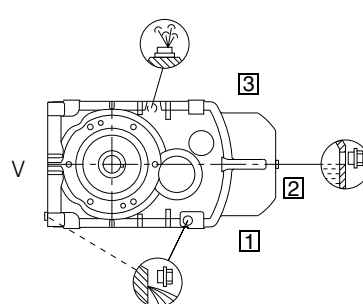
F.Z, F.F: B5-03 (IM B5-03)
Order code (output side A): **D32**
F.AZ, F.AF: H-02
Order code (output side A): **D78**



F.Z, F.F: B5-02 (IM B5-02)
Order code (output side A): **D27**
F.AZ, F.AF: H-03
Order code (output side A): **D80**



F.Z, F.F: B5-00 (IM B5-00)
Order code (output side A): **D18**
F.AZ, F.AF: H-04
Order code (output side A): **D82**



MOTOX Geared Motors

Parallel shaft geared motors

Mounting types and mounting positions

Selection and ordering data (continued)

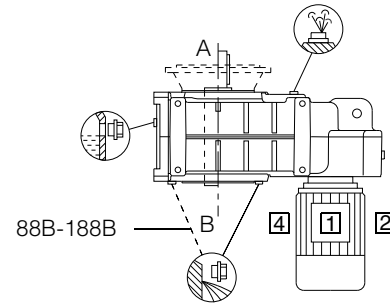
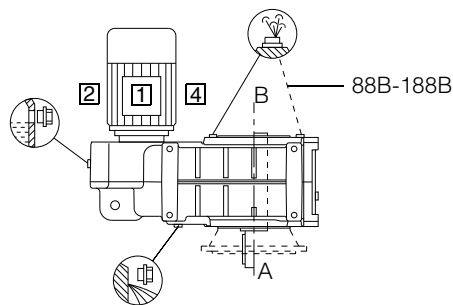
2-stage and 3-stage parallel shaft gearbox, foot-mounted design, flange-mounted design and with housing flange

Oil control valves:

- Size 28: These types are lubricated for life. No ventilation, oil level, or drain plugs are present.
- Size 38B: V Oil inlet
- From size 48B up:  Oil level  Ventilation  Oil drain  Oil dipstick - - - Alternative
- ② 2-stage gearbox ③ 3-stage gearbox * On opposite side A,B position of the customer's solid/plug-in shaft
- ① ... ④ Position of the terminal box, see Chapter 8.

F.Z, F.F: V1-00 (IM V1-00)
Order code (output side A): **D90**
F.AZ, F.AF: H-05
Order code (output side A): **D84**

F.Z, F.F: V3-00 (IM V3-00)
Order code (output side A): **D98**
F.AZ, F.AF: H-06
Order code (output side A): **D86**



2-stage and 3-stage parallel shaft gearbox with mixer flange (FZ.M/FD.M)

Mounting positions correspond to those of standard gearboxes.

2-stage and 3-stage parallel shaft gearbox with extruder flange (FZAE/FDAE)

Mounting positions correspond to those of standard gearboxes with hollow shaft.

MOTOX Geared Motors

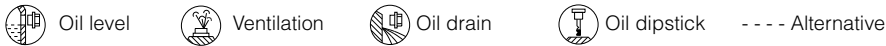
Parallel shaft geared motors

Mounting types and mounting positions

Selection and ordering data (continued)

2-stage and 3-stage parallel shaft gearbox, foot-mounted design, flange-mounted design, and with housing flange for size 208

Oil control valves:

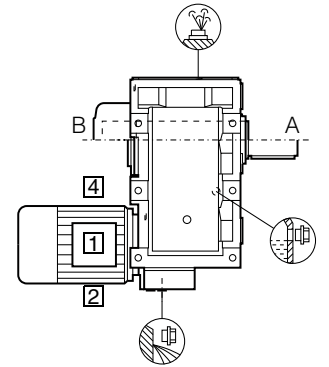
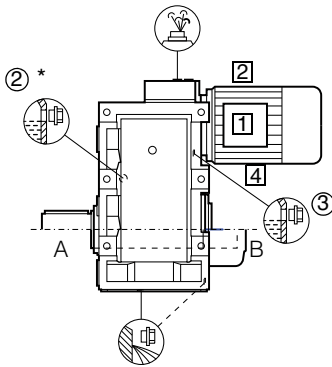


② 2-stage gearbox ③ 3-stage gearbox ④ Tandem gearbox * On opposite side A,B position of the customer's solid/plug-in shaft

① ... ④ Position of the terminal box, see Chapter 8.

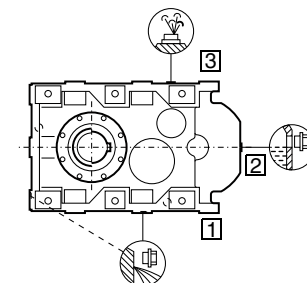
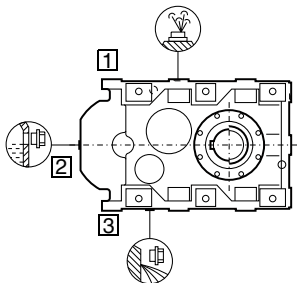
F.Z: B5-01 (IM B5-01) ¹⁾
Order code (output side A): **D22**
F.A.: H-01 ¹⁾
Order code (output side A): **D76**

F.Z: B5-03 (IM B5-03)
Order code (output side A): **D32**
F.A.: H-02
Order code (output side A): **D78**



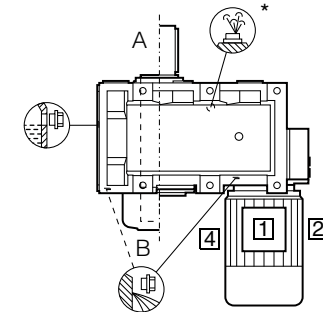
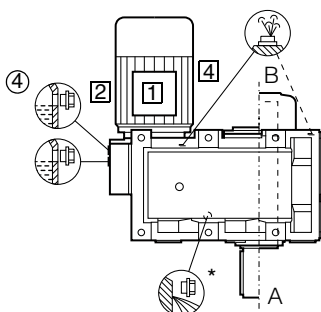
F.Z: B5-02 (IM B5-02)
Order code (output side A): **D27**
F.A.: H-03
Order code (output side A): **D80**

F.Z: B5-00 (IM B5-00)
Order code (output side A): **D18**
F.A.: H-04
Order code (output side A): **D82**



F.Z: V1-00 (IM V1-00)
Order code (output side A): **D90**
F.A.: H-05
Order code (output side A): **D84**

F.Z: V3-00 (IM V3-00)
Order code (output side A): **D98**
F.A.: H-06
Order code (output side A): **D86**



Selection and ordering data (continued)

Parallel shaft tandem gearbox

The mounting type / mounting position of the tandem gearbox corresponds to that of the main gearbox. The figures below are only designed to show the position of the oil control valves of the 2nd gearbox.

Note:

In a horizontal operating position the bulging part of the housing of the 2nd gearbox generally faces vertically downwards.

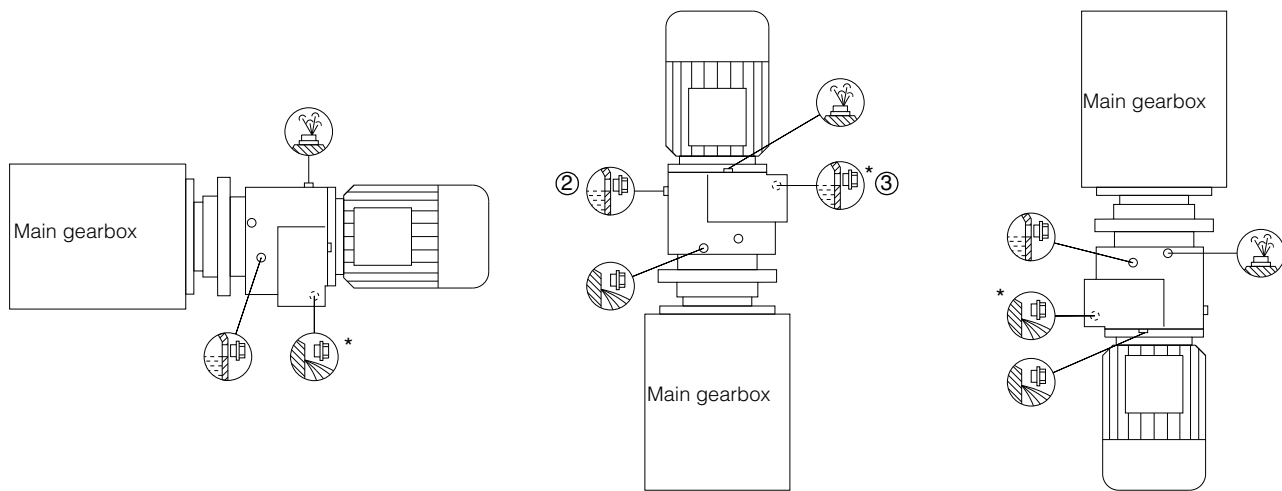
Oil control valves:

- Size 28/38 (2nd gearbox): These types are lubricated for life. No ventilation, oil level, or drain plugs are present.

- From size 48B up:  Oil level  Ventilation  Oil drain * On opposite side

② 2-stage gearbox

③ 3-stage gearbox



MOTOX Geared Motors

Parallel shaft geared motors

Special versions

Lubricants

Parallel shaft gearboxes are filled with mineral oil as standard.

If the gearbox is to be used in an application with special requirements, the lubricants listed in the table below can be used.

Area of application	Ambient temperature ¹⁾	DIN ISO designation	Order code
Standard oils			
Standard temperature	-10 ... +40 °C	CLP ISO VG220	K06
Improved oil service life	-20 ... +50 °C	CLP ISO PG VG220	K07
High temperature usage	0 ... +60 °C	CLP ISO PG VG460	K08
Low temperature usage	-40 ... +40 °C	CLP ISO PAO VG220	²⁾
Lowest temperature usage	-40 ... +10 °C	CLP ISO PAO VG68	²⁾
Physiologically safe oils (for use in the food industry) in acc. with NSF (USDA)-H1			
Standard temperature	-30 ... +40 °C	CLP ISO H1 VG460	K11
Biologically degradable oils			
Standard temperature	-20 ... +40 °C	CLP ISO E VG220	K10

¹⁾ Recommendation

²⁾ On request

Size 28 does not feature any ventilation, oil level, or drain plugs. The lubricant does not need to be changed, due to the low thermal load the gearbox is subjected to.

Parallel shaft gearboxes of size 38B have an oil screw; these gearboxes do not require ventilation or ventilation elements.

Gearboxes of sizes 48B to 188B are fitted with filler, oil level, and drain plugs as standard. The ventilation and vent filter, which is delivered loose, must be attached in place of the filler plug prior to startup.

Oil level control

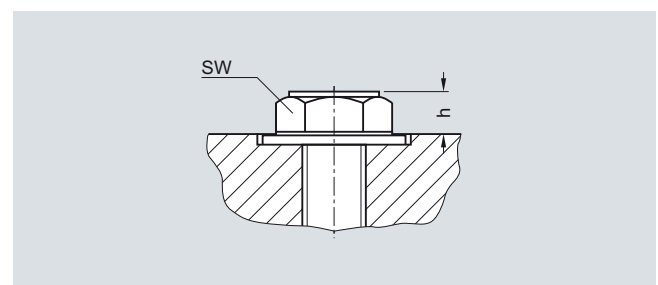
Oil sight glass

For size 48B and above, gearboxes can be equipped with a visual oil level indicator (oil sight glass) for most mounting types and mounting positions.

Order code:

Oil sight glass **G34**

Size	SW (Wrench width)	h
FD./FZ.48B ... FD./FZ.128B	19	8
FD./FZ.148B ... FD./FZ.188B	24	8
FD./FZ.208	32	11



Electrical oil level monitoring system

If required, the gearbox can be supplied with an electrical oil level monitoring system, which enables the oil level of the gearbox to be monitored remotely. The oil level is monitored by a capacitive sensor only when the gearbox starts up; it is not measured continuously.

MOTOX Geared Motors

Parallel shaft geared motors

Special versions

Gearbox ventilation

The positions of the ventilation and ventilation elements can be seen on the mounting position diagrams.

If required, a pressure ventilation valve can be used for size 48B and above.

Vent filter:

Size	d	d1	c	h
FD./FZ.48B ... FD./FZ.128B	27	22	4.0	20.0
FD./FZ.148B ... FD./FZ.188B	32	32	4.0	24.0
FD./FZ.208	45	40	6.5	23.5

Pressure ventilation valve:

Size	SW (Wrench width)	d2	h1
FD./FZ.48B ... FD./FZ.128B	17	11	15.0
FD./FZ.148B ... FD./FZ.188B	24	11	14.0
FD./FZ.208	27	11	29.5

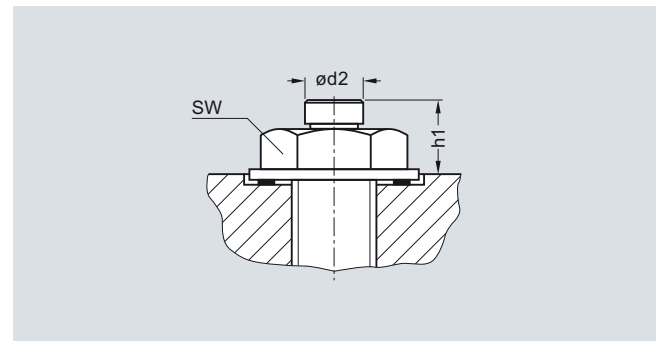
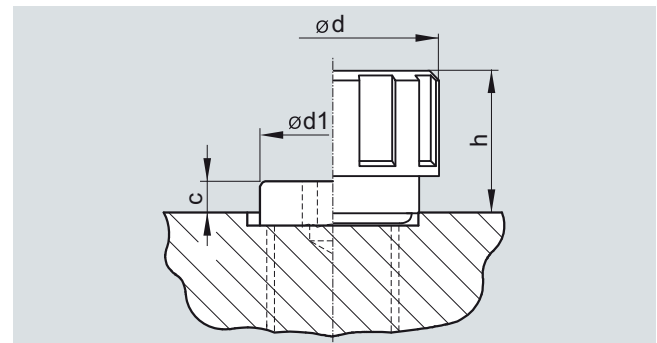
Order code:

Vent filter

G44

Pressure ventilation valve

G45



Oil drain

Magnetic oil drain plug

A magnetic oil drain plug for inserting in the oil drainage hole is available on request for parallel shaft gearboxes of size 48B and above. This serves to collect any grit contained in the gear lubricant.

Order code:

Magnetic oil drain plug

G53

Oil drain valve

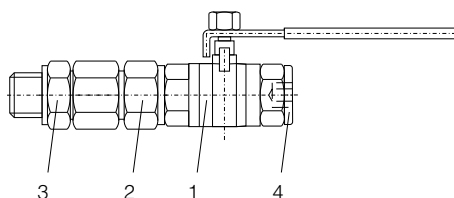
An oil drain valve is available on request for parallel shaft gearboxes of size 48B and above.

The oil drain valve may be designed as a complete unit featuring a screw plug, depending on the corresponding mounting position.

Order code:

Oil drain valve, straight **G54**

An angled oil drain valve is also available on request.

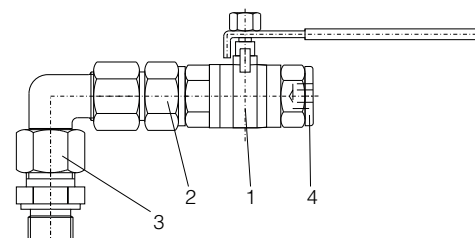


Item 1 Oil drain valve

Item 2 Screwed connection EGE

Item 3 Screwed connection GE

Item 4 Screw plug



Item 1 Oil drain valve

Item 2 Screwed connection EGE

Item 3 Screwed connection GE

Item 4 Screw plug

MOTOX Geared Motors

Parallel shaft geared motors

Special versions

Sealing

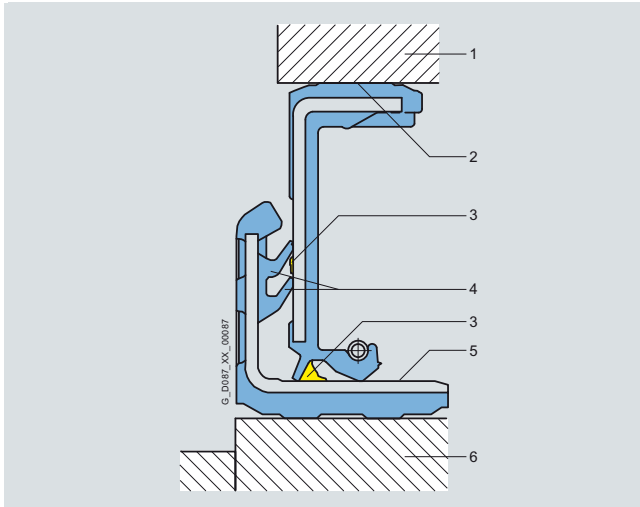
Combination shaft sealing

A combination shaft sealing, which helps to prevent oil from leaking, is available for parallel shaft gearboxes of sizes 38B to 168B.

A combination shaft sealing is particularly well suited to external use.

Order code:

Combination shaft sealing **G24**



- 1 • Housing
- 2 • Rubberized inner and outer diameter
- 3 • Grease filling prevents dry running of the sealing lips
- 4 • Additional sealing lips to protect against dirt
 - Decoupled sealing system prevents scoring of the shaft as a result of corrosion or dirt
- 5 • Protected running surface for radial shaft sealing ring
 - No damage when mounting
- 6 • Shaft

Double sealing

Double sealing is possible for parallel shaft gearboxes of sizes 28 and 188B. Double sealing is particularly well suited to external use.

Order code:

Double sealing MSS1 (size 28)

G23

Double radial shaft seal (sizes 188B)

G22+G31

High temperature resistant sealing

High temperature resistant sealings (Viton/fluorinated rubber) for high operating and ambient temperatures of +60 °C and above are available for parallel shaft gearboxes.

Order code:

High temperature resistant sealing **G25**

Hollow shaft cover (protection cover)

Gearboxes with hollow shaft are delivered with a plastic sealing cap as standard.

If required, they can be fitted with a fixed protection cover. Gearboxes of size 28 are fitted with a steel protection cover as standard.

The steel protection cover can only be used for gearboxes with hollow shaft and shrink disk.

For outdoor applications we recommend the ATEX versions.

Order codes:

Protection cover

G62

Protection cover (ATEX)

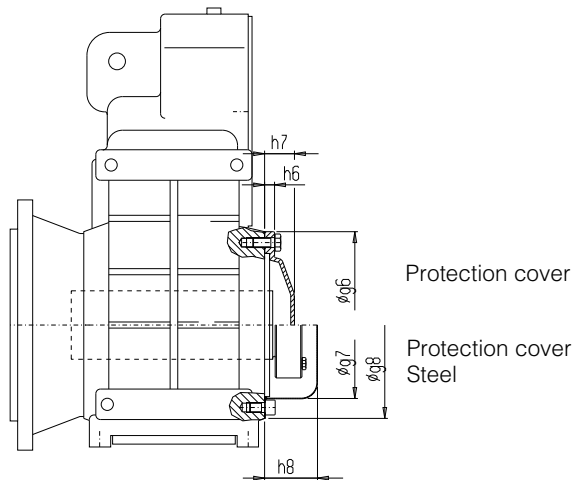
G63

Steel protection cover

G60

Steel protection cover (ATEX)

G61



F.A, F.AF, F.AZ, F.AS ¹⁾, F.AFS ¹⁾, F.AZS ¹⁾, F.AT, F.AFT, F.AZT

¹⁾ Only a steel protection cover is available for F.AS, F.ADS, F.AFS, and F.AZS

Gearbox type	Steel protection cover			Protection cover		
	g7	g8	h8	g6	h6	h7
F.28	58.0	102	33.5	–	–	–
F.38B	82.2	115	40.0	120	10	33
F.48B	99.0	130	44.0	132	10	33
F.68B	115.0	150	62.5	150	10	37
F.88B	137.0	190	70.0	190	13	50
F.108B	187.0	240	80.0	245	13	55
F.128B	233.0	292	85.0	295	16	48
F.148B	257.5	334	100.0	335	13	50
F.168B	309.5	390	129.5	400	13	50
F.188B	309.5	390	129.5	400	13	50
F.208	373.0	373	179.0	–	–	–

Radially reinforced output shaft bearings

The bearings of the MOTOX gearboxes are dimensioned such that they are strong enough to withstand most application cases.

However, the gearboxes can be fitted with a reinforced output shaft bearing arrangement for applications with particularly high radial forces.

Order code:

Radially reinforced output shaft bearings **G20**

MOTOX Geared Motors

Parallel shaft geared motors

Special versions

Mixer flange in dry-well design

The agitator flange can be fitted with an additional "V" ring (1) in mounting position V1-00 in order to drain off any leak oil to a safety chamber and protect the equipment against the effects of leakages.

The oil can either be viewed through a sight glass, or its presence indicated by an electrical oil sensor (2).

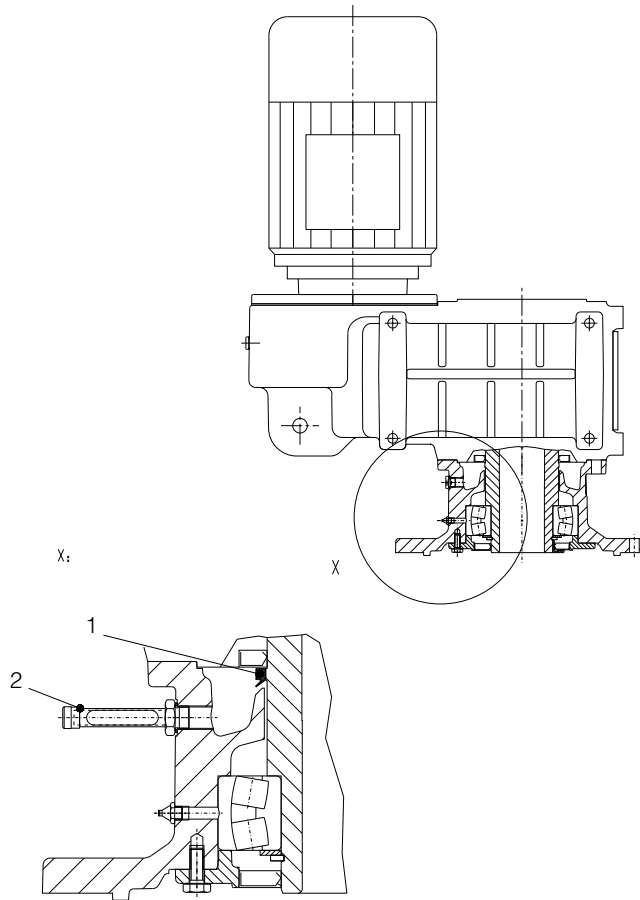
Order codes:

Dry-well design with sight glass

G89

Dry-well design with sensor

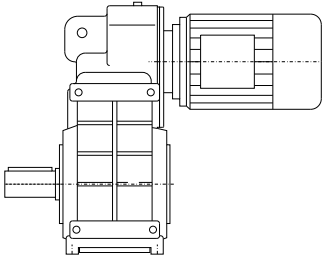
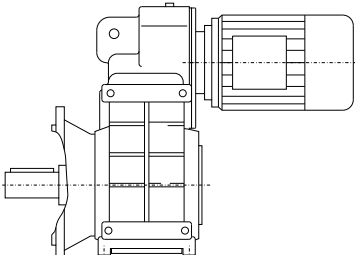
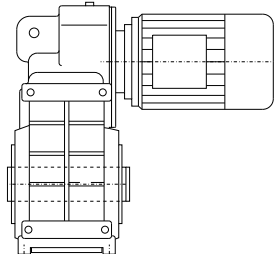
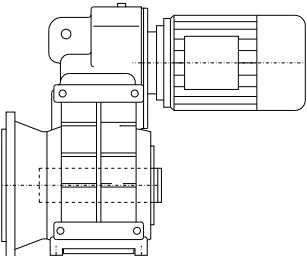
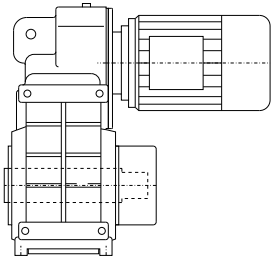
G90



Regreasing device for the mixer flange

The mixer gearbox can be fitted with a regreasing device on request.

Dimension drawing overview

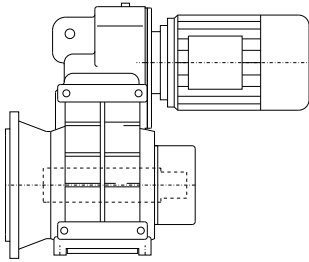
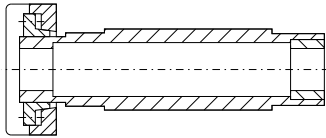
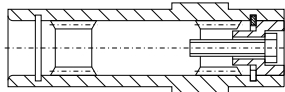
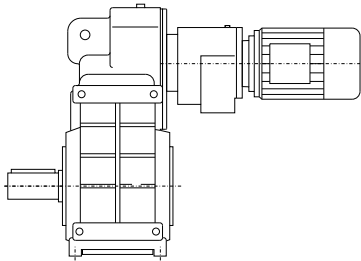
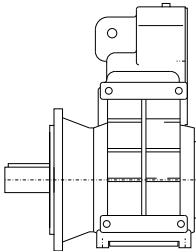
Representation	Gearbox type	Dimension drawing on page
	F.Z28	3/104
	F.Z38B	3/110
	F.Z48B	3/116
	F.Z68B	3/122
	F.Z88B	3/128
	F.Z108B	3/134
	F.Z128B	3/140
	F.Z148B	3/146
	F.Z168B	3/152
	F.Z188B	3/158
	F.Z208	3/164
	F.F28	3/105
	F.F38B	3/111
	F.F48B	3/117
	F.F68B	3/123
	F.F88B	3/129
	F.F108B	3/135
	F.F128B	3/141
	F.F148B	3/147
	F.F168B	3/153
	F.F188B	3/159
	F.A28 / F.AZ28	3/106
	F.A38B / F.AZ38B	3/112
	F.A48B / F.AZ48B	3/118
	F.A68B / F.AZ68B	3/124
	F.A88B / F.AZ88B	3/130
	F.A108B / F.AZ108B	3/136
	F.A128B / F.AZ128B	3/142
	F.A148B / F.AZ148B	3/148
	F.A168B / F.AZ168B	3/154
	F.A188B / F.AZ188B	3/160
	F.AF28	3/107
	F.AF38B	3/113
	F.AF48B	3/119
	F.AF68B	3/125
	F.AF88B	3/131
	F.AF108B	3/137
	F.AF128B	3/143
	F.AF148B	3/149
	F.AF168B	3/155
	F.AF188B	3/161
	F.AS28 / F.AZS28	3/108
	F.AS38B / F.AZS38B	3/114
	F.AS48B / F.AZS48B	3/120
	F.AS68B / F.AZS68B	3/126
	F.AS88B / F.AZS88B	3/132
	F.AS108B / F.AZS108B	3/138
	F.AS128B / F.AZS128B	3/144
	F.AS148B / F.AZS148B	3/150
	F.AS168B / F.AZS168B	3/156
	F.AS188B / F.AZS188B	3/162
	F.AS1208 / F.AZS208	3/165

MOTOX Geared Motors

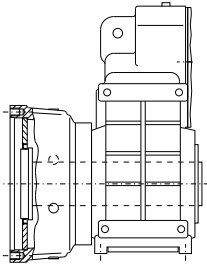
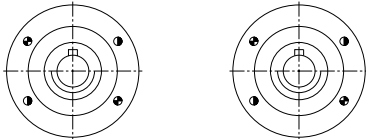
Parallel shaft geared motors

Dimensions

Dimension drawing overview (continued)

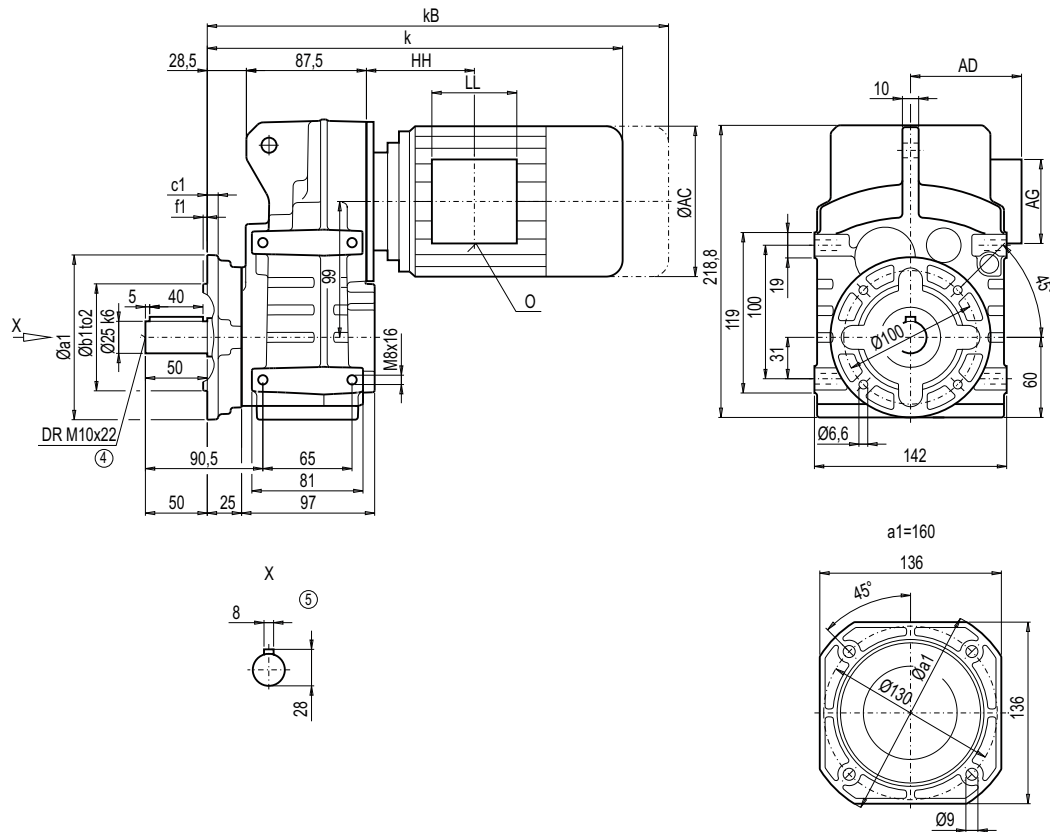
Representation	Gearbox type	Dimension drawing on page
	F.AFS28	3/109
	F.AFS38B	3/115
	F.AFS48B	3/121
	F.AFS68B	3/127
	F.AFS88B	3/133
	F.AFS108B	3/139
	F.AFS128B	3/145
	F.AFS148B	3/151
	F.AFS168B	3/157
	F.AFS188B	3/163
	F.A.S38B ... F.A.S188B	3/168
	F.A.T38B ... F.A.T188B	3/169
	F.38B-Z28 ... F.188B-Z68	3/170
	F.M88B ... F.M168B	3/174

Dimension drawing overview (continued)

Representation	Gearbox type	Dimension drawing on page
	F.E88B ... F.E168B	3/176
	Additional flange-mounted design	3/178
	Pin holes	3/179

Gearbox FDF/FZF28 (3- / 2-stage), flange-mounted design (A-type)

FF012



Flange	a1	b1	to2	c1	f1
A120	120	80	j6	8	3.0
A160	160	110	j6	9	3.5

Motor	F.F28								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDF28	FZF28
LA71	318.5	373.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	11	10
LA71Z	337.5	392.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	11	10
LA80	420.5	474.0	156.5	155	90	90	123.0	M20x1.5/M25x1.5	15	15
LA80Z	443.0	506.5	156.5	155	90	90	196.0	M20x1.5/M25x1.5	19	19
LA90S/L	415.5	486.5	174.0	163	90	90	87.0	M20x1.5/M25x1.5	20	20
LA90ZL	460.5	531.5	174.0	163	90	90	211.0	M20x1.5/M25x1.5	29	29
LA100L	497.5	578.5	195.0	168	120	120	163.5	2xM32x1.5	–	29
LA100ZL	567.5	648.5	195.0	168	120	120	295.5	2xM32x1.5	–	39

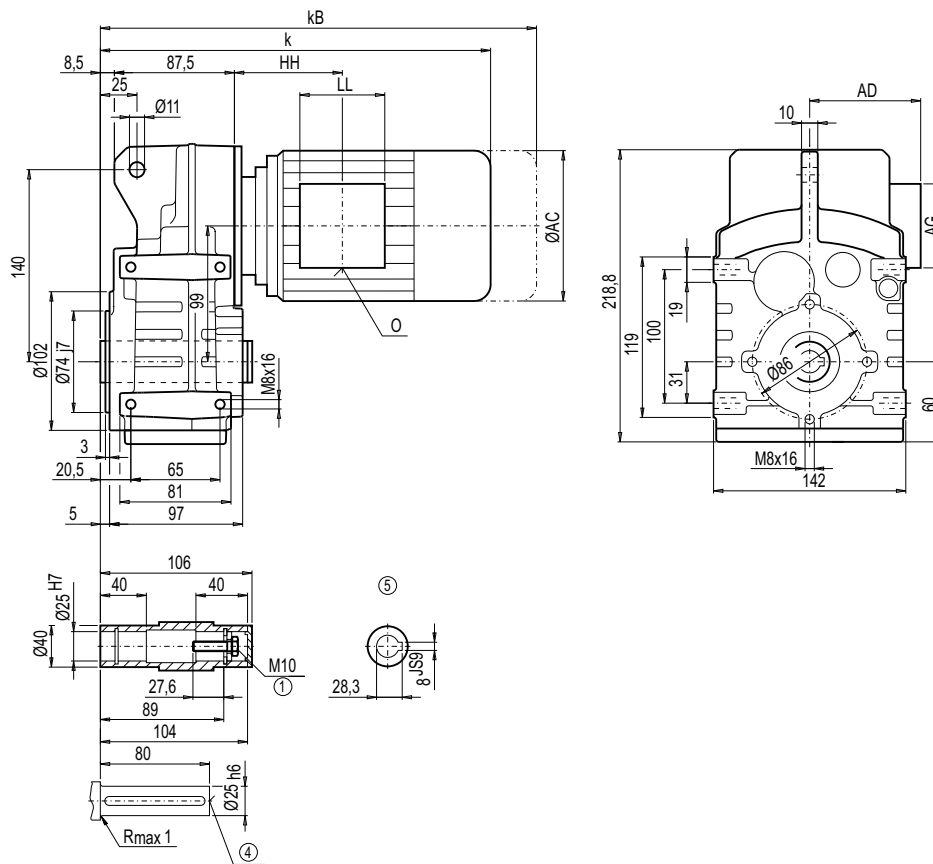
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDA/FZA28, FDAZ/FZAZ28 (3- / 2-stage), housing-flange-mounted design (C-type)

FA012
FAZ012



Motor	F.A.28								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDA.28	FZA.28
LA71	299.5	354.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	9
LA71Z	318.5	373.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	9
LA80	401.5	465.0	156.5	155	90	90	123.0	M20x1.5/M25x1.5	15	15
LA80Z	424.0	487.5	156.5	155	90	90	196.0	M20x1.5/M25x1.5	19	19
LA90S/L	396.5	467.5	174.0	163	90	90	87.0	M20x1.5/M25x1.5	19	19
LA90ZL	441.5	512.5	174.0	163	90	90	211.0	M20x1.5/M25x1.5	28	28
LA100L	478.5	559.5	195.0	168	120	120	163.5	2xM32x1.5	–	28
LA100ZL	548.5	629.5	195.0	168	120	120	295.5	2xM32x1.5	–	38

① DIN EN ISO 4017

④ DIN 332

⑤ Feather key / keyway DIN 6885

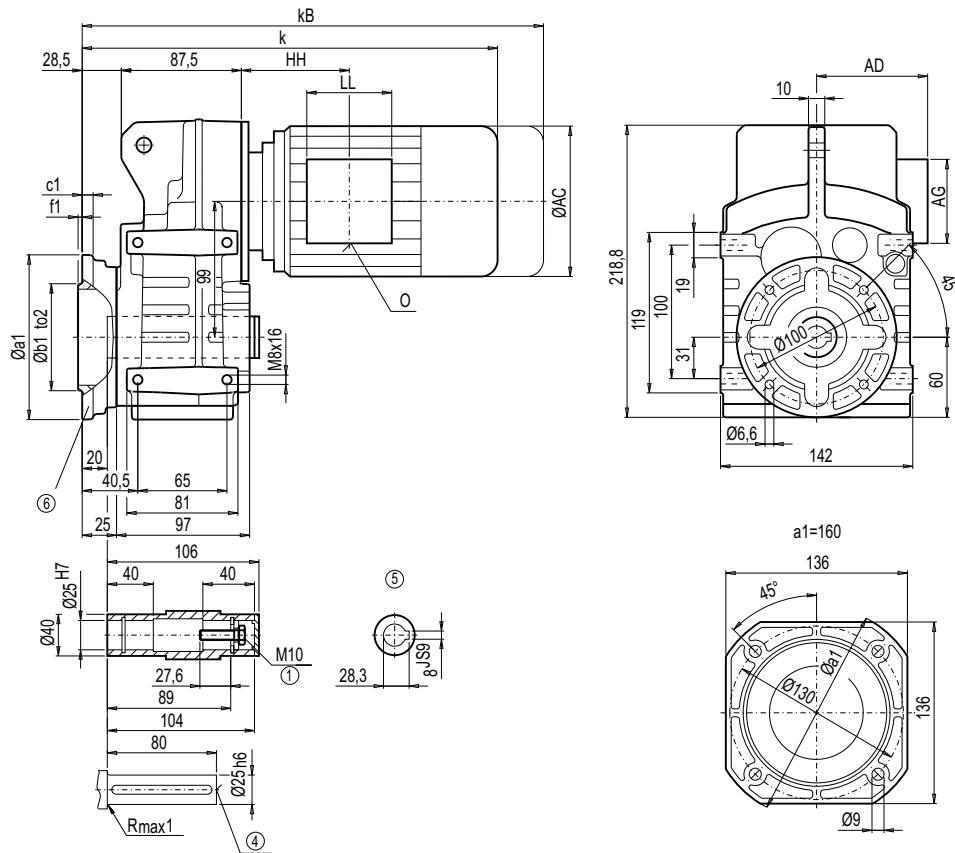
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAF/FZAF28 (3- / 2-stage), flange-mounted design

FAF012



Flange	a1	b1	to2	c1	f1
A120	120	80	j6	8	3.0
A160	160	110	j6	9	3.5

F.AF28									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF28	FZAF28
LA71	318.5	373.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	9
LA71Z	337.5	392.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	9
LA80	420.5	474.0	156.5	155	90	90	123.0	M20x1.5/M25x1.5	15	15
LA80Z	443.0	506.5	156.5	155	90	90	196.0	M20x1.5/M25x1.5	19	19
LA90S/L	415.5	486.5	174.0	163	90	90	87.0	M20x1.5/M25x1.5	19	19
LA90ZL	460.5	531.5	174.0	163	90	90	211.0	M20x1.5/M25x1.5	28	28
LA100L	497.5	578.5	195.0	168	120	120	163.5	2xM32x1.5	—	28
LA100ZL	567.5	648.5	195.0	168	120	120	295.5	2xM32x1.5		38

① DIN EN ISO 4017

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

Gearbox FDAS/FZAS28, FDAZS/FZAZS28 (3- / 2-stage) shaft-mounted design with shrink disk

[illegible]3/108

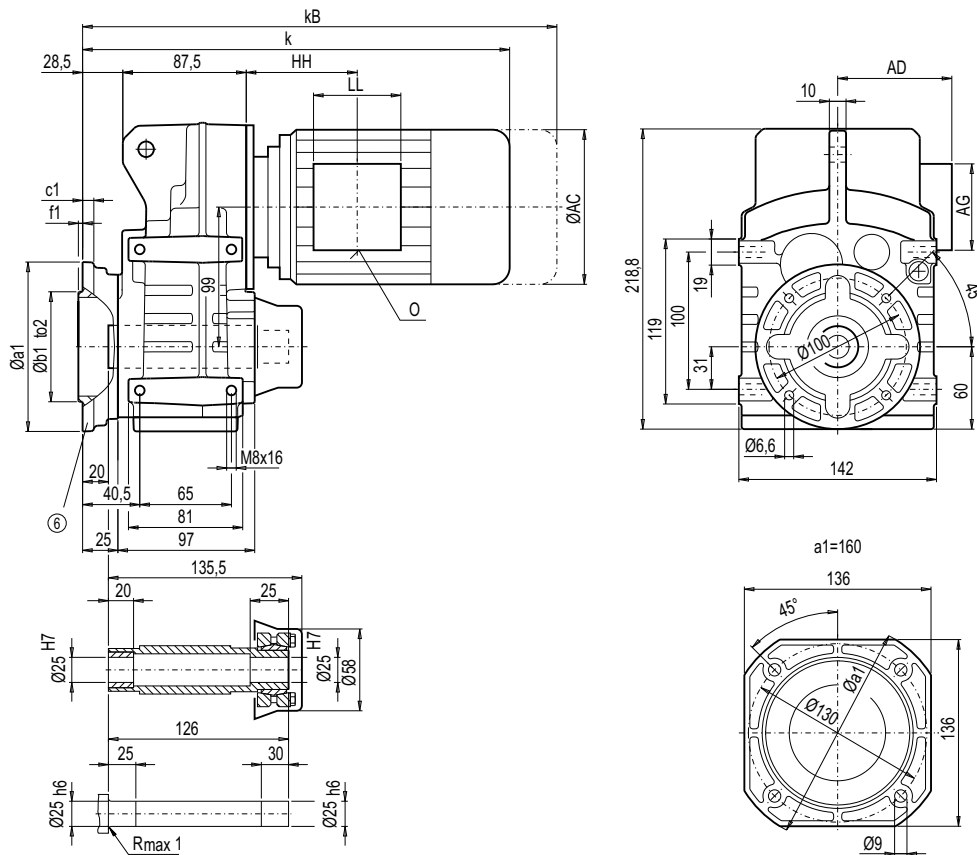
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAFS/FZAFS28 (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



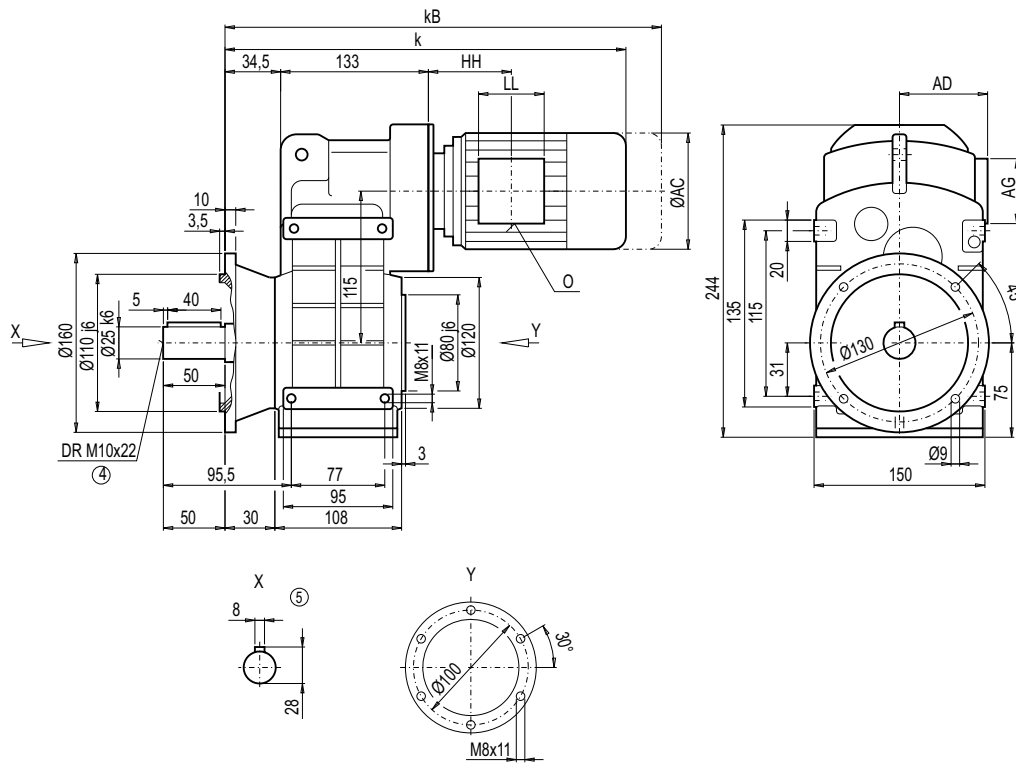
Flange	a1	b1	to2	c1	f1
A120	120	80	j6	8	3.0
A160	160	110	j6	9	3.5

F.AFS28									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS28	FZAFS28
LA71	318.5	373.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	10
LA71Z	337.5	392.5	139.0	146	90	90	58.5	M20x1.5/M25x1.5	10	10
LA80	420.5	474.0	156.5	155	90	90	123.0	M20x1.5/M25x1.5	15	15
LA80Z	443.0	506.5	156.5	155	90	90	196.0	M20x1.5/M25x1.5	19	19
LA90S/L	415.5	486.5	174.0	163	90	90	87.0	M20x1.5/M25x1.5	20	19
LA90ZL	460.5	531.5	174.0	163	90	90	211.0	M20x1.5/M25x1.5	29	28
LA100L	497.5	578.5	195.0	168	120	120	163.5	2xM32x1.5	–	29
LA100ZL	567.5	648.5	195.0	168	120	120	295.5	2xM32x1.5	–	39

© For note, see page 3/178

Gearbox FDF/FZF38B (3- / 2-stage), flange-mounted design (A-type)

FF012



Motor	F.F38B								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDF38B	FZF38B
LA71	401.0	456.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	20	19
LA71Z	420.0	475.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	20	19
LA80	438.0	501.5	156.5	155	90	90	89.0	M20x1.5/M25x1.5	25	24
LA80Z	460.5	524.0	156.5	155	90	90	162.0	M20x1.5/M25x1.5	29	28
LA90S/L	469.0	540.0	174.0	163	90	90	89.0	M20x1.5/M25x1.5	29	29
LA90ZL	514.0	585.0	174.0	163	90	90	213.0	M20x1.5/M25x1.5	35	35
LA100L	515.0	596.0	195.0	168	120	120	129.5	2xM32x1.5	–	38
LA100ZL	585.0	666.0	195.0	168	120	120	261.5	2xM32x1.5	–	48

④ DIN 332

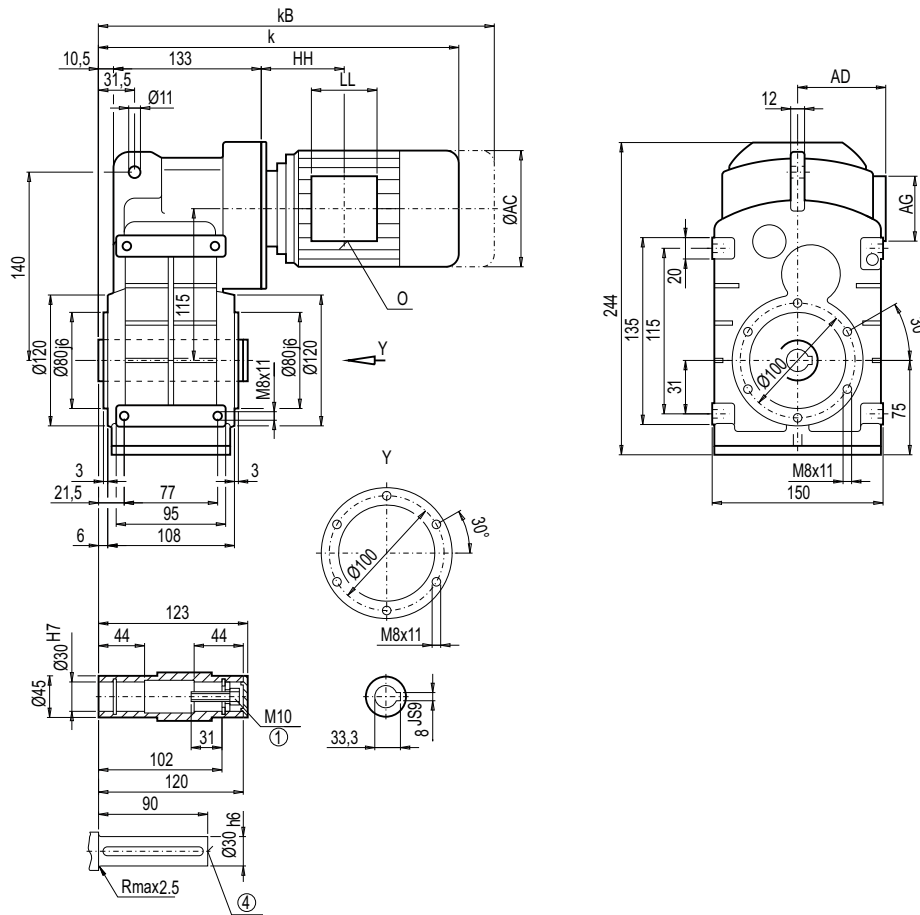
⑤ Feather key / keyway DIN 6885

MOTOX Geared Motors

Dimensions

Gearbox FDA/FZA38B, FDAZ/FZAZ38B (3- / 2-stage), housing-flange-mounted design (C-type)

FA012
FAZ012



Motor	F.A.38B								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDA.38B	FZA.38B
LA71	377.0	432.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	16	16
LA71Z	396.0	451.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	16	16
LA80	414.0	477.5	156.5	155	90	90	89.0	M20x1.5/M25x1.5	21	21
LA80Z	436.5	500.0	156.5	155	90	90	162.0	M20x1.5/M25x1.5	25	25
LA90S/L	445.0	516.0	174.0	163	90	90	89.0	M20x1.5/M25x1.5	26	26
LA90ZL	490.0	561.0	174.0	163	90	90	213.0	M20x1.5/M25x1.5	32	32
LA100L	491.0	572.0	195.0	168	120	120	129.5	2xM32x1.5	–	35
LA100ZL	561.0	642.0	195.0	168	120	120	261.5	2xM32x1.5	–	45

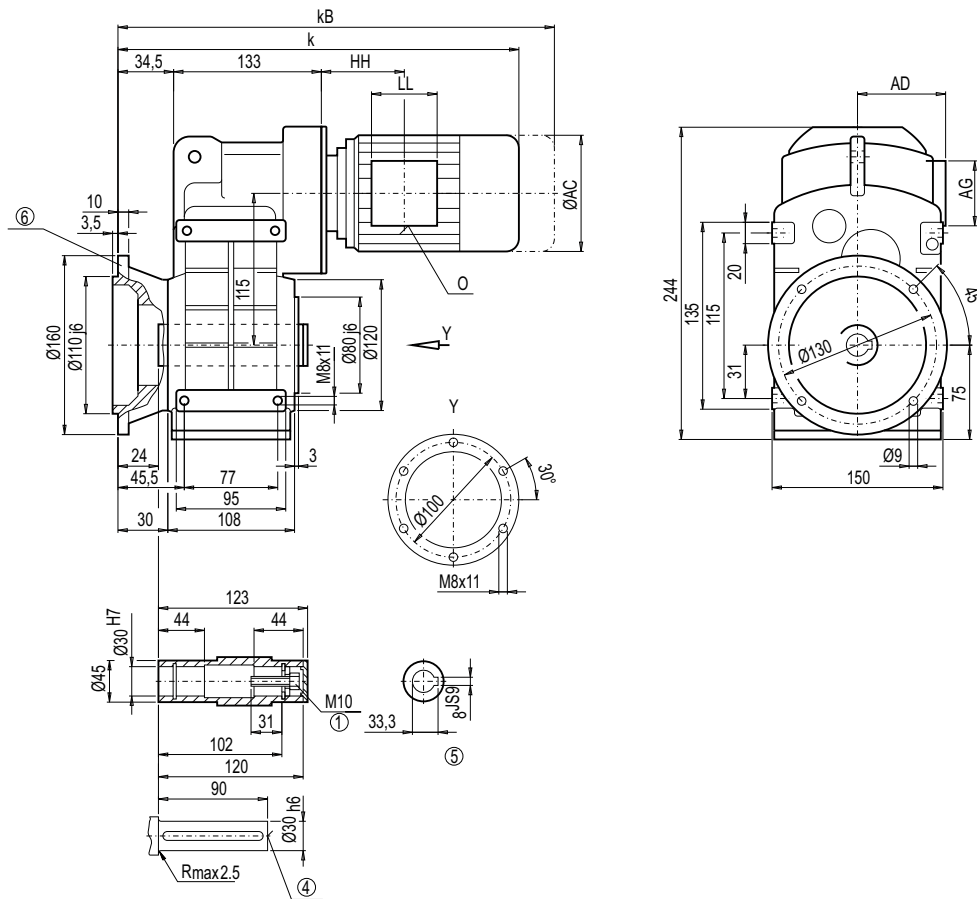
① DIN 6912

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF38B (3- / 2-stage), flange-mounted design

FAF012



Motor	F.AF38B								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDAF38B	FZAF38B
LA71	401.0	456.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	18	18
LA71Z	420.0	475.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	18	18
LA80	438.0	501.5	156.5	155	90	90	89.0	M20x1.5/M25x1.5	23	23
LA80Z	460.5	524.0	156.5	155	90	90	162.0	M20x1.5/M25x1.5	27	27
LA90S/L	469.0	540.0	174.0	163	90	90	89.0	M20x1.5/M25x1.5	28	28
LA90ZL	514.0	585.0	174.0	163	90	90	213.0	M20x1.5/M25x1.5	34	34
LA100L	515.0	596.0	195.0	168	120	120	129.5	2xM32x1.5	–	37
LA100ZL	585.0	666.0	195.0	168	120	120	261.5	2xM32x1.5	–	47

① DIN 6912

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

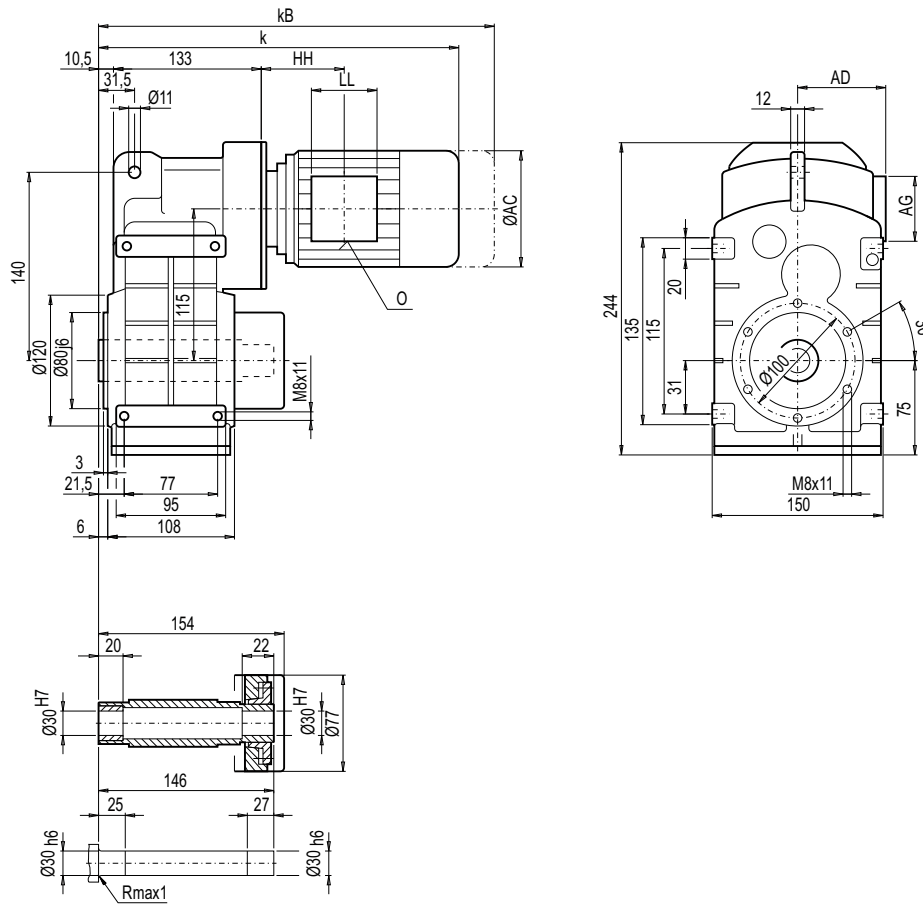
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAS/FZAS38B, FDAZS/FZAZS38B (3- / 2-stage), shaft-mounted design with shrink disk

FAS012
FAZS012



F.A.S38B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDA.S38B	FZA.S38B
LA71	377.0	432.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	17	17
LA71Z	396.0	451.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	17	17
LA80	414.0	477.5	156.5	155	90	90	89.0	M20x1.5/M25x1.5	22	22
LA80Z	436.5	500.0	156.5	155	90	90	162.0	M20x1.5/M25x1.5	26	26
LA90S/L	445.0	516.0	174.0	163	90	90	89.0	M20x1.5/M25x1.5	27	26
LA90ZL	490.0	561.0	174.0	163	90	90	213.0	M20x1.5/M25x1.5	33	32
LA100L	491.0	572.0	195.0	168	120	120	129.5	2xM32x1.5	–	35
LA100ZL	561.0	642.0	195.0	168	120	120	261.5	2xM32x1.5	–	45

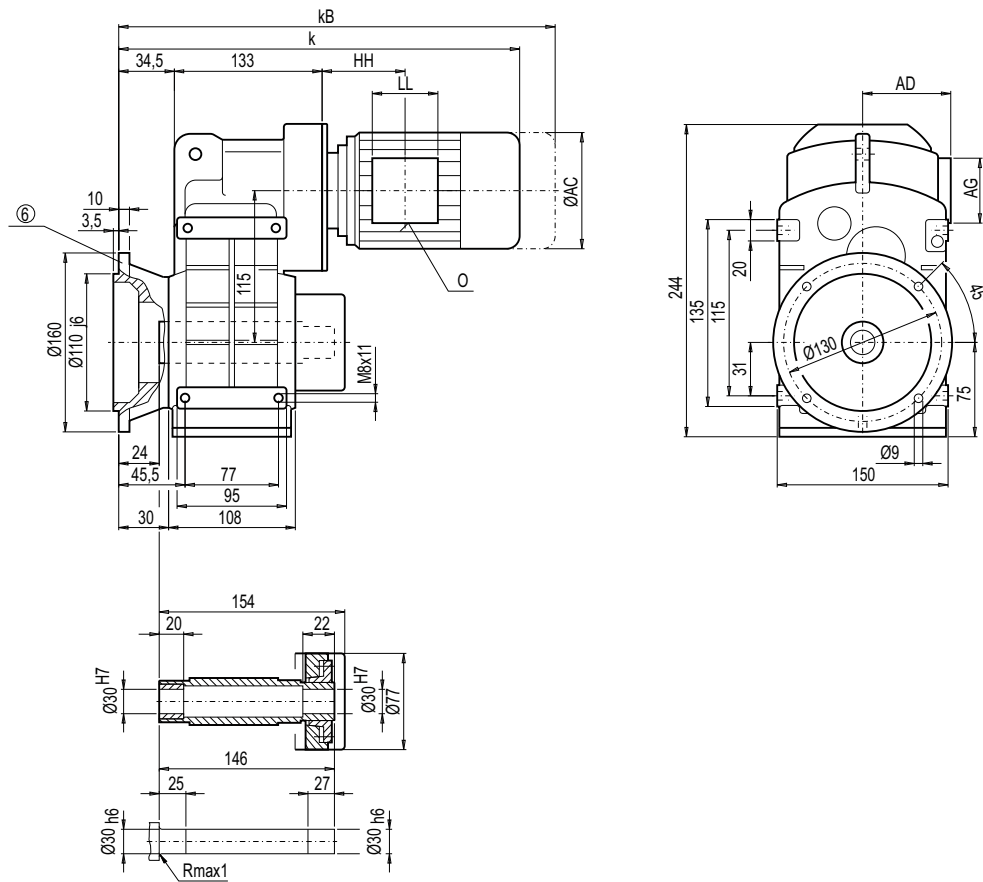
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAFS/FZAFS38B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



3

F.AFS38B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS38B	FZAFS38B
LA71	401.0	456.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	19	19
LA71Z	420.0	475.0	139.0	146	90	90	89.5	M20x1.5/M25x1.5	19	19
LA80	438.0	501.5	156.5	155	90	90	89.0	M20x1.5/M25x1.5	24	24
LA80Z	460.5	524.0	156.5	155	90	90	162.0	M20x1.5/M25x1.5	28	28
LA90S/L	469.0	540.0	174.0	163	90	90	89.0	M20x1.5/M25x1.5	29	28
LA90ZL	514.0	585.0	174.0	163	90	90	213.0	M20x1.5/M25x1.5	35	34
LA100L	515.0	596.0	195.0	168	120	120	129.5	2xM32x1.5	–	37
LA100ZL	585.0	666.0	195.0	168	120	120	261.5	2xM32x1.5	–	47

© For note, see page 3/178

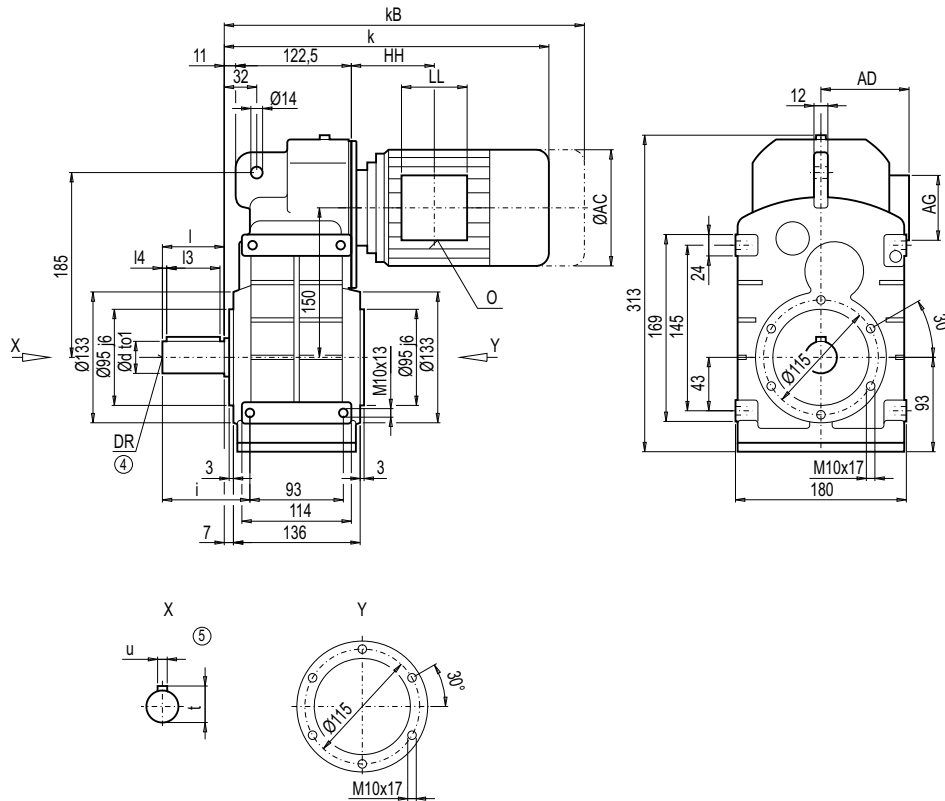
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDZ/FZZ48B (3- / 2-stage), housing-flange-mounted design (C-type)

FZ012



d	to1	l	l3	l4	t	u	i	DR
30 ^{*)}	k6	60	50	3.5	33	8	88.5	M10x22
40	k6	80	70	5.0	43	12	108.5	M16x36

^{*)} Preferred series

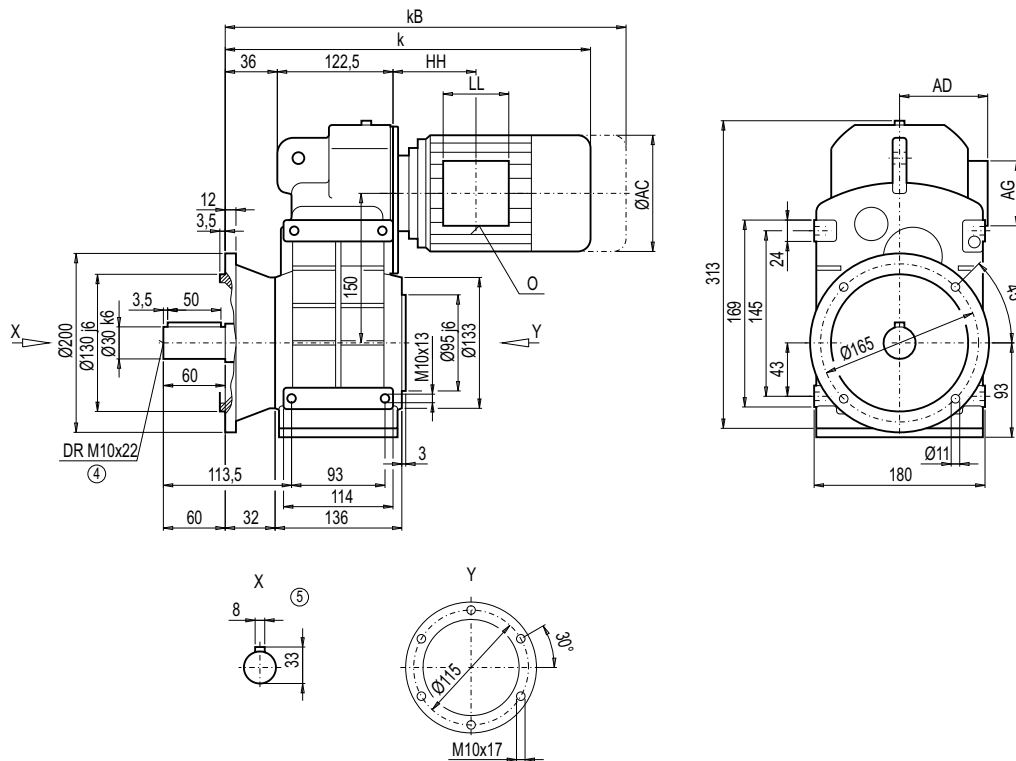
F.Z48B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDZ48B	FZZ48B
LA71	392.0	447.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	25	25
LA71Z	411.0	466.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	25	25
LA80	429.0	492.5	156.5	155	90	90	114.0	M20x1.5/M25x1.5	30	30
LA80Z	451.5	515.0	156.5	155	90	90	187.0	M20x1.5/M25x1.5	34	34
LA90S/L	460.0	531.0	174.0	163	90	90	114.0	M20x1.5/M25x1.5	35	35
LA90ZL	505.0	576.0	174.0	163	90	90	238.0	M20x1.5/M25x1.5	41	41
LA100L	506.0	587.0	195.0	168	120	120	154.5	2xM32x1.5	44	44
LA100ZL	576.0	657.0	195.0	168	120	120	286.5	2xM32x1.5	54	54
LA112M	535.5	616.5	219.0	181	120	120	160.0	2xM32x1.5	—	54
LA112ZM	563.5	644.5	219.0	181	120	120	264.0	2xM32x1.5	—	61

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDF/FZF48B (3- / 2-stage), flange-mounted design (A-type)

FF012



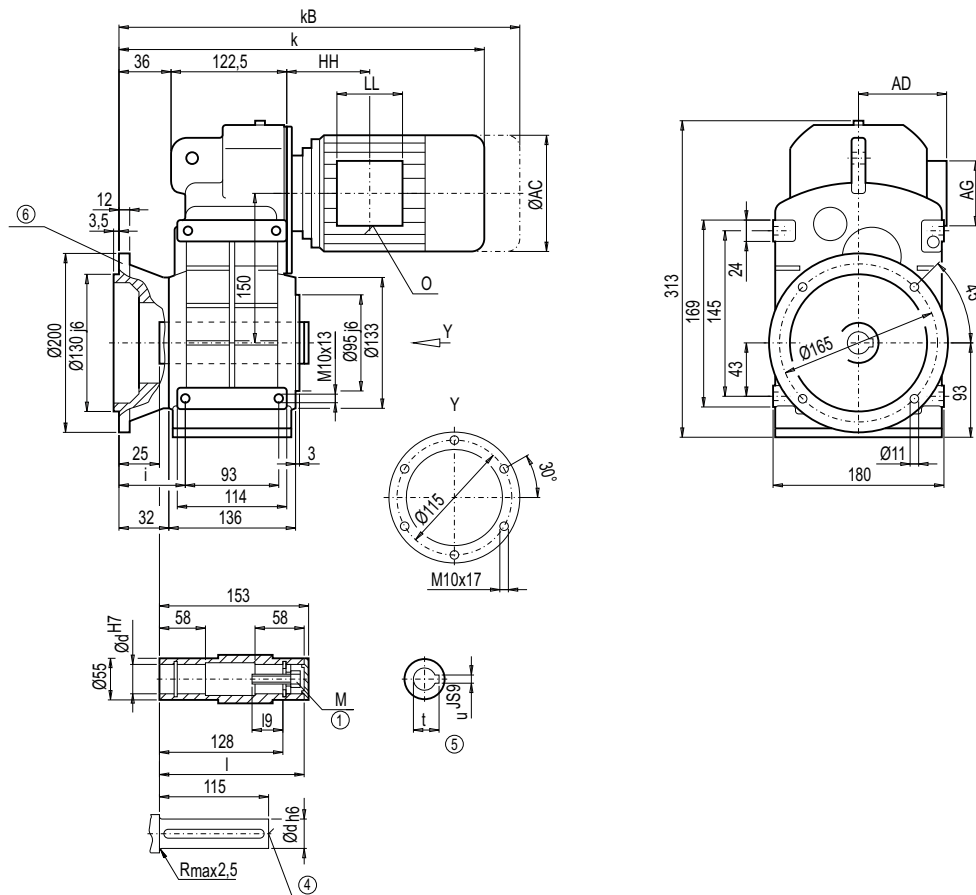
Motor	F.F48B								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDF48B	FZF48B
LA71	417.0	472.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	28	28
LA71Z	436.0	491.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	28	28
LA80	454.0	517.5	156.5	155	90	90	114.0	M20x1.5/M25x1.5	33	33
LA80Z	476.5	540.0	156.5	155	90	90	187.0	M20x1.5/M25x1.5	37	37
LA90S/L	485.0	556.0	174.0	163	90	90	114.0	M20x1.5/M25x1.5	38	38
LA90ZL	530.0	601.0	174.0	163	90	90	238.0	M20x1.5/M25x1.5	44	44
LA100L	531.0	612.0	195.0	168	120	120	154.5	2xM32x1.5	47	47
LA100ZL	601.0	682.0	195.0	168	120	120	286.5	2xM32x1.5	57	57
LA112M	560.5	641.5	219.0	181	120	120	160.0	2xM32x1.5	–	57
LA112ZM	588.5	669.5	219.0	181	120	120	264.0	2xM32x1.5	–	64

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF48B (3- / 2-stage), flange-mounted design

FAF012



d	l	l ₉	M	t	u	i
35 ^{*)}	150	40	M12	38.3	10	53.5
40	150	48	M16	43.3	12	53.5

*) Preferred series

F.AF48B									Weight	
Motor	k	k _B	AC	AD	AG	LL	HH	O	FDAF48B	FZAF48B
LA71	417.0	472.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	27	27
LA71Z	436.0	491.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	27	27
LA80	454.0	517.5	156.5	155	90	90	114.0	M20x1.5/M25x1.5	32	32
LA80Z	476.5	540.0	156.5	155	90	90	187.0	M20x1.5/M25x1.5	36	36
LA90S/L	485.0	556.0	174.0	163	90	90	114.0	M20x1.5/M25x1.5	37	36
LA90ZL	530.0	601.0	174.0	163	90	90	238.0	M20x1.5/M25x1.5	43	42
LA100L	531.0	612.0	195.0	168	120	120	154.5	2xM32x1.5	46	45
LA100ZL	601.0	682.0	195.0	168	120	120	286.5	2xM32x1.5	56	55
LA112M	560.5	641.5	219.0	181	120	120	160.0	2xM32x1.5	—	56
LA112ZM	588.5	669.5	219.0	181	120	120	264.0	2xM32x1.5	—	63

① DIN 6912

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

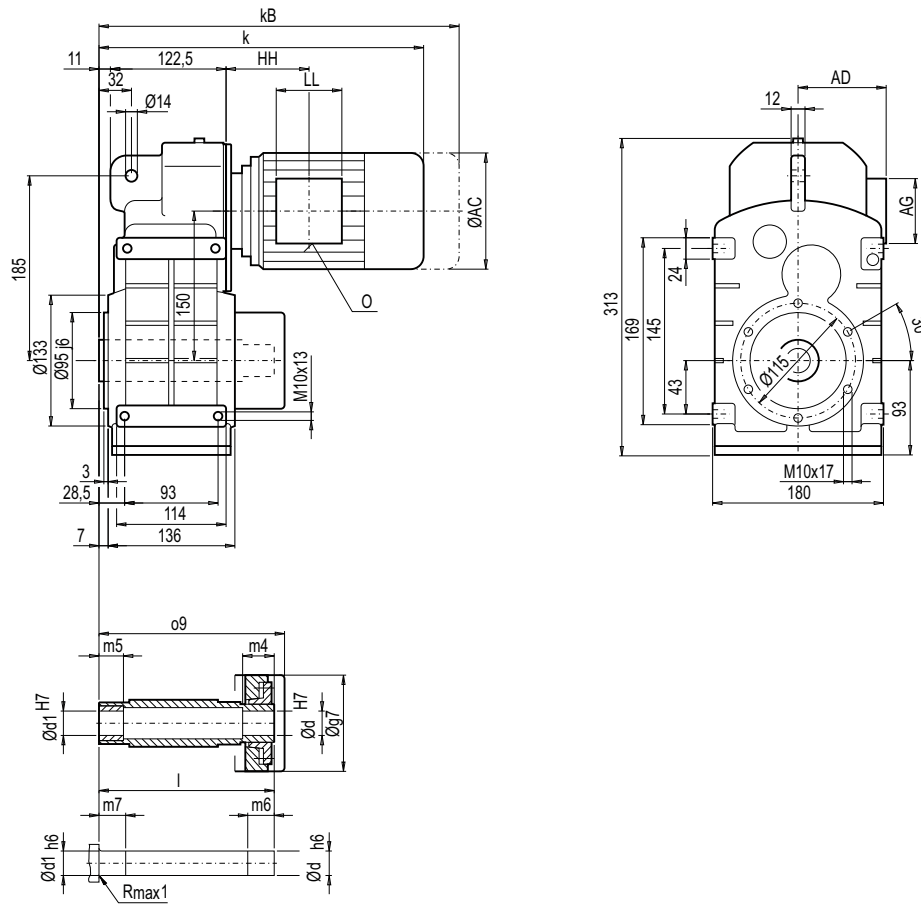
MOTEX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAS/FZAS48B, FDAZS/FZAZS48B (3- / 2-stage), shaft-mounted design with shrink disk

FAS012
FAZS012



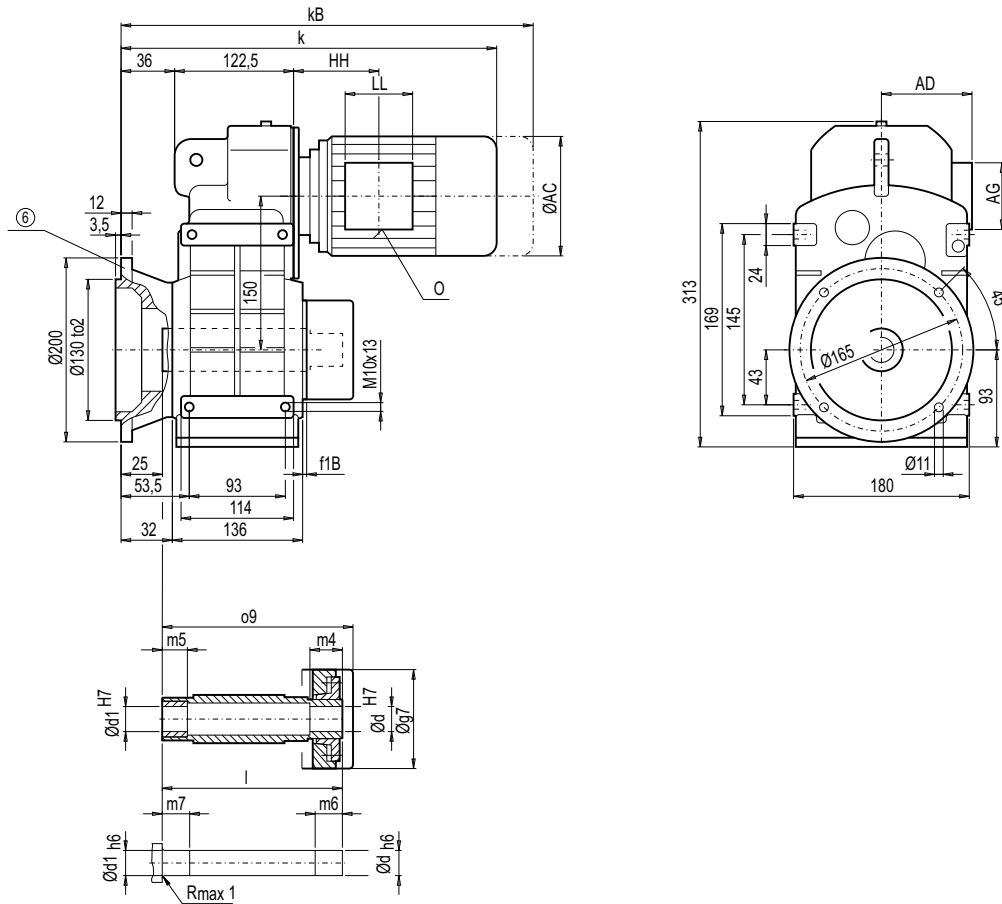
d	d1	l	o9	m4	m5	m6	m7	g7
35 *)	35	177	184	32	20	37	25	93
40	50	177	184	25	20	30	25	93

*) Preferred series

Motor	F.A.S48B								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDA.S48B	FZA.S48B
LA71	392.0	447.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	25	25
LA71Z	411.0	466.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	25	25
LA80	429.0	492.5	156.5	155	90	90	114.0	M20x1.5/M25x1.5	30	30
LA80Z	451.5	515.0	156.5	155	90	90	187.0	M20x1.5/M25x1.5	34	34
LA90S/L	460.0	531.0	174.0	163	90	90	114.0	M20x1.5/M25x1.5	34	34
LA90ZL	505.0	576.0	174.0	163	90	90	238.0	M20x1.5/M25x1.5	40	40
LA100L	506.0	587.0	195.0	168	120	120	154.5	2xM32x1.5	43	43
LA100ZL	576.0	657.0	195.0	168	120	120	286.5	2xM32x1.5	53	53
LA112M	535.5	616.5	219.0	181	120	120	160.0	2xM32x1.5	–	54
LA112ZM	563.5	644.5	219.0	181	120	120	264.0	2xM32x1.5	–	61

Gearbox FDAFS/FZAFS48B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



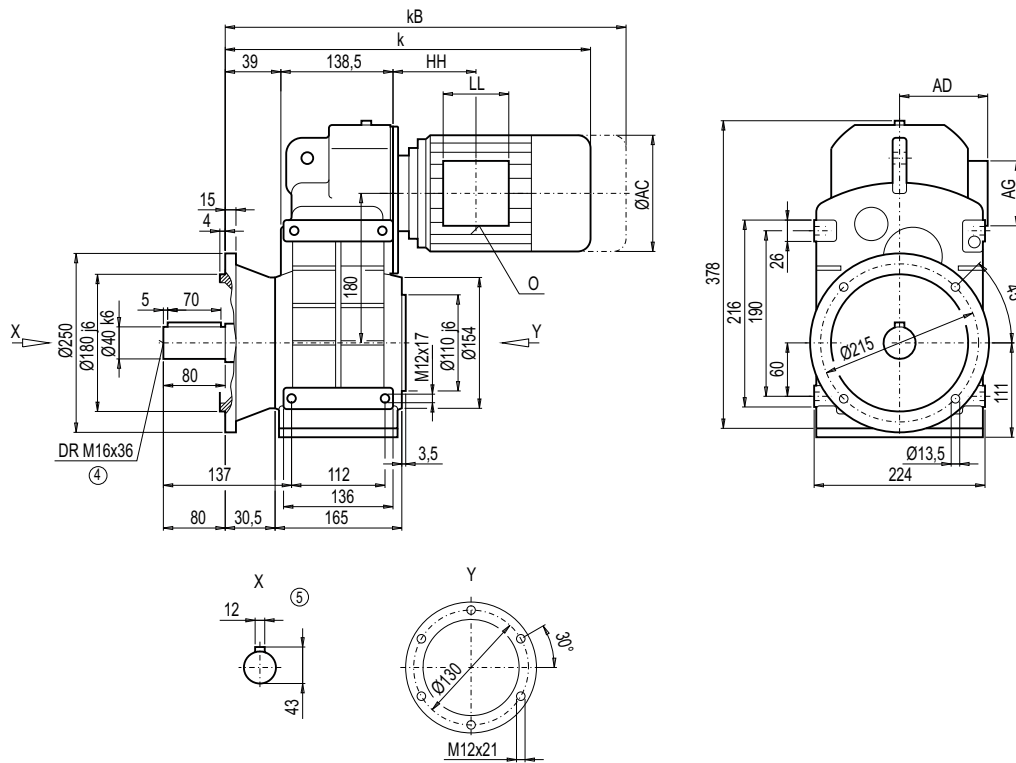
d	d1	l	o9	m4	m5	m6	m7	g7
35 *)	35	177	184	32	20	37	25	93
40	40	177	184	25	20	30	25	93

*¹) Preferred series

F.AFS48B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS48B	FZAFS48B
LA71	417.0	472.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	28	28
LA71Z	436.0	491.0	139.0	146	90	90	114.5	M20x1.5/M25x1.5	28	28
LA80	454.0	517.5	156.5	155	90	90	114.0	M20x1.5/M25x1.5	33	33
LA80Z	476.5	540.0	156.5	155	90	90	187.0	M20x1.5/M25x1.5	37	37
LA90S/L	485.0	556.0	174.0	163	90	90	114.0	M20x1.5/M25x1.5	37	37
LA90ZL	530.0	601.0	174.0	163	90	90	238.0	M20x1.5/M25x1.5	43	43
LA100L	531.0	612.0	195.0	168	120	120	154.5	2xM32x1.5	46	46
LA100ZL	601.0	682.0	195.0	168	120	120	286.5	2xM32x1.5	56	56
LA112M	560.5	641.5	219.0	181	120	120	160.0	2xM32x1.5	–	57
LA112ZM	588.5	669.5	219.0	181	120	120	264.0	2xM32x1.5	–	64

Gearbox FDF/FZF68B (3- / 2-stage), flange-mounted design (A-type)

FF012



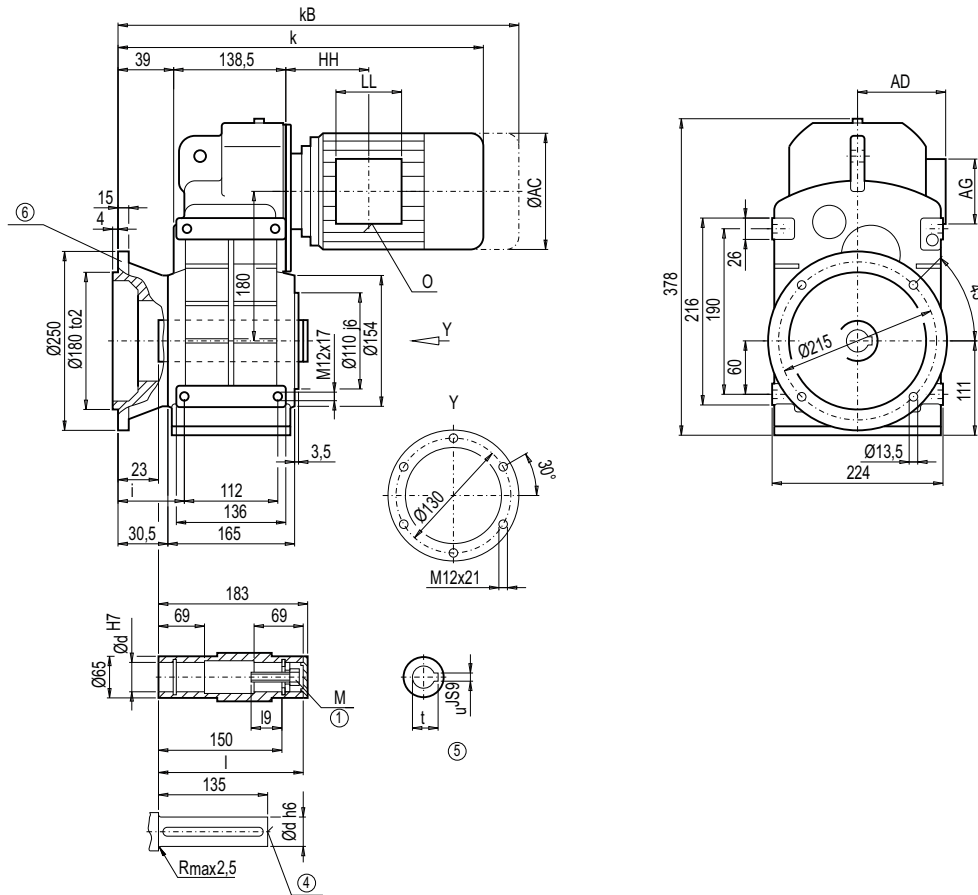
	F.F68B								Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF68B	FZF68B
LA71	430.5	485.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	48	—
LA71Z	449.5	504.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	48	—
LA80	467.5	531.0	156.5	155	90	90	108.5	M20x1.5/M25x1.5	53	53
LA80Z	490.0	553.5	156.5	155	90	90	181.5	M20x1.5/M25x1.5	57	57
LA90S/L	498.5	569.5	174.0	163	90	90	108.5	M20x1.5/M25x1.5	57	57
LA90ZL	543.5	614.5	174.0	163	90	90	232.5	M20x1.5/M25x1.5	63	63
LA100L	544.5	625.5	195.0	168	120	120	149.0	2xM32x1.5	67	67
LA100ZL	614.5	695.5	195.0	168	120	120	281.0	2xM32x1.5	77	77
LA112M	573.5	654.5	219.0	181	120	120	154.0	2xM32x1.5	—	78
LA112ZM	601.5	682.5	219.0	181	120	120	258.0	2xM32x1.5	—	85
LA132S/M	635.5	737.5	259.0	195	140	140	196.5	2xM32x1.5	—	88
LA132ZM	681.5	783.5	259.0	195	140	140	304.5	2xM32x1.5	—	109

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF68B (3- / 2-stage), flange-mounted design

FAF012



d	l	l ₉	M	t	u	i
40 ^{*)}	180	48	M16	43.3	12	57
45	180	47	M16	48.8	14	57

^{*)} Preferred series

F.AF68B									Weight	
Motor	k	k _B	AC	AD	AG	LL	HH	O	FDAF68B	FZAF68B
LA71	430.5	485.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	45	–
LA71Z	449.5	504.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	45	–
LA80	467.5	531.0	156.5	155	90	90	108.5	M20x1.5/M25x1.5	50	50
LA80Z	490.0	553.5	156.5	155	90	90	181.5	M20x1.5/M25x1.5	54	54
LA90S/L	498.5	569.5	174.0	163	90	90	108.5	M20x1.5/M25x1.5	54	54
LA90ZL	543.5	614.5	174.0	163	90	90	232.5	M20x1.5/M25x1.5	60	60
LA100L	544.5	625.5	195.0	168	120	120	149.0	2xM32x1.5	63	63
LA100ZL	614.5	695.5	195.0	168	120	120	281.0	2xM32x1.5	73	73
LA112M	573.5	654.5	219.0	181	120	120	154.0	2xM32x1.5	–	75
LA112ZM	601.5	682.5	219.0	181	120	120	258.0	2xM32x1.5	–	82
LA132S/M	635.5	737.5	259.0	195	140	140	196.5	2xM32x1.5	–	85
LA132ZM	681.5	783.5	259.0	195	140	140	304.5	2xM32x1.5	–	106

① DIN 6912

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

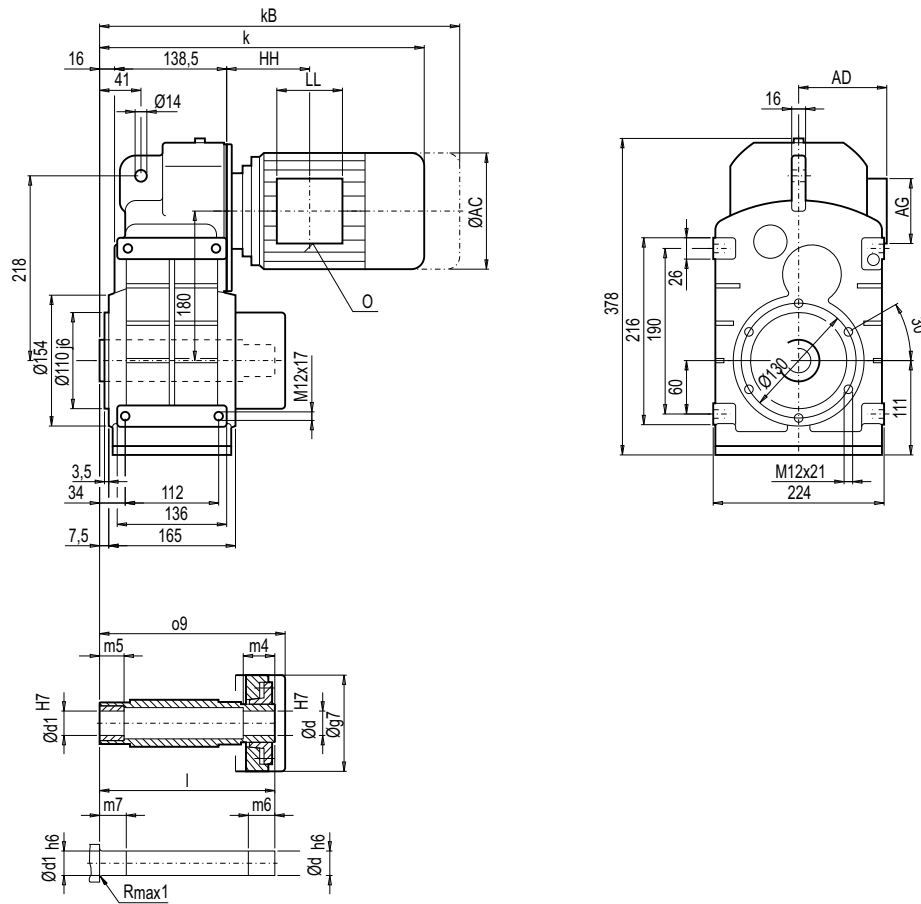
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAS/FZAS68B, FDAZS/FZAZS68B (3- / 2-stage), shaft-mounted design with shrink disk

FAS012
FAZS012



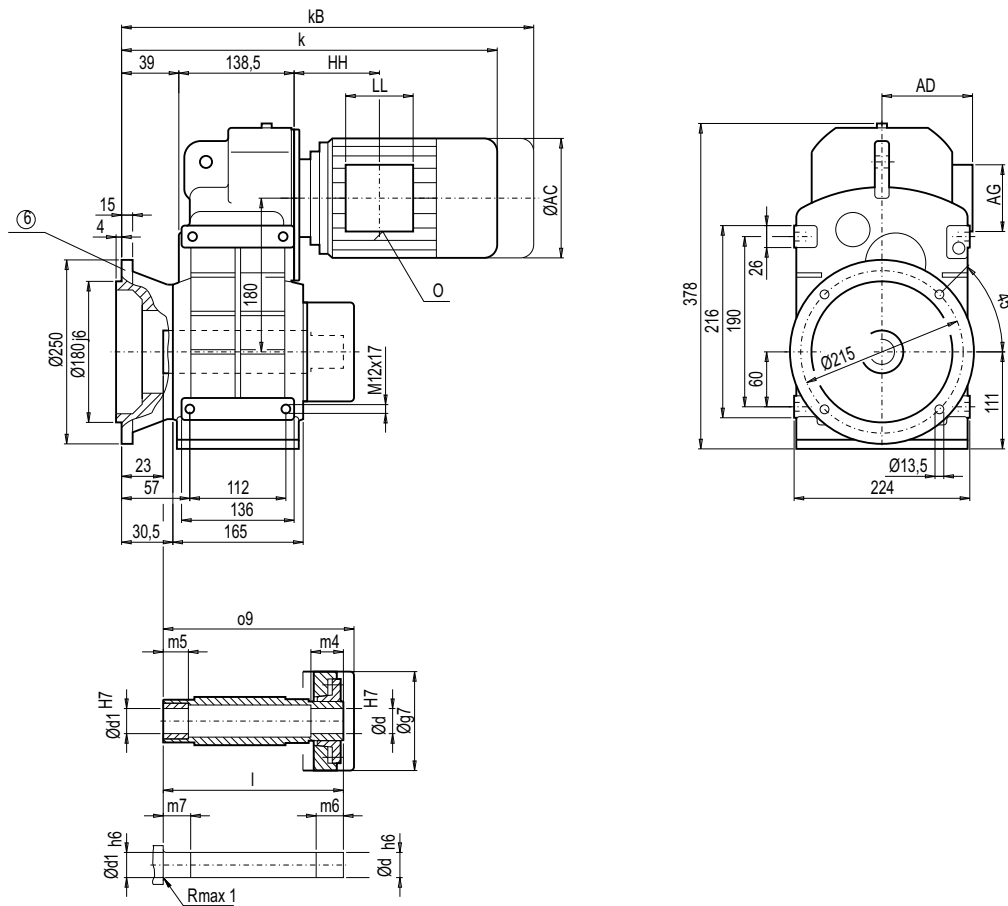
d	d1	l	o9	m4	m5	m6	m7	g7
40 *)	40	209	216	35	20	40	25	112
50	50	209	216	27	20	32	25	112

*) Preferred series

F.A.S68B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDA.S68B	FZA.S68B
LA71	407.5	462.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	38	–
LA71Z	426.5	481.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	38	–
LA80	444.5	508.0	156.5	155	90	90	108.5	M20x1.5/M25x1.5	43	43
LA80Z	467.0	530.5	156.5	155	90	90	181.5	M20x1.5/M25x1.5	47	47
LA90S/L	475.5	546.5	174.0	163	90	90	108.5	M20x1.5/M25x1.5	48	48
LA90ZL	520.5	591.5	174.0	163	90	90	232.5	M20x1.5/M25x1.5	54	54
LA100L	521.5	602.5	195.0	168	120	120	149.0	2xM32x1.5	57	57
LA100ZL	591.5	672.5	195.0	168	120	120	281.0	2xM32x1.5	67	67
LA112M	550.5	631.5	219.0	181	120	120	154.0	2xM32x1.5	–	68
LA112ZM	578.5	659.5	219.0	181	120	120	258.0	2xM32x1.5	–	75
LA132S/M	612.5	714.5	259.0	195	140	140	196.5	2xM32x1.5	–	78
LA132ZM	658.5	760.5	259.0	195	140	140	304.5	2xM32x1.5	–	99

Gearbox FDAFS/FZAFS68B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



d	d1	l	o9	m4	m5	m6	m7	g7
40 *)	40	209	216	35	20	40	25	112
50	50	209	216	27	20	32	25	112

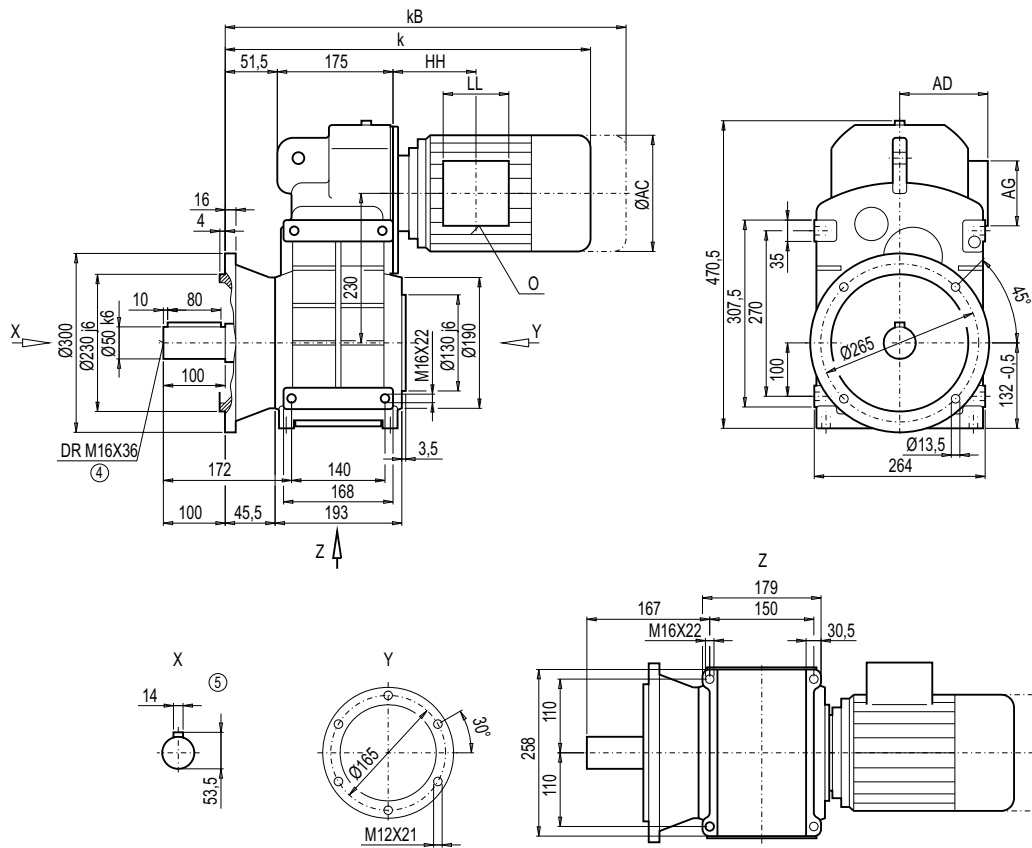
*) Preferred series

F.AFS68B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS68B	FZAFS68B
LA71	430.5	485.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	46	–
LA71Z	449.5	504.5	139.0	146	90	90	109.0	M20x1.5/M25x1.5	46	–
LA80	467.5	531.0	156.5	155	90	90	108.5	M20x1.5/M25x1.5	51	51
LA80Z	490.0	553.5	156.5	155	90	90	181.5	M20x1.5/M25x1.5	55	55
LA90S/L	498.5	569.5	174.0	163	90	90	108.5	M20x1.5/M25x1.5	55	55
LA90ZL	543.5	614.5	174.0	163	90	90	232.5	M20x1.5/M25x1.5	61	61
LA100L	544.5	625.5	195.0	168	120	120	149.0	2xM32x1.5	65	65
LA100ZL	614.5	695.5	195.0	168	120	120	281.0	2xM32x1.5	75	75
LA112M	573.5	654.5	219.0	181	120	120	154.0	2xM32x1.5	–	76
LA112ZM	601.5	682.5	219.0	181	120	120	258.0	2xM32x1.5	–	83
LA132S/M	635.5	737.5	259.0	195	140	140	196.5	2xM32x1.5	–	86
LA132ZM	681.5	783.5	259.0	195	140	140	304.5	2xM32x1.5	–	107

⑥ For note, see page 3/178

Gearbox FDF/FZF88B (3- / 2-stage), flange-mounted design (A-type)

FF012



F.F88B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF88B	FZF88B
LA71	473.5	528.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	81	–
LA71Z	492.5	547.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	81	–
LA80	510.5	574.0	156.5	155	90	90	102.5	M20x1.5/M25x1.5	86	86
LA80Z	533.0	596.5	156.5	155	90	90	175.5	M20x1.5/M25x1.5	90	90
LA90S/L	541.5	612.5	174.0	163	90	90	102.5	M20x1.5/M25x1.5	91	91
LA90ZL	586.5	657.5	174.0	163	90	90	226.5	M20x1.5/M25x1.5	97	97
LA100L	587.5	668.5	195.0	168	120	120	143.0	2xM32x1.5	100	100
LA100ZL	657.5	738.5	195.0	168	120	120	275.0	2xM32x1.5	110	110
LA112M	614.5	695.5	219.0	181	120	120	146.0	2xM32x1.5	111	112
LA112ZM	642.5	723.5	219.0	181	120	120	250.0	2xM32x1.5	118	119
LA132S/M	674.5	776.5	259.0	195	140	140	186.5	2xM32x1.5	124	125
LA132ZM	720.5	822.5	259.0	195	140	140	294.5	2xM32x1.5	146	146
LA160M/L	777.0	895.5	313.5	227	165	165	212.0	2xM40x1.5	–	158
LA160ZL	825.0	943.5	313.5	227	165	165	365.0	2xM40x1.5	–	197

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDA/FZA88B, FDAZ/FZAZ88B (3- / 2-stage), housing-flange-mounted design (C-type)

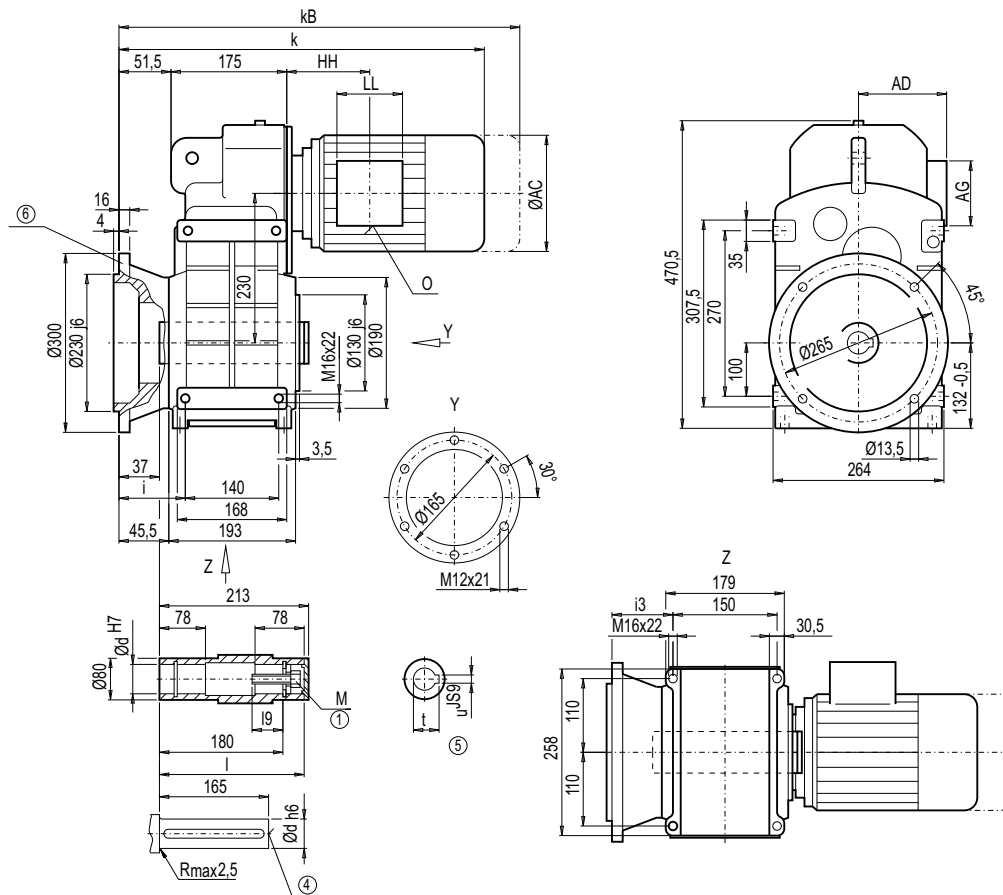
[illegible]

*) Preferred series

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF88B (3- / 2-stage), flange-mounted design

FAF012



d	l	l9	M	t	u	i	i3
50 ^{*)}	210	44.5	M16	53.8	14	72	67
60	210	54.0	M20	64.4	18	72	67

*) Preferred series

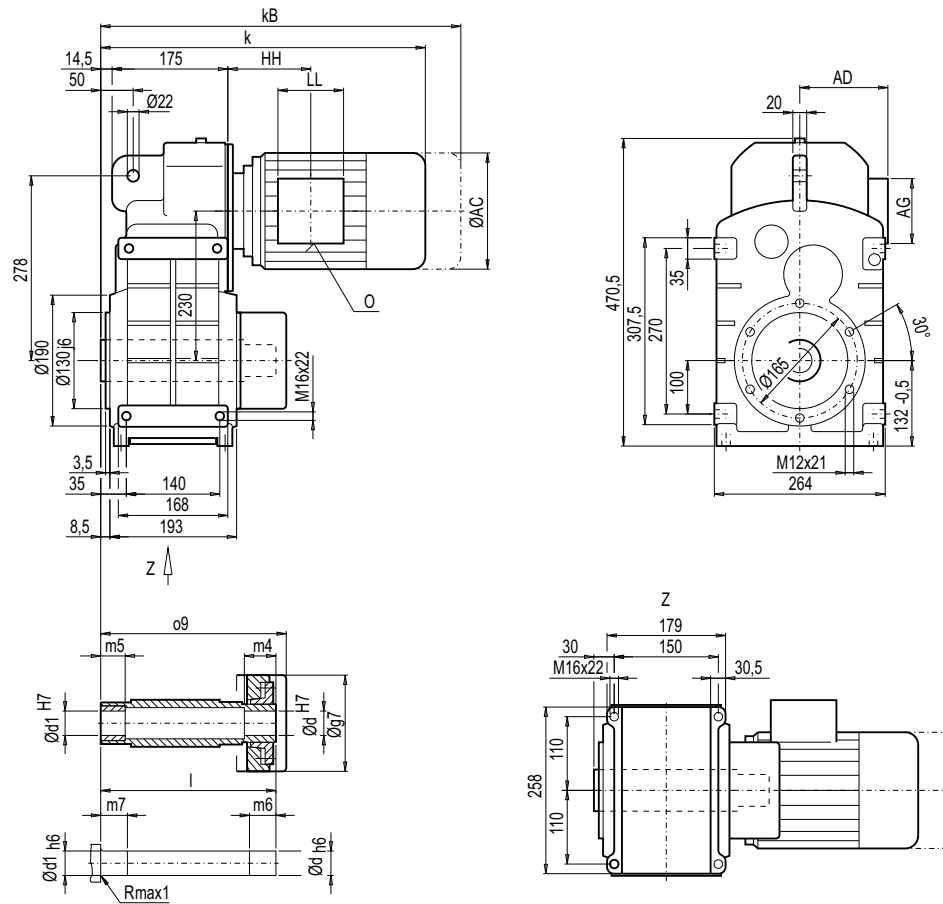
F.AF88B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF88B	FZAF88B
LA71	473.5	528.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	73	–
LA71Z	492.5	547.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	73	–
LA80	510.5	574.0	156.5	155	90	90	102.5	M20x1.5/M25x1.5	78	78
LA80Z	533.0	596.5	156.5	155	90	90	175.5	M20x1.5/M25x1.5	82	82
LA90S/L	541.5	612.5	174.0	163	90	90	102.5	M20x1.5/M25x1.5	82	83
LA90ZL	586.5	657.5	174.0	163	90	90	226.5	M20x1.5/M25x1.5	88	89
LA100L	587.5	668.5	195.0	168	120	120	143.0	2xM32x1.5	92	92
LA100ZL	657.5	738.5	195.0	168	120	120	275.0	2xM32x1.5	102	102
LA112M	614.5	695.5	219.0	181	120	120	146.0	2xM32x1.5	103	104
LA112ZM	642.5	723.5	219.0	181	120	120	250.0	2xM32x1.5	110	111
LA132S/M	674.5	776.5	259.0	195	140	140	186.5	2xM32x1.5	116	117
LA132ZM	720.5	822.5	259.0	195	140	140	294.5	2xM32x1.5	137	138
LA160M/L	777.0	895.5	313.5	227	165	165	212.0	2xM40x1.5	–	150
LA160ZL	825.0	943.5	313.5	227	165	165	365.0	2xM40x1.5	–	189

MOTOX Geared Motors

Dimensions

Gearbox FDAS/FZAS88B, FDAZS/FZAZS88B (3- / 2-stage), shaft-mounted design with shrink disk

FAS012
FAZS012



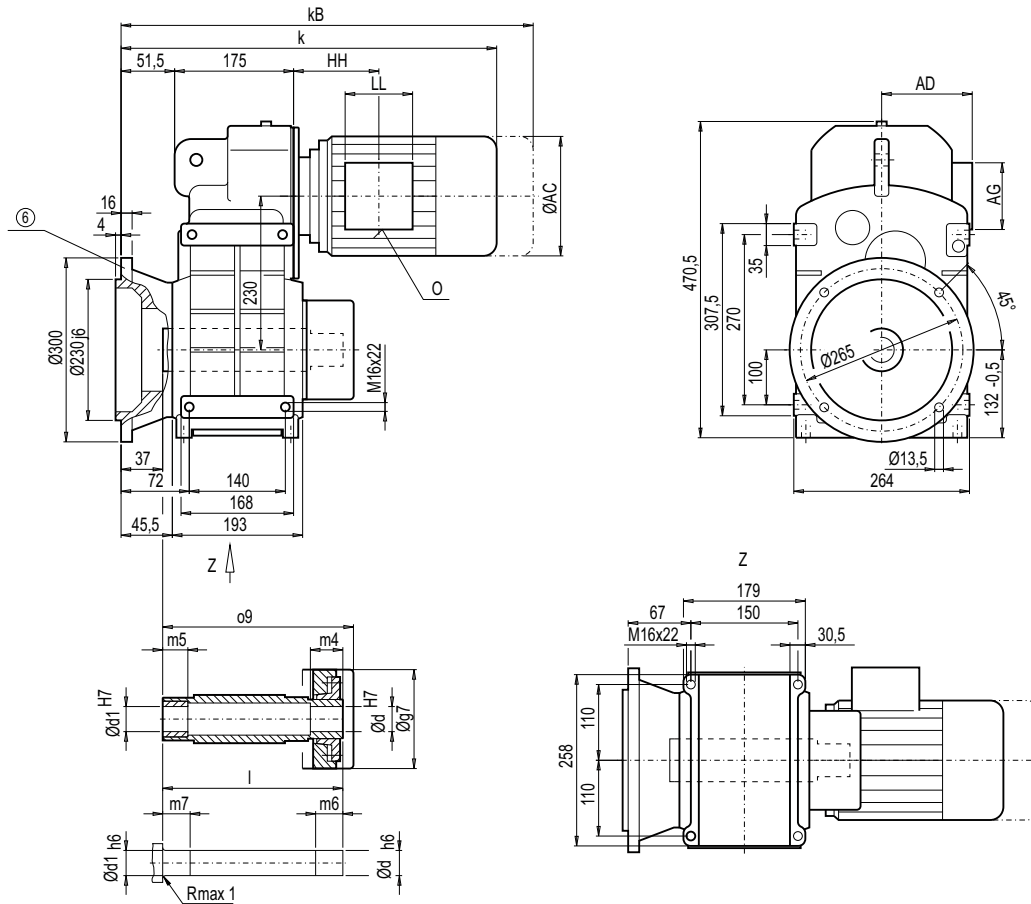
d	d1	l	o9	m4	m5	m6	m7	g7
50 *)	50	241	249	29	30	34	35	132
60	60	241	249	29	30	34	35	132

*¹) Preferred series

F.A.S88B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDA.S88B	FZA.S88B
LA71	436.5	491.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	64	–
LA71Z	455.5	510.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	64	–
LA80	473.5	537.0	156.5	155	90	90	102.5	M20x1.5/M25x1.5	69	69
LA80Z	496.0	559.5	156.5	155	90	90	175.5	M20x1.5/M25x1.5	73	73
LA90S/L	504.5	575.5	174.0	163	90	90	102.5	M20x1.5/M25x1.5	73	74
LA90ZL	549.5	620.5	174.0	163	90	90	226.5	M20x1.5/M25x1.5	79	80
LA100L	550.5	631.5	195.0	168	120	120	143.0	2xM32x1.5	82	83
LA100ZL	620.5	701.5	195.0	168	120	120	275.0	2xM32x1.5	92	93
LA112M	577.5	658.5	219.0	181	120	120	146.0	2xM32x1.5	94	95
LA112ZM	605.5	686.5	219.0	181	120	120	250.0	2xM32x1.5	101	102
LA132S/M	637.5	739.5	259.0	195	140	140	186.5	2xM32x1.5	107	108
LA132ZM	683.5	785.5	259.0	195	140	140	294.5	2xM32x1.5	128	129
LA160M/L	740.0	858.5	313.5	227	165	165	212.0	2xM40x1.5	–	140
LA160ZL	788.0	906.5	313.5	227	165	165	365.0	2xM40x1.5	–	179

Gearbox FDAFS/FZAFS88B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



d	d1	l	o9	m4	m5	m6	m7	g7
50 ^{a)}	50	241	249	29	30	34	35	132
60	60	241	249	29	30	34	35	132

*) Preferred series

F.AFS88B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS88B	FZAFS88B
LA71	473.5	528.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	75	–
LA71Z	492.5	547.5	139.0	146	90	90	103.0	M20x1.5/M25x1.5	75	–
LA80	510.5	574.0	156.5	155	90	90	102.5	M20x1.5/M25x1.5	80	80
LA80Z	533.0	596.5	156.5	155	90	90	175.5	M20x1.5/M25x1.5	84	84
LA90S/L	541.5	612.5	174.0	163	90	90	102.5	M20x1.5/M25x1.5	84	85
LA90ZL	586.5	657.5	174.0	163	90	90	226.5	M20x1.5/M25x1.5	90	91
LA100L	587.5	668.5	195.0	168	120	120	143.0	2xM32x1.5	93	94
LA100ZL	657.5	738.5	195.0	168	120	120	275.0	2xM32x1.5	103	104
LA112M	614.5	695.5	219.0	181	120	120	146.0	2xM32x1.5	105	106
LA112ZM	642.5	723.5	219.0	181	120	120	250.0	2xM32x1.5	112	113
LA132S/M	674.5	776.5	259.0	195	140	140	186.5	2xM32x1.5	118	119
LA132ZM	720.5	822.5	259.0	195	140	140	294.5	2xM32x1.5	139	140
LA160M/L	777.0	895.5	313.5	227	165	165	212.0	2xM40x1.5	–	151
LA160ZL	825.0	943.5	313.5	227	165	165	365.0	2xM40x1.5	–	190

⑥ For note, see page 3/178

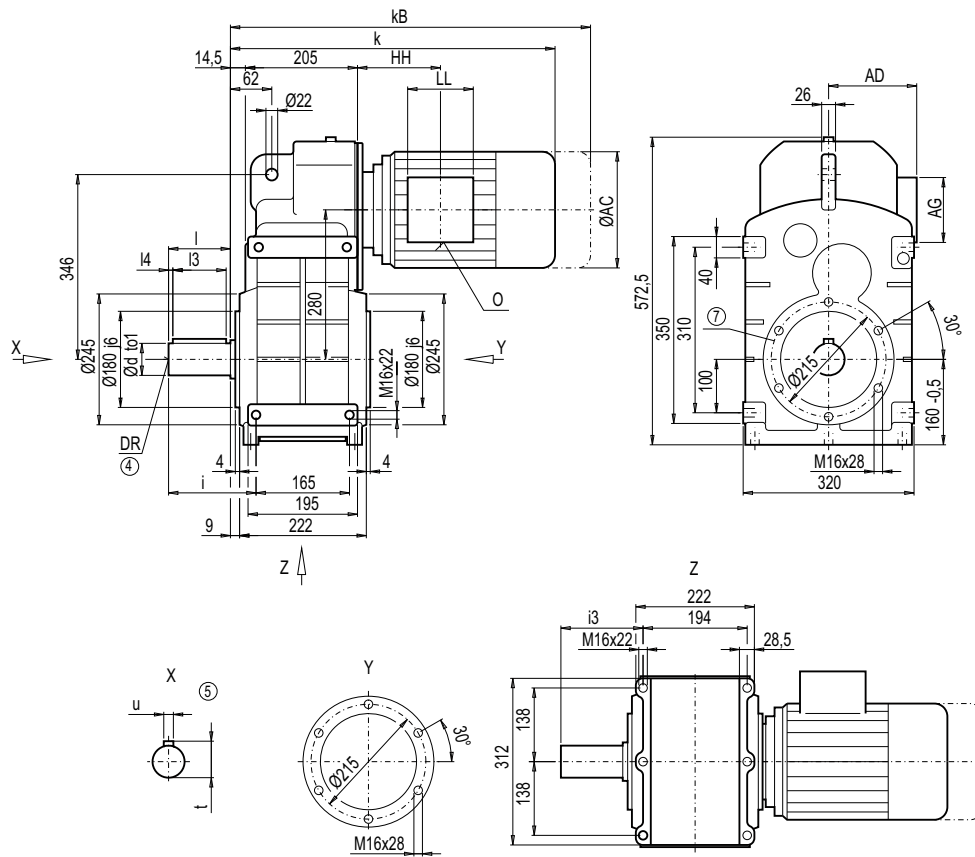
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDZ/FZZ108B (3- / 2-stage), housing-flange-mounted design (C-type)

FZ012



d	to1	l	i3	i4	t	u	i	i3	DR
60 ^{*)}	m6	120	110	5	64	18	157.5	143	M20x42
80	m6	170	125	20	85	22	207.5	193	M20x42

*) Preferred series

F.Z108B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDZ108B	FZZ108B
LA80	488.5	552.0	156.5	155.0	90	90	87.5	M20x1.5/M25x1.5	121	–
LA80Z	511.0	574.5	156.5	155.0	90	90	160.5	M20x1.5/M25x1.5	125	–
LA90S/L	519.5	590.5	174.0	163.0	90	90	87.5	M20x1.5/M25x1.5	126	–
LA90ZL	564.5	635.5	174.0	163.0	90	90	211.5	M20x1.5/M25x1.5	132	–
LA100L	563.0	644.0	195.0	168.0	120	120	125.5	2xM32x1.5	134	134
LA100ZL	633.0	714.0	195.0	168.0	120	120	257.5	2xM32x1.5	144	144
LA112M	589.0	670.0	219.0	181.0	120	120	127.5	2xM32x1.5	146	146
LA112ZM	617.0	698.0	219.0	181.0	120	120	231.5	2xM32x1.5	153	153
LA132S/M	649.0	751.0	259.0	195.0	140	140	168.0	2xM32x1.5	157	158
LA132ZM	695.0	797.0	259.0	195.0	140	140	276.0	2xM32x1.5	179	179
LA160M/L	753.5	872.0	313.5	227.0	165	165	195.5	2xM40x1.5	192	193
LA160ZL	801.5	920.0	313.5	227.0	165	165	348.5	2xM40x1.5	231	232
LG180M/L	813.0	935.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	285
LG180ZM/ZL	864.0	986.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	315

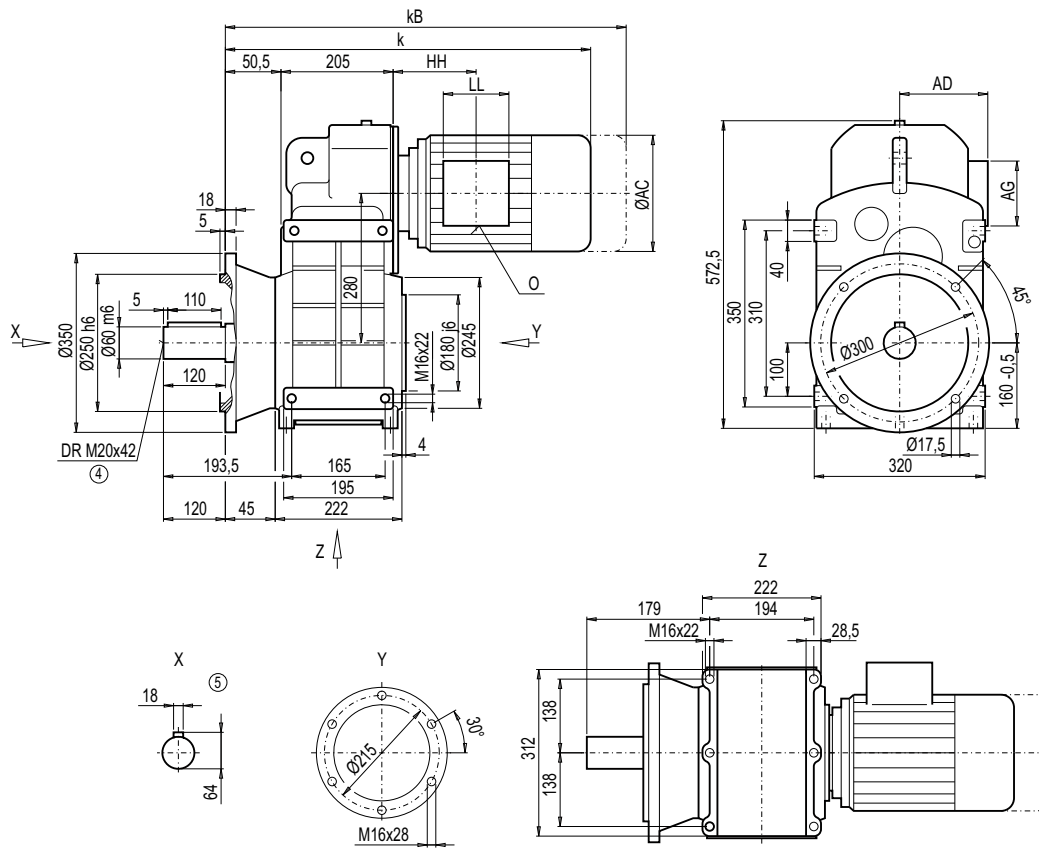
④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 3/179

Gearbox FDF/FZF108B (3- / 2-stage), flange-mounted design (A-type)

FF012



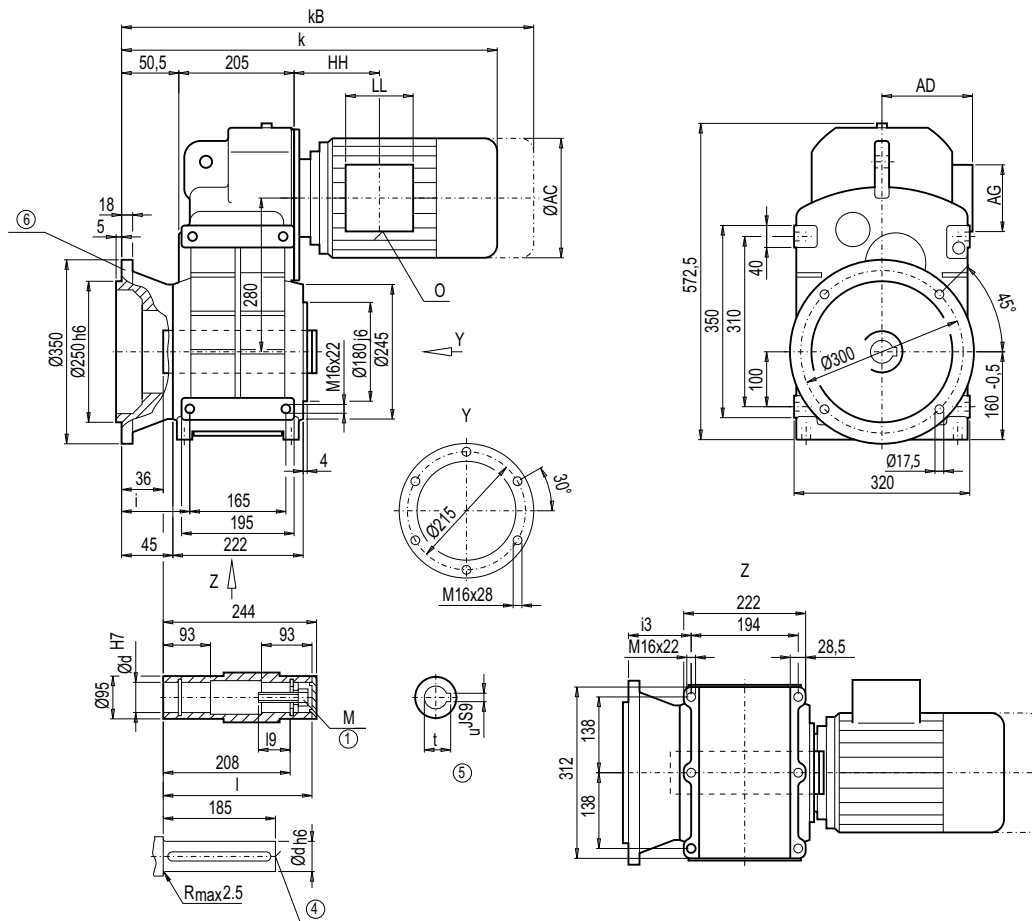
F.F108B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF108B	FZF108B
LA80	524.5	588.0	156.5	155.0	90	90	87.5	M20x1.5/M25x1.5	134	–
LA80Z	547.0	610.5	156.5	155.0	90	90	160.5	M20x1.5/M25x1.5	138	–
LA90S/L	555.5	626.5	174.0	163.0	90	90	87.5	M20x1.5/M25x1.5	139	–
LA90ZL	600.5	671.5	174.0	163.0	90	90	211.5	M20x1.5/M25x1.5	145	–
LA100L	599.0	680.0	195.0	168.0	120	120	125.5	2xM32x1.5	147	147
LA100ZL	669.0	750.0	195.0	168.0	120	120	257.5	2xM32x1.5	157	157
LA112M	625.0	706.0	219.0	181.0	120	120	127.5	2xM32x1.5	159	159
LA112ZM	653.0	734.0	219.0	181.0	120	120	231.5	2xM32x1.5	166	166
LA132S/M	685.0	787.0	259.0	195.0	140	140	168.0	2xM32x1.5	170	171
LA132ZM	731.0	833.0	259.0	195.0	140	140	276.0	2xM32x1.5	192	192
LA160M/L	789.5	908.0	313.5	227.0	165	165	195.5	2xM40x1.5	205	206
LA160ZL	837.5	956.0	313.5	227.0	165	165	348.5	2xM40x1.5	244	245
LG180M/L	849.0	971.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	298
LG180ZM/ZL	900.0	1 022.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	328

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF108B (3- / 2-stage), flange-mounted design

FAF012



d	l	l9	M	t	u	i	i3
60 ^{*)}	240	63.5	M20	64.4	18	73.5	59
70	240	63.5	M20	74.9	20	73.5	59

*¹) Preferred series

F.AF108B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF108B	FZAF108B
LA80	524.5	588.0	156.5	155.0	90	90	87.5	M20x1.5/M25x1.5	121	–
LA80Z	547.0	610.5	156.5	155.0	90	90	160.5	M20x1.5/M25x1.5	125	–
LA90S/L	555.5	626.5	174.0	163.0	90	90	87.5	M20x1.5/M25x1.5	126	–
LA90ZL	600.5	671.5	174.0	163.0	90	90	211.5	M20x1.5/M25x1.5	132	–
LA100L	599.0	680.0	195.0	168.0	120	120	125.5	2xM32x1.5	134	134
LA100ZL	669.0	750.0	195.0	168.0	120	120	257.5	2xM32x1.5	144	144
LA112M	625.0	706.0	219.0	181.0	120	120	127.5	2xM32x1.5	146	147
LA112ZM	653.0	734.0	219.0	181.0	120	120	231.5	2xM32x1.5	153	154
LA132S/M	685.0	787.0	259.0	195.0	140	140	168.0	2xM32x1.5	158	158
LA132ZM	731.0	833.0	259.0	195.0	140	140	276.0	2xM32x1.5	179	179
LA160M/L	789.5	908.0	313.5	227.0	165	165	195.5	2xM40x1.5	192	193
LA160ZL	837.5	956.0	313.5	227.0	165	165	348.5	2xM40x1.5	231	232
LG180M/L	849.0	971.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	285
LG180ZM/ZL	900.0	1 022.0	348.0	322.5	260	192	212.5	2xM40x1.5	–	315

① DIN EN ISO 4014

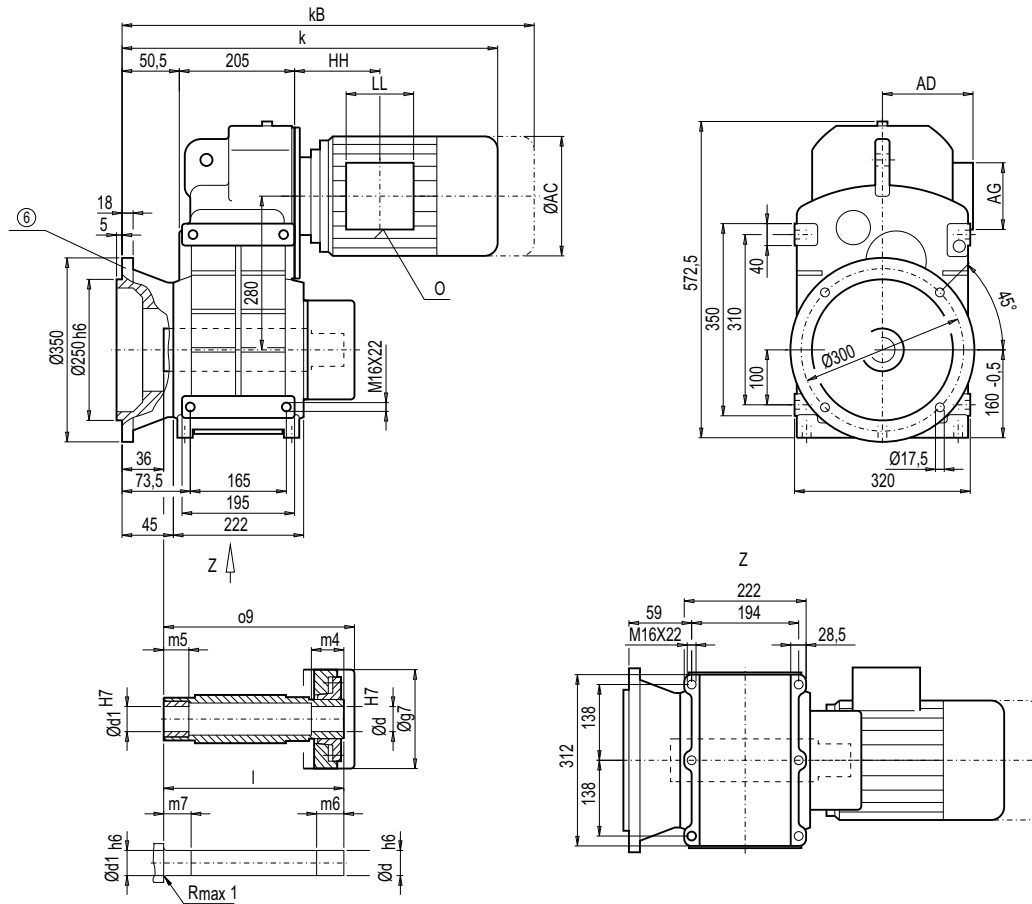
④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

Gearbox FDAFS/FZAFS108B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



d	d1	l	o9	m4	m5	m6	m7	g7
65 *)	65	280	288	30	40	35	45	144
70	70	280	288	30	40	35	45	144

*) Preferred series

F.AFS108B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS108B	FZAFS108B
LA80	524.5	588.0	156.5	155.0	90	90	87.5	M20x1.5/M25x1.5	128	—
LA80Z	547.0	610.5	156.5	155.0	90	90	160.5	M20x1.5/M25x1.5	132	—
LA90S/L	555.5	626.5	174.0	163.0	90	90	87.5	M20x1.5/M25x1.5	133	—
LA90ZL	600.5	671.5	174.0	163.0	90	90	211.5	M20x1.5/M25x1.5	139	—
LA100L	599.0	680.0	195.0	168.0	120	120	125.5	2xM32x1.5	141	141
LA100ZL	669.0	750.0	195.0	168.0	120	120	257.5	2xM32x1.5	151	151
LA112M	625.0	706.0	219.0	181.0	120	120	127.5	2xM32x1.5	153	153
LA112ZM	653.0	734.0	219.0	181.0	120	120	231.5	2xM32x1.5	160	160
LA132S/M	685.0	787.0	259.0	195.0	140	140	168.0	2xM32x1.5	164	165
LA132ZM	731.0	833.0	259.0	195.0	140	140	276.0	2xM32x1.5	186	186
LA160M/L	789.5	908.0	313.5	227.0	165	165	195.5	2xM40x1.5	199	200
LA160ZL	837.5	956.0	313.5	227.0	165	165	348.5	2xM40x1.5	238	239
LG180M/L	849.0	971.0	348.0	322.5	260	192	212.5	2xM40x1.5	—	292
LG180ZM/ZL	900.0	1 022.0	348.0	322.5	260	192	212.5	2xM40x1.5	—	322

© For note, see page 3/178

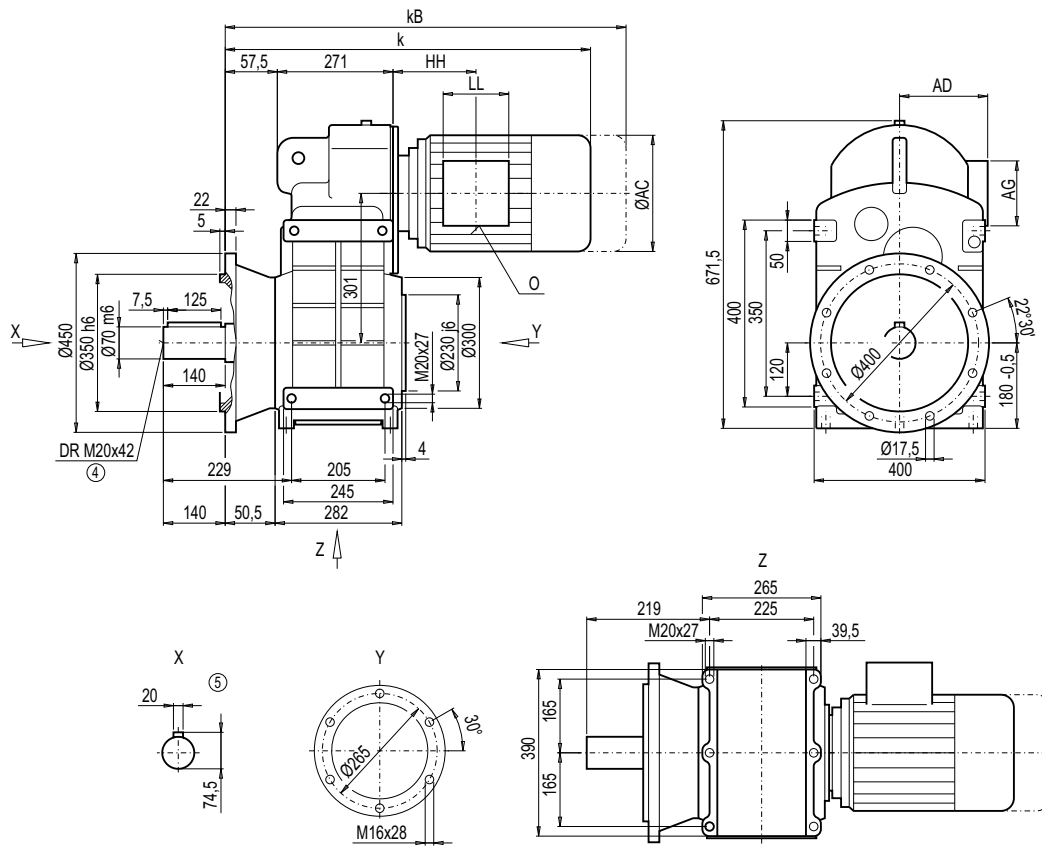
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDF/FZF128B (3- / 2-stage), flange-mounted design (A-type)

FF012



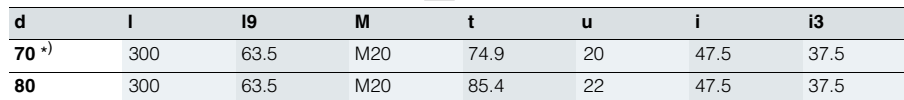
F.F128B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF128B	FZF128B
LA90S/L	617.0	688.0	174.0	163.0	90	90	76.0	M20x1.5/M25x1.5	226	—
LA90ZL	662.0	733.0	174.0	163.0	90	90	200.0	M20x1.5/M25x1.5	232	—
LA100L	660.0	741.0	195.0	168.0	120	120	113.5	2xM32x1.5	234	—
LA100ZL	730.0	811.0	195.0	168.0	120	120	245.5	2xM32x1.5	244	—
LA112M	686.5	767.5	219.0	181.0	120	120	116.0	2xM32x1.5	246	244
LA112ZM	714.5	795.5	219.0	181.0	120	120	220.0	2xM32x1.5	253	251
LA132S/M	745.5	847.5	259.0	195.0	140	140	155.5	2xM32x1.5	256	255
LA132ZM	791.5	893.5	259.0	195.0	140	140	263.5	2xM32x1.5	278	276
LA160M/L	851.0	969.5	313.5	227.0	165	165	184.0	2xM40x1.5	291	289
LA160ZL	899.0	1 017.5	313.5	227.0	165	165	337.0	2xM40x1.5	230	328
LG180M/L	907.5	1 029.5	348.0	322.5	260	192	198.0	2xM40x1.5	387	386
LG180ZM/ZL	958.5	1 080.5	348.0	322.5	260	192	198.0	2xM40x1.5	417	416
LG200L	963.5	1 089.5	385.0	301.0	260	192	228.0	2xM50x1.5	467	466
K4-LGI225S	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	621
K4-LGI225M	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	609
K4-LGI225ZM	1 284.0	1 523.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	667

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDA/FZA128B, FDAZ/FZAZ128B (3- / 2-stage), housing-flange-mounted design (C-type)

3



⑦ For note, see page 3/179

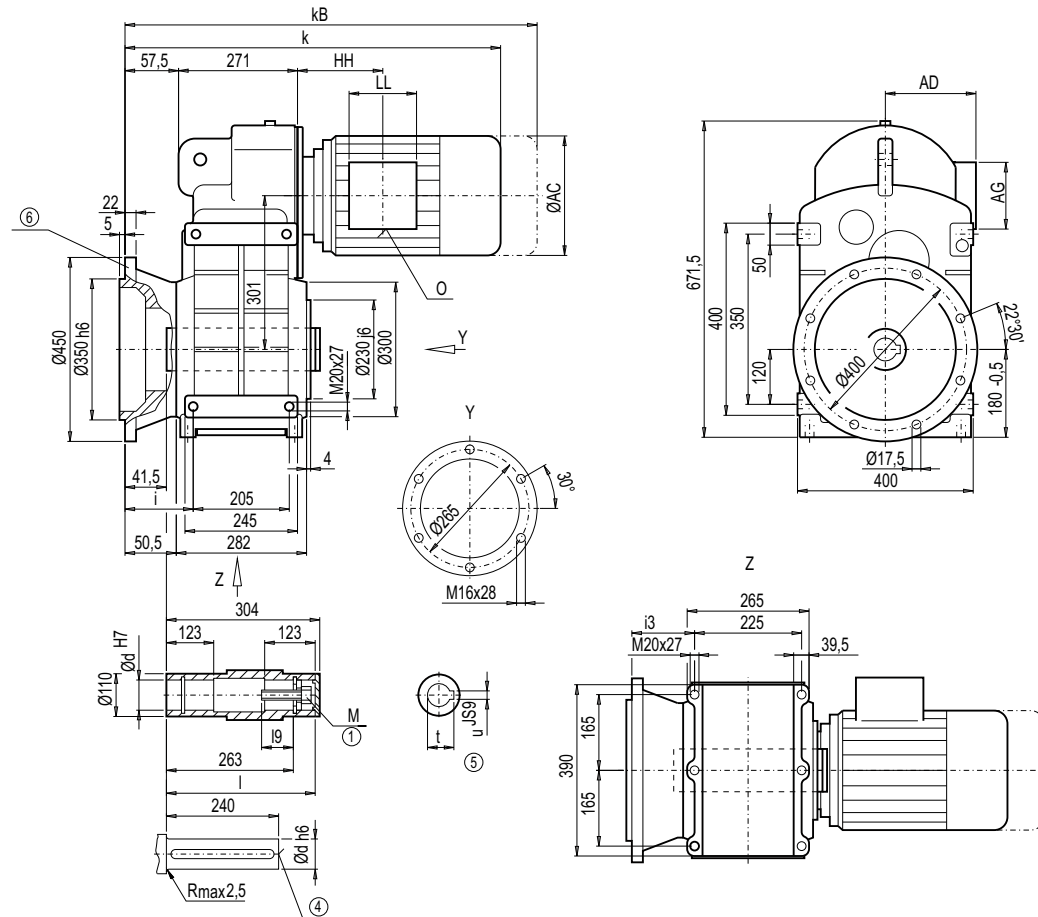
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAF/FZAF128B (3- / 2-stage), flange-mounted design

FAF012



d	l	l9	M	t	u	i	i3
70 *)	300	63.5	M20	74.9	20	89	79
80	300	63.5	M20	85.4	22	89	79

*) Preferred series

F.AF128B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF128B	FZAF128B
LA90S/L	617.0	688.0	174.0	163.0	90	90	76.0	M20x1.5/M25x1.5	206	—
LA90ZL	662.0	733.0	174.0	163.0	90	90	200.0	M20x1.5/M25x1.5	212	—
LA100L	660.0	741.0	195.0	168.0	120	120	113.5	2xM32x1.5	214	—
LA100ZL	730.0	811.0	195.0	168.0	120	120	245.5	2xM32x1.5	224	—
LA112M	686.5	767.5	219.0	181.0	120	120	116.0	2xM32x1.5	226	224
LA112ZM	714.5	795.5	219.0	181.0	120	120	220.0	2xM32x1.5	238	236
LA132S/M	745.5	847.5	259.0	195.0	140	140	155.5	2xM32x1.5	237	235
LA132ZM	791.5	893.5	259.0	195.0	140	140	263.5	2xM32x1.5	258	256
LA160M/L	851.0	969.5	313.5	227.0	165	165	184.0	2xM40x1.5	271	269
LA160ZL	899.0	1 017.5	313.5	227.0	165	165	337.0	2xM40x1.5	310	308
LG180M/L	907.5	1 029.5	348.0	322.5	260	192	198.0	2xM40x1.5	368	366
LG180ZM/ZL	958.5	1 080.5	348.0	322.5	260	192	198.0	2xM40x1.5	398	396
LG200L	963.5	1 089.5	385.0	301.0	260	192	228.0	2xM50x1.5	448	446
K4-LGI225S	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	601
K4-LGI225M	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	589
K4-LGI225ZM	1 284.0	1 523.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	647

① DIN EN ISO 4014

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

Gearbox FDAS/FZAS128B, FDAZS/FZAZS128B (3- / 2-stage), shaft-mounted design with shrink disk

*¹) Preferred series

⑦ For note, see page 3/179

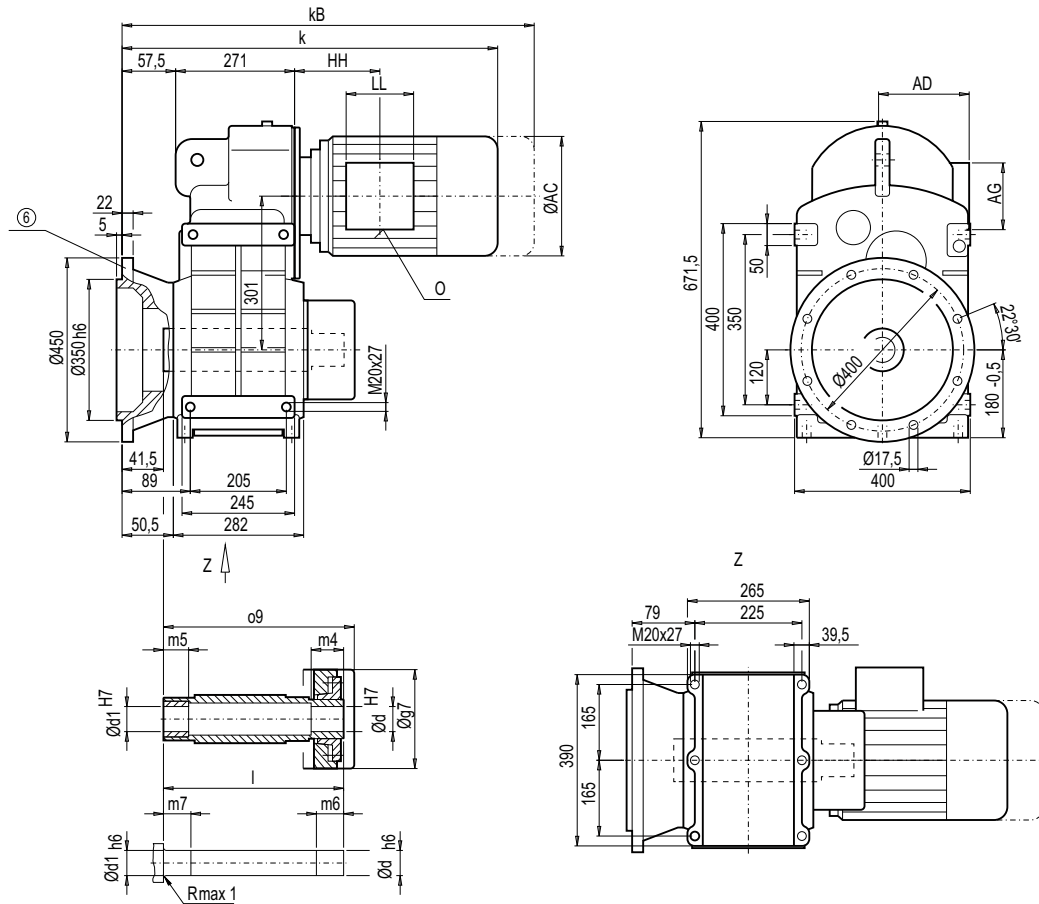
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAFS/FZAFS128B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



d	d1	l	o9	m4	m5	m6	m7	g7
75 ^{*)}	75	345	357	44	50	49	55	180
80	80	345	357	40	50	45	55	180

^{*)} Preferred series

F.AFS128B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS128B	FZAFS128B
LA90S/L	617.0	688.0	174.0	163.0	90	90	76.0	M20x1.5/M25x1.5	210	—
LA90ZL	662.0	733.0	174.0	163.0	90	90	200.0	M20x1.5/M25x1.5	216	—
LA100L	660.0	741.0	195.0	168.0	120	120	113.5	2xM32x1.5	218	—
LA100ZL	730.0	811.0	195.0	168.0	120	120	245.5	2xM32x1.5	228	—
LA112M	686.5	767.5	219.0	181.0	120	120	116.0	2xM32x1.5	230	228
LA112ZM	714.5	795.5	219.0	181.0	120	120	220.0	2xM32x1.5	237	235
LA132S/M	745.5	847.5	259.0	195.0	140	140	155.5	2xM32x1.5	241	239
LA132ZM	791.5	893.5	259.0	195.0	140	140	263.5	2xM32x1.5	262	260
LA160M/L	851.0	969.5	313.5	227.0	165	165	184.0	2xM40x1.5	275	274
LA160ZL	899.0	1 017.5	313.5	227.0	165	165	337.0	2xM40x1.5	314	313
LG180M/L	907.5	1 029.5	348.0	322.5	260	192	198.0	2xM40x1.5	372	370
LG180ZM/ZL	958.5	1 080.5	348.0	322.5	260	192	198.0	2xM40x1.5	402	400
LG200L	963.5	1 089.5	385.0	301.0	260	192	228.0	2xM50x1.5	452	450
K4-LGI225S	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	615
K4-LGI225M	1 224.0	1 463.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	593
K4-LGI225ZM	1 284.0	1 523.0	442.0	325.0	260	192	443.0	2xM50x1.5	—	651

© For note, see page 3/178

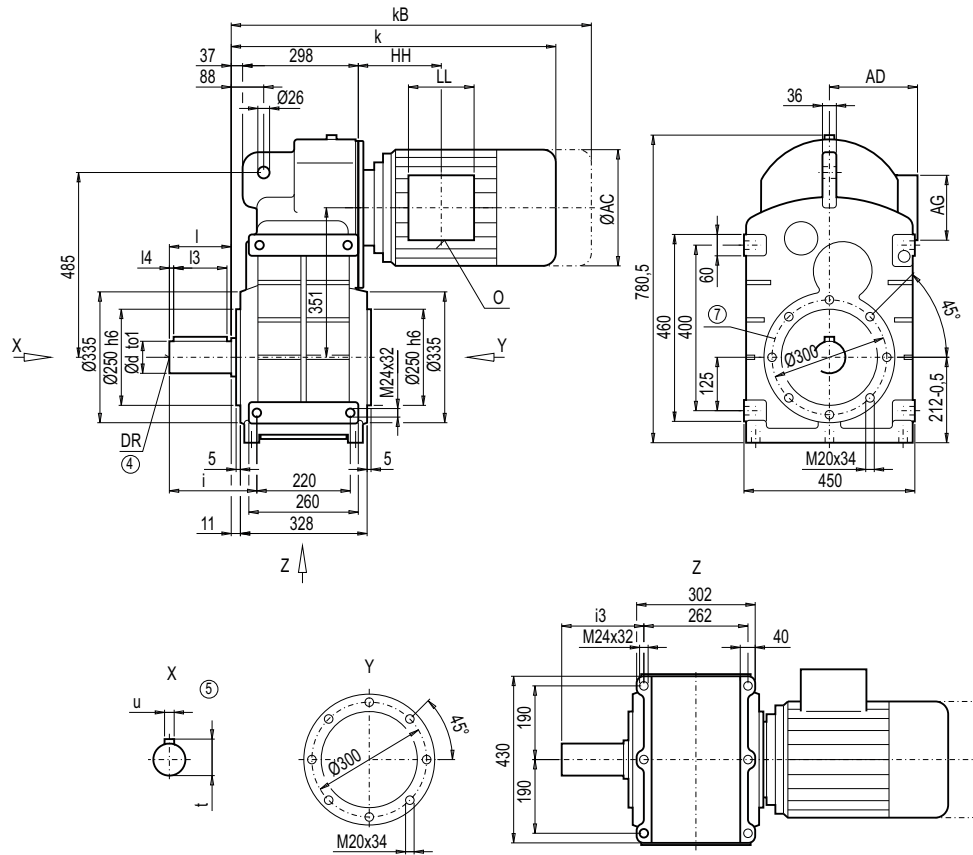
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDZ/FZZ148B (3- / 2-stage), housing-flange-mounted design (C-type)

FZ012



d	to1	l	i3	i4	t	u	i	i3	DR
90 ^{*)}	m6	170	140	15	95	25	235	214	M24x50
100	m6	210	180	15	106	28	275	254	M24x50

*) Preferred series

F.Z148B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDZ148B	FZZ148B
LA100L	657.0	738.0	195.0	168.0	120	120	104.0	2xM32x1.5	311	–
LA100ZL	727.0	808.0	195.0	168.0	120	120	236.0	2xM32x1.5	321	–
LA112M	682.5	763.5	219.0	181.0	120	120	105.5	2xM32x1.5	322	–
LA112ZM	710.5	791.5	219.0	181.0	120	120	209.5	2xM32x1.5	329	–
LA132S/M	741.5	843.5	259.0	195.0	140	140	145.0	2xM32x1.5	331	327
LA132ZM	787.5	889.5	259.0	195.0	140	140	253.0	2xM32x1.5	352	348
LA160M/L	841.0	959.5	313.5	227.0	165	165	167.5	2xM40x1.5	371	366
LA160ZL	889.0	1 007.5	313.5	227.0	165	165	320.5	2xM40x1.5	410	405
LG180M/L	900.5	1 022.5	348.0	322.5	260	192	184.5	2xM40x1.5	462	457
LG180ZM/ZL	951.5	1 073.5	348.0	322.5	260	192	184.5	2xM40x1.5	492	487
LG200L	956.5	1 082.5	385.0	301.0	260	192	214.5	2xM50x1.5	542	537
LG225S	1 027.5	1 266.5	442.0	325.0	260	192	250.5	2xM50x1.5	618	614
LG225M	1 027.5	1 266.5	442.0	325.0	260	192	250.5	2xM50x1.5	606	602
LG225ZM	1 087.5	1 326.5	442.0	325.0	260	192	250.5	2xM50x1.5	664	660
K4-LGI250M	1 314.5	1 539.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	781
K4-LGI250ZM	1 384.5	1 609.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	884

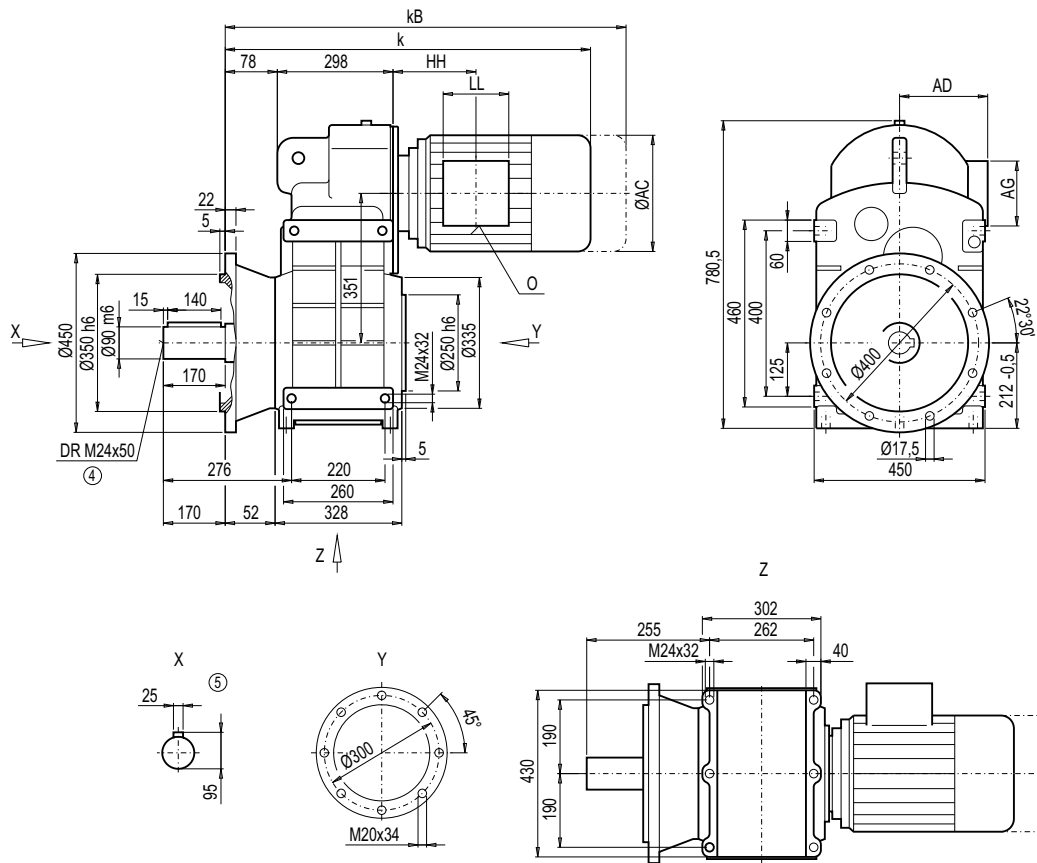
④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 3/179

Gearbox FDF/FZF148B (3- / 2-stage), flange-mounted design (A-type)

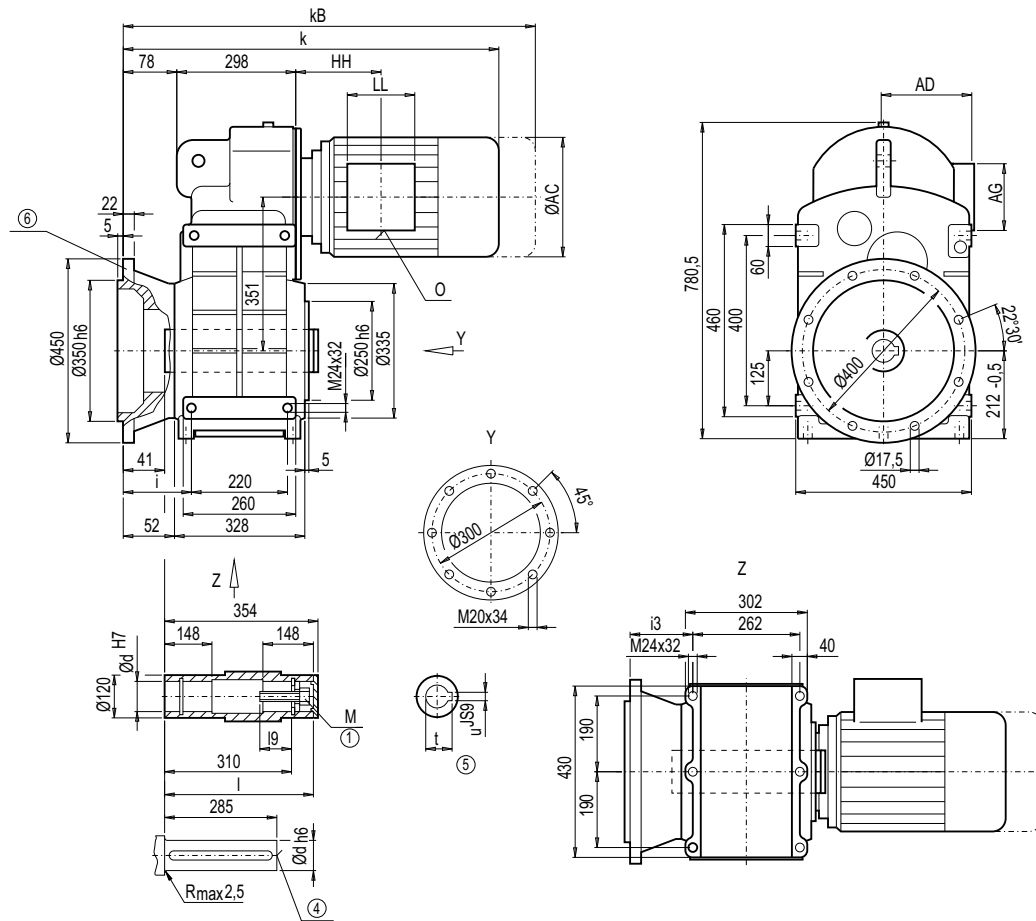
FF012



F.F148B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF148B	FZF148B
LA100L	698.0	779.0	195.0	168.0	120	120	104.0	2xM32x1.5	333	–
LA100ZL	768.0	849.0	195.0	168.0	120	120	236.0	2xM32x1.5	343	–
LA112M	723.5	804.5	219.0	181.0	120	120	105.5	2xM32x1.5	345	–
LA112ZM	751.5	832.5	219.0	181.0	120	120	209.5	2xM32x1.5	352	–
LA132S/M	782.5	884.5	259.0	195.0	140	140	145.0	2xM32x1.5	354	350
LA132ZM	828.5	930.5	259.0	195.0	140	140	253.0	2xM32x1.5	375	371
LA160M/L	882.0	1 000.5	313.5	227.0	165	165	167.5	2xM40x1.5	393	389
LA160ZL	930.0	1 048.5	313.5	227.0	165	165	320.5	2xM40x1.5	432	428
LG180M/L	941.5	1 063.5	348.0	322.5	260	192	184.5	2xM40x1.5	484	480
LG180ZM/ZL	992.5	1 114.5	348.0	322.5	260	192	184.5	2xM40x1.5	514	510
LG200L	997.5	1 123.5	385.0	301.0	260	192	214.5	2xM50x1.5	564	560
LG225S	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	638	637
LG225M	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	626	625
LG225ZM	1 128.5	1 367.5	442.0	325.0	260	192	250.5	2xM50x1.5	684	683
K4-LGI250M	1 355.5	1 580.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	804
K4-LGI250ZM	1 425.5	1 650.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	907

Gearbox FDAF/FZAF148B (3- / 2-stage), flange-mounted design

FAF012



d	l	l9	M	t	u	i	i3
80 ^{*)}	350	63.5	M20	85.4	22	106	85
90	350	72.0	M24	95.4	25	106	85

*) Preferred series

F.AF148B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF148B	FZAF148B
LA100L	698.0	779.0	195.0	168.0	120	120	104.0	2xM32x1.5	305	–
LA100ZL	768.0	849.0	195.0	168.0	120	120	236.0	2xM32x1.5	315	–
LA112M	723.5	804.5	219.0	181.0	120	120	105.5	2xM32x1.5	317	–
LA112ZM	751.5	832.5	219.0	181.0	120	120	209.5	2xM32x1.5	324	–
LA132S/M	782.5	884.5	259.0	195.0	140	140	145.0	2xM32x1.5	326	322
LA132ZM	828.5	930.5	259.0	195.0	140	140	253.0	2xM32x1.5	347	343
LA160M/L	882.0	1 000.5	313.5	227.0	165	165	167.5	2xM40x1.5	365	361
LA160ZL	930.0	1 048.5	313.5	227.0	165	165	320.5	2xM40x1.5	404	400
LG180M/L	941.5	1 063.5	348.0	322.5	260	192	184.5	2xM40x1.5	456	452
LG180ZM/ZL	992.5	1 114.5	348.0	322.5	260	192	184.5	2xM40x1.5	486	482
LG200L	997.5	1 123.5	385.0	301.0	260	192	214.5	2xM50x1.5	536	532
LG225S	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	610	609
LG225M	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	598	597
LG225ZM	1 128.5	1 367.5	442.0	325.0	260	192	250.5	2xM50x1.5	656	655
K4-LGI250M	1 355.5	1 580.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	776
K4-LGI250ZM	1 425.5	1 650.5	495.0	392.0	300	236	469.5	2xM63x1.5	–	879

① DIN EN ISO 4014

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

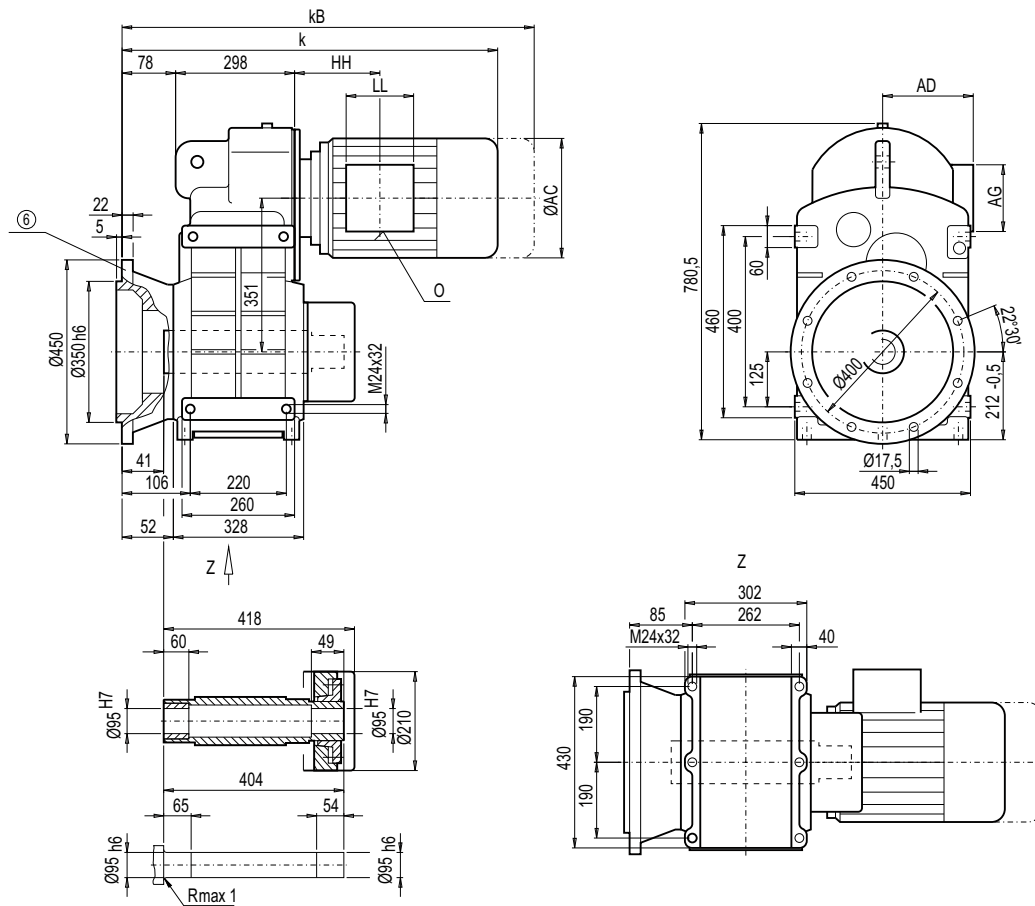
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAFS/FZAFS148B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



F.AFS148B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS148B	FZAFS148B
LA100L	698.0	779.0	195.0	168.0	120	120	104.0	2xM32x1.5	312	—
LA100ZL	768.0	849.0	195.0	168.0	120	120	236.0	2xM32x1.5	322	—
LA112M	723.5	804.5	219.0	181.0	120	120	105.5	2xM32x1.5	324	—
LA112ZM	751.5	832.5	219.0	181.0	120	120	209.5	2xM32x1.5	331	—
LA132S/M	782.5	884.5	259.0	195.0	140	140	145.0	2xM32x1.5	333	329
LA132ZM	828.5	930.5	259.0	195.0	140	140	253.0	2xM32x1.5	354	350
LA160M/L	882.0	1 000.5	313.5	227.0	165	165	167.5	2xM40x1.5	372	368
LA160ZL	930.0	1 048.5	313.5	227.0	165	165	320.5	2xM40x1.5	411	407
LG180M/L	941.5	1 063.5	348.0	322.5	260	192	184.5	2xM40x1.5	463	459
LG180ZM/ZL	992.5	1 114.5	348.0	322.5	260	192	184.5	2xM40x1.5	493	489
LG200L	997.5	1 123.5	385.0	301.0	260	192	214.5	2xM50x1.5	543	539
LG225S	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	617	616
LG225M	1 068.5	1 307.5	442.0	325.0	260	192	250.5	2xM50x1.5	605	604
LG225ZM	1 128.5	1 367.5	442.0	325.0	260	192	250.5	2xM50x1.5	663	662
K4-LGI250M	1 355.5	1 580.5	495.0	392.0	300	236	469.5	2xM63x1.5	—	783
K4-LGI250ZM	1 425.5	1 650.5	495.0	392.0	300	236	469.5	2xM63x1.5	—	886

© For note, see page 3/178

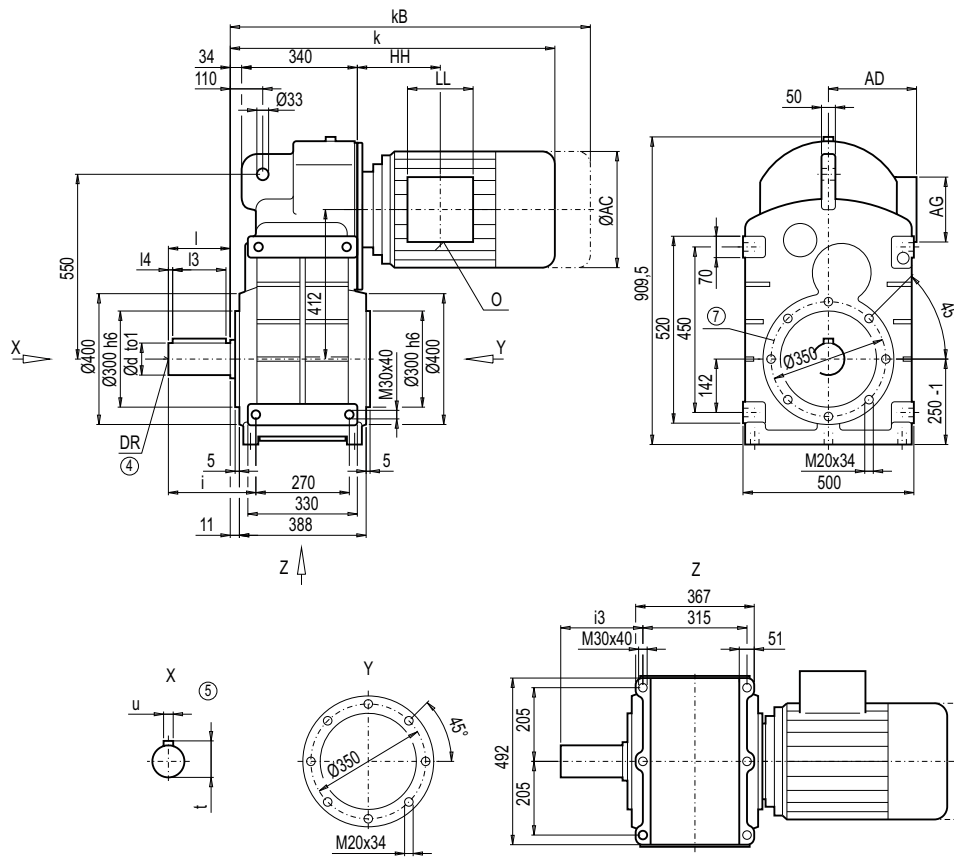
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDZ/FZZ168B (3- / 2-stage), housing-flange-mounted design (C-type)

FZ012



d	to1	l	l3	l4	t	u	i	i3	DR
110 *)	m6	210	180	15	116	28	280	252	M24x50
120	m6	210	180	15	127	32	280	252	M24x50

*¹) Preferred series

F.Z168B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDZ168B	FZZ168B
LA132S/M	772.5	874.5	259.0	195.0	140	140	137.0	2xM32x1.5	496	485
LA132ZM	818.5	920.5	259.0	195.0	140	140	245.0	2xM32x1.5	517	506
LA160M/L	872.5	991.0	313.5	227.0	165	165	160.0	2xM40x1.5	530	519
LA160ZL	920.5	1 039.0	313.5	227.0	165	165	313.0	2xM40x1.5	569	558
LG180M/L	932.0	1 054.0	348.0	322.5	260	192	177.0	2xM40x1.5	626	614
LG180ZM/ZL	983.0	1 105.0	348.0	322.5	260	192	177.0	2xM40x1.5	656	644
LG200L	988.0	1 114.0	385.0	301.0	260	192	207.0	2xM50x1.5	706	694
LG225S	1 059.0	1 298.0	442.0	325.0	260	192	243.0	2xM50x1.5	779	768
LG225M	1 059.0	1 298.0	442.0	325.0	260	192	243.0	2xM50x1.5	767	756
LG225ZM	1 119.0	1 358.0	442.0	325.0	260	192	243.0	2xM50x1.5	825	814
LG250M	1 152.5	1 377.5	495.0	392.0	300	236	278.5	2xM63x1.5	869	858
LG250ZM	1 222.5	1 448.0	495.0	392.0	300	236	278.5	2xM63x1.5	972	961
K4-LGI280S	1 431.5	1 658.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 089
K4-LGI280M	1 431.5	1 658.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 101
K4-LGI280ZM	1 541.5	1 768.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 189

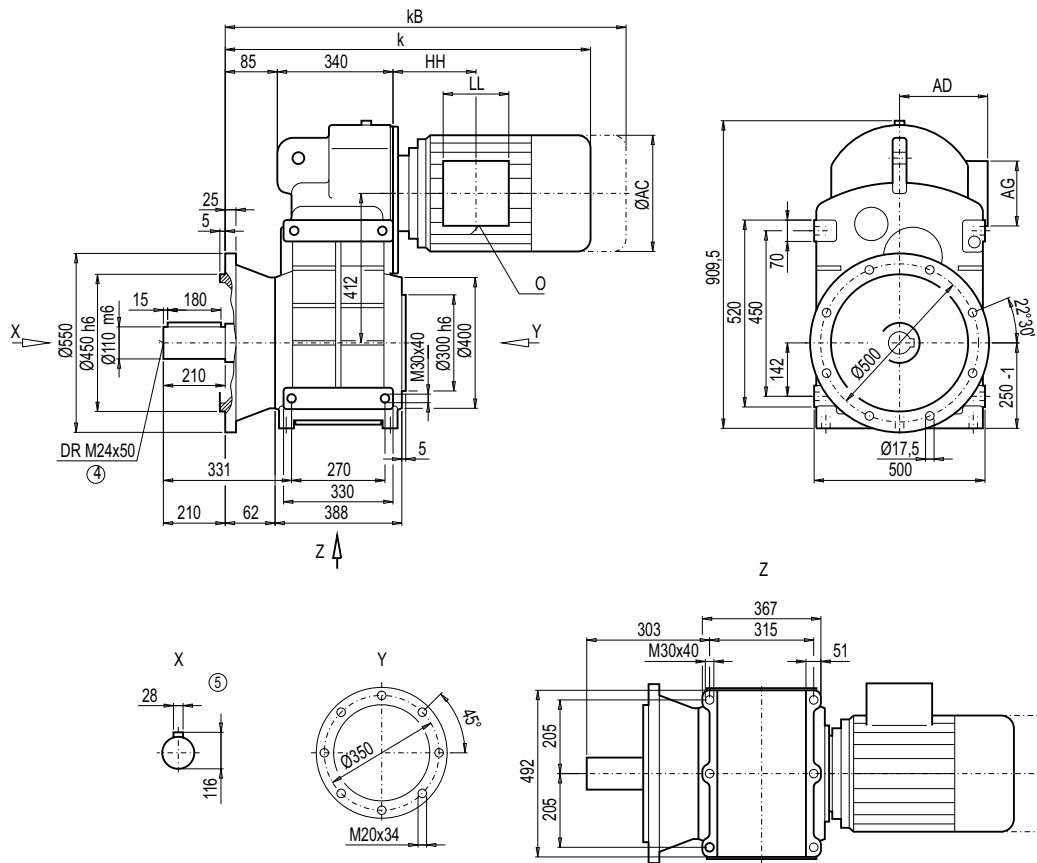
④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 3/179

Gearbox FDF/FZF168B (3- / 2-stage), flange-mounted design (A-type)

FF012



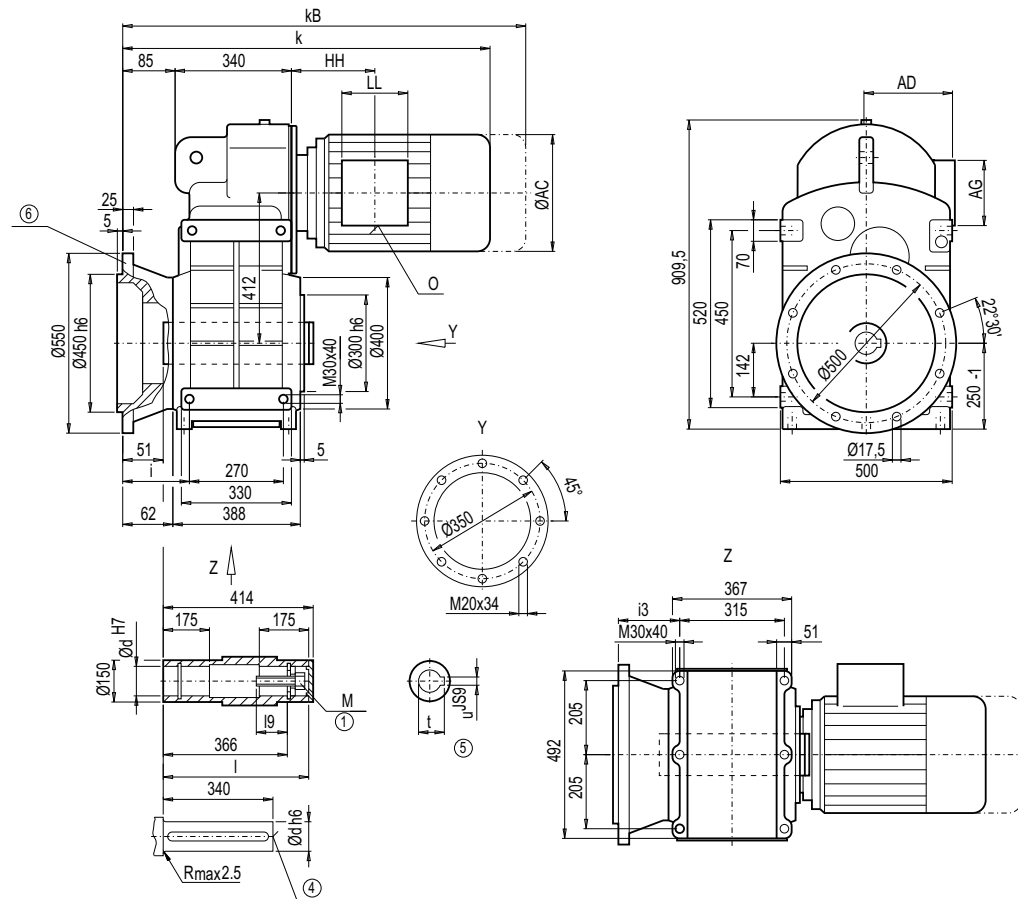
F.F168B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF168B	FZF168B
LA132S/M	823.5	925.5	259.0	195.0	140	140	137.0	2xM32x1.5	533	522
LA132ZM	869.5	971.5	259.0	195.0	140	140	245.0	2xM32x1.5	554	543
LA160M/L	923.5	1 042.0	313.5	227.0	165	165	160.0	2xM40x1.5	567	556
LA160ZL	971.5	1 090.0	313.5	227.0	165	165	313.0	2xM40x1.5	606	595
LG180M/L	983.0	1 105.0	348.0	322.5	260	192	177.0	2xM40x1.5	663	651
LG180ZM/ZL	1 034.0	1 156.0	348.0	322.5	260	192	177.0	2xM40x1.5	693	681
LG200L	1 039.0	1 165.0	385.0	301.0	260	192	207.0	2xM50x1.5	743	731
LG225S	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	816	805
LG225M	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	804	793
LG225ZM	1 170.0	1 409.0	442.0	325.0	260	192	243.0	2xM50x1.5	862	851
LG250M	1 203.5	1 428.5	495.0	392.0	300	236	278.5	2xM63x1.5	906	895
LG250ZM	1 273.5	1 499.0	495.0	392.0	300	236	278.5	2xM63x1.5	1 009	998
K4-LGI280S	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 125
K4-LGI280M	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 138
K4-LGI280ZM	1 592.5	1 819.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 226

④ DIN 332

⑤ Feather key / keyway DIN 6885

Gearbox FDAF/FZAF168B (3- / 2-stage), flange-mounted design

FAF012



d	l	l9	M	t	u	i	i3
100 ^{*)}	410	72	M24	106.4	28	121	93
110	410	73	M24	116.4	28	121	93

*) Preferred series

F.AF168B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF168B	FZAF168B
LA132S/M	823.5	925.5	259.0	195.0	140	140	137.0	2xM32x1.5	488	477
LA132ZM	869.5	971.5	259.0	195.0	140	140	245.0	2xM32x1.5	509	498
LA160M/L	923.5	1 042.0	313.5	227.0	165	165	160.0	2xM40x1.5	522	511
LA160ZL	971.5	1 090.0	313.5	227.0	165	165	313.0	2xM40x1.5	561	550
LG180M/L	983.0	1 105.0	348.0	322.5	260	192	177.0	2xM40x1.5	618	606
LG180ZM/ZL	1 034.0	1 156.0	348.0	322.5	260	192	177.0	2xM40x1.5	648	636
LG200L	1 039.0	1 165.0	385.0	301.0	260	192	207.0	2xM50x1.5	698	686
LG225S	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	771	760
LG225M	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	759	748
LG225ZM	1 170.0	1 409.0	442.0	325.0	260	192	243.0	2xM50x1.5	817	806
LG250M	1 203.5	1 428.5	495.0	392.0	300	236	278.5	2xM63x1.5	861	850
LG250ZM	1 273.5	1 499.0	495.0	392.0	300	236	278.5	2xM63x1.5	964	953
K4-LGI280S	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	—	1 081
K4-LGI280M	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	—	1 093
K4-LGI280ZM	1 592.5	1 819.5	555.0	432.0	300	236	489.5	2xM63x1.5	—	1 181

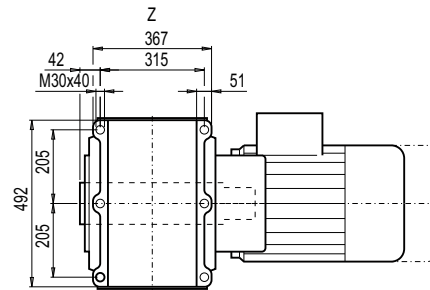
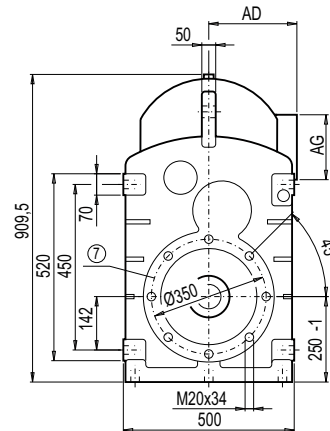
① DIN EN ISO 4014

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

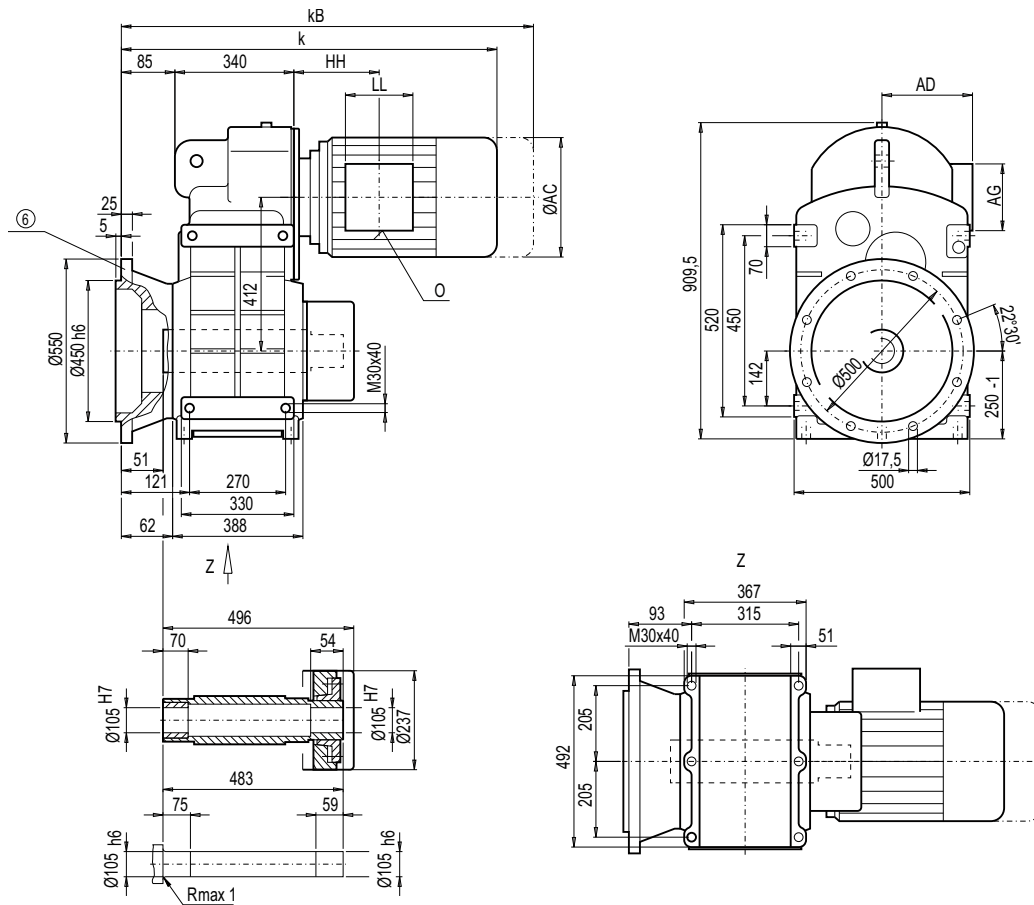
Gearbox FDAS/FZAS168B, FDAZS/FZAZS168B (3- / 2-stage), shaft-mounted design with shrink disk



⑦ For note, see page 3/179

Gearbox FDAFS/FZAFS168B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



F.AFS168B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS168B	FZAFS168B
LA132S/M	823.5	925.5	259.0	195.0	140	140	137.0	2xM32x1.5	498	487
LA132ZM	869.5	971.5	259.0	195.0	140	140	245.0	2xM32x1.5	519	508
LA160M/L	923.5	1 042.0	313.5	227.0	165	165	160.0	2xM40x1.5	532	521
LA160ZL	971.5	1 090.0	313.5	227.0	165	165	313.0	2xM40x1.5	571	560
LG180M/L	983.0	1 105.0	348.0	322.5	260	192	177.0	2xM40x1.5	628	616
LG180ZM/ZL	1 034.0	1 156.0	348.0	322.5	260	192	177.0	2xM40x1.5	658	646
LG200L	1 039.0	1 165.0	385.0	301.0	260	192	207.0	2xM50x1.5	708	696
LG225S	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	781	770
LG225M	1 110.0	1 349.0	442.0	325.0	260	192	243.0	2xM50x1.5	769	758
LG225ZM	1 170.0	1 409.0	442.0	325.0	260	192	243.0	2xM50x1.5	827	816
LG250M	1 203.5	1 428.5	495.0	392.0	300	236	278.5	2xM63x1.5	871	860
LG250ZM	1 273.5	1 499.0	495.0	392.0	300	236	278.5	2xM63x1.5	974	963
K4-LGI280S	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 091
K4-LGI280M	1 482.5	1 709.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 103
K4-LGI280ZM	1 592.5	1 819.5	555.0	432.0	300	236	489.5	2xM63x1.5	–	1 191

⑥ For note, see page 3/178

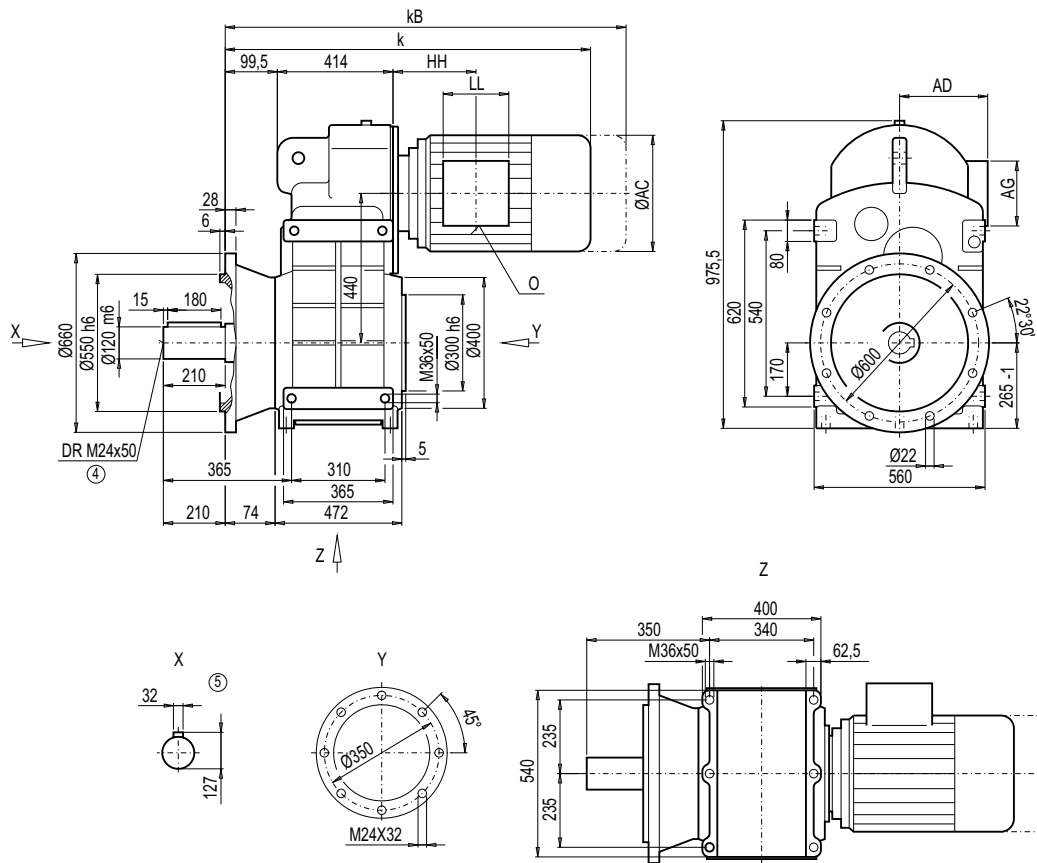
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDF/FZF188B (3- / 2-stage), flange-mounted design (A-type)

FF012



F.F188B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDF188B	FZF188B
LA132S/M	897.5	999.5	259.0	195.0	140	140	122.5	2xM32x1.5	740	–
LA132ZM	943.5	1 045.5	259.0	195.0	140	140	230.5	2xM32x1.5	761	–
LA160M/L	997.5	1 116.0	313.5	227.0	165	165	145.5	2xM40x1.5	773	759
LA160ZL	1 045.5	1 164.0	313.5	227.0	165	165	298.5	2xM40x1.5	812	798
LG180M/L	1 057.0	1 179.0	348.0	322.5	260	192	162.5	2xM40x1.5	869	854
LG180ZM/ZL	1 108.0	1 230.0	348.0	322.5	260	192	162.5	2xM40x1.5	899	884
LG200L	1 113.0	1 239.0	385.0	301.0	260	192	192.5	2xM50x1.5	949	934
LG225S	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 022	1 007
LG225M	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 010	995
LG225ZM	1 244.0	1 483.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 068	1 053
LG250M	1 277.5	1 502.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 112	1 097
LG250ZM	1 347.5	1 573.0	495.0	392.0	300	236	264.0	2xM63x1.5	1 215	1 200
K4-LGI280S	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 240	1 226
K4-LGI280M	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 346	1 331
K4-LGI280ZM	1 667.0	1 894.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 434	1 419
K2-LGI315S/M	1 745.0	2 010.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 461
K2-LGI315ZM	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 616
K2-LGI315L	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 756
K2-LGI315ZL	2 045.0	2 310.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 956

④ DIN 332

⑤ Feather key / keyway DIN 6885

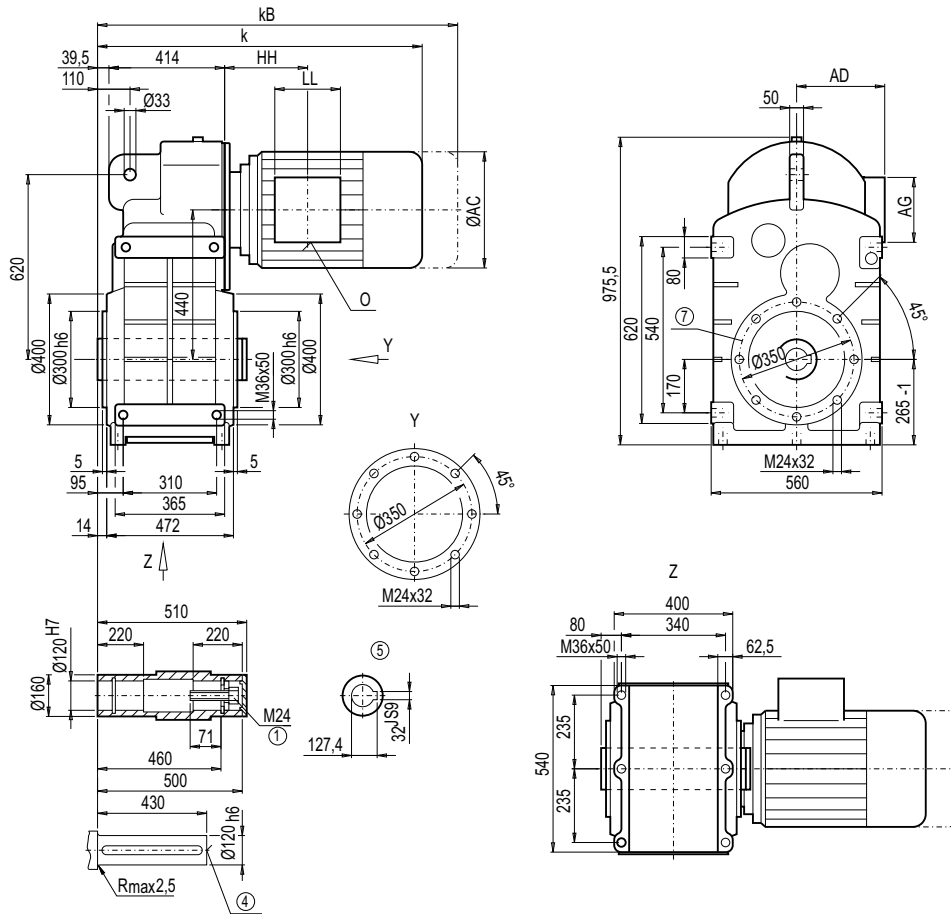
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDA/FZA188B, FDAZ/FZAZ188B (3- / 2-stage), housing-flange-mounted design (C-type)

FA012
FAZ012



F.A.188B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDA.188B	FZA.188B
LA132S/M	837.5	939.5	259.0	195.0	140	140	122.5	2xM32x1.5	622	—
LA132ZM	883.5	985.5	259.0	195.0	140	140	230.5	2xM32x1.5	643	—
LA160M/L	937.5	1 056.0	313.5	227.0	165	165	145.5	2xM40x1.5	655	641
LA160ZL	985.5	1 104.0	313.5	227.0	165	165	298.5	2xM40x1.5	694	680
LG180M/L	997.0	1 119.0	348.0	322.5	260	192	162.5	2xM40x1.5	751	736
LG180ZM/ZL	1 048.0	1 170.0	348.0	322.5	260	192	162.5	2xM40x1.5	781	766
LG200L	1 053.0	1 179.0	385.0	301.0	260	192	192.5	2xM50x1.5	831	816
LG225S	1 124.0	1 363.0	442.0	325.0	260	192	228.5	2xM50x1.5	904	889
LG225M	1 124.0	1 363.0	442.0	325.0	260	192	228.5	2xM50x1.5	892	877
LG225ZM	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	950	935
LG250M	1 217.5	1 442.5	495.0	392.0	300	236	264.0	2xM63x1.5	994	979
LG250ZM	1 287.5	1 513.0	495.0	392.0	300	236	264.0	2xM63x1.5	1 097	1 082
K4-LGI280S	1 497.0	1 724.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 122	1 108
K4-LGI280M	1 497.0	1 724.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 228	1 213
K4-LGI280ZM	1 607.0	1 834.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 316	1 301
K2-LGI315S/M	1 685.0	1 950.0	610.0	500.0	380	307	584.5	2xM63x1.5	—	1 343
K2-LGI315ZM	1 845.0	2 110.0	610.0	500.0	380	307	584.5	2xM63x1.5	—	1 498
K2-LGI315L	1 845.0	2 110.0	610.0	500.0	380	307	584.5	2xM63x1.5	—	1 643
K2-LGI315ZL	1 985.0	2 250.0	610.0	500.0	380	307	584.5	2xM63x1.5	—	1 838

① DIN EN ISO 4014

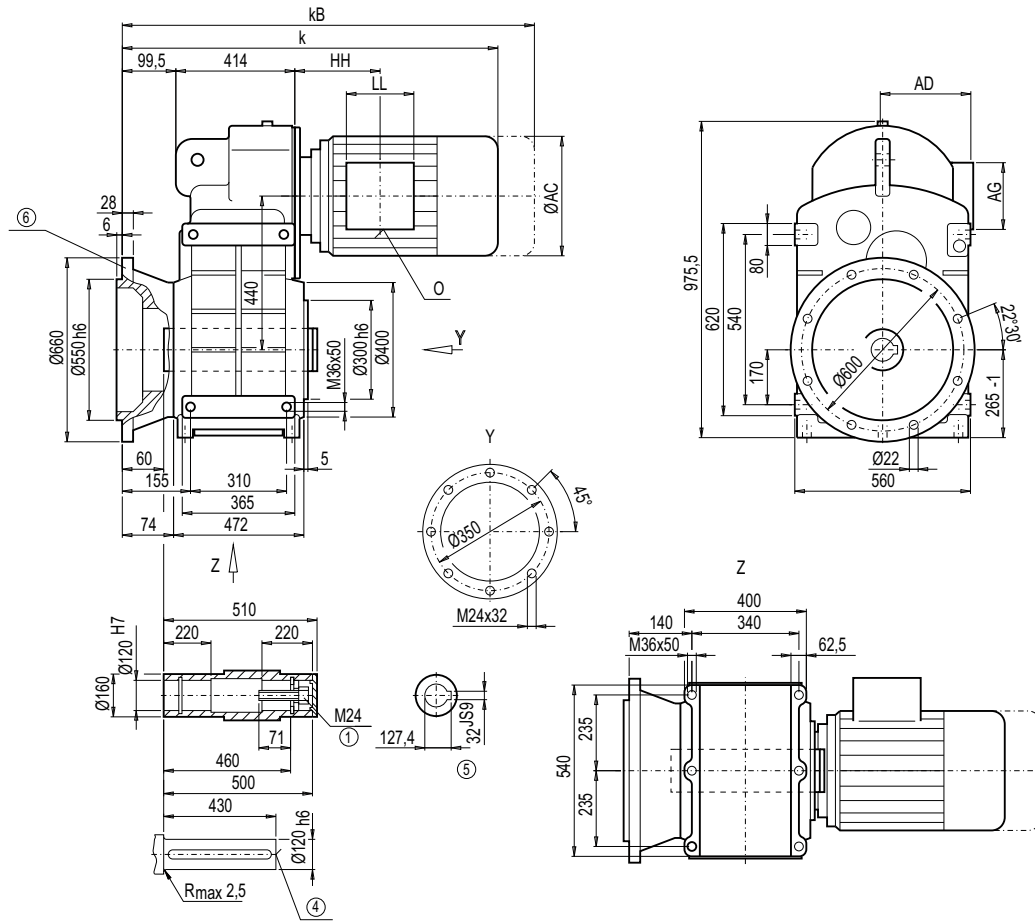
④ DIN 332

⑤ Feather key / keyway DIN 6885

⑦ For note, see page 3/179

Gearbox FDAF/FZAF188B (3- / 2-stage), flange-mounted design

FAF012



F.AF188B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAF188B	FZAF188B
LA132S/M	897.5	999.5	259.0	195.0	140	140	122.5	2xM32x1.5	677	–
LA132ZM	943.5	1 045.5	259.0	195.0	140	140	230.5	2xM32x1.5	698	–
LA160M/L	997.5	1 116.0	313.5	227.0	165	165	145.5	2xM40x1.5	710	696
LA160ZL	1 045.5	1 164.0	313.5	227.0	165	165	298.5	2xM40x1.5	749	735
LG180M/L	1 057.0	1 179.0	348.0	322.5	260	192	162.5	2xM40x1.5	806	791
LG180ZM/ZL	1 108.0	1 230.0	348.0	322.5	260	192	162.5	2xM40x1.5	836	821
LG200L	1 113.0	1 239.0	385.0	301.0	260	192	192.5	2xM50x1.5	886	871
LG225S	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	959	944
LG225M	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	947	932
LG225ZM	1 244.0	1 483.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 005	990
LG250M	1 277.5	1 502.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 051	1 034
LG250ZM	1 347.5	1 573.0	495.0	392.0	300	236	264.0	2xM63x1.5	1 152	1 137
K4-LGI280S	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 177	1 163
K4-LGI280M	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 283	1 268
K4-LGI280ZM	1 667.0	1 894.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 371	1 356
K2-LGI315S/M	1 745.0	2 010.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 398
K2-LGI315ZM	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 553
K2-LGI315L	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 693
K2-LGI315ZL	2 045.0	2 310.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 893

① DIN EN ISO 4014

④ DIN 332

⑤ Feather key / keyway DIN 6885

⑥ For note, see page 3/178

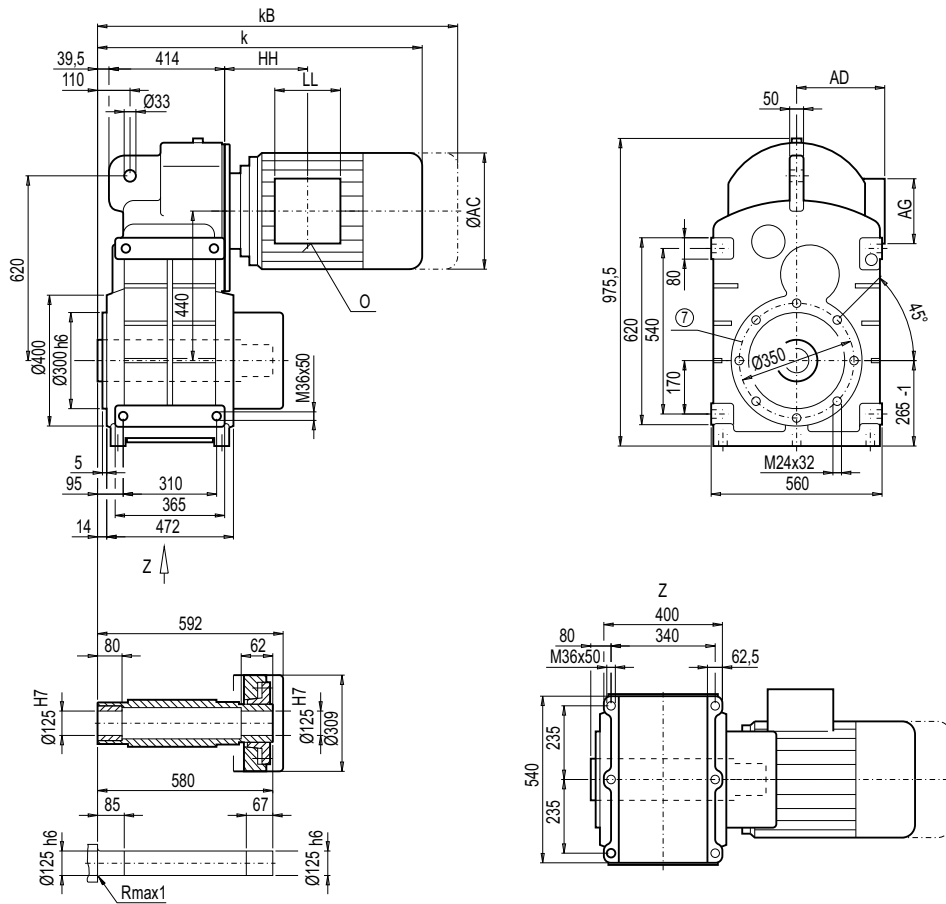
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAS/FZAS188B, FDAZS/FZAZS188B (3- / 2-stage), shaft-mounted design with shrink disk

FAS012
FAZS012



F.A.S188B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDA.S188B	FZA.S188B
LA132S/M	837.5	939.5	259.0	195.0	140	140	122.5	2xM32x1.5	738	–
LA132ZM	883.5	985.5	259.0	195.0	140	140	230.5	2xM32x1.5	759	–
LA160M/L	937.5	1 056.0	313.5	227.0	165	165	145.5	2xM40x1.5	771	757
LA160ZL	985.5	1 104.0	313.5	227.0	165	165	298.5	2xM40x1.5	810	796
LG180M/L	997.0	1 119.0	348.0	322.5	260	192	162.5	2xM40x1.5	867	852
LG180ZM/ZL	1 048.0	1 170.0	348.0	322.5	260	192	162.5	2xM40x1.5	897	882
LG200L	1 053.0	1 179.0	385.0	301.0	260	192	192.5	2xM50x1.5	947	932
LG225S	1 124.0	1 363.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 020	1 005
LG225M	1 124.0	1 363.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 008	993
LG225ZM	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 066	1 051
LG250M	1 217.5	1 442.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 110	1 095
LG250ZM	1 287.5	1 513.0	495.0	392.0	300	236	264.0	2xM63x1.5	1 213	1 198
K4-LGI280S	1 497.0	1 724.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 238	1 224
K4-LGI280M	1 497.0	1 724.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 344	1 329
K4-LGI280ZM	1 607.0	1 834.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 432	1 417
K2-LGI315S/M	1 685.0	1 950.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 459
K2-LGI315ZM	1 845.0	2 110.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 614
K2-LGI315L	1 845.0	2 110.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 754
K2-LGI315ZL	1 985.0	2 250.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 954

⑦ For note, see page 3/179

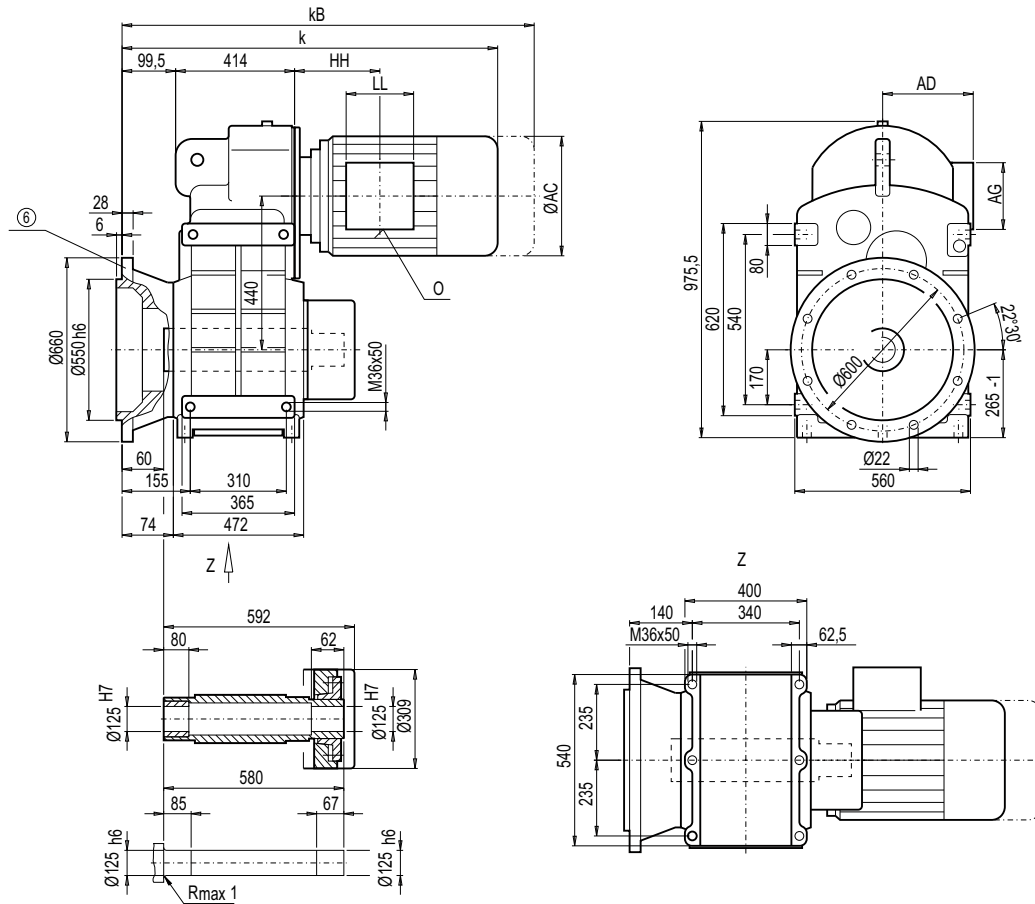
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAFS/FZAFS188B (3- / 2-stage), flange-mounted design and shrink disk

FAFS012



F.AFS188B									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FDAFS188B	FZAFS188B
LA132S/M	897.5	999.5	259.0	195.0	140	140	122.5	2xM32x1.5	687	–
LA132ZM	943.5	1 045.5	259.0	195.0	140	140	230.5	2xM32x1.5	708	–
LA160M/L	997.5	1 116.0	313.5	227.0	165	165	145.5	2xM40x1.5	721	706
LA160ZL	1 045.5	1 164.0	313.5	227.0	165	165	298.5	2xM40x1.5	760	745
LG180M/L	1 057.0	1 179.0	348.0	322.5	260	192	162.5	2xM40x1.5	816	802
LG180ZM/ZL	1 108.0	1 230.0	348.0	322.5	260	192	162.5	2xM40x1.5	846	832
LG200L	1 113.0	1 239.0	385.0	301.0	260	192	192.5	2xM50x1.5	896	882
LG225S	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	969	954
LG225M	1 184.0	1 423.0	442.0	325.0	260	192	228.5	2xM50x1.5	957	942
LG225ZM	1 244.0	1 483.0	442.0	325.0	260	192	228.5	2xM50x1.5	1 015	1 000
LG250M	1 277.5	1 502.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 059	1 044
LG250ZM	1 347.5	1 573.0	495.0	392.0	300	236	264.0	2xM63x1.5	1 162	1 147
K4-LGI280S	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 187	1 173
K4-LGI280M	1 557.0	1 784.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 293	1 278
K4-LGI280ZM	1 667.0	1 894.0	555.0	432.0	300	236	475.5	2xM63x1.5	1 381	1 366
K2-LGI315S/M	1 745.0	2 010.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 408
K2-LGI315ZM	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 563
K2-LGI315L	1 905.0	2 170.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 703
K2-LGI315ZL	2 045.0	2 310.0	610.0	500.0	380	307	584.5	2xM63x1.5	–	1 903

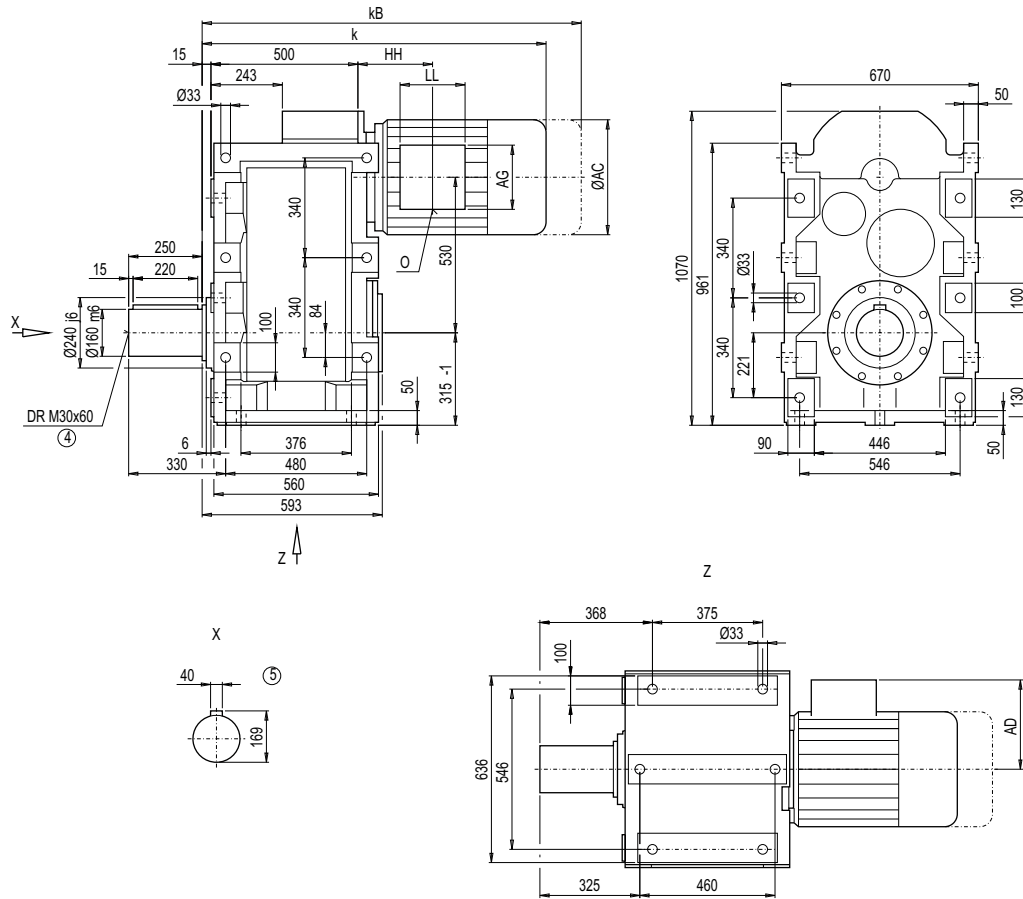
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FD/FZ208, FDZ/FZZ208 (3- / 2-stage), housing-flange-mounted design (C-type)

F012
FZ012



F.208									Weight	
Motor	k	kB	AC	AD	AG	LL	HH	O	FD.208	FZ.208
LA132S/M	899.0	1 001.0	259.0	195.0	140	140	122.5	2xM32x1.5	1 124	–
LA132ZM	945.0	1 047.0	259.0	195.0	140	140	230.5	2xM32x1.5	1 145	–
LA160M/L	999.0	1 117.5	313.5	227.0	165	165	145.5	2xM40x1.5	1 158	1 128
LA160ZL	1 047.0	1 165.5	313.5	227.0	165	165	298.5	2xM40x1.5	1 197	1 166
LG180M/L	1 058.5	1 180.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 253	1 223
LG180ZM/ZL	1 109.5	1 231.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 283	1 253
LG200L	1 114.5	1 240.5	385.0	301.0	260	192	192.5	2xM50x1.5	1 333	1 303
LG225S	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 406	1 376
LG225M	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 394	1 364
LG225ZM	1 245.5	1 484.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 452	1 422
LG250M	1 279.0	1 504.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 496	1 466
LG250ZM	1 349.0	1 574.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 599	1 569
K4-LGI280S	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 625	1 595
K4-LGI280M	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 730	1 700
K4-LGI280ZM	1 668.5	1 895.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 818	1 788
K2-LGI315S/M	1 746.5	2 011.5	610.0	500.0	380	307	584.5	2xM63x1.5	1 860	1 830
K2-LGI315ZM	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 015	1 985
K2-LGI315L	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 155	2 125
K2-LGI315ZL	2 046.5	2 311.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 355	2 325

④ DIN 332

⑤ Feather key / keyway DIN 6885

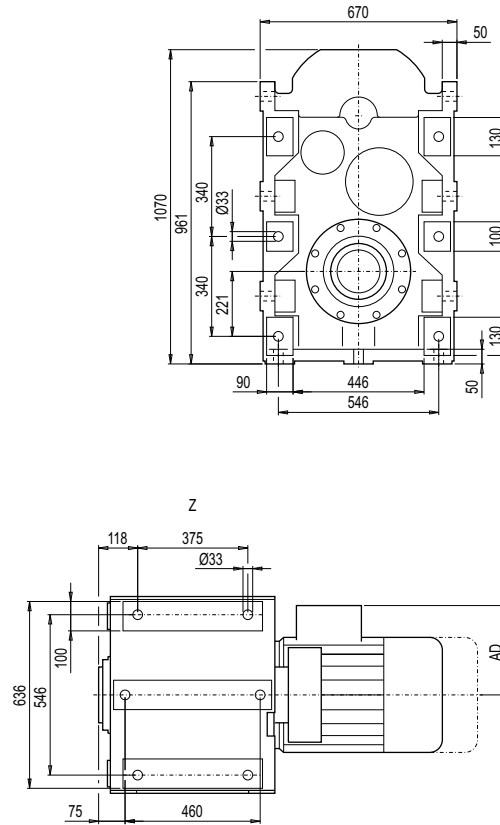
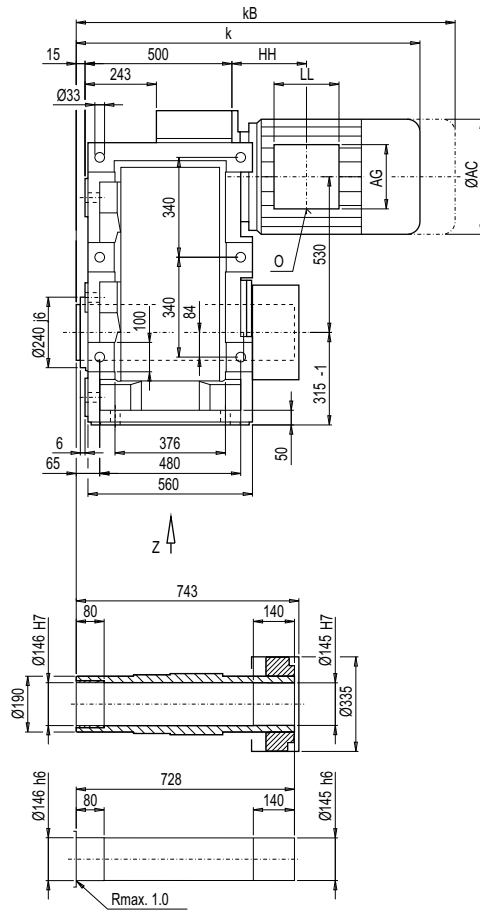
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDAS/FZAS208, FDAZS/FZAS208 (3- / 2-stage) shaft-mounted design with shrink disk

FAS012
FAZS012



F.A.S208B									Weight	
Motor	k	k _B	AC	AD	AG	LL	HH	O	FDA.S208	FZA.S208
LA132S/M	899.0	1 001.0	259.0	195.0	140	140	122.5	2xM32x1.5	1 054	–
LA132ZM	945.0	1 047.0	259.0	195.0	140	140	230.5	2xM32x1.5	1 075	–
LA160M/L	999.0	1 117.5	313.5	227.0	165	165	145.5	2xM40x1.5	1 088	1 060
LA160ZL	1 047.0	1 165.5	313.5	227.0	165	165	298.5	2xM40x1.5	1 127	1 099
LG180M/L	1 058.5	1 180.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 183	1 155
LG180ZM/ZL	1 109.5	1 231.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 213	1 185
LG200L	1 114.5	1 240.5	385.0	301.0	260	192	192.5	2xM50x1.5	1 263	1 235
LG225S	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 336	1 308
LG225M	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 324	1 296
LG225ZM	1 245.5	1 484.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 382	1 354
LG250M	1 279.0	1 504.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 426	1 398
LG250ZM	1 349.0	1 574.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 529	1 501
K4-LGI280S	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 555	1 527
K4-LGI280M	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 660	1 632
K4-LGI280ZM	1 668.5	1 895.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 748	1 720
K2-LGI315S/M	1 746.5	2 011.5	610.0	500.0	380	307	584.5	2xM63x1.5	1 790	1 762
K2-LGI315ZM	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	1 945	1 917
K2-LGI315L	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 085	2 057
K2-LGI315ZL	2 046.5	2 311.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 285	2 257

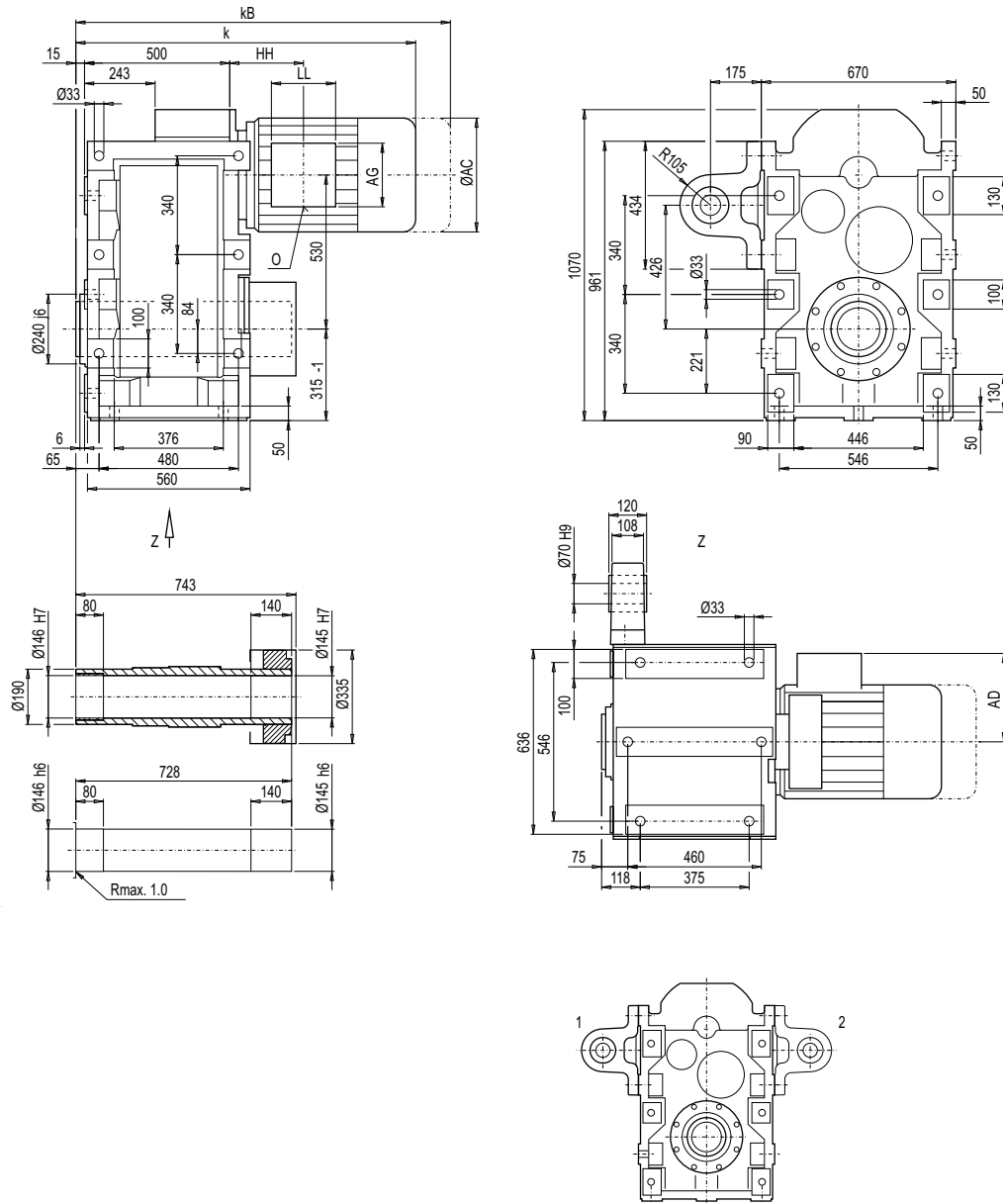
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDADS/FZADS208 (3- / 2-stage), shaft-mounted design with torque arm

FADS012



MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Gearbox FDADS/FZADS208 (3- / 2-stage), shaft-mounted design with torque arm (continued)

FADS012

Motor	F.ADS208								Weight	
	k	kB	AC	AD	AG	LL	HH	O	FDADS208	FZADS208
LA132S/M	899.0	1 001.0	259.0	195.0	140	140	122.5	2xM32x1.5	1 077	–
LA132ZM	945.0	1 047.0	259.0	195.0	140	140	230.5	2xM32x1.5	1 098	–
LA160M/L	999.0	1 117.5	313.5	227.0	165	165	145.5	2xM40x1.5	1 111	1 083
LA160ZL	1 047.0	1 165.5	313.5	227.0	165	165	298.5	2xM40x1.5	1 150	1 122
LG180M/L	1 058.5	1 180.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 206	1 178
LG180ZM/ZL	1 109.5	1 231.5	348.0	322.5	260	192	162.5	2xM40x1.5	1 236	1 208
LG200L	1 114.5	1 240.5	385.0	301.0	260	192	192.5	2xM50x1.5	1 286	1 258
LG225S	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 359	1 331
LG225M	1 185.5	1 424.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 348	1 319
LG225ZM	1 245.5	1 484.5	442.0	325.0	260	192	228.5	2xM50x1.5	1 405	1 377
LG250M	1 279.0	1 504.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 449	1 421
LG250ZM	1 349.0	1 574.5	495.0	392.0	300	236	264.0	2xM63x1.5	1 552	1 524
K4-LGI280S	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 578	1 550
K4-LGI280M	1 558.5	1 785.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 683	1 655
K4-LGI280ZM	1 668.5	1 895.5	555.0	432.0	300	236	475.5	2xM63x1.5	1 771	1 743
K2-LGI315S/M	1 746.5	2 011.5	610.0	500.0	380	307	584.5	2xM63x1.5	1 813	1 785
K2-LGI315ZM	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	1 968	1 940
K2-LGI315L	1 906.5	2 171.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 108	2 080
K2-LGI315ZL	2 046.5	2 311.5	610.0	500.0	380	307	584.5	2xM63x1.5	2 308	2 280

MOTOX Geared Motors

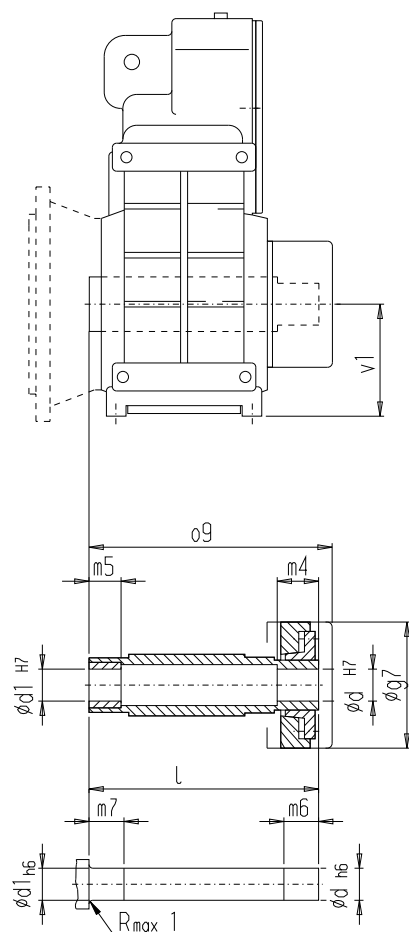
Parallel shaft geared motors

Dimensions

Offset hollow shafts with shrink disk

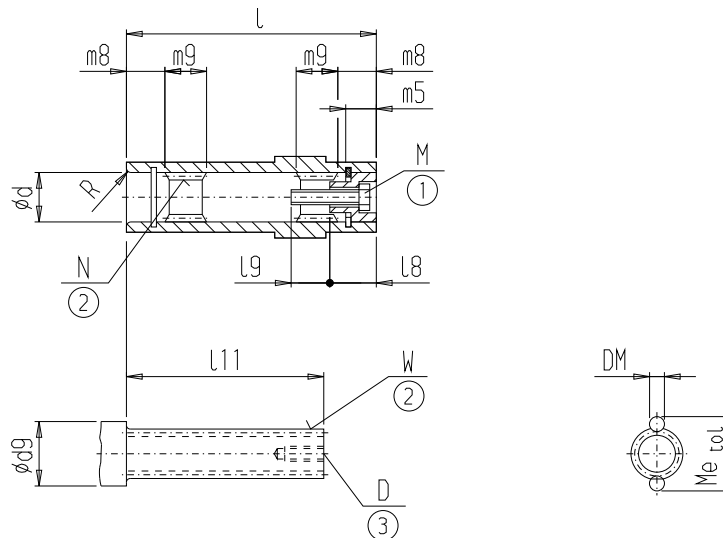
Optional hollow shafts for parallel shaft gearbox with shrink disk

FA.S



Gearbox	d	d1	l	o9	m4	m5	m6	m7	g7	v1
F.AS/F.AFS38B	30	31	146	154	22	20	27	25	77	75
F.AS/F.AFS48B	40	41	177	184	25	20	30	25	93	93
F.AS/F.AFS68B	40	42	209	216	35	20	40	25	112	111
	50	51	209	216	27	20	32	25	112	111
F.AS/F.AFS88B	50	52	214	249	29	30	34	35	132	132
	60	61	241	249	29	30	34	35	132	132
F.AS/F.AFS108B	65	66	280	288	30	40	35	45	144	160
	70	71	280	288	30	40	35	45	144	160
F.AS/F.AFS128B	75	76	345	357	44	50	49	55	180	180
	80	81	345	357	40	50	45	55	180	180
F.AS/F.AFS148B	95	96	404	418	49	60	54	65	210	212
F.AS/F.AFS168B	105	106	483	496	54	70	59	75	237	250
F.AS/F.AFS188B	125	126	580	592	61	80	67	85	263	265

Shaft-mounted design with splined shaft in acc. with DIN 5480



3

Gearbox	d	l	d9 min.	l11	W	D	R	m8	m9
F.A.T28	30	104	36	72	W25x1.25x30x18 8f	M10	R1.6	17.0	25
F.A.T38B	35	120	45	95	W35x1.25x30x26 8f	M10	R2	17.0	27
F.A.T48B	40	150	52	120	W40x2x30x18 8f	M12	R3	22.0	34
F.A.T68B	55	180	65	142	W50x2x30x24 8f	M16	R2	21.0	40
F.A.T88B	65	210	80	172	W60x2x30x28 8f	M16	R2	22.5	49
F.A.T108B	72	240	85	201	W70x2x30x34 8f	M20	R2	22.5	56
F.A.T128B	90	300	105	257	W80x3x30x25 8f	M20	R2	24.0	71
F.A.T148B	90	350	110	306	W90x3x30x28 8f	M20	R3	25.0	88
F.A.T168B	110	410	130	350	W110x3x30x35 8f	M24	R3	32.0	99
F.A.T188B	135	500	145	445	W130x5x30x24 8f	M24	R4	42.0	120

Gearbox	N	m5	l8	l9	M	DM	Me	tol
F.A.T28	N25x1.25x30x18 9H	9.0	17	31.8	M10x40	2.75	28.023	-0.049
F.A.T38B	N35x1.25x30x26 9H	12.0	18	27.0	M10x35	2.50	37.423	-0.041
F.A.T48B	N40x2x30x18 9H	14.0	20	37.0	M12x45	4.50	45.083	-0.043
F.A.T68B	N50x2x30x24 9H	16.0	23	49.5	M16x55	4.00	54.156	-0.049
F.A.T88B	N60x2x30x28 9H	16.5	26	46.5	M16x55	4.00	63.918	-0.053
F.A.T108B	N70x2x30x34 9H	16.5	28	51.0	M20x60	4.00	74.181	-0.057
F.A.T128B	N80x3x30x25 9H	17.0	31	46.0	M20x60	6.00	85.856	-0.053
F.A.T148B	N90x3x30x28 9H	17.0	31	51.0	M20x60	6.00	95.911	-0.053
F.A.T168B	N110x3x30x35 9H	20.0	41	65.5	M24x80	6.00	115.998	-0.061
F.A.T188B	N130x5x30x24 9H	20.0	50	35.5	M24x60	10.00	139.848	-0.061

① DIN 912

② DIN 5480

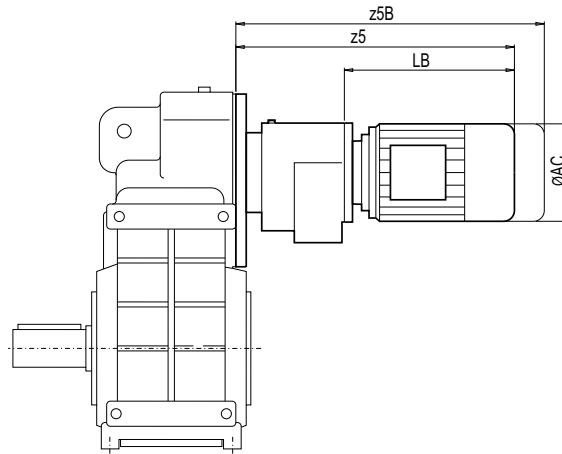
③ DIN 332-D

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Parallel shaft tandem gearbox



Gearbox	Motor	AC	z5	z5B	LB
FZ.38B-Z28	LA71	139.0	338.0	393.0	202.5
	LA71Z	139.0	357.0	412.0	221.5
	LA80	156.5	440.0	503.5	304.5
	LA80Z	156.5	462.5	526.0	327.0
	LA90S/L	174.0	435.0	506.0	299.5
	LA90ZL	174.0	480.0	551.0	344.5
	LA100L	195.0	517.0	598.0	381.5
	LA100ZL	195.0	587.0	668.0	451.5
FZ.38B-D28	LA71	139.0	338.0	393.0	202.5
	LA71Z	139.0	357.0	412.0	221.5
	LA80	156.5	440.0	503.5	304.5
	LA80Z	156.5	462.5	526.0	327.0
	LA90S/L	174.0	435.0	506.0	299.5
	LA90ZL	174.0	480.0	551.0	344.5
FD.48B-Z28	LA71	139.0	363.0	418.0	202.5
	LA71Z	139.0	382.0	437.0	221.5
	LA80	156.5	465.0	528.5	304.5
	LA80Z	156.5	487.5	551.0	327.0
	LA90S/L	174.0	460.0	531.0	299.5
	LA90ZL	174.0	505.0	576.0	344.5
	LA100L	195.0	542.0	623.0	381.5
	LA100ZL	195.0	612.0	693.0	451.5
FD.48B-D28	LA71	139.0	363.0	418.0	202.5
	LA71Z	139.0	382.0	437.0	221.5
	LA80	156.5	465.0	528.5	304.5
	LA80Z	156.5	487.5	551.0	327.0
	LA90S/L	174.0	460.0	531.0	299.5
	LA90ZL	174.0	505.0	576.0	344.5
FD.68B-Z28	LA71	139.0	357.5	412.5	202.5
	LA71Z	139.0	376.5	431.5	221.5
	LA80	156.5	459.5	523.0	304.5
	LA80Z	156.5	482.0	545.5	327.0
	LA90S/L	174.0	454.5	525.5	299.5
	LA90ZL	174.0	499.5	570.5	344.5
	LA100L	195.0	536.5	617.5	381.5
	LA100ZL	195.0	606.5	687.5	451.5

Gearbox	Motor	AC	z5	z5B	LB
FD.68B-D28	LA71	139.0	357.5	412.5	202.5
	LA71Z	139.0	376.5	431.5	221.5
	LA80	156.5	459.5	523.0	304.5
	LA80Z	156.5	482.0	545.5	327.0
	LA90S/L	174.0	454.5	525.5	299.5
	LA90ZL	174.0	499.5	570.5	344.5
FD.88B-Z28	LA71	139.0	351.5	406.5	202.5
	LA71Z	139.0	370.5	425.5	221.5
	LA80	156.5	453.5	517.0	304.5
	LA80Z	156.5	476.0	539.5	327.0
	LA90S/L	174.0	448.5	519.5	299.5
	LA90ZL	174.0	493.5	564.5	344.5
FD.88B-D28	LA71	139.0	351.5	406.5	202.5
	LA71Z	139.0	370.5	425.5	221.5
	LA80	156.5	453.5	517.0	304.5
	LA80Z	156.5	476.0	539.5	327.0
	LA90S/L	174.0	448.5	519.5	299.5
	LA90ZL	174.0	493.5	564.5	344.5
FD.108B-Z38	LA71 ¹⁾	139.0	465.5	520.5	258.5
	LA71Z ¹⁾	139.0	484.5	539.5	277.5
	LA80 ¹⁾	156.5	502.5	566.0	295.5
	LA80Z ¹⁾	156.5	525.0	588.5	318.0
	LA90S/L ¹⁾	174.0	533.5	604.5	326.5
	LA90ZL ¹⁾	174.0	578.5	649.5	371.5
	LA100L ¹⁾	195.0	579.5	660.5	372.5
	LA100ZL ¹⁾	195.0	649.5	730.5	442.5
	LA112M ¹⁾	219.0	609.0	690.0	402.0
	LA112ZM ¹⁾	219.0	637.0	718.0	430.0
	LA71 ²⁾	139.0	476.0	531.0	258.5
	LA71Z ²⁾	139.0	495.0	550.0	277.5
FD.108B-D38	LA80 ²⁾	156.5	513.0	576.5	295.5
	LA80Z ²⁾	156.5	535.5	599.0	318.0
	LA90S/L ²⁾	174.0	544.0	615.0	326.5
	LA90ZL ²⁾	174.0	589.0	660.0	371.5

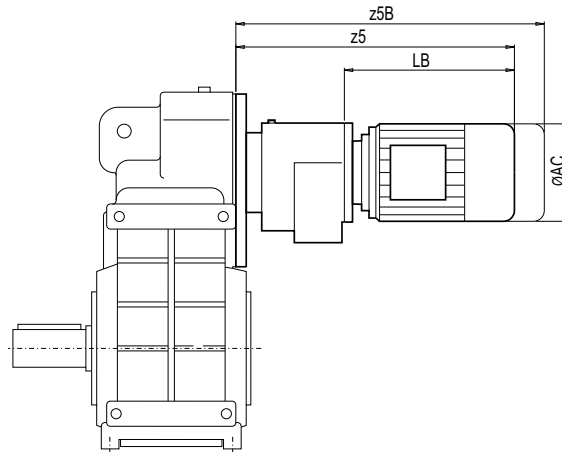
1) $i_{tot} \geq 1647$ 2) $i_{tot} < 1647$

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Parallel shaft tandem gearbox (continued)



Gearbox	Motor	AC	z5	zB5	LB
FD.108B-Z38	LA100L ²⁾	195.0	590.0	671.0	372.5
	LA100ZL ²⁾	195.0	660.0	741.0	442.5
	LA112M ²⁾	219.0	619.5	700.5	402.0
	LA112ZM ²⁾	219.0	647.5	728.5	430.0
FD.108B-D38	LA71	139.0	480.5	535.5	273.5
	LA71Z	139.0	499.5	554.5	292.5
	LA80	156.5	517.5	581.0	310.5
	LA80Z	156.5	540.0	603.5	333.0
	LA90S/L	174.0	548.5	619.5	341.5
	LA90ZL	174.0	593.5	664.5	386.5
FD.128B-Z38	LA71	139.0	458.5	513.5	258.5
	LA71Z	139.0	477.5	532.5	277.5
	LA80	156.5	495.5	559.0	295.5
	LA80Z	156.5	518.0	581.5	318.0
	LA90S/L	174.0	526.5	597.5	326.5
	LA90ZL	174.0	571.5	642.5	371.5
	LA100L	195.0	572.5	653.5	372.5
	LA100ZL	195.0	642.5	723.5	442.5
	LA112M	219.0	602.0	683.0	402.0
	LA112ZM	219.0	630.0	711.0	430.0
FD.128B-D38	LA71	139.0	473.5	528.5	273.5
	LA71Z	139.0	492.5	547.5	292.5
	LA80	156.5	510.5	574.0	310.5
	LA80Z	156.5	533.0	596.5	333.0
	LA90S/L	174.0	541.5	612.5	341.5
	LA90ZL	174.0	586.5	657.5	386.5
FD.128B-Z48	LA71	139.0	532.0	587.0	253.0
	LA71Z	139.0	551.0	606.0	272.0
	LA80	156.5	569.0	632.5	290.0
	LA80Z	156.5	591.5	655.0	312.5
	LA90S	174.0	600.0	671.0	321.0
	LA90L	174.0	600.0	671.0	321.0
	LA90ZL	174.0	645.0	716.0	366.0
	LA100L	195.0	646.0	727.0	367.0
	LA100ZL	195.0	716.0	797.0	437.0

Gearbox	Motor	AC	z5	zB5	LB
FD.128B-Z48	LA112M	219.0	675.0	756.0	396.0
	LA112ZM	219.0	703.0	784.0	424.0
	LA132S	259.0	737.0	839.0	458.0
	LA132M	259.0	737.0	839.0	458.0
	LA132ZM	259.0	783.0	885.0	504.0
FD.148B-Z38	LA71	139.0	454.0	509.0	258.5
	LA71Z	139.0	473.0	528.0	277.5
	LA80	156.5	491.0	554.5	295.5
	LA80Z	156.5	513.5	577.0	318.0
	LA90S	174.0	522.0	593.0	326.5
	LA90L	174.0	522.0	593.0	326.5
	LA90ZL	174.0	567.0	638.0	371.5
	LA100L	195.0	568.0	649.0	372.5
	LA100ZL	195.0	638.0	719.0	442.5
	LA112M	219.0	597.5	678.5	402.0
FD.148B-D38	LA112ZM	219.0	625.5	706.5	430.0
	LA71	139.0	469.0	524.0	273.5
	LA71Z	139.0	488.0	543.0	292.5
	LA80	156.5	506.0	569.5	310.5
	LA80Z	156.5	528.5	592.0	333.0
	LA90S	174.0	537.0	608.0	341.5
	LA90L	174.0	537.0	608.0	341.5
	LA90ZL	174.0	582.0	653.0	386.5
FD.148B-Z48	LA71	139.0	521.5	576.5	253.0
	LA71Z	139.0	540.5	595.5	272.0
	LA80	156.5	558.5	622.0	290.0
	LA80Z	156.5	581.0	644.5	312.5
	LA90S/L	174.0	589.5	660.5	321.0
	LA90ZL	174.0	634.5	705.5	366.0
	LA100L	195.0	635.5	716.5	367.0
	LA100ZL	195.0	705.5	786.5	437.0
	LA112M	219.0	664.5	745.5	396.0
	LA112ZM	219.0	692.5	773.5	424.0
	LA132S/M	259.0	726.5	828.5	458.0
	LA132ZM	259.0	772.5	874.5	504.0

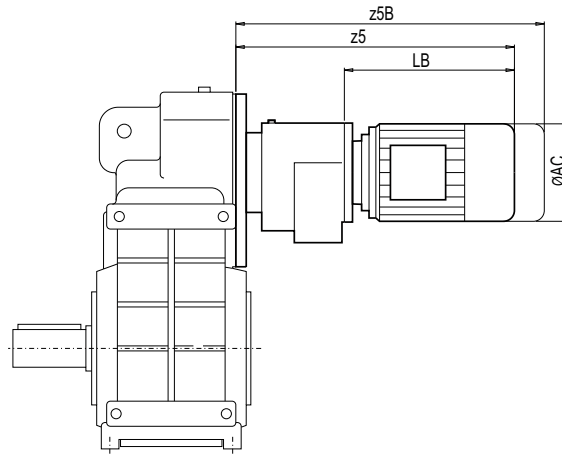
2) $i_{tot} < 1647$

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Parallel shaft tandem gearbox (continued)



Gearbox	Motor	AC	z5	zB5	LB
FD.168B-Z48	LA71	139.0	513.5	568.5	253.0
	LA71Z	139.0	532.5	587.5	272.0
	LA80	156.5	550.5	614.0	290.0
	LA80Z	156.5	573.0	636.5	312.5
	LA90S	174.0	581.5	652.5	321.0
	LA90L	174.0	581.5	652.5	321.0
	LA90ZL	174.0	626.5	697.5	366.0
	LA100L	195.0	627.5	708.5	367.0
	LA100ZL	195.0	697.5	778.5	437.0
	LA112M	219.0	656.5	737.5	396.0
	LA112ZM	219.0	684.5	765.5	424.0
	LA132S	259.0	718.5	820.5	458.0
	LA132M	259.0	718.5	820.5	458.0
FD.168B-D48	LA71	139.0	530.5	585.5	270.0
	LA71Z	139.0	549.5	604.5	289.0
	LA80	156.5	567.5	631.0	307.0
	LA80Z	156.5	590.0	653.5	329.5
	LA90S/L	174.0	598.5	669.5	338.0
	LA90ZL	174.0	643.5	714.5	383.0
	LA100L	195.0	644.5	725.5	384.0
	LA100ZL	195.0	714.5	795.5	454.0
FD.168B-Z68	LA71	139.0	583.0	638.0	247.0
	LA71Z	139.0	602.0	657.0	266.0
	LA80	156.5	620.0	683.5	284.0
	LA80Z	156.5	642.5	706.0	306.5
	LA90S/L	174.0	651.0	722.0	315.0
	LA90ZL	174.0	696.0	767.0	360.0
	LA100L	195.0	697.0	778.0	361.0
	LA100ZL	195.0	767.0	848.0	431.0
	LA112M	219.0	724.0	805.0	388.0
	LA112ZM	219.0	752.0	833.0	416.0
	LA132S/M	259.0	784.0	886.0	448.0
	LA132ZM	259.0	830.0	932.0	494.0
	LA160M/L	313.5	886.5	1 005.0	550.5
	LA160ZL	313.5	934.5	1 053.0	598.5

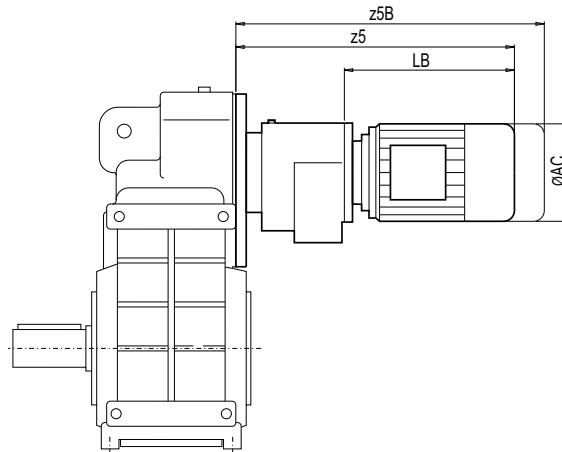
Gearbox	Motor	AC	z5	zB5	LB
FD.188B-Z48	LA71	139.0	499.0	554.0	253.0
	LA71Z	139.0	518.0	573.0	272.0
	LA80	156.5	536.0	599.5	290.0
	LA80Z	156.5	558.5	622.0	312.5
	LA90S/L	174.0	567.0	638.0	321.0
	LA90ZL	174.0	612.0	683.0	366.0
	LA100L	195.0	613.0	694.0	367.0
	LA100ZL	195.0	683.0	764.0	437.0
	LA112M	219.0	642.0	723.0	396.0
	LA112ZM	219.0	670.0	751.0	424.0
	LA132S/M	259.0	704.0	806.0	458.0
	LA132ZM	259.0	750.0	852.0	504.0
FD_188B-D48	LA71	139.0	516.0	571.0	270.0
	LA71Z	139.0	535.0	590.0	289.0
	LA80	156.5	553.0	616.5	307.0
	LA80Z	156.5	575.5	639.0	329.5
	LA90S	174.0	584.0	655.0	338.0
	LA90L	174.0	584.0	655.0	338.0
	LA90ZL	174.0	629.0	700.0	383.0
	LA100L	195.0	630.0	711.0	384.0
	LA100ZL	195.0	700.0	781.0	454.0
FD_188B-Z68	LA71	139.0	585.0	640.0	247.0
	LA71Z	139.0	604.0	659.0	266.0
	LA80	156.5	622.0	685.5	284.0
	LA80Z	156.5	644.5	708.0	306.5
	LA90S/L	174.0	653.0	724.0	315.0
	LA90ZL	174.0	698.0	769.0	360.0
	LA100L	195.0	699.0	780.0	361.0
	LA100ZL	195.0	769.0	850.0	431.0
	LA132S/M	259.0	786.0	888.0	448.0
	LA132ZM	259.0	832.0	934.0	494.0
	LA160M/L	313.5	888.5	1 007.0	550.5
	LA160ZL	313.5	936.5	1 055.0	598.5

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

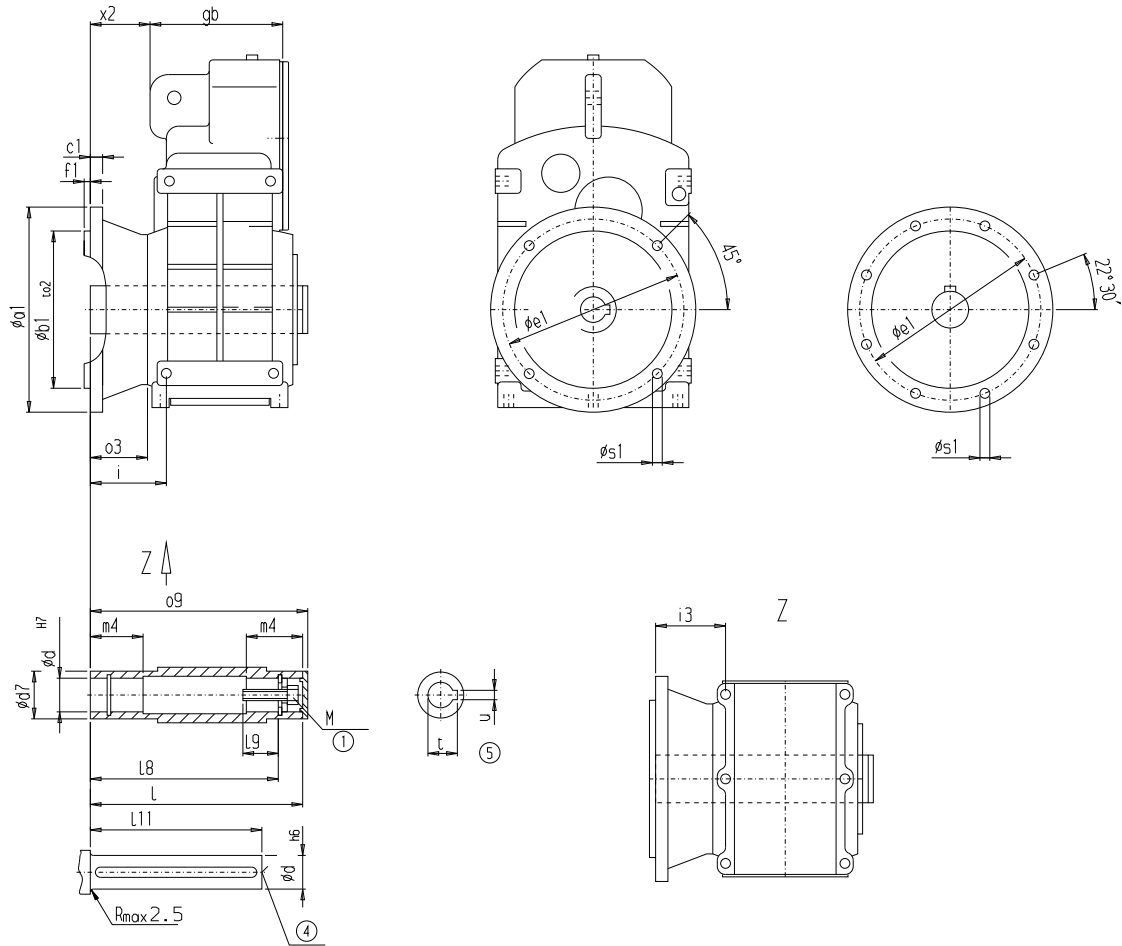
Parallel shaft tandem gearbox (continued)



Gearbox	Motor	AC	z5	zB5	LB
FD.208-Z68	LA71	139.0	585.0	640.0	247.0
	LA71Z	139.0	604.0	659.0	266.0
	LA80	156.5	622.0	685.5	284.0
	LA80Z	156.5	644.5	708.0	306.5
	LA90S/L	174.0	653.0	724.0	315.0
	LA90ZL	174.0	698.0	769.0	360.0
	LA100L	195.0	699.0	780.0	361.0
	LA100ZL	195.0	769.0	850.0	431.0
	LA132S/M	259.0	786.0	888.0	448.0
	LA132ZM	259.0	832.0	934.0	494.0
	LA160M/L	313.5	888.5	1 007.0	550.5
	LA160ZL	313.5	936.5	1 055.0	598.5
FD_208-D68	LA71	139.0	603.5	658.5	265.5
	LA71Z	139.0	622.5	677.5	284.5
	LA80	156.5	640.5	704.0	302.5
	LA80Z	156.5	663.0	726.5	325.0
	LA90S/L	174.0	671.5	742.5	333.5
	LA90ZL	174.0	716.5	787.5	378.5
	LA100L	195.0	717.5	798.5	379.5
	LA100ZL	195.0	787.5	868.5	449.5

Gearbox	Motor	AC	z5	zB5	LB
FD.208-Z88	LA90S/L	174.0	776.5	847.5	300.0
	LA90ZL	174.0	821.5	892.5	345.0
	LA100L	195.0	820.0	901.0	343.5
	LA100ZL	195.0	890.0	971.0	413.5
	LA112M	219.0	846.0	927.0	369.5
	LA112ZM	219.0	874.0	955.0	397.5
	LA132S/M	259.0	906.0	1 008.0	429.5
	LA132ZM	259.0	952.0	1 054.0	475.5
	LA160M/L	313.5	1 010.5	1 129.0	534.0
	LA160ZL	313.5	1 058.5	1 177.0	582.0
	LG180M/L	348.0	1 070.0	1 192.0	593.5
	LG180ZM/ZL	348.0	1 121.0	1 243.0	644.5

Shaft-mounted design with mixer flanges



Gearbox	a1	b1	to2	c1	e1	f1	s1	o3	i	i3	x2	gb
FDAM88B FZAM88B	300	230	j6	20	265	4	13.5	120	146.5	141.5	126.0	175
FDAM108B FZAM108B	350	250	h6	20	300	5	17.5	135	163.5	149.0	140.5	205
FDAM128B FZAM128B	450	350	h6	25	400	5	17.5	165	203.5	193.5	172.0	271
FDAM148B FZAM148B	450	350	h6	25	400	5	17.5	185	239.0	218.0	211.0	298
FDAM168B FZAM168B	550	450	h6	28	500	5	17.5	210	269.0	241.0	237.0	336

Gearbox	o9	d	d7	l	m4	l8	l9	l11	t	u	M	Weights	
												FDAM	FZAM
FDAM88B FZAM88B	324.0	60	80	321	78	291	54.0	275	64.4	18	M20	72	73
FDAM108B FZAM108B	369.5	70	95	366	93	334	63.5	310	74.9	20	M20	122	122
FDAM128B FZAM128B	458.0	80	110	456	123	419	63.5	395	85.4	22	M20	216	214
FDAM148B FZAM148B	526.0	90	120	524	148	484	72.0	460	95.4	25	M24	309	305
FDAM168B FZAM168B	611.0	110	150	609	175	565	73.0	540	116.4	28	M24	495	484

④ DIN 332

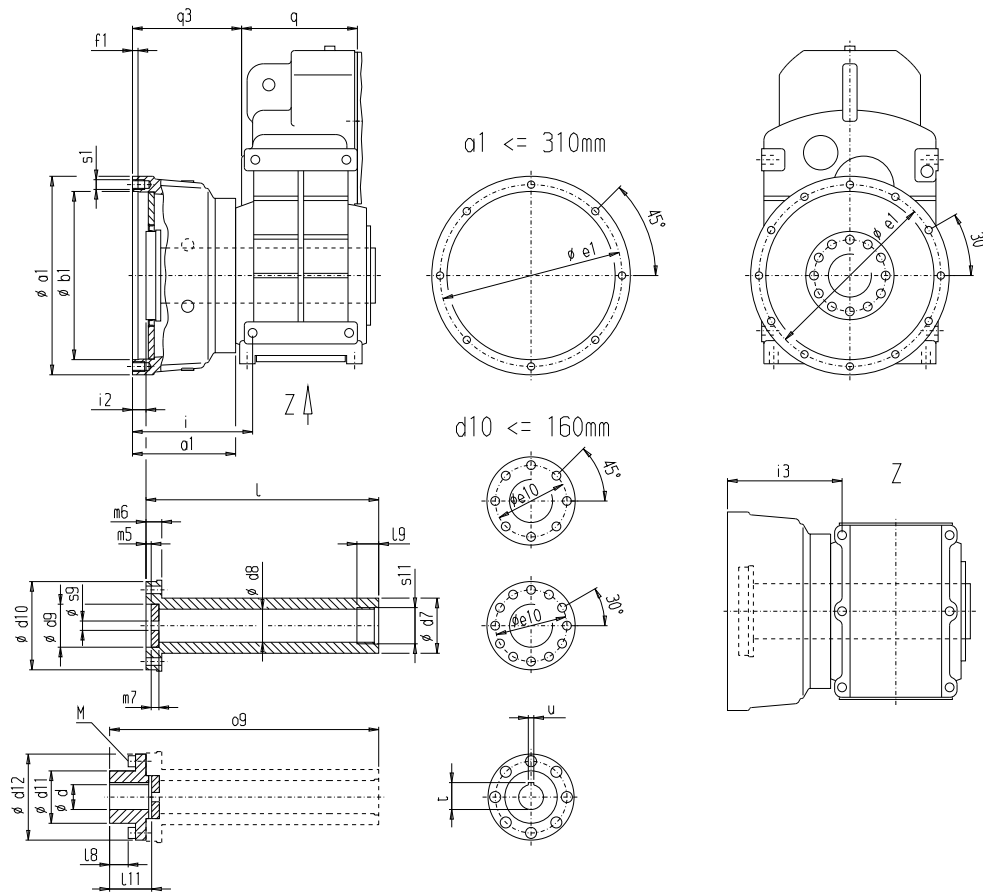
⑤ Feather key / keyway DIN 6885

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Flange design for extruder drives



Gearbox	a1	b1	e1	f1	s1	q1	i	i3	i2	q3	q	
FDAE/FZAE68B	260	220	+0.046 / 0	236	10	M12x17	147.5	174.0	–	15.0	156.0	138.5
FDAE/FZAE88B	310	255	+0.052 / 0	280	10	M16x22	171.0	197.5	192.5	15.5	177.0	175.0
FDAE/FZAE108B	360	305	+0.052 / 0	330	10	M16x22	188.0	216.5	202.0	23.0	193.5	205.0
FDAE/FZAE128B	420	345	+0.057 / 0	380	10	M20x27	206.0	244.5	234.5	25.0	213.0	271.0
FDAE/FZAE148B	450	360	+0.057 / 0	400	10	M24x32	225.0	279.0	258.0	27.0	251.0	298.0
FDAE/FZAE168B	510	420	+0.063 / 0	460	15	M24x32	262.0	321.0	293.0	38.0	285.0	340.0

MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Flange design for extruder drives (continued)

Gearbox	d	l11	d7	d8	l9	s11	o9	d10	m6	e10
							l	d12		
FDAE/FZAE68B	20	48	65	38	30	M42x2	349.0	105	14	88
	25									
	30						305.0	104		
FDAE/FZAE88B	30	58	80	49	39	M56x2	410.5	130	23	110
	35									
	40						357.0	129		
FDAE/FZAE108B	40	71	95	60	39	M64x2	462.0	160	25	130
	45									
	50						396.0	156		
FDAE/FZAE128B	45	87	110	71	49	M80x3	554.0	175	31	150
	50									
	60						472.0	174		
FDAE/FZAE148B	60	95	120	88	52	M95x3	626.0	190	33	160
	70									
	75						537.0	189		
FDAE/FZAE168B	70	105	150	104	57	M110x3	722.0	230	42	195
	80									
	90						623.0	229		

Gearbox	d	d9	s9	m7	d11	m5	l8	M	t	u	
FDAE/FZAE68B	20	48	+0.025 / 0	11	11	65	4.0	20.0	M10x25	22.8	6
	25									28.3	8
	30									33.3	8
FDAE/FZAE88B	30	63	+0.030 / 0	17	12	80	4.5	23.5	M12x35	33.3	8
	35									38.3	10
	40									43.3	12
FDAE/FZAE108B	40	78	+0.030 / 0	17	14	95	5.0	31.0	M16x40	43.3	12
	45									48.8	14
	50									53.8	14
FDAE/FZAE128B	45	88	+0.035 / 0	22	17	110	5.0	42.0	M16x45	48.8	14
	50									53.8	14
	60									64.4	18
FDAE/FZAE148B	60	105	+0.035 / 0	22	20	120	6.0	45.0	M16x55	64.4	18
	70									74.9	20
	75									79.9	20
FDAE/FZAE168B	70	125	+0.040 / 0	25	22	150	6.0	49.0	M20x55	74.9	20
	80									85.4	22
	90									95.4	25

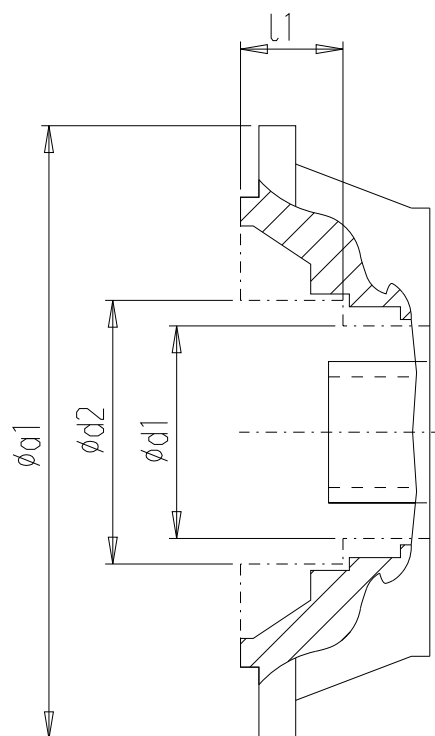
MOTOX Geared Motors

Parallel shaft geared motors

Dimensions

Inside contour of the flange-mounted design (A-type)

Design notes for the customer's interface, e.g. plug-in shaft for hollow shaft design.



Gearbox	a1	d1	d2	l1
F.F.28	120	70	72	24.0
F.F.28	160	70	103	8.5
F.F.38B	160	70	77	20.0
F.F.48B	200	84	90	22.5
F.F.68B	250	96	96	–
F.F.88B	300	126	138	31.0
F.F.108B	350	176	185	32.0
F.F.128B	450	226	234	38.5
F.F.148B	450	246	262	34.0
F.F.168B	550	296	313	39.0
F.F.188B	660	296	296	–

Pin holes

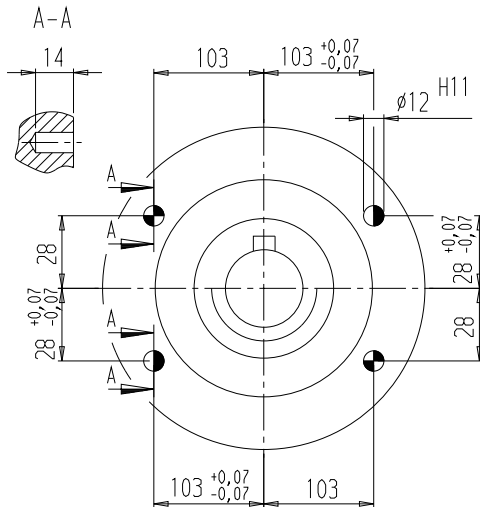
In the case of sizes F.Z.108B to 188B, the customer's interface can be pinned on the housing flange (C-type).

The output flanges have been designed to ensure the reliable transmission of the permissible torques and radial forces by the bolt connections.

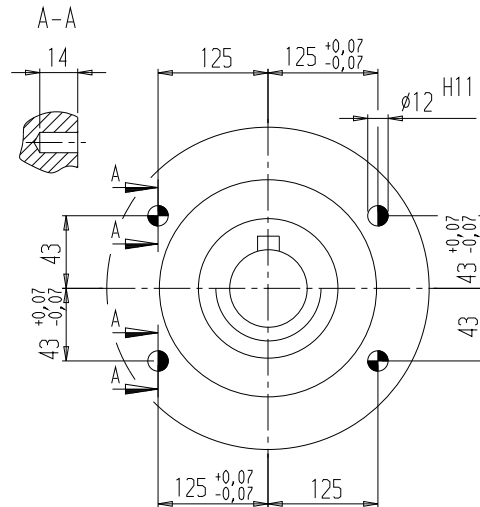
If an additional fuse, e. g. for high shock loads, is required, the existing pin holes can be used.

The gearbox and the machine can be drilled and pinned together. To do so, the provided dimensions must be observed.

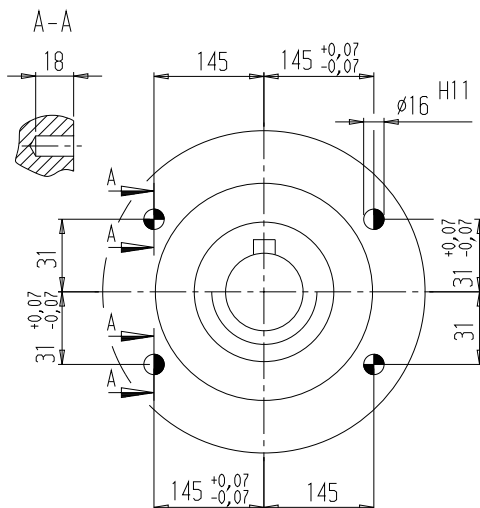
F.Z.108B



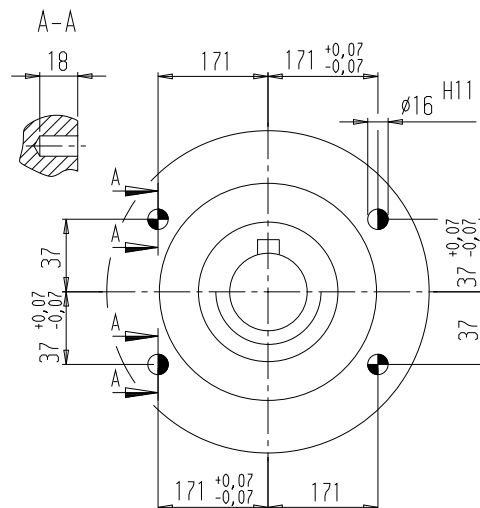
F.Z.128B



F.Z.148B



F.Z.168B, F.Z.188B



- Spring pins, heavy-duty design, to DIN 1481: Use pin holes provided in the housing flange.
- ⊙ Grooved cylindrical pins with chamfer to DIN EN 28740/ISO 8740: Drill connecting component together with housing.

MOTOX Geared Motors

Parallel shaft geared motors

Notes

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